vSphere Supervisor Platform

Activate and manage **vSphere Supervisor** as a VI administrator to provide a platform for running Kubernetes workloads in vSphere.

In This Chapter

- vSphere Supervisor Concepts
- Deploying Supervisor with VCF Networking with VPC
- Deploying vSphere Supervisor with Foundation Load Balancer
- Deploying Supervisor with a Simplified Deployment Flow
- Deploying vSphere Supervisor with NSX and Avi Load Balancer
- Deploying vSphere Supervisor with vDS and Avi Load Balancer
- Deploying vSphere Supervisor from an Exported Configuration
- Configuring and Managing vSphere Namespaces
- Configuring and Managing vSphere Zones
- Connecting to Supervisor and VKS Clusters
- Configuring and Managing a Supervisor
- Monitoring vSphere Supervisor
- Updating vSphere Supervisor
- Backing Up and Restoring vSphere Supervisor
- Troubleshooting vSphere Supervisor

Getting to Know VCF

To apply this documentation, you must be acquainted with the VCF Product Overview, VCF Design, and VCF Release Notes.

SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See Administration SDKs, APIs, and CLI.

vSphere Supervisor Concepts

Learn the key concepts and architecture of vSphere Supervisor.

vSphere Supervisor Concepts

vSphere Supervisor is a control plane that is built in vSphere that integrates desired state and a single consistent API directly in the hypervisor. vSphere Supervisor enables native deployment and management of Kubernetes clusters, declarative virtual machines, and vSphere Pods alongside traditional workloads in VCF. It transforms vSphere into an extendable platform by providing a declarative API for self-service provisioning of modern infrastructure services in a resource boundaries enabled by vSphere Namespaces.

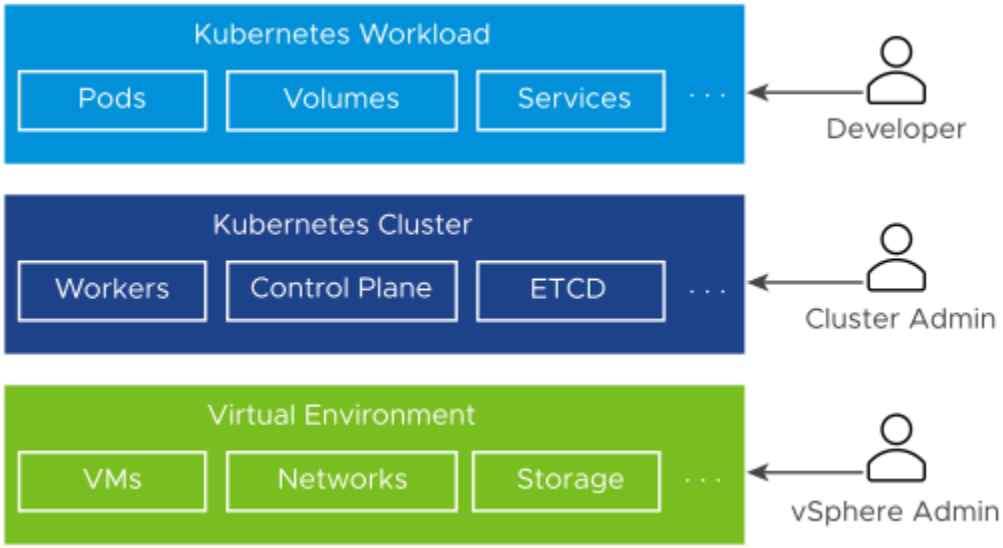
What Is vSphere Supervisor?

You can use vSphere Supervisor to enable a consistent declarative API in VCF and a platform for building a modern cloud experience with VCF Automation. vSphere Supervisor is a workload domain capability that you can activate during the workload domain creation or separately through the vSphere Client. When activated on vSphere clusters, vSphere Supervisor provides self-service cloud consumption capabilities for running and managing workloads, including upstream Kubernetes clusters, virtual machines, and vSphere Pods . The resource boundaries where workloads run in a vSphere Supervisor are called vSphere Namespaces.

The Challenges of Today's Application Stack

Today's distributed systems are constructed of multiple microservices usually running a large number of Kubernetes pods and virtual machines. A typical stack that is not based on vSphere Supervisor consists of an underlying virtual environment, with Kubernetes infrastructure that is deployed inside VMs and Kubernetes pods also running in these VMs. There are three separate operational roles to administer each part of the stack, application developers, Kubernetes cluster administrators, and vSphere or Cloud administrators.

Figure 490: Today's Application Stack



The different roles do not have visibility or control over each other's environments:

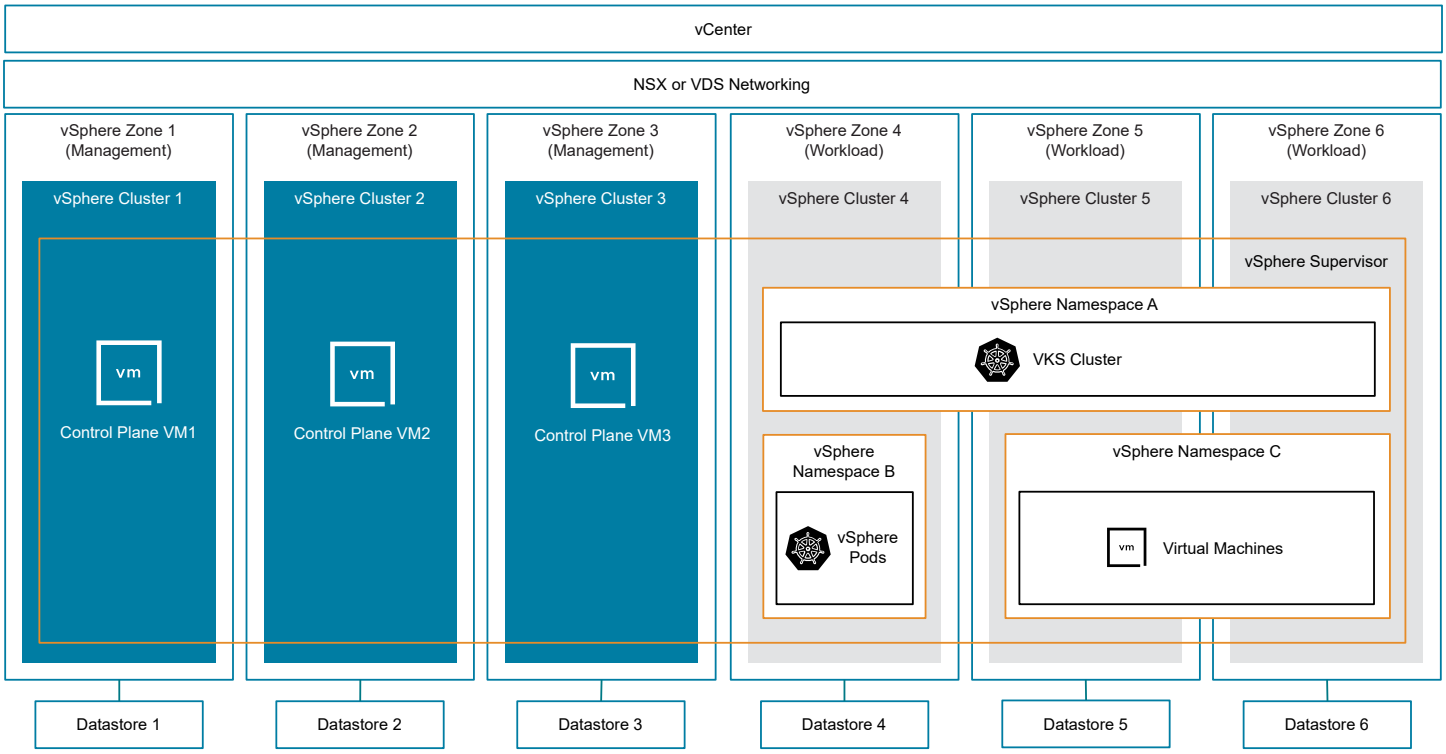
- As an application developer, you can run Kubernetes pods, and deploy and manage Kubernetes based applications. You do not have visibility over the entire stack that is running hundreds of applications.
- As a DevOps engineer or cluster admin, you only have control over the Kubernetes infrastructure, without the tools to manage or monitor the virtual environment and resolve any resource-related and other problems.
- As a vSphere or Cloud administrator, you have full control over the underlying virtual environment, but you do not have visibility over the Kubernetes infrastructure, the placement of the different Kubernetes objects in the virtual environment, and how they consume resources.

Operations on the full stack can be challenging, because they require communication between all three roles. The lack of integration between the different layers of the stack can also introduce challenges. For example, the Kubernetes scheduler does not have visibility over the vCenter inventory and it cannot place pods intelligently.

How Does vSphere Supervisor Help?

vSphere Supervisor provides a declarative API and desired state by deploying a Kubernetes control plane directly on vSphere clusters. As a VI administrator, you activate existing vSphere clusters for vSphere Supervisor, thus creating a Kubernetes layer within the clusters.

Figure 491: vSphere Supervisor



Having a Kubernetes control plane on the vSphere clusters enables the following capabilities in vSphere:

- As a VI administrator, you can create namespaces on the Supervisor, called vSphere Namespaces, and configure them with specified amount of memory, CPU, and storage. You create user assignments to vSphere Namespaces for DevOps engineers.
- As a DevOps engineer, you can run workloads on the same platform with shared resource pools within a vSphere Namespace. You can deploy and manage multiple upstream Kubernetes clusters created by using vSphere Kubernetes Service. You can also deploy Kubernetes containers directly on the Supervisor inside a special type of VM called vSphere Pod. You can also deploy regular VMs.
- As a DevOps engineer, you can use VCF Automation to deploy and manage workloads on the Supervisor, when it is deployed with full on-premises cloud experience using the NSX VPC networking stack.
- As a VI administrator, you can manage and monitor vSphere Pods, VMs, and VKS clusters using the vSphere Client.
- As a Platform operator, you can leverage the capabilities in VCF Operations to monitor the workloads deployed in the Supervisor instances in the VCF fleet.
- As a VI administrator, you have full visibility over vSphere Pods, VMs, and VKS clusters running within different namespaces, their placement in the environment, and how they consume resources.

Having Kubernetes running on vSphere clusters also eases the collaboration between VI administrators and DevOps teams, because both roles are working with the same objects.

What Is a Workload?

In vSphere Supervisor, workloads are applications that are deployed in one of the following ways:

- Applications that consist of containers running inside vSphere Pods.
- Virtual Machines provisioned through the VM service.
- Kubernetes clusters deployed by using vSphere Kubernetes Service (VKS)
- Application pods that run inside VKS clusters.

What are vSphere Zones?

vSphere Zones protect against cluster-level failure by providing high-availability to workloads deployed on vSphere Supervisor. As a VI administrator, you create vSphere Zones in the vSphere Client and then map vSphere clusters to

zones. You can dynamically add and remove vSphere Clusters from vSphere Zones in case you must scale up or down the resources in a zone.

You can use zones to provide isolation between the management plane and workloads in the Supervisor in the following way:

- Deploy the Supervisor on one or three vSphere Zones, which you reserve for running the Supervisor control plane. These zones are marked as management in the vSphere Client. Deploying the Supervisor on three vSphere Zones spreads the control plane VMs across these three zones, which provide cluster-level high availability to the Supervisor control plane. Deploying the Supervisor on a single vSphere Zone only provides host level high availability. For more information, see Supervisor Architecture and Deployment Options.
- Assign vSphere Zones to vSphere Namespaces and use them for running workloads only. You can have multiple vSphere Clusters assigned to a zone.

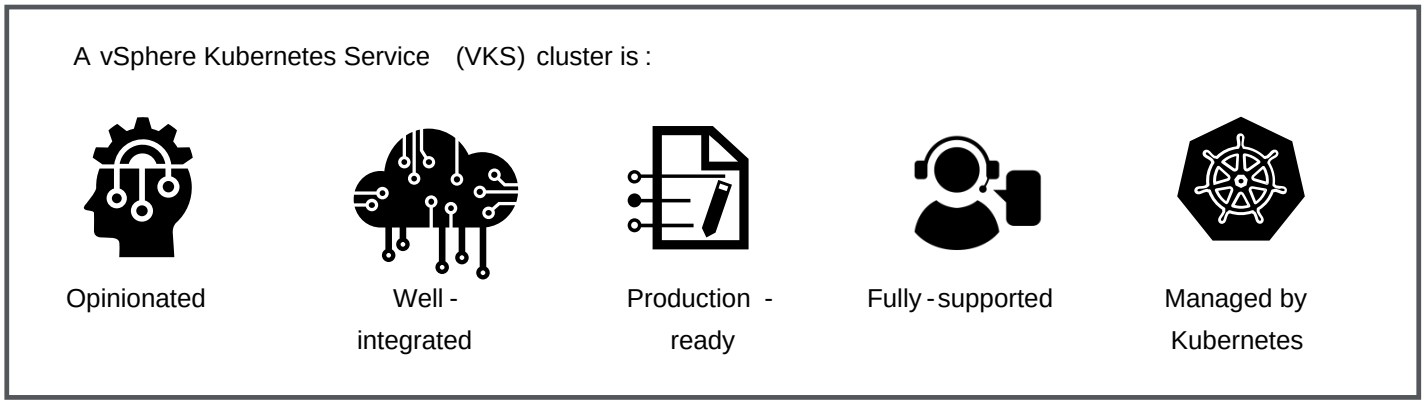
For more information about the Supervisor architectural options with vSphere Zones and different deployment designs in VCF, see vSphere Supervisor Models and vSphere Supervisor Detailed Design.

What Is vSphere Kubernetes Service or VKS Cluster?

You can use the vSphere Kubernetes Service to deploy upstream Kubernetes clusters on vSphere Supervisor. Kubernetes clusters deployed by the vSphere Supervisor service are also called VKS clusters. A VKS cluster is a full distribution of Kubernetes that is built, signed, and supported by VMware by Broadcom.

A VKS cluster has the following characteristics:

Figure 492: VKS characteristics



- Opinionated installation of Kubernetes. vSphere Kubernetes Service provides well-thought-out defaults that are optimized for vSphere to provision VKS clusters. By using vSphere Kubernetes Service, you can reduce the amount of time and effort that you typically spend deploying and running an enterprise-grade Kubernetes cluster.
- Integrated with the VCF and vSphere Infrastructure. A VKS cluster is integrated with the VCF SDDC stack, including storage, networking, and authentication. In addition, a VKS cluster is built on a Supervisor that maps to vSphere clusters. Because of the tight integration, running a VKS cluster is a unified product experience.
- Production Ready. VKS provisions production-ready VKS clusters. You can run production workloads any additional configuration. In addition, you can ensure availability and support rolling Kubernetes software upgrades and run different versions of Kubernetes in separate clusters.
- High-availability for Kubernetes workloads. VKS clusters deployed on more than one vSphere Zones reserved for workloads are protected against failure on vSphere cluster level. The workload and control plane nodes of VKS

clusters are distributed across all vSphere Zones, which makes the Kubernetes workloads running inside them highly available. VKS clusters running on one vSphere Zone are protected against failure at ESX host level, through vSphere HA.

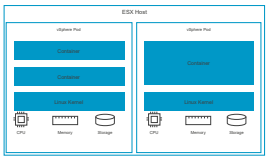
- Fully Supported by VMware by Broadcom. VKS clusters use open source based Linux., are deployed on VCF infrastructure, and run on ESX hosts. If you experience problems with any layer of the stack, from the hypervisor to the Kubernetes cluster, VMware is the only vendor you need to contact.
- Managed by Kubernetes. VKS clusters are built on top of the Supervisor, which itself is a Kubernetes cluster. A VKS cluster is defined in the vSphere Namespace using a custom resource. You provision VKS clusters in a self-service way using familiar kubectl commands and the VCF CLI. There is consistency across the toolchain, whether you are provisioning a cluster or deploying workloads, you use the same commands, familiar YAML, and common workflows.

For more information see vSphere Kubernetes Service and Components.

What Is a vSphere Pod?

vSphere Supervisor introduces a construct that is called vSphere Pod, which is the equivalent of a Kubernetes pod. A vSphere Pod is a VM with a small footprint that runs one or more Linux containers. Each vSphere Pod is sized precisely for the workload that it accommodates and has explicit resource reservations for that workload. It allocates the exact amount of storage, memory, and CPU resources required for the workload to run. vSphere Pods are only supported with Supervisors that are configured with NSX as the networking stack.

Figure 493: vSphere Pods

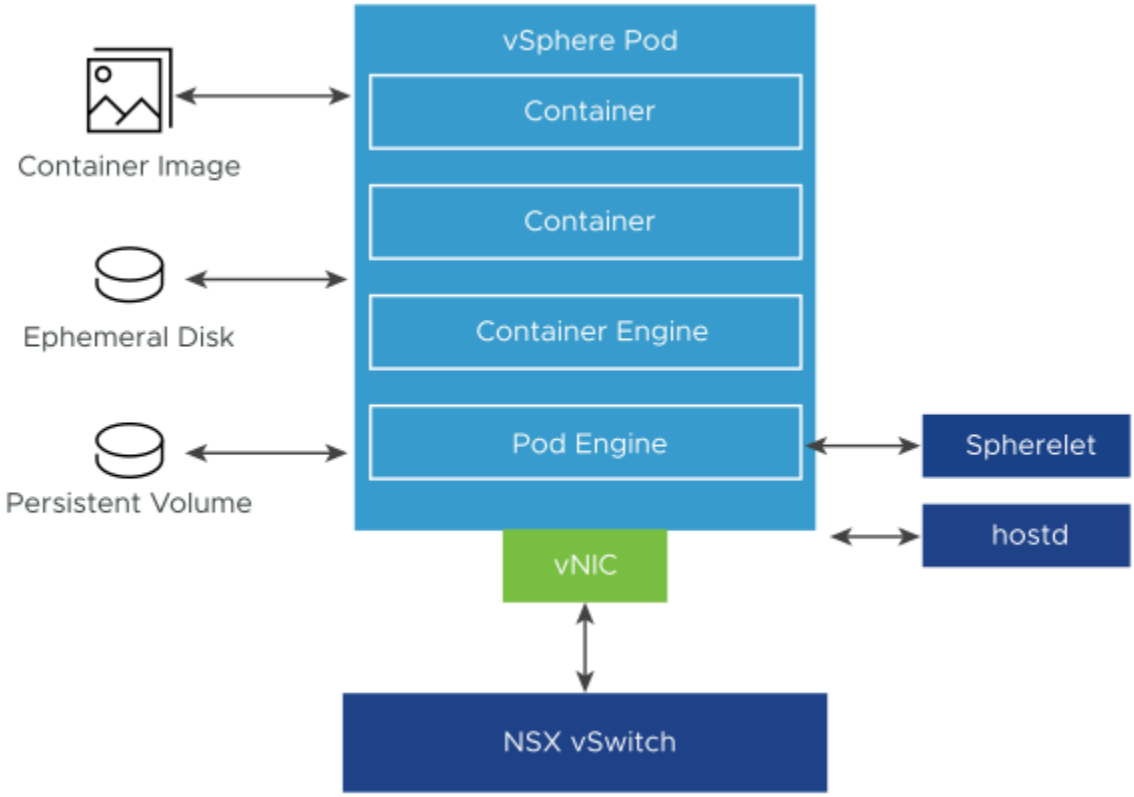


vSphere Pods are objects in vCenter and enable the following capabilities for workloads:

- Strong isolation. A vSphere Pod is isolated in the same manner as a virtual machine. Each vSphere Pod has its own unique Linux kernel that is based on the kernel used in Photon OS. Rather than many containers sharing a kernel, as in a bare metal configuration, in a vSphere Pod, each container has a unique Linux kernel
- Resource Management. vSphere Distributed Resource Schedular (DRS) handles the placement of vSphere Pods on the Supervisor.
- High performance. vSphere Pods get the same level of resource isolation as VMs, eliminating noisy neighbor problems while maintaining the fast start-up time and low overhead of containers.
- Diagnostics. As a VI administrator you can use all the monitoring and introspection tools that are available with vSphere on workloads.

vSphere Pods are Open Container Initiative (OCI) compatible and can run containers from any operating system as long as these containers are also OCI compatible.

Figure 494: vSphere Pod Networking and Storage



vSphere Pods use three types of storage depending on the objects that are stored, that are ephemeral VMDKs, persistent volume VMDKs, and containers image VMDKs. As a VI administrator, you configure storage policies for placement of container image cache and ephemeral VMDKs on the Supervisor level. On a vSphere Namespace level, you configure storage policies for placement of persistent volumes. See Storage Policies for Supervisor and VKS Clusters for details about the storage requirements and concepts with vSphere Supervisor.

For networking, vSphere Pods and the VMs of the VKS clusters use the topology provided by NSX. For details, see Supervisor Networking.

Spherelet is an additional process that is created on each host. It is a kubelet that is ported natively to ESX and allows the ESX host to become part of the Kubernetes cluster.

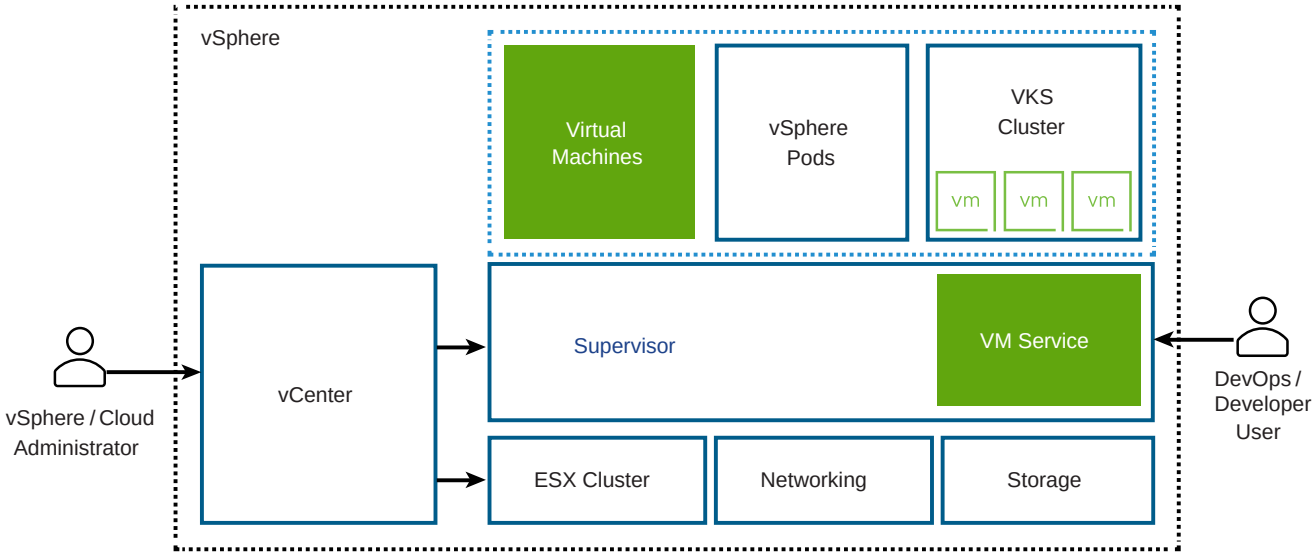
For information about using vSphere Pods on Supervisors, see Deploying Workloads to vSphere Pods.

Using Virtual Machines in vSphere Supervisor

vSphere Supervisor offers a VM Service functionality that enables DevOps engineers to deploy and run VMs in a declarative manner, in addition to containers, in a common, shared Kubernetes environment. Both, containers and VMs, share the same vSphere Namespace resources and can be managed through a single vSphere Supervisor interface.

The VM Service addresses the needs of DevOps teams that use Kubernetes, but have existing VM-based workloads that cannot be easily containerized. It also helps users reduce the overhead of managing a non-Kubernetes platform alongside a container platform. When running containers and VMs on a Kubernetes platform, DevOps teams can consolidate their workload footprint to just one platform.

> **Note:**
>
> In addition to stand-alone VMs, the VM Service manages the VMs that make up VKS clusters.



Each VM deployed through the VM Service functions as a complete machine running all the components, including its own operating system, on top of the vSphere Supervisor infrastructure. The VM has access to networking and storage that a Supervisor provides, and is managed using the standard Kubernetes `kubectl` command. The VM runs as a fully isolated system that is immune to interference from other VMs or workloads in the Kubernetes environment.

When to Use Virtual Machines on a Kubernetes Platform?

Generally, a decision to run workloads in a container or in a VM depends on your business needs and goals. Among the reasons to use VMs are the following:

- Your applications cannot be containerized.
- Applications are designed for a custom kernel or custom operating system.
- Applications are better suited to running in a VM.
- You want a consistent Kubernetes experience and avoid overhead. Rather than running separate sets of infrastructure for your non-Kubernetes and container platforms, you can consolidate these stacks and manage them with a familiar `kubectl` command.
- Running VMs that are deployed and managed by VCF Automation within the All Applications Organization setup.

For information about deploying and managing stand-alone virtual machines in a Supervisor, see Deploying and Managing Virtual Machines.

Supervisor Services in vSphere Supervisor

Supervisor Services are vSphere certified Kubernetes operators that deliver Infrastructure-as-a-Service components and tightly-integrated Independent Software Vendor services to developers. You can install and manage Supervisor Services on the vSphere Supervisor environment so that to make them available for use with Kubernetes workloads. When Supervisor Services are installed on Supervisors, DevOps engineers can use the service APIs to create instances on Supervisors in their user namespaces. These instances can then be consumed in vSphere Pods and VKS clusters.

Learn more about the supported Supervisor Services and how to download their service YAML files at http://vmware.com/go/supervisor-service.

For information about how to use Supervisor Services, see Managing Supervisor Services.

What is a vSphere Namespace?

A vSphere Namespace sets the resource boundaries where vSphere Pods, VMs, and VKS clusters, and Supervisor Services can run. As a VI administrator, you create and configure vSphere Namespaces through the vSphere Client. As an Organization administrator, you createvSphere Namespaces through VCF Automation

When initially created, a vSphere Namespace has unlimited resources within the Supervisor. As a VI administrator, you can set limits for CPU, memory, storage, and the number of Kubernetes objects that can run within the vSphere Namespace. Storage limitations are represented as storage quotas in Kubernetes. A resource pool is created in vSphere per each vSphere Namespace on the Supervisor

You can map up to three vSphere Zones to a vSphere Namespace for running workloads on them. If a vSphere Zone has more than one vSphere cluster associated with it, the Supervisor selects a specific cluster in the vSphere Zone to run the workloads that are associated with the vSphere Namespace:

- The Supervisor filters the vSphere clusters in the zones that are mapped to the vSphere Namespace based on their resource availability and datastore assignments. The datastore connectedto the vSphere clusters must match the storage policy assigned the vSphere Namespace.
- After the Supervisor filters the vSphere clusters in the vSphere Zones, it places the vSphere Namespace to the clusters that meet the requirements for resources reservations and assigned datastores.

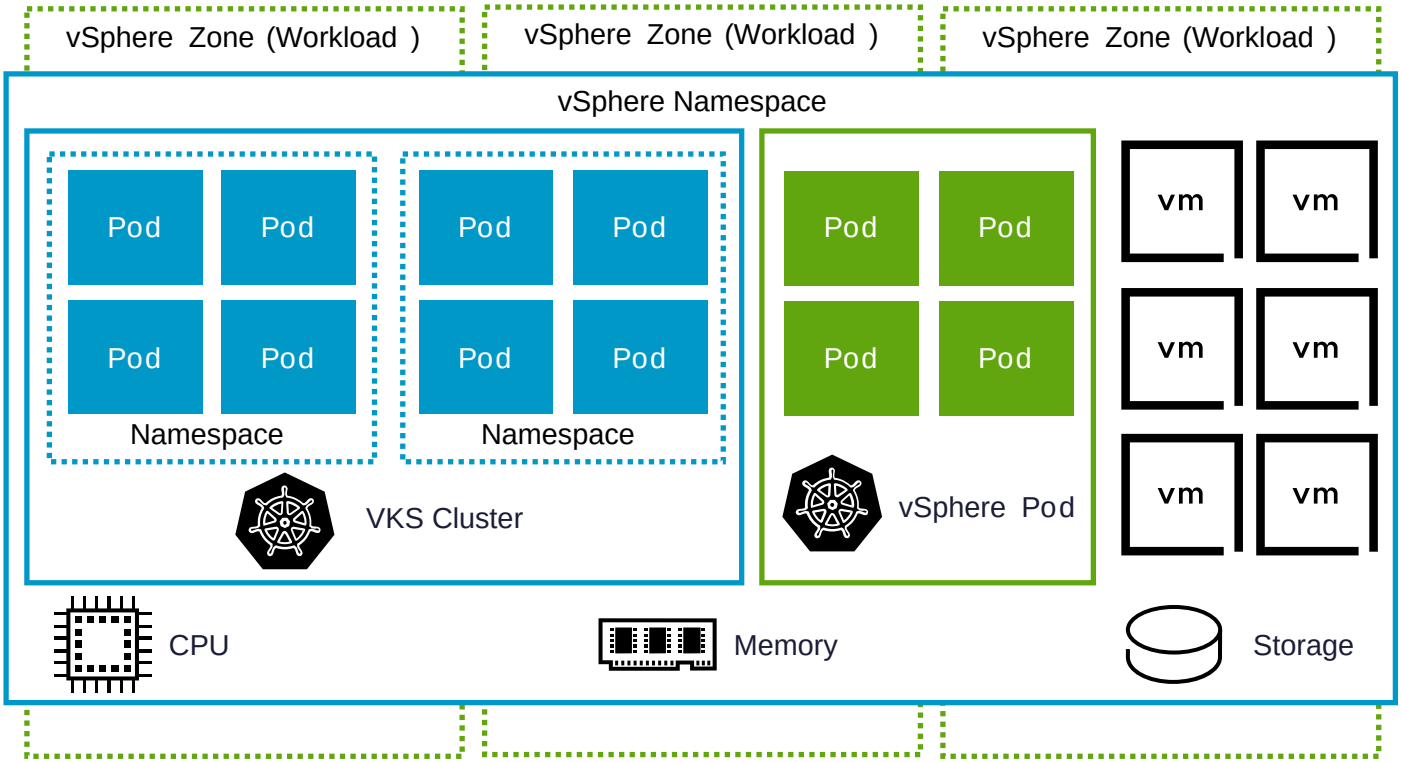
  Note:

  Only one cluster per vSphere Zone can host workloads of a vSphere Namespace that spans acrross several zones.

The resources utilized in a vSphere Namespace are taken from all the underlying vSphere clusters where the vSphere Namespace is placed with the vSphere Zone on equal parts. For example, if you dedicate 300 MHz of CPU, 100 MHz are taken from each vSphere cluster.

To provide workload isolation, you should use vSphere Zones marked as management only for running the Supervisor control plane on these zones.

Figure 495: vSphere Namespace Architecture



To provide access to vSphere Namespaces to a DevOps engineer, as a VI administrator, you assign permissions to users or user groups available within an identity source that is associated with vCenter Single Sign-On or from an OIDC provider that is registered with the Supervisor. For more information, see Understanding Authentication and Authorization in vSphere Supervisor.

After a vSphere Namespace is created and configured with resource and object limits as well as with permissions and storage policies, as a DevOps engineer, you can access the vSphere Namespace to run workloads such as VKS clusters, vSphere Pods ,and VMs created through the VM service.

Differences Between a vSphere Namespace and Kubernetes Namespace

Although in its core a vSphere Namespace serves the same function as a Kubernetes namespace, a vSphere Namespace is specific to the vSphere Supervisor. You should not confuse a vSphere Namespace with a Kubernetes namespace.

A vSphere Namespace is implemented as an extension to a vSphere resource pool and its function is to provide resources to workloads running in the Supervisor. A vSphere Namespace has a direct mapping to a Kubernetes namespace through which object and storage quotas are enforced upon workloads.

Another difference with a regular Kubernetes namespace is that the VI administrator manages the user access to vSphere Namespaces, as mentioned from above. The VI administrator can also associate VM classes and Content Libraries containing VM templates that DevOps engineers can use to self-service VMs. For more information, see Deploying and Managing Virtual Machines in vSphere Supervisor.

vSphere Supervisor Inventory

The vSphere Supervisor inventory is a representation of the Supervisor folder and its associated resources.

vSphere configures separate Supervisor folders for each Supervisor and all the associated resources. The vSphere Namespaces and control plane VMs are mapped under the respective Supervisor folder isolating them from the resources of other Supervisors, even though names of namespaces must be unique within vCenter.

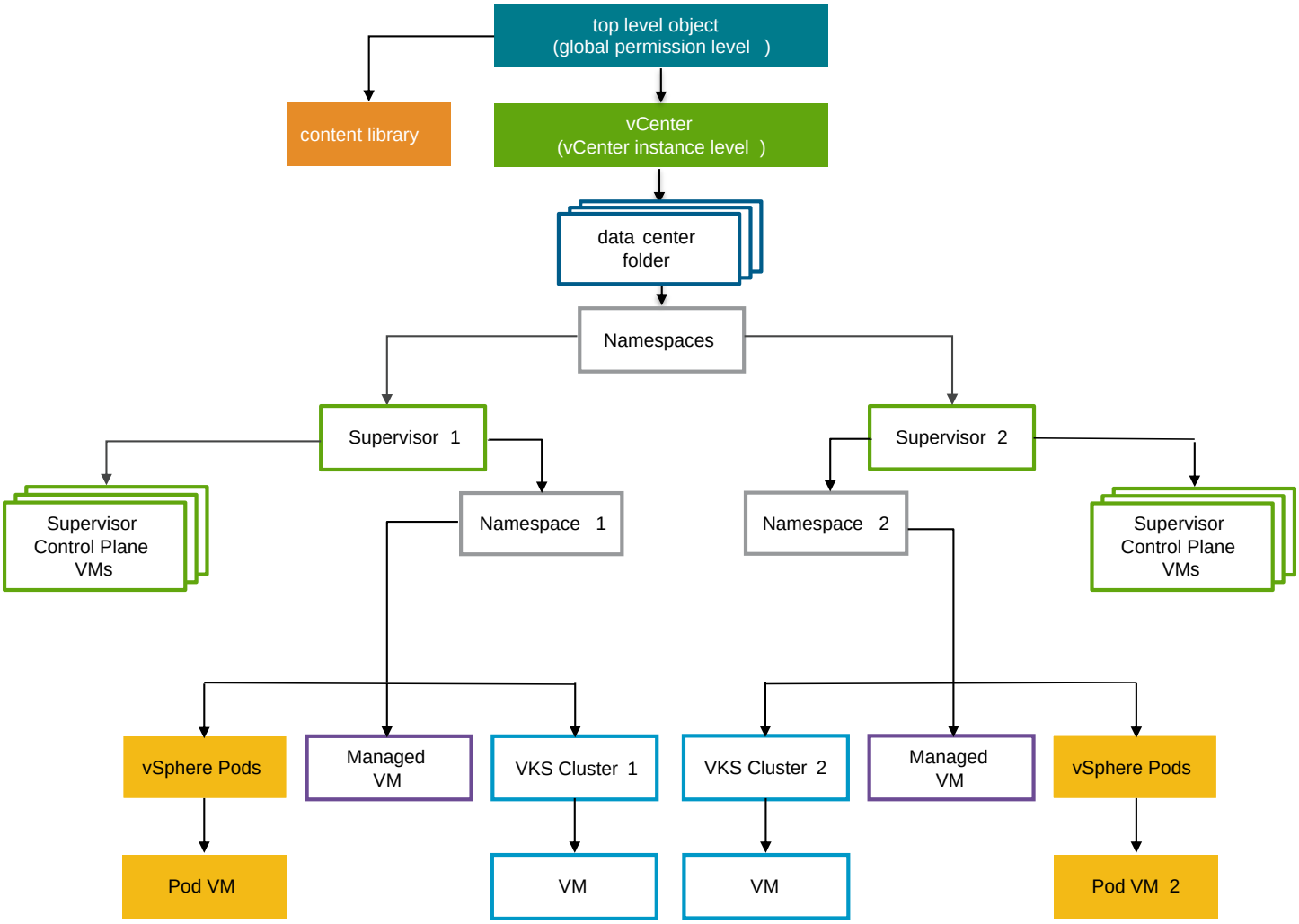
The vSphere Pods within each namespace are segregated into a folder called vSphere Pods. This ensures their isolation from other VMs and VKS clusters that are in the same namespace and allows distinct and precise privilege allocation.

Note:

This section describes the Supervisor identity management starting from the VCF 9.0 release. For Supervisors based on previous releases, the hierarchy and permissions assignment remain the same as the 8.0 version.

The following figure illustrates the inventory hierarchy and the paths by which permissions can propagate.

Figure 496: Inventory Hierarchy

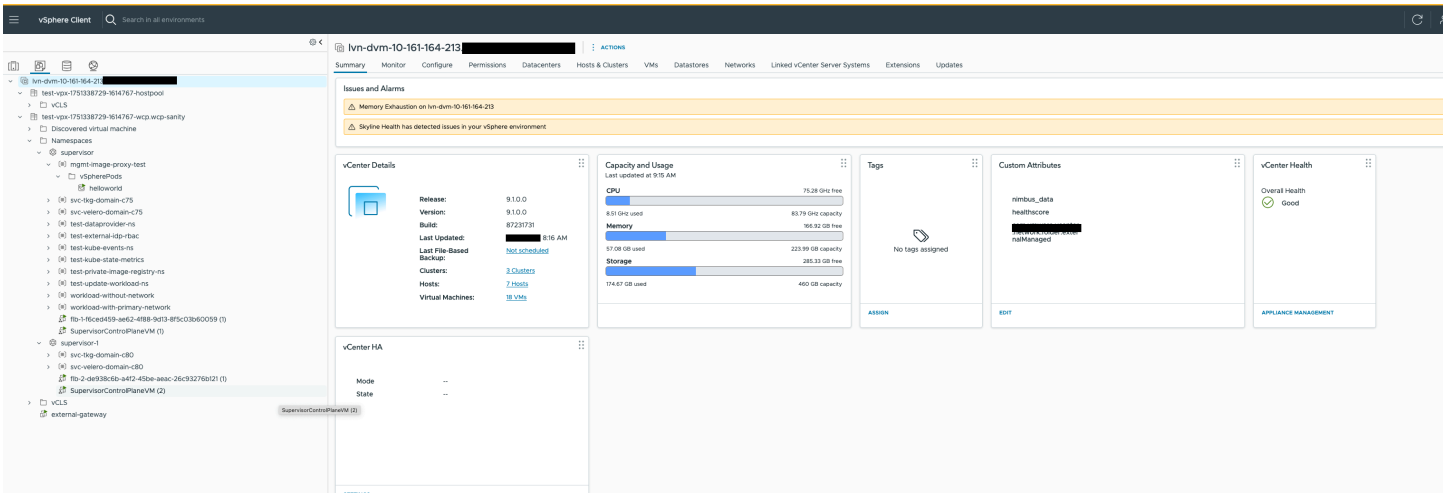


Starting with VCF 9.0, every newly deployed Supervisor is represented by a dedicated *Supervisor* folder under the **Folders** tab. The *Supervisor* folders are created under the *Namespaces* folder which is the root folder for workload management. This *Namespaces* folder represents the resource pool for CPU/memory allocation for the vSphere Namespaces created under it. The control plane VMs for the cluster are placed within this folder as shown in the screenshot below. User-created namespaces appear as child folders within the **Supervisor** folder. Each **Namespace** folder includes all workload entities created under it, such as VMs deployed through VM Service and VKS clusters. Additionally, each **Namespace** folder contains a subfolder named *vSpherePods* which contains all the vSphere Pods associated with the Namespace. Since Pod VMs primarily serve infrastructure purposes, the *vSpherePods* folder is configured with restrictive permissions to prevent users from performing any direct operations on the pods.

For environments upgrading to VCF 9.0 from earlier versions, the existing **Supervisor** folder structure remains unchanged. The new folder hierarchy is applied only when the Supervisor is upgraded to version 9.0.

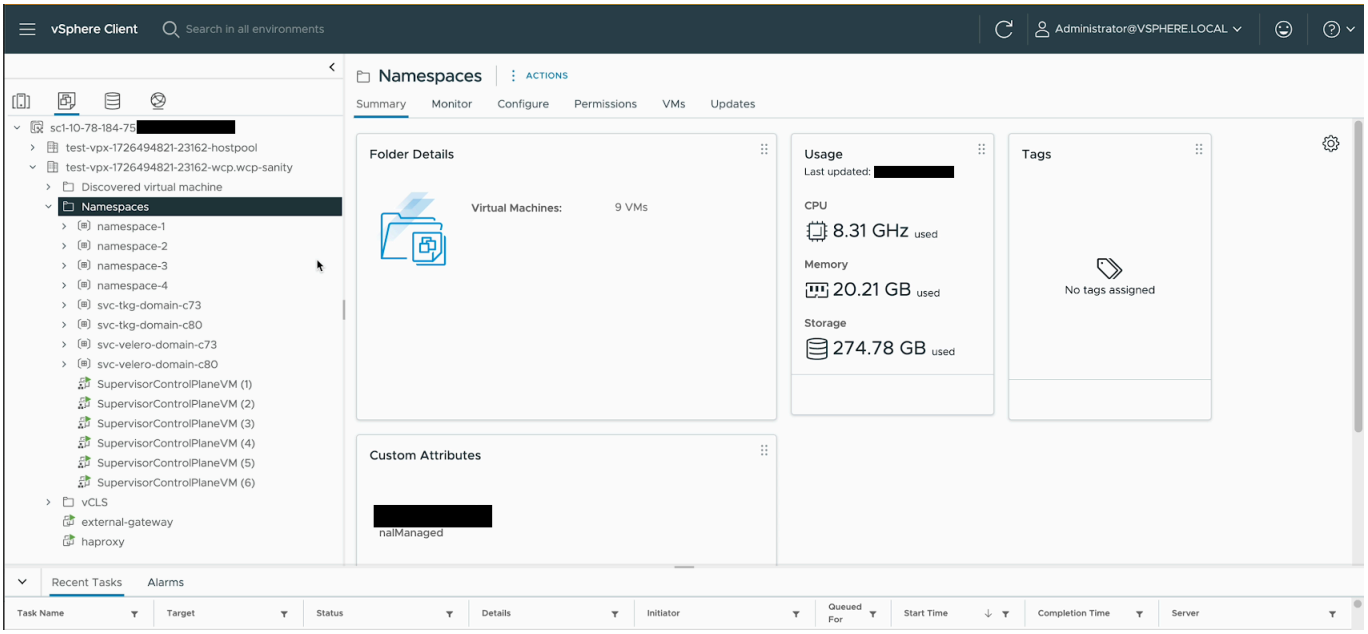
The following screenshot displays the folder structure in VCF 9.0:

Figure 497: Folder Structure in 9.0



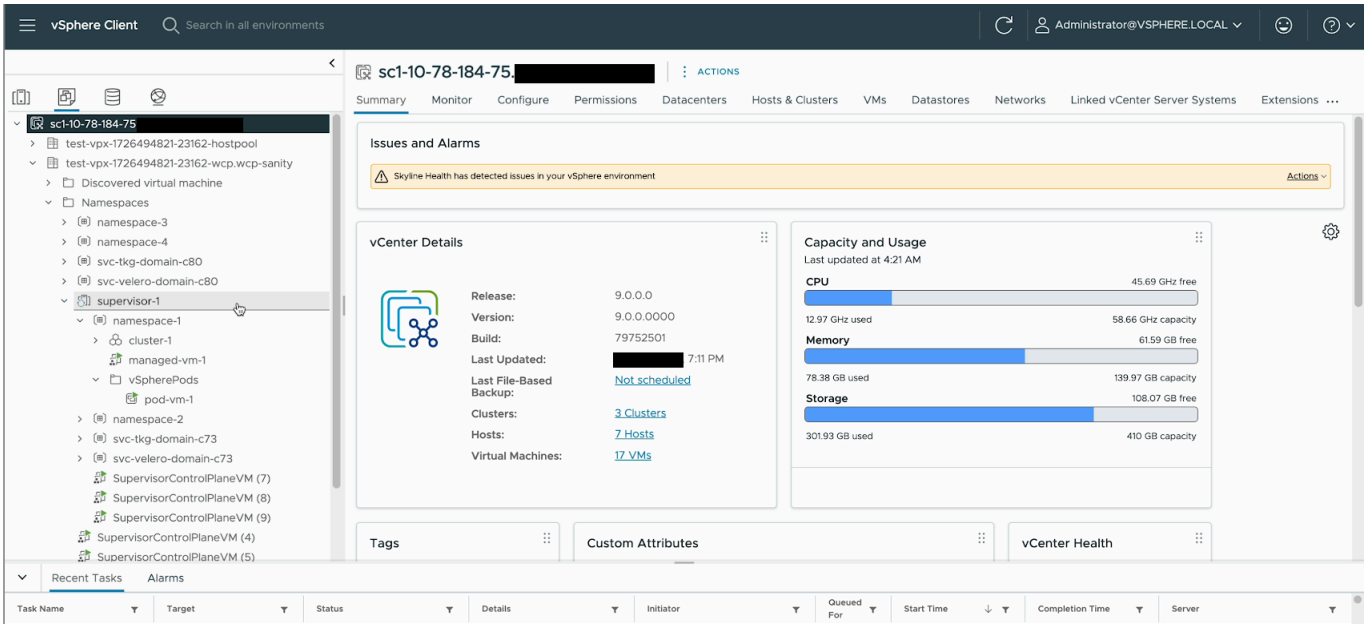
The following screenshot displays the folder structure with two Supervisors in the 8.0 release:

Figure 498: Folder Structure in 8.0



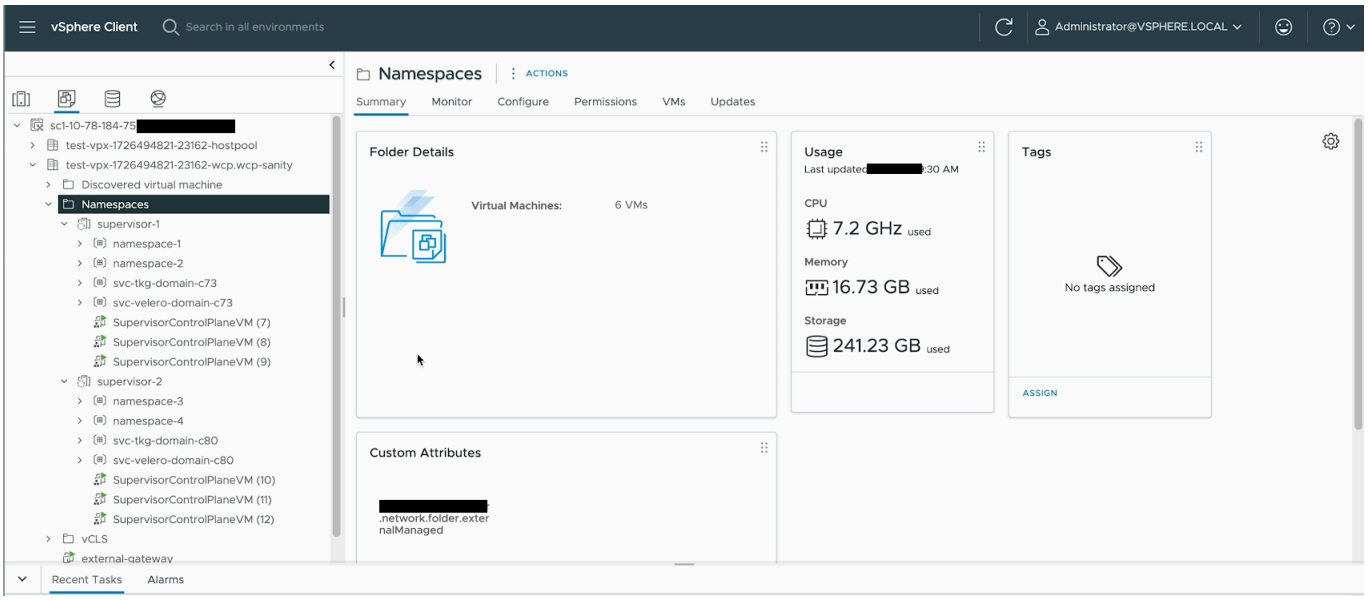
The following screenshot displays a folder structure on 9.x with one Supervisor that has been upgraded to 9.0 and one Supervisor on 8.0:

Figure 499:



The following screenshot displays a folder structure on 9.x with two Supervisors upgraded to 9.0:

Figure 500:



Understanding Authentication and Authorization in vSphere Supervisor

Authentication controls who can access to the vSphere environment and authorization controls what resources the users can access.

Authentication and authorization govern access to your vSphere environment. vCenter Single Sign-On supports authentication, which means it determines whether a user can log in to vSphere components at all. Each user must also be authorized to view or manipulate vSphere objects.

vCenter allows fine-grained control over authorization with permissions and roles. When you grant a permission to an object in the vCenter object hierarchy, you associate a role to a user or group on that object. To specify the privileges, you use roles, which are sets of privileges.

For more information on authorization in vSphere, see https://techdocs.broadcom.com/bin/gethidpage?ux-context-string=vsphclient_048&appid=vsphere-9-1&language=&format=rendered.

Authenticating to Supervisor using vCenter Single Sign-On or External Identity Provider

vCenter Single Sign-On is an authentication broker and security token exchange infrastructure. vCenter Single Sign-On issues a token when a principal, such as a user, group, or service account, authenticates. The principal can use the token to authenticate to vCenter services. The principal can then perform the actions that the principal has privileges for.

A user or group with the necessary privileges can be authenticated through vCenter Single Sign-On or an external identity provider (IdP) integrated with the Supervisor, such as Okta or Azure AD and can access the vSphere Namespace using VCF CLI. For information on how on register external IdP with Supervisor, see Register an External Identity Provider with Supervisor.

When you log in to the Supervisor using VCF CLI, the authentication proxy forwards your request to vCenter Single Sign-On if the Supervisor has not been configured with an external IdP. The VCF CLI then initiates a session with vCenter, obtains an authentication token, and retrieves the list of vSphere Namespaces you have access to and updates the kubeconfig accordingly. For more information, see https://techdocs.broadcom.com/bin/gethidpage?ux-context-string=vks-2&appid=sv-async&language=&format=rendered.

If you use an external identity provider (IdP), the Pinniped Supervisor and Concierge components facilitate federated authentication for users and groups. Any changes to your account permissions are reflected the next time you log in, and the list of accessible vSphere Namespaces is refreshed.

The VCF CLI command for logging in to vSphere Namespace is below. After logging in, you can set context to the vSphere Namespaces the user has privileges on and then run `kubectl` commands on the Supervisor.

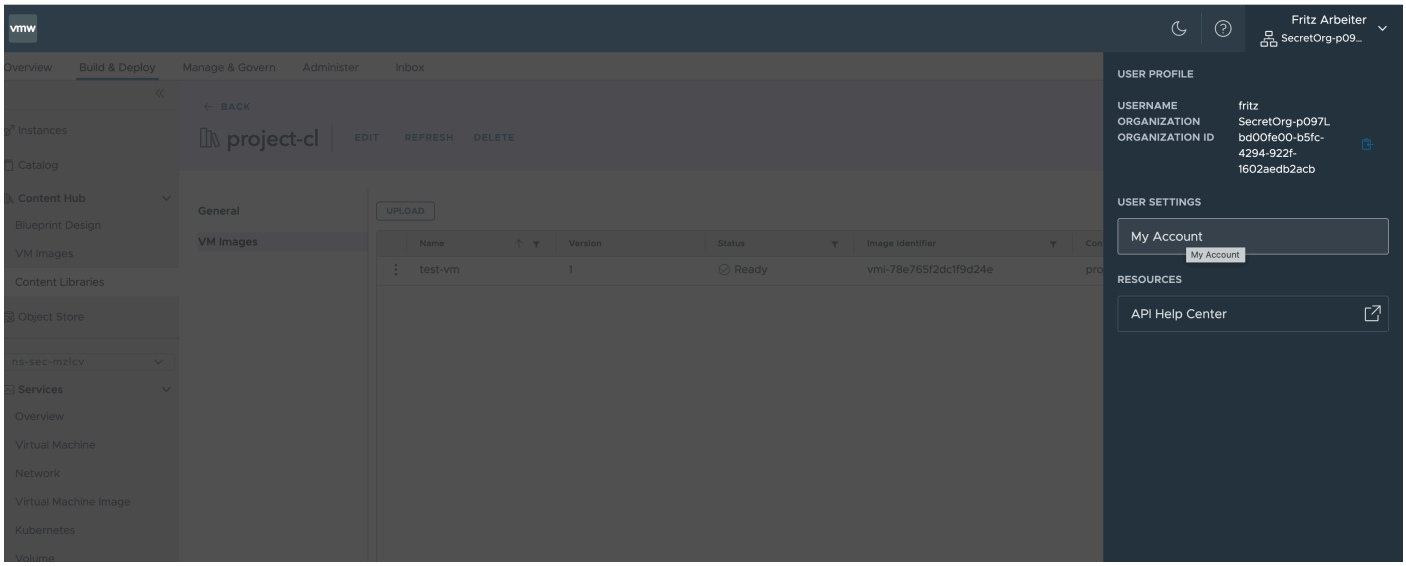
VCF CLI command to connect with Supervisor for SSO or external IdP user	`export SUPERVISOR_ENDPOINT=<supervisor cluster IP address or FQDN>`
	
	`vcf context create <context_name> --endpoint=<SUPERVISOR_ENDPOINT> --type=k8s --username=<user_name>`

Authenticating VCF Automation Using a Token

When a Supervisor is registered with VCF Automation, then it is configured to use VCF Automation as its identity provider. Authenticating with the vSphere Namespace requires you to log in with VCF Automation context and get an API token.

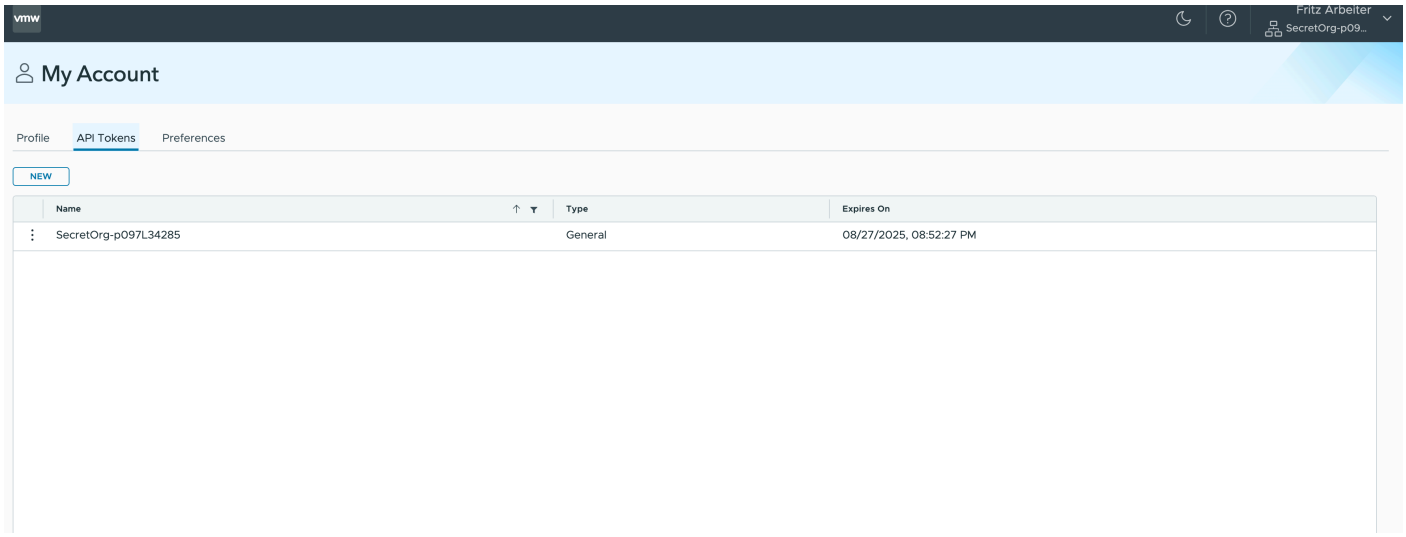
1. To get an API token from VCF Automation, click your profile on the top right corner and select **My Account**.

   Figure 501:


vmw
Fritz Arbeiter
SecretOrg-p09...
Overview
Build & Deploy
Manage & Govern
Administer
Inbox
BACK
project-cl
EDIT
REFRESH
DELETE
Instances
Catalog
Content Hub
Blueprint Design
VM Images
Content Libraries
Object Store
ns-sec-mzlcv
Services
Overview
Virtual Machine
Network
Virtual Machine Image
Kubernetes
General
VM Images
UPLOAD
Name
Version
Status
Image Identifier
test-vm
1
Ready
vmi-78e765f2dc1f9d24e
USER PROFILE
USERNAME
fritz
ORGANIZATION
SecretOrg-p097L
ORGANIZATION ID
bd00fe00-b5fc-4294-922f-1602aedb2acb
USER SETTINGS
My Account
My Account
RESOURCES
API Help Center

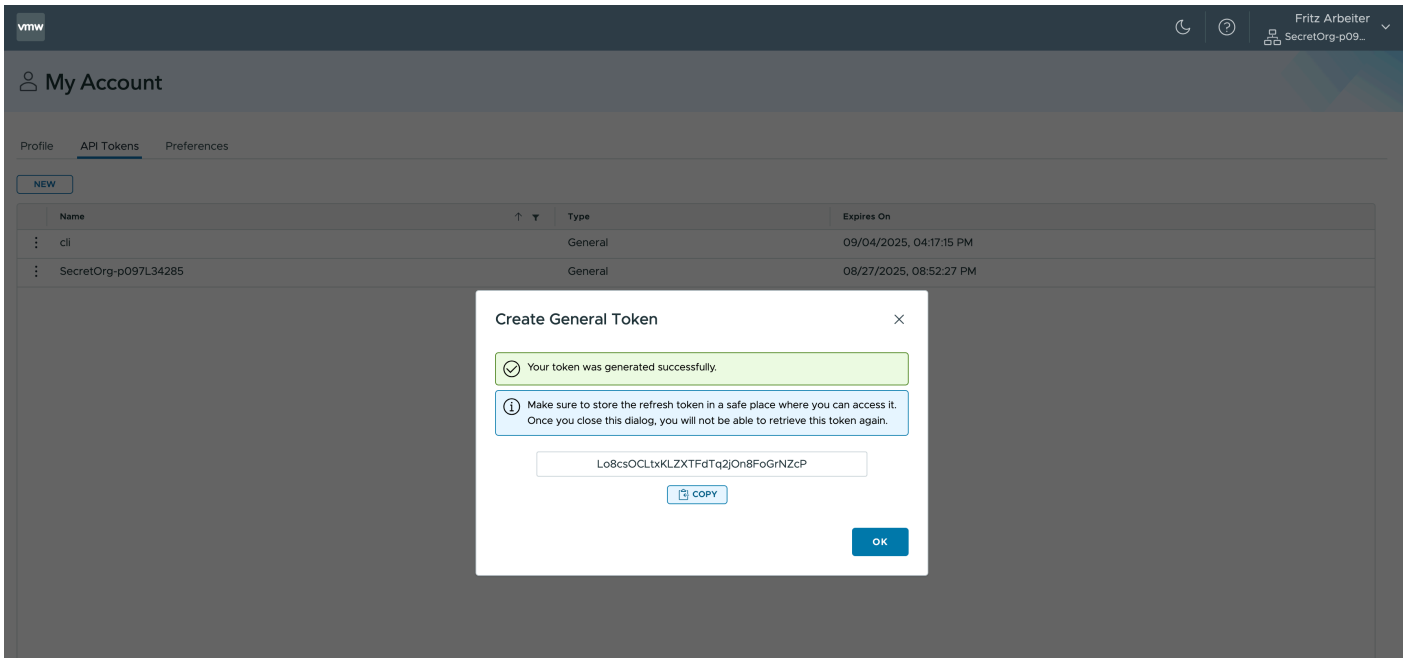
2. Go to the **API Tokens** section and click **New** to generate a new API token.

   Figure 502:



3. Copy and save the token.

   Figure 503:



4. Export the `VCF_CLI_VCFA_API_TOKEN` variable to the token from the above step using the following command:

   ```
   export VCF_CLI_VCFA_API_TOKEN=<api token from above step>
   ```

5. Import the certificate. Run the following to import the certificate:

   ```
   openssl s_client -connect $VCFA_ENDPOINT:443 -showcerts </dev/null 2>/dev/null | openssl x509 -outform PEM
   > vcfa.cert
   ```

6. Create the VCFA context.

   ```
   vcf context create vcfa_ctx -e $VCFA_ENDPOINT --api-token $VCF_CLI_VCFA_API_TOKEN --tenant-name
   $TENANT_NAME --ca-certificate vcfa.cert
   ```

7. Use the VCFA context created above.

   ```
   vcf context use vcfa_ctx:namespace1:project1
   ```

8. You can delete the current context with the following command:

   ```
   vcf context delete vcfa_ctx
   ```

How Permissions Work in Supervisor

The permission model for Supervisor relies on assigning permissions to objects in the object hierarchy.

Users get permissions to Supervisor in the following ways.

- From a specific permission for the user or from the groups that the user is a member of.
- From a permission on the object.

Permissions give a specific role to a user or group, granting them corresponding privileges on selected inventory objects. You can use the vSphere Client to add permissions. That role gives those users the corresponding privileges on the objects in the vCenter inventory.

Object-Level Permission Model

The following define the permission model:

- Permissions - Each object in the vCenter object hierarchy has associated permissions. Each permission defines the privileges that a specific group or user has on the object. Permissions can propagate to child objects.
- Users and Groups - On vCenter systems, you can assign privileges only to authenticated users or groups of authenticated users. Users are authenticated through vCenter Single Sign-On. Users and groups must be defined in the identity source that vCenter Single Sign-On uses to authenticate. Define users and groups using the tools in your identity source, for example, Active Directory.
- Privileges - Privileges are fine-grained access controls. You can group those privileges into roles, which you can then map to users or groups.
- Roles - Roles are sets of privileges. Roles allow you to assign permissions on an object based on a typical set of tasks that users perform. System roles, such as Administrator, are predefined on vCenter and cannot be changed. vCenter also provides some default sample roles, such as Resource Pool Administrator, that you can modify. You can create custom roles either from scratch or by cloning and modifying sample roles. See https://techdocs.broadcom.com/bin/gethidpage?ux-context-string=vsphclient_045&appid=vsphere-9-1&language=&format=rendered.

For more information, https://techdocs.broadcom.com/bin/gethidpage?ux-context-string=vsphclient_046&appid=vsphere-9-1&language=&format=rendered.

vCenter Roles for the vSphere Namespaces

VCF 9.0 introduces a new role called *Supervisor Administrator*. The *Supervisor Administrator* role is granted by default to members of the Administrators group on the *vSphere Namespace* resource pool and folder in newly deployed Supervisors instead of the Read-only role in the prior versions.

Based on the vCenter role assigned within the Supervisor inventory, a user or group can be assigned one of the following roles:

Administrator Role	As a VI administrator, you can view and manage the vSphere infrastructure, including VKS clusters, vSphere Pods, and VM service VMs. The VI administrator has the membership of the "Administrators SSO Group". This Administrators group is automatically assigned the *Supervisor Administrator* role on the Supervisor hierarchy.
Supervisor Administrator Role	A *Supervisor Administrator* role can only be assigned on the *Supervisor* folder. You will be able to view the namespaces inventory only after assigning it the `Read-only` role on the Cluster. When the *Supervisor Administrator* role is assigned on the *Supervisor* folder, a context is created that gives access only to cluster level resources in the *Supervisor* and not the vSphere Namespace. That is, you cannot list or view vSphere Namespaces. To be able to view *vSphere Namespaces* when you log in through VCF CLI, you must have permissions on the specific vSphere Namespace.
Namespace Owner/Edit/View Role	In this role, you are a consumer of the vSphere Namespace and can provision VKS clusters and vSphere Pods using kubectl.

Given the Supervisor cluster IP address, you can log in and start deploying and managing workloads on the vSphere Namespace you have been granted access to.

Depending on the resource, you have the following access:

Roles ->	VI administrator	Supervisor Administrator	Namespace Owner/Edit/View
Resource			
vSphere Cluster	Yes	No	No
vSphere VM Inventory	Yes	Yes	No
vSphere Namespace Inventory	Yes	Yes (After assigning the `Read-only` role on the vSphere cluster)	No
vSphere Namespace	Yes	Yes	No

vCenter Privileges on Supervisor and vSphere Namespaces

vCenter privileges control how you interact with vSphere Supervisor. You can set a privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

The Supervisor Administrator role has the following privileges:

Table 1776: Supervisor Administrator Privileges

vCenter Privileges	Required On	Access	
		VC Inventory	Supervisor Resources

`Alarms.Acknowledge`
`Alarms.Create`
`Alarms.Edit`
`Alarms.Delete`
`Alarms.ToggleEnableOnEntity`
`Alarms.DisableActions`
`Alarms.SetStatus`
	Supervisor folder	As a user with `Alarms` privileges on the Supervisor folder you can perform the following actions on the inventory.
		• Acknowledge Alarm: Suppresses alarm actions on triggered alarms.
		• Create Alarm: Allows creating new alarm definitions.
		• Modify Alarm: Allows changing the settings of an existing alarm.
		• Disable Alarm Action: Prevents alarm actions from occurring after an alarm is triggered, but the alarm remains active.
		• Disable or Enable Alarm on Entity: Allows toggling the activation of a specific alarm on a target object.
		• Set Alarm Status: Allows changing the status of an alarm, for example, from triggered to cleared.	

`VirtualMachine.GuestOperations.Query` `VirtualMachine.GuestOperations.QueryAliases` `VirtualMachine.GuestOperations.ModifyAliases` `VirtualMachine.GuestOperations.Modify` `VirtualMachine.GuestOperations.Execute`	Supervisor folder	As a user with `GuestOperations` privileges, you can perform the following actions on the inventory: • Guest Operation Queries: Allows virtual machine guest operations that involve querying the guest operating system, such as listing files in the guest operating system. • Guest Operation Alias query: Allows virtual machine guest operations that involve querying the alias for the virtual machine. • Guest Operation Alias modifications: Allows virtual machine guest operations that involve modifying the alias for the virtual machine. • Guest Operation Modifications: Allows virtual machine guest operations that involve modifications to a guest operating system in a virtual machine, such as transferring a file to the virtual machine. • Guest Operation Program Execution: Allows virtual machine guest operations that involve running an application in the virtual machine. With these set of privileges user can perform troubleshooting or support bundle collection when network is not available.	

vCenter Privileges	Required On	Access	
		VC Inventory	Supervisor Resources
Authorization.ModifyPermissions	Supervisor folder	As a user with ModifyPermissions privileges, you can control the assigning of roles and permissions to other users. Modify permissions: Allows defining one or more permission rules on an entity, or updating rules if rules are already present for the given user or group on the entity.	
InventoryService.Tagging. InventoryService.Tagging.ObjectAttachable	Supervisor folder	As a user with Tagging privileges, you can control the ability to assign and unassign tags to selected objects in the Supervis or inventory. • Assign or Unassign vSphere Tag:Allows assignment or unassignment of a tag for an object in the Supervisor inventory. • Assign or Unassign vSphere Tag on Object: Allows objects to have tags assigned or unassigned. Use this privilege to limit which objects users are able to assign or unassign tags to.	
SupervisorServices.InstallSupervisorServices.Manage	Supervisor folder	As a user with Supervisor Services privileges, you can control who can create and manage Supervisor Services. • Manage Supervisor Services: Allows deployment, upgrade, and management of Supervisor Services on vCenter. • Install Supervisor Services: Allows installation of Supervisor Services.	
Namespaces.Manage	Supervisor folder	Manage: Allows modifying the cluster-wide configuration and activating and deactivating Super visor.	
Namespaces.ManageDisks	Supervisor folder	Manage Disks: Allows for decommissioning operations of data stores associated with a particular vSphere Namespace.	
Namespaces.Upgrade	Supervisor folder	Upgrade Supervisor: Allows initiation of the Supervisor upgrade.	
Namespaces.Configure	Supervisor folder	Configure Namespaces: Allows creating, deleting, and managing vSphere Namespaces.	

vCenter Privileges	Required On	Access	
		VC Inventory	Supervisor Resources
`Namespaces.ListAccess`	Supervisor folder	List Namespaces: Allows listing of the accessible vSphere Namespaces within a vSphere environment.	
`Namespaces.Backup`	Supervisor folder	Backup Workloads component files: Allows backing up the contents of the `etcd` cluster.	
`Namespaces.SelfServiceManage`	Supervisor folder	Modify cluster-wide namespace self-service configuration: Allows modifying the namespace self-service configuration.	
`Namespaces.Observe`	Supervisor folder		Manage Observability Services: As an user with this privilege, you can modify configuration of observability services such as, `telegraf` and `fluentbit` services running on the Supervisor.
`Namespaces.ConfigureAdmissionPolicy` `Namespaces.ViewAdmissionPolicy`	Supervisor folder		Configure Validating Admission Policy: Allows performing CRUD operations on Validation Admission Policies and bindings across all the namespaces on the S upervisor. View Validating Admission Policy: Allows viewing and listing Validation Admission Policies and bindings on all the namespaces on the Supe rvisor.
`Namespaces.Owner/Edit/View`	Supervisor folder		The Supervisor Administrator is an uber role which will also have the Namespaces privileges listed below.

The Namespaces Owner role has the following privileges:

Table 1777: Namespaces Owner Privileges

vCenter Privileges	Required On	Access	
		VC Inventory	Supervisor VKS Cluster Resources
`Namespaces.Owner Authorization.ModifyPermissions`	User created Namespace folder		A user with the `Owner` privilege can perform all operations as per the `Namespace.Edit` role but has the below privileges: Modify permissions: Allows defining one or more permission rules on an entity, or updating rules if rules are already present for the given user or group on the entity. Delete Namespace: Allows deleting the namespace on which user has `Owner` privilege.

The Namespaces View has the following privileges:

Table 1778: Namespaces View Privileges

vCenter Privileges	Required On	Access	
		VC Inventory	Supervisor Resources
`Namespaces.view`	User created Namespace folder		View Supervisor resources: A user with the `View` privilege can perform get, list, and watch operations on various Kubernetes API resources within the Supervisor .

The Namespaces Edit has the following privileges:

Table 1779: Namespaces Edit Privileges

vCenter Privileges	Required On	Access	
		VC Inventory	Supervisor VKS Cluster Resources
`Namespaces.Edit`	User created Namespace folder		Edit Supervisor resources: A user with the `Edit` privilege can perform CRUD operations such as, create, patch, update, delete, and watch, get, and list actions on various Kubernetes API resources in the Supervisor.

How Does vSphere Supervisor Change the VCF Environment?

A Supervisor adds objects to the vCenter inventory, such as namespaces, vSphere Pods, and VKS clusters, and VMs.

Under each Supervisor, you can view:

- vSphere Namespaces that represent logical grouping of resources running in the Supervisor.
- Resource pools for each vSphere Namespace on the Supervisor. In a multi-zone deployment, a resource pool for each vSphere Namespace is created on each vSphere compute cluster part of a vSphere Zone. When a vSphere Zone has multiple compute clusters that are associated with it, a vSphere Namespace is placed on the clusters that best fit the resource and datastore requirements of the nameapce. See What are vSphere Zones?For review regarding VGL-43945 - Supervisor on wide range of topologies

Within every namespace, you can view:

- vSphere Pods.
- vSphere Pods of Supervisor Services and other objects that are brought by Supervisor Services.
- VKS clusters.
- Stand-alone VMs created by using the VM service.
- Supervisor control plane VMs found in the system namespace.
- User permissions for that namespace.

Licensing for vSphere Supervisor

Starting with vSphere 9, product components are licensed automatically after you license the vCenter instance connected to the component. You no longer license vSphere Supervisor independently. For more information, see *Licensing Overview in the Licensing* guide.

vSphere Supervisor Security

vSphere Supervisor leverages vSphere security features and provisions VKS clusters that are secure by default.

vSphere Supervisor is an extension to vSphere that is able to leverage the security features that are built into vCenter and ESX. For more information, see the *vSphere Security* documentation.

The Supervisor encrypts all secrets stored in the database (etcd). The secrets are encrypted via a local encryption key file, which is provided at boot by vCenter. The decryption key is stored in memory (tempfs) on the Supervisor nodes and on disk in encrypted form within vCenter database. The key is available in clear text to the root users of each system. Secrets held within the database of each workload cluster are stored in clear text. All etcd connections are authenticated with certificates that are generated at installation and rotated during upgrades. The same encryption model applies to the data in the database (etcd) that is installed on the control plane for each VKS cluster.

In Supervisor, you can run confidential vSphere Pods on compatible systems. You can create confidential vSphere Pods by adding Secure Encrypted Virtualization-Encrypted State (SEV-ES) as a security enhancement. For more information, see Deploy a Confidential vSphere Pod.

A VKS cluster is secure by default. Restrictive Pod Security Admission (PSA) and PodSecurityPolicy (PSP) are available for VKS clusters. If developers need to run privileged pods or root containers, at a minimum a cluster administrator must create a RoleBinding that grants user access to the default privileged PSP. For more information, see Security for VKS Clusters.

A VKS cluster does not have infrastructure credentials. The credentials that are stored within a VKS cluster are only sufficient to access the vSphere Namespace where the VKS cluster has tenancy. As a result, there is no privilege escalation avenue for cluster operators or users.

The authentication token used to access a VKS cluster is scoped such that the token cannot be used to access the Supervisor or other VKS clusters. This prevents cluster operators, or individuals who might try to compromise a cluster, from using their root-level access to capture a VI administrator's token when they log in to a VKS cluster.

How Supervisor Artifacts are Delivered and Consumed in a VCF Environment

In VCF 9.1 online deployments, vSphere Supervisor and related components use a unified artifact delivery model. After VCF is deployed, Supervisor artifacts are no longer pulled directly from public repositories. Instead, compatible artifacts are downloaded automatically from the online VCF Unified Depots to a local Software Depot and consumed from there.

This model applies to all components related to Supervisor, including:

- vSphere Supervisor
- vSphere Kubernetes release (VKr)
- Supervisor Services
- vSphere Kubernetes Service (VKS) add-ons
- VCF CLI and CLI plugins

Note:

Offline deployments in VCF 9.1 continue to use existing workflows.

Artifact Delivery Overview

In VCF 9.1 online deployments, the Software Depot performs these actions:

- Acts as a proxy for artifact download requests to online depots.
- Handles authentication and authorization to the online depots.
- Might cache downloaded artifacts locally for a period of time.

No manual configuration of external repositories or registries is required.

Artifact Consumption by Component Type

- Non-OCI artifacts, for example, OVF templates used by Supervisor and VKr, are consumed through vCenter Content Library by subscribing to the endpoint of the local Software Depot.
  Another example is VCF CLI, which is downloaded by proxying through the local Software Depot on the Supervisor landing page.
- OCI artifacts used by Supervisor Services, VKS add-ons, and CLI plugins are pulled from a regional Harbor registry. The regional Harbor registry is configured as a proxy cache to the local Software Depot.

This model ensures consistent, validated, and protected access to artifacts across Supervisors and clusters.

Asynchronous Releases

Supervisor components can be released asynchronously, outside of the main VCF release cycle. When an installation or upgrade workflow requires an asynchronous release artifact, the workflow can request the artifact through the local Software Depot.

VMware vSphere Foundation Deployments

In VVF 9.1 deployments, refer to the Software Depot documentation for VVF deployments for details about required configuration and workflow differences.

vSphere Supervisor Architecture and Components

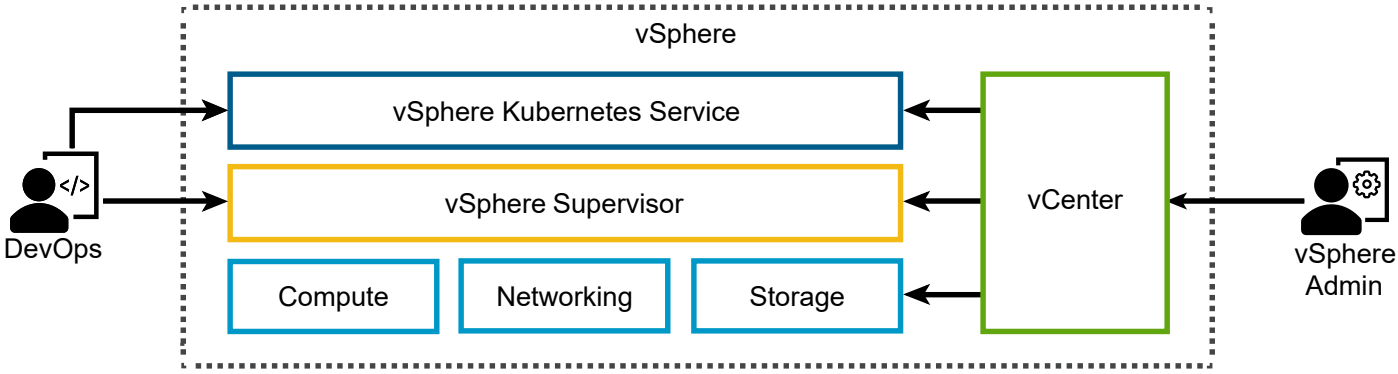
vSphere Supervisor provides a platform for running workloads in a declarative fashion by enabling a Kubernetes control plane in vSphere. You can deploy the Supervisor on one vSphere cluster mapped to a vSphere Zone, or three vSphere Clusters mapped to vSphere Zones. You can also select to deploy one Supervisor control plane VM or three control plane VMs for high availability.

Supervisor Architecture and Deployment Options

When you activate a Supervisor on vSphere Zones, a Kubernetes control plane is created. This control plane enables the capability to run workloads in a declarative fashion on vSphere clusters.

Figure 504: Supervisor General Architecture

The diagram shows the vSphere Supervisor high-level architecture, with vSphere Kubernetes Service on top, the Supervisor in the middle, then ESX hosts, networking, and storage at the bottom, and vCenter managing them.



A Supervisor runs on top of an SDDC layer that consists of ESX for compute, NSX or vSphere Distributed Switch networking, and vSAN or another shared storage solution. Shared storage is used for persistent volumes for vSphere Pods and VMs running inside the Supervisor, and pods in a VKS cluster. After a Supervisor is created, as a vSphere administrator you can create vSphere Namespaces within the Supervisor. As a DevOps engineer, you can run workloads consisting of containers running inside vSphere Pods, deploy VMs through the VM service, and create VKS clusters.

Supervisor Components

The Supervisor has the following components:

- Supervisor control plane VM. You can select to deploy the Supervisor with either one or three control plane VMs. For control plane HA, three VMs are recommended. The three Supervisor control plane VMs are load balanced as each one of them has its own IP address. Additionally, a floating IP address is assigned to one of the VMs and a 5th IP address is reserved for patching purposes. vSphere DRS determines the exact placement of the control plane VMs on the ESX hosts part of the Supervisor and migrates them when needed.
- VKS and Cluster API. Modules running on the Supervisor and enable the provisioning and management of VKS clusters.
- Virtual Machine Service. A module that is responsible for deploying and running stand-alone VMs and VMs that make up VKS clusters.
- Spherelet. An additional process called Spherelet is created on each host. It is a kubelet that is ported natively to ESX and allows the ESX host to become part of the Kubernetes cluster.
- Container Runtime Executive (CRX) component. CRX is similar to a VM from the perspective of Hostd and vCenter. CRX includes a paravirtualized Linux kernel that works together with the hypervisor. CRX uses the same hardware virtualization techniques as VMs and it has a VM boundary around it. A direct boot technique is used, which allows the Linux guest of CRX to initiate the main init process without passing through kernel initialization. This allows vSphere Pods to boot nearly as fast as containers.

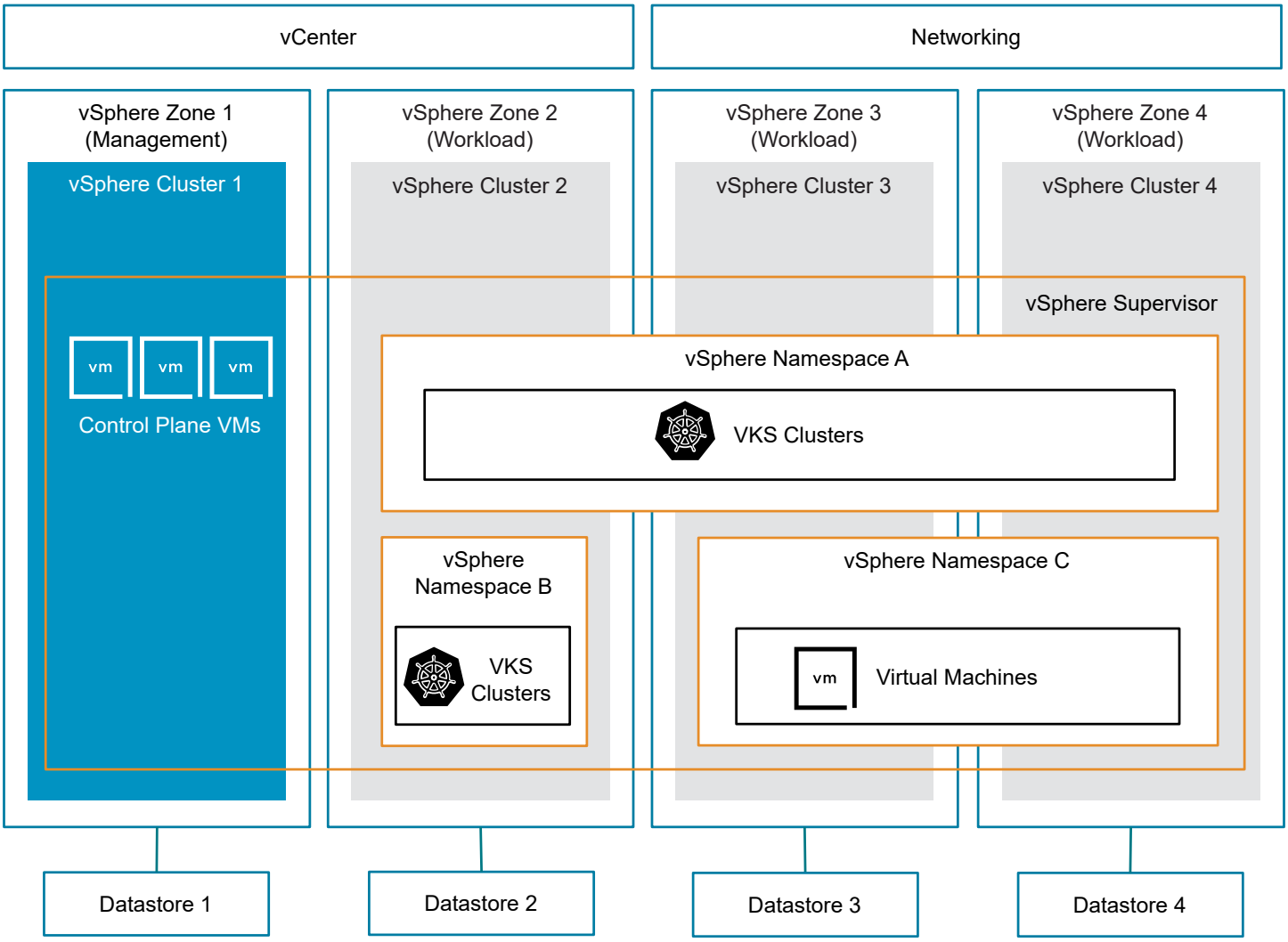
Workload and Management Isolation

You can deploy a Supervisor on three vSphere Zones, that become marked for management and host the Supervisor control plane. In a three-zone deployment, all three vSphere clusters mapped to the vSphere Zones become one Supervisor. A three-zone deployment provides cluster-level high-availability that protects the control plane VMs against a single cluster-level failure. Each control pane VM is placed in a different vSphere Zone. A vSphere Zone maps to one vSphere cluster that you can setup as an independent failure domain. The resources utilized to a vSphere Namespace on a three-zone Supervisor are taken from all three underlying vSphere clusters on equal parts. For example, if you dedicate 300 MHz of CPU, 100 MHz are taken from each vSphere cluster.

To provide isolation between the management components and workloads running on the Supervisor, you can reserve the vSphere Zones where the Supervisor control plane VMs are deployed as a management plane only. Respectively, you can assign different vSphere Zones to vSphere Namespaces and use them for running workloads. A vSphere Namespace requires at least one vSphere Zone, and you can configure up to three vSphere Zones to a vSphere Namespace.

On single-zone deployment, as Supervisor control plane VMs run on a single vSphere Cluster, providing cluster-level HA.

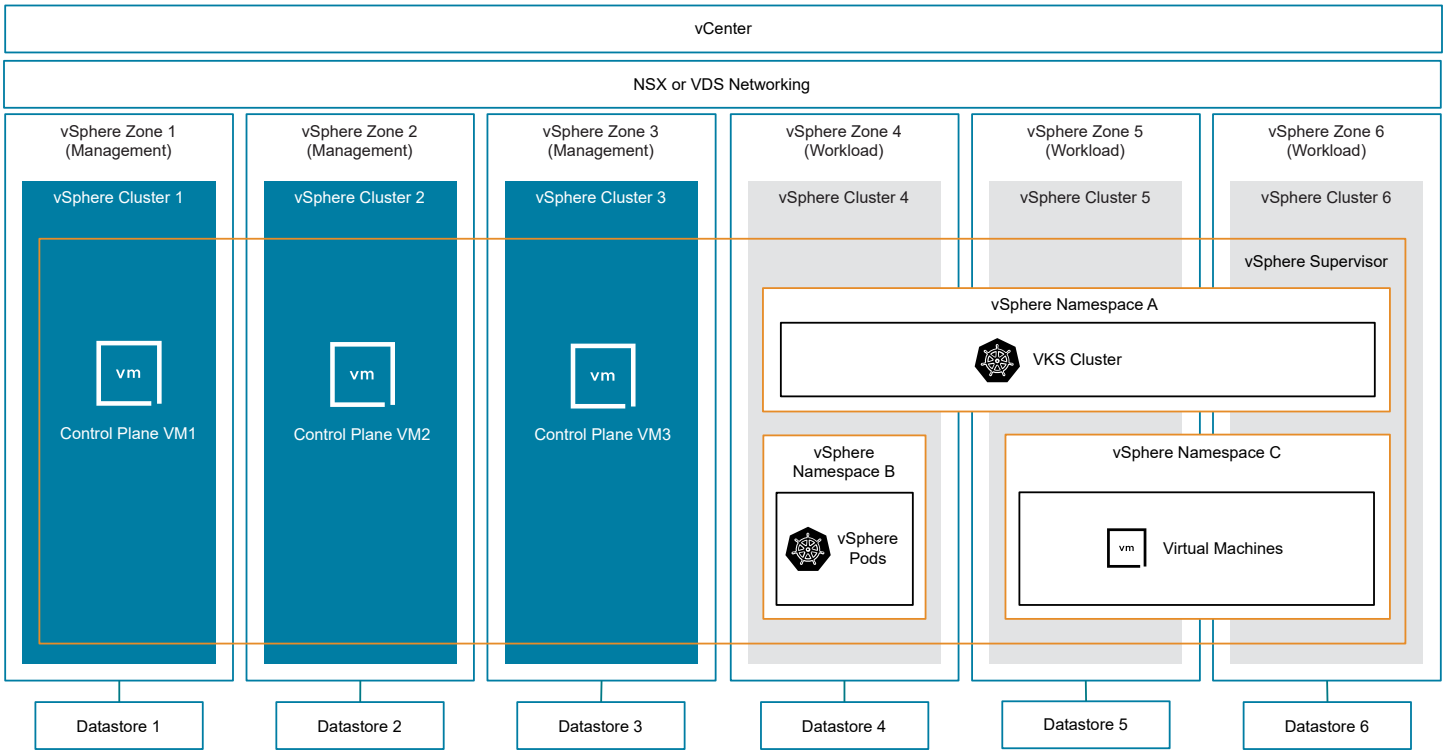
Figure 505: Single-Zone Supervisor



Control Plane High Availability

You can deploy the Supervisor with one control plane VM, or with three control plane VMs to provide high availability. When you deploy the Supervisor on three vSphere Zones, three control plane VMs are deployed for the Supervisor that will be distributed across the vSphere Zones. You must always configure control plane HA for a Supervisor activated on three vSphere Zones.

Figure 506: Hightly-Available Supervisor



Supervisor Networking

A Supervisor can either use a vSphere networking stack or NSX to provide connectivity to Supervisor control plane VMs, services, and workloads.

Depending on your requirements, you can configure a networking stack and load balancer to provide connectivity to external services and DevOps users.

The table summarizes the different networking stacks and load balancers that you can configure for your Supervisor environment.

Table 1780: Supervisor Networking Options

Networking Stack	Load Balancer
NSX Virtual Private Clouds (VPC)	• NSX Load Balancer • Avi Load Balancer • Virtual Network Appliance Load Balancer
vSphere Networking and vSphere Distributed Switch	• Foundational Load Balancer • Avi Load Balancer
NSX Segment Networking	• NSX Load Balancer • Avi Load Balancer **Note:** Starting with VCF 9.1, the vSphere Client UI no longer supports deploying a Supervisor with classic NSX Segment Networking. You can deploy Supervisor through the API.

Note:

Starting with VCF 9.1, it is recommended that you deploy Supervisors with VPC for greenfield deployments. Brownfield deployments of Supervisors configured with NSX are supported. For NSX configuration through the API, see vSphere Automation API.

Note:

When you plan your network infrastructure, make sure that downstream VKS clusters do not use overlapping IP addresses. For the specific non-overlap requirements across different VCF versions, see the CIDR rules in the v1beta1/v1beta2 Example: Default Cluster topic.

When a Supervisor is configured with the vSphere networking stack, all hosts from the Supervisor are connected to a VDS that provides connectivity to workloads and Supervisor control plane VMs. A Supervisor that uses the vSphere networking stack requires a load balancer on the vCenter management network to provide connectivity to DevOps users and external services.

A Supervisor that is configured with NSX, uses virtualized networks of the solution and NSX Load Balancer, Virtual Network Appliance Load Balancer, or the Avi Load Balancer to provide connectivity to external services and DevOps users. You can configure the Avi Load Balancer on NSX if your environment meets the following conditions:

- NSX version is VCF 9.1.x.
- The Avi Load Balancer version is 32.1.1 or 32.1.3.
- The Avi Load Balancer Controller you plan to configure is registered on NSX.
- An NSX load balancer or a Virtual Network Appliance Load Balancer is not already configured on the Supervisor.

Supervisor with VCF Networking with VPC

For networking and security of vSphere Supervisor, you can use Virtual Private Cloud (VPC). VPC is the recommended workload network for deploying the VMware Cloud Foundation (VCF) stack as it is the supported network stack for VCF Automation environments. The Supervisor can be activated as an optional configuration during workload domain creation and by default, it uses VCF Networking with VPC. If you do not activate the Supervisor during the workload domain creation, you can deploy the Supervisor later with VPC from vCenter to gain the full cloud experience with VCF Automation.

For more information, see Deploying Supervisor with VCF Networking with VPC.

For more information about the Virtual Network Appliance Load Balancer, see https://techdocs.broadcom.com/bin/gethidpage?ux-context-string=design-library-002&appid=vcf-9-1&language=en&format=rendered.

Supervisor Networking with VDS

In a Supervisor that is backed by VDS as the networking stack, all hosts from the vSphere clusters backing the Supervisor must be connected to the same VDS. The Supervisor uses distributed port groups as workload networks for workloads and control plane traffic. You assign Workload Networks to vSphere Namespaces in the Supervisor.

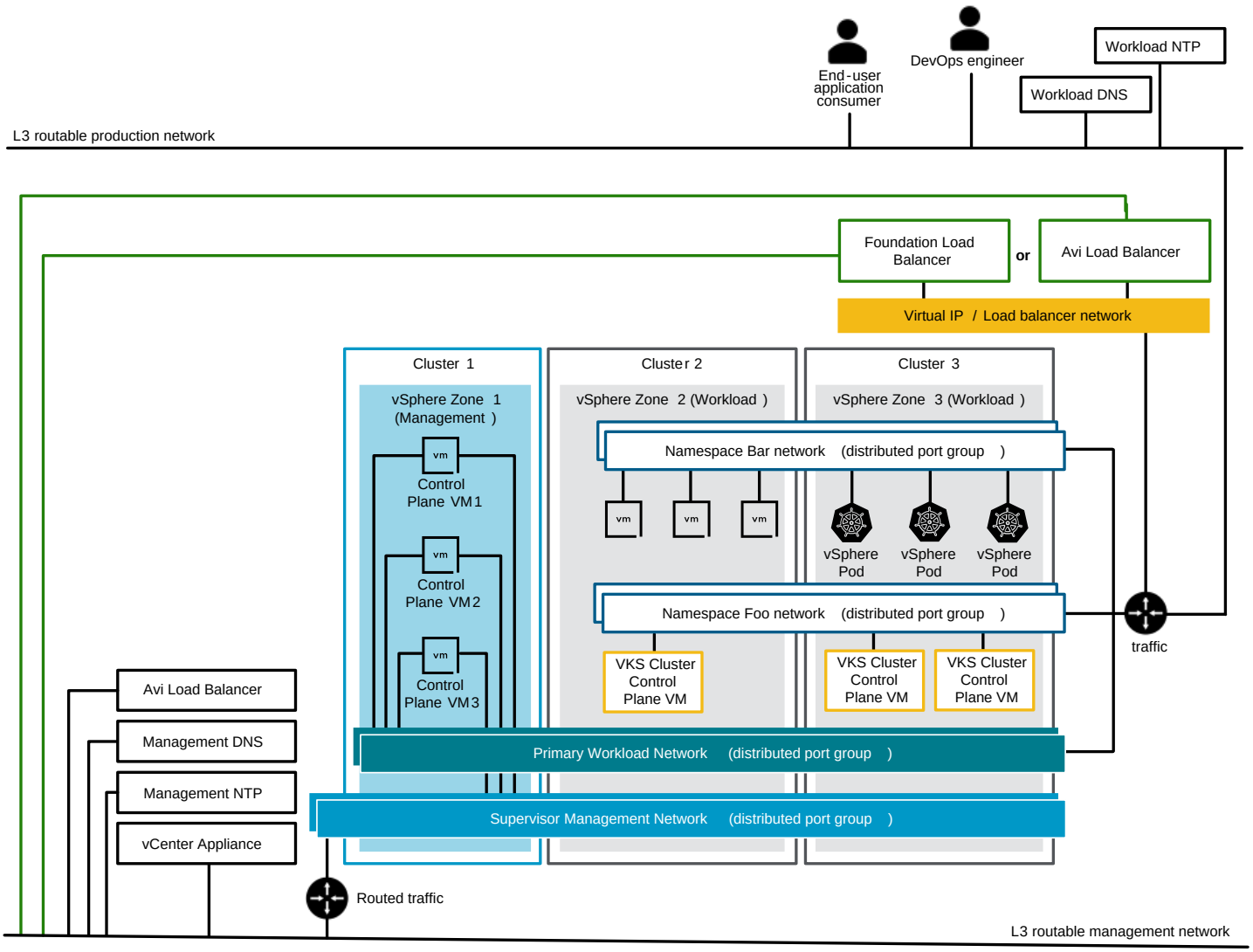
Depending on the topology that you implement for the Supervisor, you can use one or more distributed port groups as Workload Networks. The network that provides connectivity to the Supervisor control plane VMs is called Primary Workload Network. You can assign this network to all the vSphere Namespaces on the Supervisor, or you can use different networks for each namespace. VKS clusters connect to the Workload Network that is assigned to the namespace where the clusters reside.

A Supervisor that is backed by a VDS uses a load balancer for providing connectivity to DevOps users and external services. You can use Foundation Load Balancer or Avi Load Balancer.

In a single-cluster Supervisor setup, the Supervisor is backed by only one vSphere cluster. All hosts from the cluster must be connected to the same VDS.

In a three-zone Supervisor, you deploy the Supervisor on three vSphere zones, each mapped to a vSphere cluster. All hosts from these vSphere clusters must be connected to the same VDS. All physical servers must be connected to the same physical network. Workload networks that you configure to a vSphere Namespace span across all three vSphere zones.

Figure 507: Singe Management Zone Supervisor networking with VDS

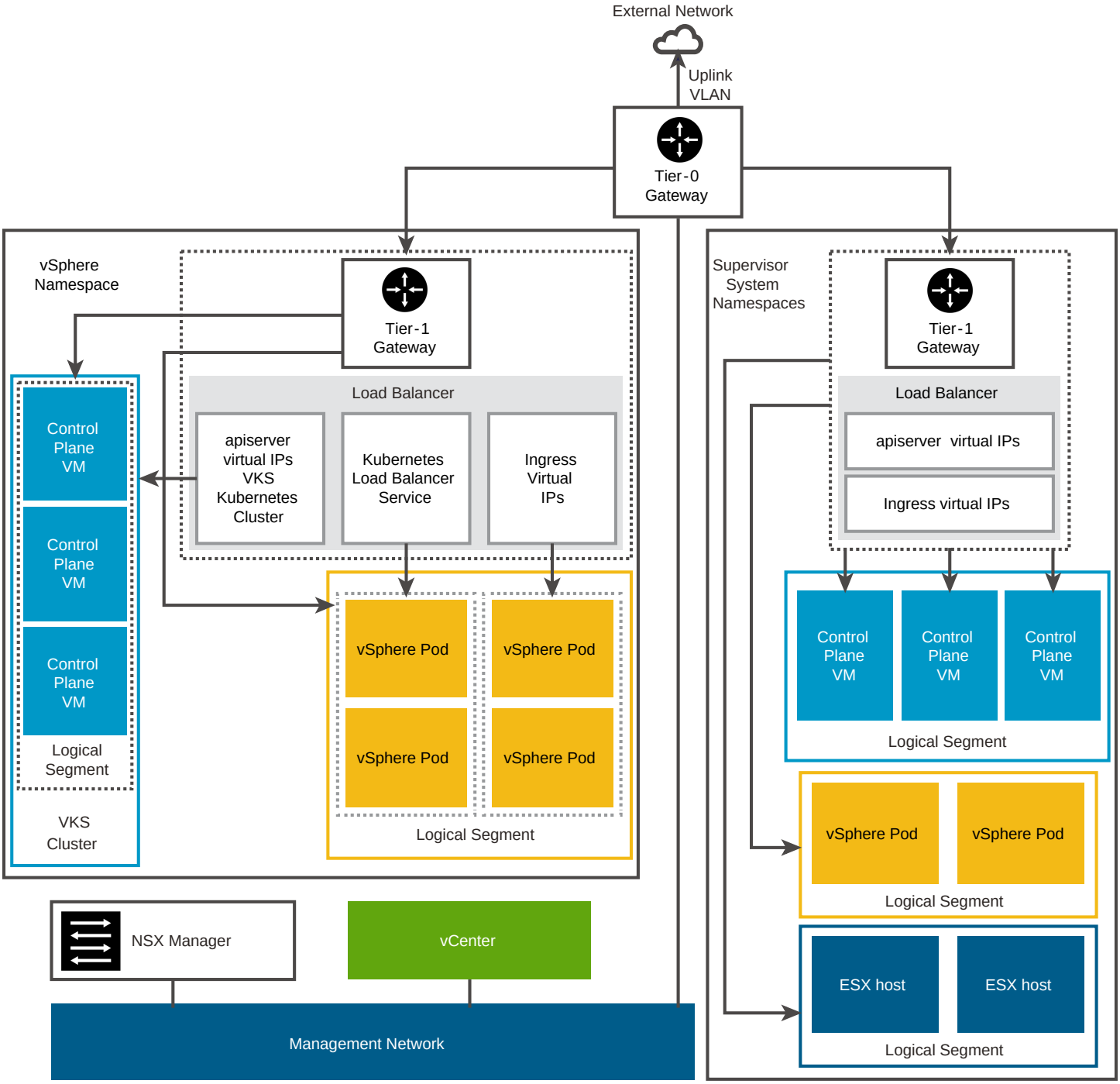


Supervisor Networking with NSX Segment Networking

NSX provides network connectivity to the objects inside the Supervisor and external networks. Connectivity to the ESX hosts comprising the cluster is handled by the standard vSphere networks.

You can also configure the Supervisor networking manually by using an existing NSX deployment or by deploying a new instance of NSX.

Figure 508: Supervisor Networking with NSX Segment Networking



- NSX Container Plugin (NCP) provides integration between NSX and Kubernetes. The main component of NCP runs in a container and communicates with NSX Manager and with the Supervisor control plane. NCP monitors changes to containers and other resources and manages networking resources such as logical ports, segments, routers, and security groups for the containers by calling the NSX API.

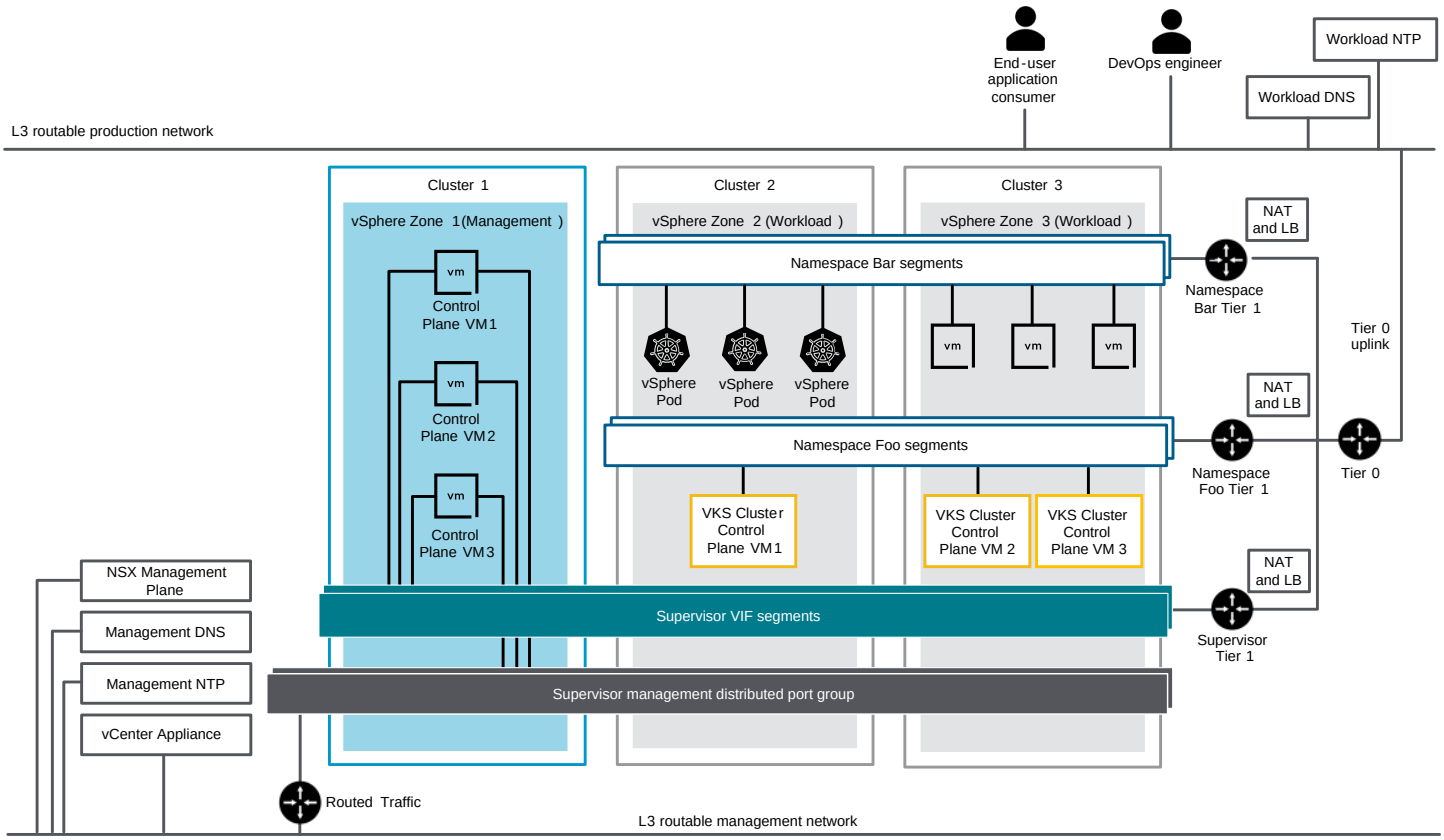
  The NCP creates one shared tier-0 gateway for system namespaces and a tier-1 gateway and load balancer for each namespace, by default. The tier-1 gateway is connected to the tier-0 gateway and a default segment.

  System namespaces are namespaces that are used by the core components that are integral to functioning of the Supervisor and VKS clusters. The shared network resources that include the tier-1 gateway, load balancer, and SNAT IP are grouped in a system namespace.

- NSX Edge provides connectivity from external networks to Supervisor objects. The NSX Edge cluster has a load balancer that provides a redundancy to the Kubernetes API servers residing on the Supervisor control plane VMs and any application that must be published and be accessible from outside the Supervisor.
- A tier-0 gateway is associated with the NSX Edge cluster to provide routing to the external network. It can use dynamic routing protocols such as BGP or static routing.
- Each vSphere Namespace has a separate network and set of networking resources shared by applications inside the namespace such as, tier-1 gateway, load balancer service, and SNAT IP address.
- Workloads running in vSphere Pods, regular VMs, or VKS clusters, that are in the same namespace, share a same SNAT IP for North-South connectivity.
- Workloads running in vSphere Pods or VKS clusters will have the same isolation rule that is implemented by the default firewall.
- A separate SNAT IP is not required for each Kubernetes namespace. East west connectivity between namespaces will not need SNAT.
- The segments for each namespace reside on the VDS functioning in Standard mode that is associated with the NSX Edge cluster. The segment provides an overlay network to the Supervisor.
- Supervisors have separate segments within the shared tier-1 gateway. For each VKS cluster, segments are defined within the tier-1 gateway of the namespace.
- The Spherelet processes on each ESX hosts communicate with vCenter through an interface on the Management Network.

In a three-zone Supervisor configured with NSX as the networking stack, all hosts from all three vSphere clusters mapped to the zones must use be connected to the same VDS and participate in the same NSX Overlay Transport Zone. All hosts must be connected to the same L2 physical device.

Figure 509: Single Management-Zone Supervisor networking with NSX Segment Network



Supervisor networking with NSX Segment Network and Avi Load Balancer

NSX provides network connectivity to the objects inside the Supervisor and external networks. A Supervisor that is configured with NSX can use the NSX Load Balancer or the Avi Load Balancer.

The components of the Avi Load Balancer include the Avi Load Balancer Controller cluster, Service Engines (data plane) VMs and the Avi Kubernetes Operator (AKO).

The Avi Load Balancer Controller interacts with the vCenter to automate the load balancing for the VKS clusters. It is responsible for provisioning service engines, coordinating resources across service engines, and aggregating service engine metrics and logging. The Controller provides a Web interface, command-line interface, and API for user operation and programmatic integration.After you deploy and configure the Controller VM, you can deploy a Controller Cluster to set up the control plane cluster for HA.

The Service Engine, is the data plane virtual machine. A Service Engine runs one or more virtual services. A Service Engine is managed by the Avi Load Balancer Controller. The Controller provisions Service Engines to host virtual services.

The Service Engine has two types of network interfaces:

- The first network interface, `vnic0` of the VM, connects to the Management Network where it can connect to the Avi Load Balancer Controller.
- The remaining interfaces, `vnic1 - 8`, connect to the Data Network where virtual services run.

The Service Engine interfaces automatically connect to correct distributed port groups. Each Service Engine can support up to 1000 virtual services.

A virtual service provides layer 4 and layer 7 load balancing services for VKS cluster workloads. A virtual service is configured with one virtual IP and multiple ports. When a virtual service is deployed, the Controller automatically selects an ESX host, spins up a Service Engine, and connects it to the correct networks (port groups).

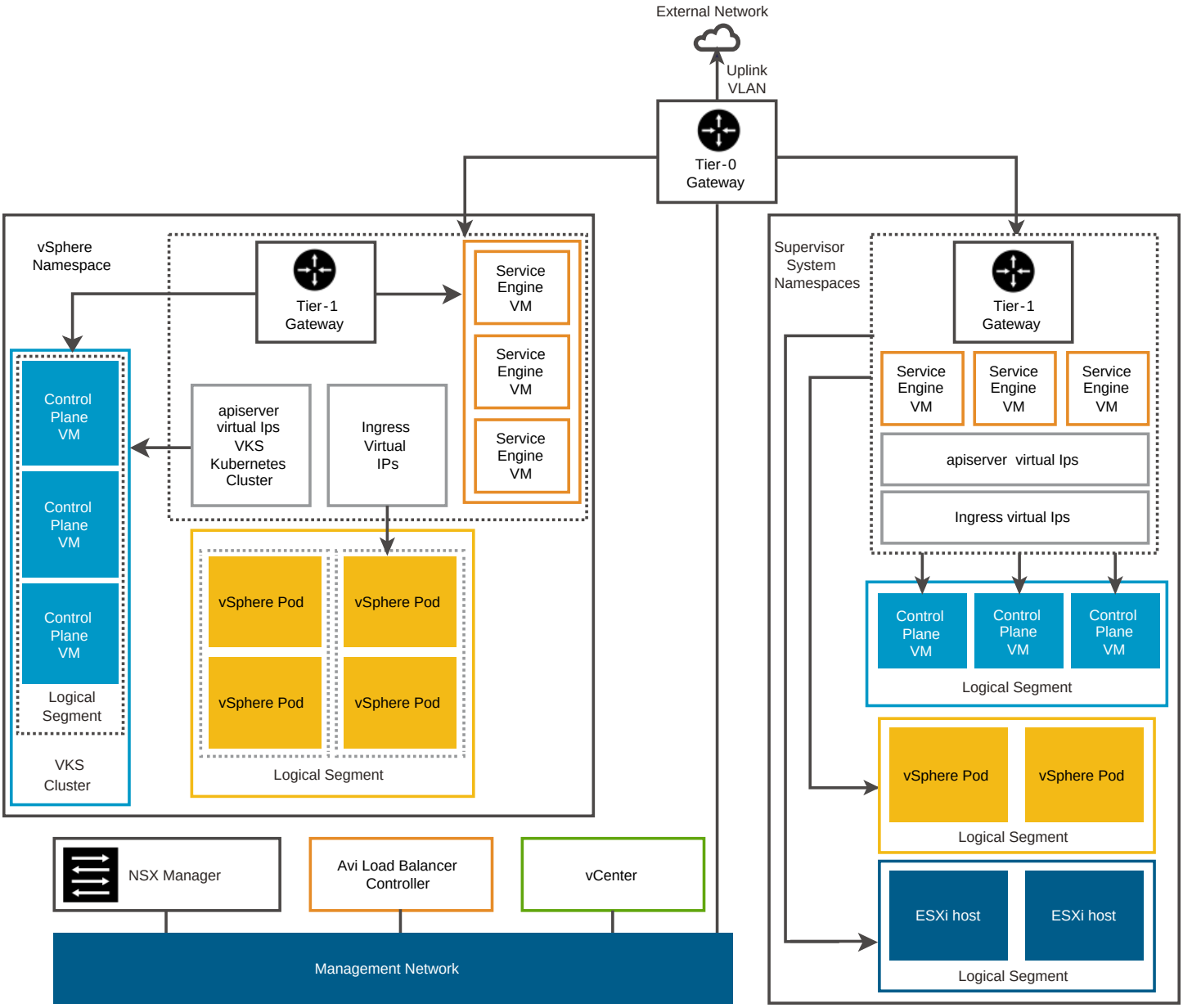
The first Service Engine is created only after the first virtual service is configured. Any subsequent virtual services that are configured use the existing Service Engine.

Each virtual server exposes a layer 4 load balancer with a distinct IP address of type load balancer for a VKS cluster. The IP address assigned to each virtual server is selected from the IP address block given to the Controller when you configure it.

The Avi Kubernetes operator (AKO) watches Kubernetes resources and communicates with the Avi Load Balancer Controller to request the corresponding load balancing resources. The Avi Kubernetes Operator is installed on the Supervisors as part of the enablement process.

For more information, see the Avi documentation.

Figure 510: Supervisor networking with NSX and Avi Load Balancer Controller



Important:

When you configure the Avi Load Balancer Controller in an NSX deployment, keep in mind the following considerations:

- You cannot deploy the Avi Load Balancer Controller in an vCenter Enhanced Linked Mode deployment. You can only deploy the Avi Load Balancer Controller in a single vCenter deployment. If more than one vCenter is linked, only one of them can be used while configuring the Avi Load Balancer Controller.
- You cannot configure the Avi Load Balancer Controller in a multi-tier tier-0 topology. If the NSX environment is set up with a multi-tier tier-0 topology, you can only use one tier-0 gateway while configuring the Avi Load Balancer Controller.

vSphere Kubernetes Service Architecture and Components

Checkout what is the architecture for VKS and how it integrates with the Supervisor and its components. Learn how networking and storage work for VKS clusters as well what is high-availability for VKS and which Supervisor deployment supports it.

VKS Architecture

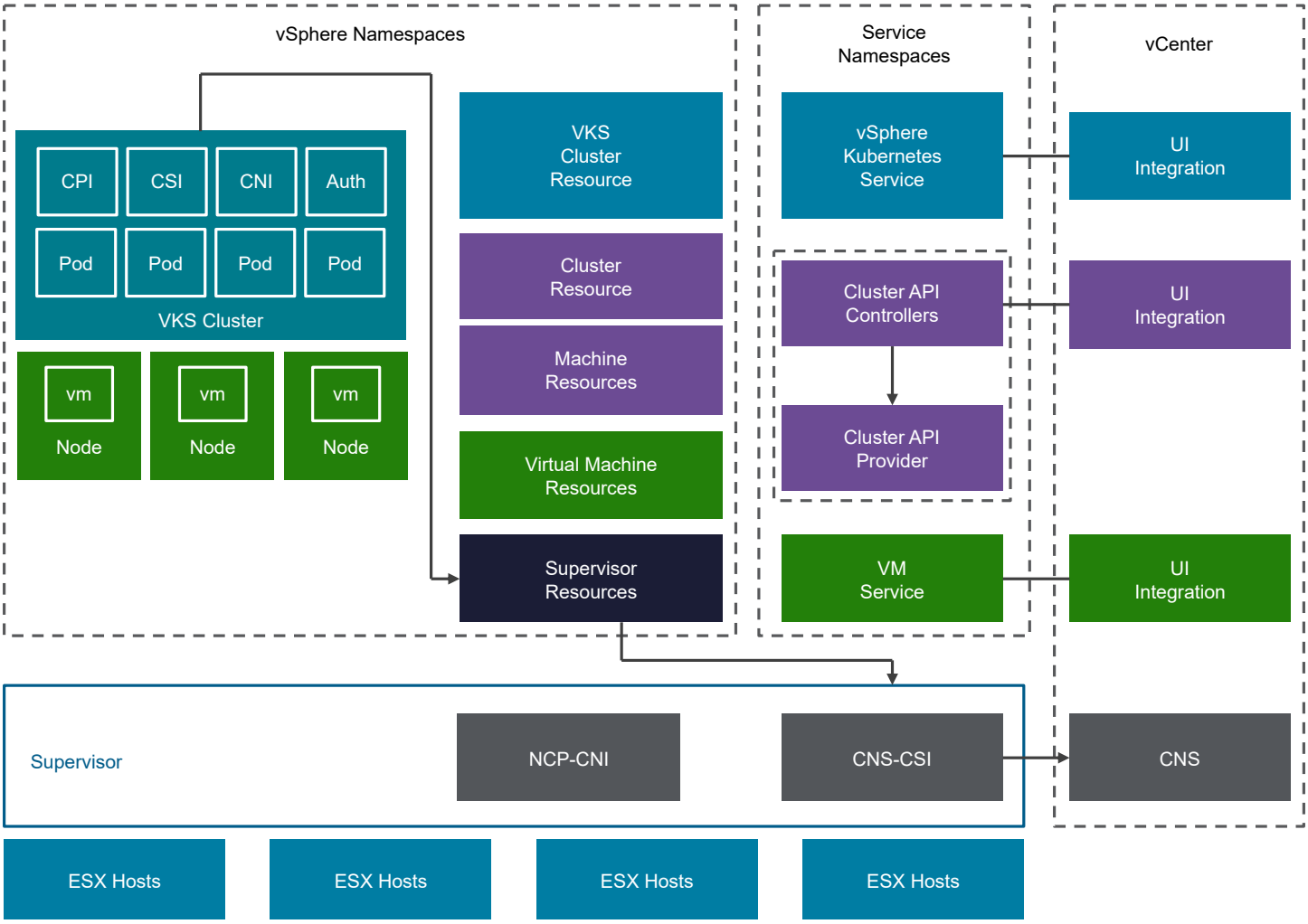
The VKS provides self-service life cycle management of VKS clusters. You use the VKS to create and manage VKS clusters in a declarative manner that is familiar to Kubernetes operators and developers.

VKS Components

The VKS exposes three layers of controllers to manage the lifecycle of a VKS cluster.

- The VKS provisions clusters that include the components necessary to integrate with the underlying vSphere Namespace resources. These components include a Cloud Provider Plug-in that integrates with the Supervisor. In addition, a VKS cluster passes requests for persistent volumes to the Supervisor, which is integrated with VMware Cloud Native Storage (CNS). See Storage Policies for Supervisor and VKS Clusters.
- The Cluster API provides declarative, Kubernetes-style APIs for cluster creation, configuration, and management. The inputs to Cluster API include a resource describing the cluster, a set of resources describing the virtual machines that make up the cluster, and a set of resources describing cluster add-ons.
- The Virtual Machine Service provides a declarative, Kubernetes-style API for management of VMs and associated vSphere resources. The Virtual Machine Service introduces the concept of a virtual machine class that represents an abstract reusable hardware configuration. The functionality provided by the Virtual Machine Service is used to manage the lifecycle of the control plane and worker node VMs hosting a VKS cluster.

Figure 511: VKS Architecture and Components



VKS Cluster Components

The components that run in a VKS cluster span four areas: Authentication and authorization, storage integration, pod networking, and load balancing.

- Authentication webhook: A webhook running as a pod inside the cluster to validate user authentication tokens.
- Container Storage Interface (CSI) Plug-in: A Paravirtual CSI plug-in that integrates with CNS through the Supervisor.
- Container Network Interface (CNI) Plug-in: A CNI plug-in that provides pod networking.
- Cloud Provider Implementation: Supports creating Kubernetes load balancer services.

VKS API

You use the VKS API to provision and manage VKS clusters. It is a declarative API that you invoke using kubectl and YAML. You can download the VMware expanded kubectl executable from the Supervisor API endpoint IP.

With a declarative API, instead of making imperative commands to the system, you specify the desired state of the VKS cluster: how many nodes, available storage, VM sizes, Kubernetes software version. The VKS does the work to provision a cluster that matches the desired state.

To call the VKS API, you invoke kubectl using a YAML file, which in turn invokes the API. After the cluster is created, you update the YAML to update the cluster.

VKS Cluster Networking

A VKS cluster provisioned by the VKS supports two CNI options: Antrea (default) and Calico. Both are open-source software that provide networking for cluster pods, services, and ingress.

VKS clusters provisioned by the VKS support the following Container Network Interface (CNI) options:

- Antrea
- Calico

Antrea is the default CNI for new VKS clusters. If you are using Antrea, you do not have to specify it as the CNI during cluster provisioning. To use Calico as the CNI you have two options:

- Specify the CNI directly in the cluster YAML. See v1alpha3 Example: VKS Cluster with Custom Network.
- Change the default CNI. See v1beta1 Example: VKS Cluster with Calico CNI.

> **Note:**
>
> The use of Antrea as the default CNI requires a minimum version of the OVA file for VKS clusters. See Updating VKS Clusters on Supervisor.

The table summarizes VKS cluster networking features and their implementation.

Table 1781: VKS Cluster Networking

Endpoint	Provider	Description
Pod connectivity	Antrea or Calico	Container network interface for pods. Antrea uses Open vSwitch. Calico uses the Linux bridge with BGP.
Service type: ClusterIP	Antrea or Calico	Default Kubernetes service type that is only accessible from within the cluster.
Service type: NodePort	Antrea or Calico	Allows external access through a port opened on each worker node by the Kubernetes network proxy.
Service type: LoadBalancer	NSX Load Balancer, Avi Load Balancer, Foundation Load Balancer	For NSX Load Balancer, one virtual server per service type definition. For Avi Load Balancer, refer to that section of this documentation.
Cluster ingress	Third-party ingress controller	Routing for inbound pod traffic; you can use any third-party ingress controller, such as Contour.
Network policy	Antrea or Calico	Controls what traffic is allowed to and from selected pods and network endpoints. Antrea uses Open vSwitch. Calico uses Linux IP tables.

High-Availability for VKS Clusters

You can provide high-availability to VKS clusters when they are deployed on vSphere Namespaces that have more than one vSphere Zones configured for running workloads.

A vSphere Zone maps to one or more vSphere clusters. When you configure a vSphere Namespace with multiple vSphere Zones (up to three), it utilizes the resources of the vSphere clusters that best fit the resource and datastore requirements of the vSphere Namespace. This protects your workloads running inside VKS clusters against failure on a vSphere cluster level. On a vSphere Namespace configured with only one vSphere Zone, high-availability for VKS clusters is provided on an ESX host level by vSphere HA.

In a multi-zonal vSphere Namespace, control plane nodes of VKS clusters are automatically placed across the vSphere Zones. However, you control how VKS cluster worker nodes are spread across zones. You must define a NodePool object

for the worker nodes of VKS clusters and map each vSphere Zone to a FailureDomain within each NodePool. This way, Cluster API takes care to spread the worker nodes across vSphere Zones accordingly.

VKS Authentication

As a DevOps engineer, you access Supervisor to provision VKS clusters. You only have access to the namespaces where you have permissions set by the VI administrator.

For information on how to connect to VKS clusters, see Connecting to vSphere Supervisor and VKS Clusters.

Storage for Supervisor and VKS Clusters

The Supervisor and VKS clusters use vSphere storage to serve two distinct needs: infrastructure storage for internal platform components, and persistent storage for workloads. Both depend on vSphere Storage Policy-Based Management (SPBM) to control placement, protection, and access to storage resources across VMFS, NFS, and vSAN datastores.

Storage Topologies for vSphere Zones

When you deploy a Supervisor across vSphere Zones, each zone acts as an independent fault domain. A failure in one zone does not affect the others. The Supervisor supports three types of datastore topologies for zonal deployments.

Zonal datastore

A zonal datastore is local to a single vSphere Zone. It is attached to all the ESX hosts that are part of the underlying cluster.

- It can replicate data to provide higher availability within the zone.
- Its availability depends on the zone. If the zone goes down, the zonal datastore also goes down.
- Examples of a zonal datatore include: non-stretched vSAN, non-stretched vSAN storage cluster, non-stretched VMFS or NFS external storage. A stretched cluster can also have a single vSphere zone.

Cross-zone datastore

A cross-zone datastore spans multiple vSphere Zones. It is attached to ESX hosts from all the zones it covers.

- A cross-zone datastore uses a single logical storage pool, such as a vSAN stretched cluster, that is accessible across all zones.
- It can be stretched storage, for example, stretched vSAN.
- It can be non-stretched storage that pools hosts from multiple zones, for example, non-stretched vSAN storage cluster.
- It typically replicates data across zones.
- It stays available if one of the zones goes down.
- Examples of a cross-zonal datastore: stretched vSAN, stretched vSAN storage cluster, metro storage cluster, non-stretched vSAN or vSAN storage cluster deployed from hosts across multiple zones.

Multi-zone datastore

A multi-zone datastore uses independent storage instances, such as a vSAN storage cluster or HCI cluster, across multiple vSphere Zones.

- Zones can be located in different racks, rooms, or buildings.
- Unlike the cross-zonal datastores, the storage layer does not provide replication or failover across zones. Applications deployed in multi-zone topologies must have built-in, application-level high availability (HA).
- It is ideal for modern, distributed applications like Cassandra, PostgreSQL, or MongoDB that manage their own data replication.
- StatefulSet replicas are distributed across zones to ensure high availability at the application layer.
- Zone-aware storage policies ensure that pods and volumes stay co-located within the same zone.
- If a zone fails, the specific datastore in that zone becomes unavailable, but the application remains available through replicas running in other zones.

When you work with zonal and cross-zonal datastores in Supervisor, avoid the following misconfigurations:

- A zonal datastore is attached to other zones.
  Example: A zonal vSAN datastore in Zone 1 is attached to Zone 2 and Zone 3 using vSAN remote datastore.
- A zonal datastore is only partially attached to ESX hosts in the zone.

Example: A VI administrator accidentally unmounts a datastore from a host.
Example: A new host is added to the zone, but the administrator forgets to attach all zonal datastores to it.

- A cross-zonal datastore is only partially attached to ESX hosts in multiple zones.
  Example: A datastore is accidentally unmounted from one of the hosts.
  Example: A new host is added to a zone, but the administrator forgets to attach all cross-zonal datastores to it.
- Editing storage policies to switch between zonal and cross-zonal compatibility.
  This action changes the storage class behavior and affects DevOps.
- Storage policies that match both zonal and cross-zonal datastores.
  DevOps cannot tell whether the volume they are getting is zonal or cross-zonal.
- A multi-zone topology is used for an application without built-in replication.
  If the zone fails, the data is lost because there is no infrastructure-level replication.

Storage Policies for vSphere Zones

When you configure storage policies for datastores to be used with zones, follow these best practices:

- Make sure that a storage policy matches either zonal datastores only or cross-zonal datastores only. Mixing both in a single policy leads to unpredictable behavior.
- Use a policy with zonal constraints to target zonal datastores.
  The system guarantees compatibility with zonal datastores and catches any misconfigurations during placement.
- If you create a policy without zonal constraints to use with zonal datastores, ensure it remains compatible with zonal datastores only.
  You must verify that no cross-zonal datastores match this policy.
- If you create a policy that matches cross-zonal datastores only, ensure it always remains compatible with cross-zonal datastores.
  Prevent any zonal datastore from matching this policy.
- If you use multi-zone datastores, ensure that datastores are attached to every host in every cluster within the zone. Inconsistent attachment causes placement failures.

Storage Policies for Supervisor and VKS Clusters

Supervisor and VKS clusters use storage policies to integrate with storage available in your vSphere environment. The policies represent datastores and manage storage placement of such components and objects as control plane VMs, vSphere Pod ephemeral disks, and container images. You might also need policies for storage placement of persistent volumes and VM content libraries. If you use VKS clusters, the storage policies also dictate how the VKS cluster nodes are deployed.

Storage policies support any shared datastores in your environment, such as VMFS, NFS, vSAN, and vSAN ESA.

Depending on your vSphere storage environment and the needs of DevOps, you can create several storage policies for different classes of storage. When you enable a Supervisor and set up vSphere Namespaces, you can assign different storage policies to be used by various objects, components, and workloads.

For information about creating storage policies, see Create Storage Policies.

For general information about storage policies, see the *Storage Policy Based Management* chapter in the *vSphere Storage* documentation.

Storage Policies for a Supervisor

A vSphere storage policy is a set of rules that specifies storage requirements, such as performance, protection, capacity, and encryption, for virtual machines and Kubernetes persistent volumes. Storage policies let administrators automate storage placement and ensure that VMs and virtual disks reside on datastores that meet their specific requirements.

The Supervisor uses storage policies to control placement of the following platform components.

Control Plane VMs	The specialized VMs that run the Supervisor control plane.

Ephemeral Virtual Disks	Temporary disks used by vSphere Pod during execution to store short-lived Kubernetes objects, such as container logs. This storage persists across container restarts but is automatically destroyed when the pod reaches the end of its lifecycle.
Image Virtual Disks	Disks that store container images that are pulled from your container registry. The Image Service transforms these images into virtual disks to be mounted by running pods, and caches them locally to accelerate subsequent pod deployments.
VM Content Libraries	Repositories used to store and distribute VM and container images.

Assign dedicated storage policies to each component type. Separating platform infrastructure storage from persistent workload storage allows you to tune performance and availability settings independently.

To change storage settings, see Change Storage Settings on the Supervisor.

Storage Policies for Persistent Workload Storage

After you define a storage policy, you can assign it to one or more vSphere Namespace. The Supervisor converts each assigned policy into a matching Kubernetes StorageClass that appears automatically in the namespace and propagates to any VKS clusters deployed within it.

Each assigned storage policy determines which datastores the namespace can access for persistent volume provisioning. Optionally, you can set a storage consumption limit per policy per namespace to prevent a single namespace from exhausting shared datastore capacity.

Persistent storage is required by certain workloads to store data permanently. It can be used by vSphere Pods, VKS clusters, VMs, and other workloads that run on the namespace.

To assign a storage policy to a namespace, see Create and Configure a vSphere Namespace on the Supervisor. To change storage settings on the namespace, see Change Storage Settings on a vSphere Namespace.

Storage Policies for VKS Cluster

To provide persistent storage to the VKS clusters, an administrator configures storage policies that describe different storage requirements. The administrator then adds the storage policies to the namespace where the VKS cluster is deployed. Storage policies visible to the namespace determine which datastores the namespace can access and use for persistent storage. They also control how the cluster nodes and workloads are placed in the vSphere storage environment.

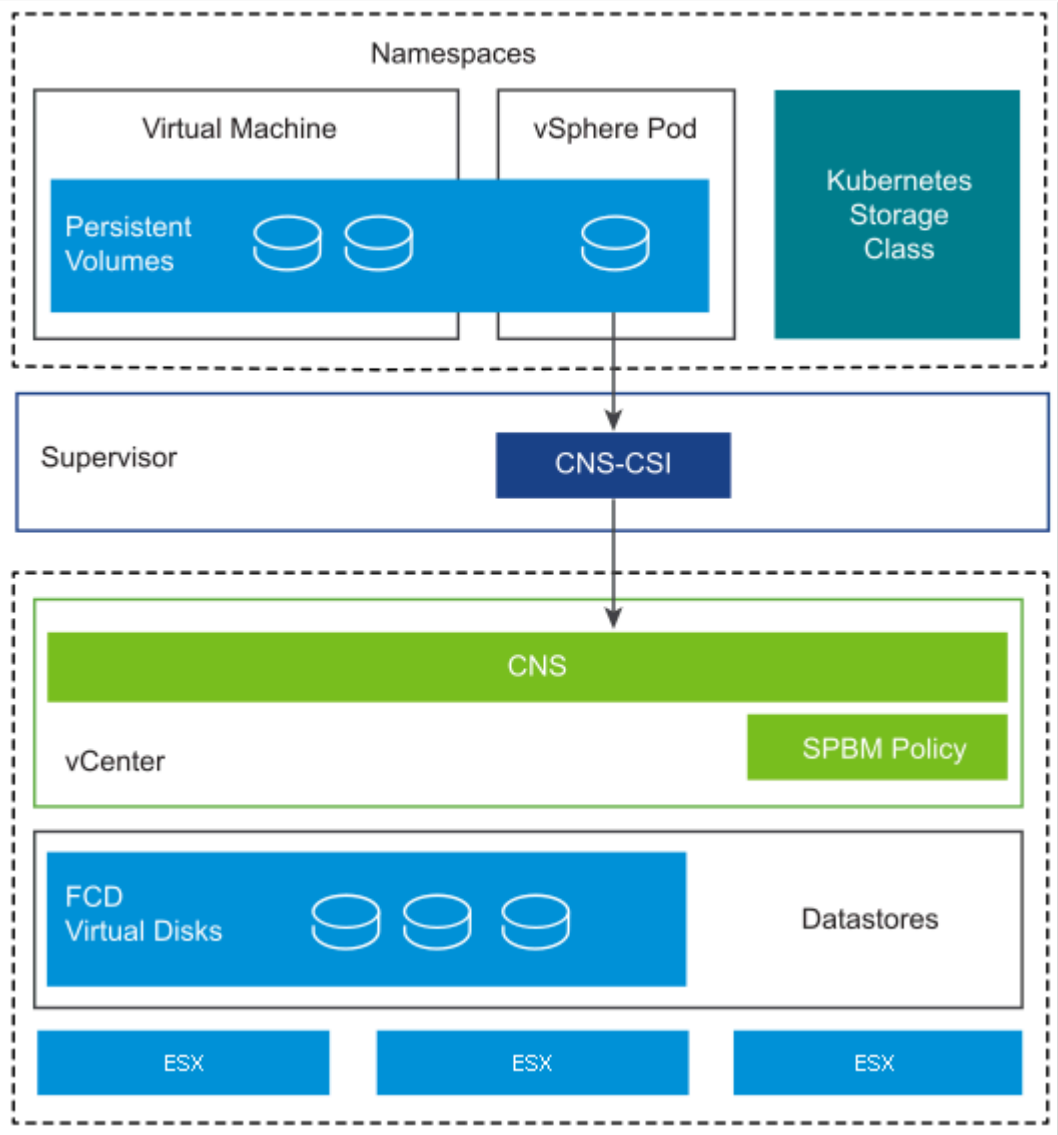
For specific details about storage policies for VKS clusters, see Create a vSphere Storage Policy for VKS Clusters.

Supervisor propagates each namespace storage policy to the VKS cluster as a Kubernetes StorageClass.

For more information about storage classes in VKS clusters, see Using Storage Classes for Persistent Volumes.

How Supervisor Integrates with vSphere Storage

Supervisor uses several components to integrate with vSphere storage.



Cloud Native Storage (CNS) on vCenter	The CNS component resides in vCenter. It is an extension of vCenter management that implements provisioning and lifecycle operations for persistent volumes. When provisioning persistent volumes, the component interacts with the vSphere First Class Disk functionality to create virtual disks that back the volumes. In addition, the CNS server component communicates with the Storage Policy Based Management to guarantee a required level of service to the disks. The CNS also performs query operations that allow VI administrators to manage and monitor persistent volumes and their backing storage objects through vCenter.
First Class Disk (FCD)	Also called Improved Virtual Disk. These disks reside on datastores and back ReadWriteOnce persistent volumes. When you use FCDs, keep in mind the following: • FCDs do not support NFS 4.x protocols. Instead, use NFS 3. • vCenter does not serialize operations on the same FCD. As a result, applications cannot simultaneously perform operations on the same FCD. Performing such operations as clone, relocate, delete, retrieve, and so on simultaneously from different threads causes unpredictable results. To avoid problems, applications must perform operations on the same FCD in a sequential order. • FCD is not a managed object and does not support a global lock protecting multiple writes to a single FCD. As a result, FCD doesn't support multiple vCenter instances managing the same FCD. If you need to use multiple vCenter instances with FCDs, you have the following options: – Multiple vCenter instances can manage different datastores. – Multiple vCenter instances do not operate on the same FCD.

Storage Policy Based Management	Storage Policy Based Management is a vCenter service that supports provisioning of persistent volumes and their backing virtual disks according to storage requirements described in a storage policy. After provisioning, the service monitors compliance of the volume with the storage policy characteristics. For more information about Storage Policy Based Management, see the *About Storage Policy Based Management* chapter in the *vSphere Storage* documentation.
vSphere CNS-CSI	The vSphere CNS-CSI component conforms to the Container Storage Interface (CSI) specification, an industry standard designed to provide an interface that container orchestrators like Kubernetes use to provision persistent storage. The CNS-CSI driver runs in the Supervisor and connects vSphere storage to Kubernetes environment on a namespace. The vSphere CNS-CSI communicates directly with the CNS component for all storage provisioning requests that originate from the namespace.

Functionality Supported by vSphere CNS-CSI

The vSphere CNS-CSI component that runs in the Supervisor supports multiple vSphere and Kubernetes storage features. However, certain limitations apply.

Supported Functionality	vSphere CNS-CSI with Supervisor
CNS Support in the vSphere Client	Yes
Enhanced Object Health in the vSphere Client	Yes (vSAN only)
Dynamic Block Persistent Volume (ReadWriteOnce Access Mode)	Yes
Dynamic File Persistent Volume (ReadWriteMany Access Mode)	No
vSphere Datastore	VMFS, NFS, vSAN (including vSAN ESA), vSAN storage cluster
vSAN Stretch Cluster	Yes
Metro Cluster	Yes **Note:** Work with external storage array vendors to get guidance on setting up Supervisor on metro clusters.
Static Persistent Volume	Yes
Encryption	Yes
Offline Volume Expansion	Yes
Online Volume Expansion	Yes
Volume Topology and Zones	Yes
WaitForFirstConsumer	Yes
Windows	No
VolumeHealth	Yes
Storage vMotion with Persistent Volumes	No for attached volumes **Note:** You can migrate detached volumes using the vSphere Client.
XFS File System	No
Raw Block Volume	No
VCHA	Yes
Multi vCenter	No
CNS Manager	Yes
Storage DRS	No
Volume Snapshots	Yes
Full Clone from Volume Snapshot	Yes
4kn Support	No

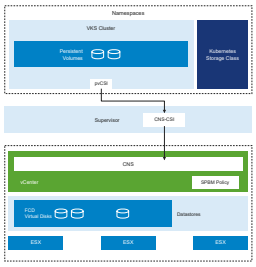
Storage for VKS Clusters

VKS clusters, as some other components and workloads that run in vSphere Namespace require persistent storage. They consume storage through the same Supervisor storage infrastructure but use a dedicated CSI driver called Paravirtual CSI (pvCSI) that runs inside each VKS cluster.

How VKS Clusters Integrate with vSphere Storage

To integrate with the Supervisor and vSphere storage, VKS clusters use Paravirtual CSI (pvCSI).

The pvCSI is a version of the vSphere CNS-CSI driver modified for VKS clusters. The pvCSI runs inside the VKS cluster and is responsible for all storage requests originating from the VKS cluster. The requests are delivered to the CNS-CSI in the Supervisor, which then propagates them to CNS in vCenter. As a result, the pvCSI does not have direct communication with the CNS component, but instead relies on the CNS-CSI for any storage provisioning operations. Unlike the CNS-CSI, the pvCSI does not require infrastructure credentials. It authenticates with a service account in the namespace.



For more details on how VKS clusters integrate with storage, see Using vSphere Namespaces with VKS Clusters.

Functionality Supported by pvCSI

The pvCSI component that runs in the VKS cluster supports a number of vSphere and Kubernetes storage features.

Supported Functionality	pvCSI with VKS Cluster
CNS Support in the vSphere Client	Yes
Enhanced Object Health in the vSphere Client	Yes (vSAN only)
Dynamic Block Persistent Volume (ReadWriteOnce Access Mode)	Yes
Dynamic File Persistent Volume (ReadWriteMany Access Mode)	Yes
vSphere Datastore	VMFS, NFS, vSAN (including vSAN ESA), vSAN storage cluster
vSAN Stretch Cluster	Yes
Metro Cluster	Yes **Note:** Work with external storage array vendors to get guidance on setting up Supervisor on metro clusters.
Static Persistent Volume	Yes
Encryption	Yes
Offline Volume Expansion	Yes
Online Volume Expansion	Yes
Volume Topology and Zones	Yes
WaitForFirstConsumer	Yes
Windows	Yes
VolumeHealth	Yes

Supported Functionality	pvCSI with VKS Cluster
Storage vMotion with Persistent Volumes	No for attached volumes **Note:** You can migrate detached volumes using the vSphere Client.
XFS File System	Yes
Raw Block Volume	Yes
VCHA	Yes
Multi vCenter	Yes
CNS Manager	Yes
Storage DRS	No
Volume Snapshots	Yes For information, see Creating Snapshots in a VKS Cluster.
Full Clone from Volume Snapshot	Yes
4kn Support	Yes

Create Storage Policies for Supervisor

Before you deploy a Supervisor, create storage policies to be used in the Supervisor and namespaces. The policies represent datastores and manage storage placement of such components and objects as Supervisor control plane VMs, vSphere Pod ephemeral disks, and container images. You might also need policies for storage placement of persistent volumes and VM content libraries. If you use VKS clusters, the storage policies also dictate how the VKS cluster nodes are deployed.

- Be familiar with information about storage policies in vSphere Supervisor, see About Storage Policies.
- Make sure that the datastore you reference in the storage policy is shared between all ESX hosts in the cluster. Any shared datastores in your environment are supported, including VMFS, NFS, or vSAN.
- For details about storage policies for VKS clusters, see Create a vSphere Storage Policy for VKS Clusters.
- For specific requirements in vSAN stretched cluster environments, see Configuring Storage Policy for vSphere Supervisor on vSAN Stretched Cluster.
- Required privileges:**VM storage policies** > **Update** and**VM storage policies** > **View**.

Depending on your vSphere storage environment and the needs of DevOps, you can create several storage policies for different classes of storage. For example, if your vSphere storage environment has three classes of datastores, Bronze, Silver, and Gold, you can create storage policies for all datastore types.

When you enable a Supervisor and set up namespaces, you can assign different storage policies to be used by various objects, components, and workloads.

> **Note:**
> Storage policies that you create for a Supervisor or for a namespace in a one-zone Supervisor do not need to be topology aware. Do not enable consumption domain for those policies.
>
> Storage policies that you create for a namespace in a three-zone Supervisor must be topology aware and have the consumption domain enabled in Step 4b. The three-zone namespace prevents you from assigning storage policies that are not topology aware.

The following example creates the storage policy for the datastore tagged as Gold.

1. Add tags to the datastore.
   a) Right-click the datastore you want to tag and select **Tags and Custom Attributes** > **Assign Tag**.
   b) Click **Add Tag** and specify the tag's properties.

Property	Description
Name	Specify the name of the datastore tag, for example, `Gold`.
Description	Add the description of the tag. For example, `Datastore for Kubernetes objects`.
Category	Select an existing category or create a new category. For example, `Storage for Kubernetes`.

2. In the vSphere Client, open the **Create VM Storage Policy** wizard.
   a) Click **Menu** > **Policies and Profiles**.
   b) Under **Policies and Profiles**, click **VM Storage Policies**.
   c) Click **Create VM Storage Policy**.
3. Enter the policy name and description.

Option	Action
vCenter	Select the vCenter instance.
Name	Enter the name of the storage policy, for example, `goldsp`.
K8s Compliant Name (optional)	Enter the name complaint with the RFC 1123 Kubernetes standard. If you do not enter this name, the system generates a unique valid Kubernetes compliant name. You cannot change this name later.
Description	Enter the description of the storage policy.

4. Follow the prompts to the **Policy structure** page.
5. On the **Policy structure** page, select the following options and click **Next**.

   Note:

   If you are creating a storage policy for a vSphere Pod and the policy has VMFS storage rules enabled, set the **Volume allocation type** to **Fully initialized**.

   a) Under **Datastore specific rules**, enable tag-based placement rules.
   b) To create a topology aware policy, under **vSphere Kubernetes Service Storage topology**, select **Enable Zonal topology for Supervisor**.

      This step is necessary only if you are creating topology aware policies to be used for persistent storage on a namespace in a three-zone Supervisor.
6. On the **Tag based placement** page, create the tag rules.

   Select the options using the following example.

Option	Description
Tag category	From the drop-down menu, select the tag's category, such as **Storage for Kubernetes**.
Usage option	Select **Use storage tagged with**.
Tags	Click **Browse Tags**, and select the datastore tag, for example, **Gold**.

7. On the **Storage compatibility** page, review the list of datastores that match this policy.

   In this example, only the datastore that is tagged as Gold is displayed.

8. On the **Review and finish** page, review the storage policy settings and click **Finish**.

The new storage policy for the datastore tagged as Gold appears on the list of existing storage policies.

After creating storage policies, a VI administrator can perform the following tasks:

- Assign the storage policies to the Supervisor. The storage policies configured on the Supervisor ensure that the control plane VMs, pod ephemeral disks, and container images are placed on the datastores that the policies represent.
- Assign the storage policies to the vSphere Namespace. Storage policies visible to the namespace determine which datastores the namespace can access and use for persistent volumes. The storage policies appear as matching Kubernetes storage classes in the namespace. They are also propagated to the VKS cluster on this namespace. DevOps engineers can use the storage classes in their persistent volume claim specifications. See Create and Configure a vSphere Namespace.

Best Practices for Storage Configuration and Policies in Supervisor and VKS Environments

Follow these best practices when you configure storage and create custom storage policies for your Supervisor and VKS clusters.

General Storage Policy Best Practices

Name storage policies clearly and use Kubernetes-compatible names	Use names that reflect the policy purpose and service level. For example, `prod-gold-raid5` or `dev-silver-noreplication`. When a single vCenter manages multiple Supervisor clusters, include a cluster identifier to distinguish policies across environments. Storage policy names become Kubernetes StorageClass names directly. Names must follow RFC 1123 requirements: • Lowercase alphanumeric characters and hyphens (-) only • Maximum 80 characters • No consecutive hyphens • No leading or trailing hyphens • Unique within vCenter Server
Create separate policies for each workload class	Define distinct policies per performance tier, availability level, redundancy requirement, encryption need, and replication tier. A single general-purpose policy cannot meet the differing SLA requirements of production, development, and test workloads.
Match every policy definition to actual datastore capabilities	Verify that the capabilities defined in a storage policy, such as performance tier, encryption, replication, exist on the target datastores. A policy referencing capabilities a datastore cannot satisfy causes provisioning failures. For VMFS and NFS datastores, apply tags based on performance type or replication capability, then build placement rules from those tags.
Add a zonal topology attribute to all policies when vSphere Zones are in use	Set the Storage topology type to Zonal on every storage policy assigned to a zonal Supervisor or three-zone namespace. Zonal topology ensures the system schedules a pod's persistent volume in the same availability zone as the pod itself. Without this attribute, volume attachment fails when a pod lands in a different zone from its volume. Three-zone namespaces block assignment of any policy that is not topology-aware.
Assign at least one policy as the default StorageClass for each namespace	Mark one storage policy per namespace as the default Kubernetes StorageClass. Any PersistentVolumeClaim (PVC) submitted without a storageClassName automatically use this default. Select a policy that reflects the preferred performance and availability requirements for that namespace.
Set a storage consumption limit per policy per namespace	Apply a consumption limit to every storage policy assigned to a namespace. Without a limit, one namespace can exhaust a shared datastore and degrade other workloads. Limits enforce predictable capacity allocation and surface unexpected growth early.
Run compliance checks after every storage infrastructure change	After any infrastructure change, such as adding hosts, extending datastores, modifying policy definitions, verify that persistent volumes and virtual disks still comply with their assigned policies. Non-compliant objects mean the underlying storage no longer satisfies the policy. Reapply or update the policy to restore compliance.
Remove storage policies only after	Before removing a storage policy from vCenter or from a namespace, confirm that no PVC in the namespace uses the corresponding StorageClass. Removing a policy leaves the StorageClass in

confirming no PVCs reference them	Kubernetes, but it remains non-functional. New provisioning requests against that StorageClass fail, and existing workloads can behave unpredictably.
Configure anti-affinity rules for control plane VMs on non-vSAN datastores	On vSAN datastores, storage policy FTT settings distribute control plane VM data across hosts automatically. On VMFS or NFS datastores, no such placement control exists at the storage layer. Create DRS VM-VM anti-affinity rules to ensure that Supervisor control plane VMs distribute across different hosts in the cluster. Without these rules, all control plane VMs can land on a single host, creating a single point of failure for the entire Supervisor. **Note:** This step is not required for vSAN-backed Supervisor deployments.
Use late-binding storage classes in a namespace configured across a multi-zone Supervisor	If a namespace is configured across a multi-zone Supervisor, enforce the use of late-binding storage classes for all stateful workloads, including VKS worker node pools, guest cluster applications, and VMs deployed via the VM Service. Delaying disk provisioning ensures perfect data locality. It also prevents cross-zone mounting failures where an immediate volume is accidentally provisioned in Zone A, but its consuming compute resource is forced to run in Zone B due to infrastructure capacity constraints. See Using Storage Classes for Persistent Volumes.

vSAN-Specific Storage Policy Best Practices

vSAN Original Storage Architecture (OSA) uses a two-tier model with dedicated cache and capacity devices organized into disk groups. This architecture is designed for environments with mixed flash and magnetic disk hardware. vSAN Express Storage Architecture (ESA) uses a single-tier pool where all NVMe devices contribute to both capacity and performance, delivering higher throughput and more efficient erasure coding.

Use RAID-1 mirroring for vSAN OSA clusters	vSAN Original Storage Architecture performs best with RAID-1 mirroring. Size the cluster to support automatic rebuild after a host failure: • Minimum 4 hosts for automatic rebuild after 1 host failure • Minimum 7 hosts for automatic rebuild after 2 simultaneous host failures
Use erasure coding for vSAN ESA clusters	vSAN Express Storage Architecture is optimized for erasure coding. Enable Auto Policy Management where available — it automatically adjusts protection as the cluster changes and removes the need for manual policy tuning. When Auto Policy Management is unavailable, choose the erasure coding level based on your failure tolerance requirement: • RAID-5: minimum 5 hosts for automatic rebuild after 1 failure • RAID-6: minimum 8 hosts for automatic rebuild after 2 failures
Define a dedicated storage policy for the Improved Virtual Disk (FCD) home	Assign a highly resilient storage policy to first class disk (FCD) home objects. The FCD home stores metadata and configuration files for the independent virtual disks that back persistent volumes. Leaving FCD home objects on the default VM storage policy can expose persistent volume metadata to data loss if a host or disk fails. In vSAN ESA environments with Auto Policy Management enabled, the recommended policy is applied automatically.
Enable Force Provisioning on stretch cluster policies	On vSAN stretch cluster deployments, set **Force Provisioning** on all Supervisor storage policies. Force Provisioning ensures that volume creation succeeds during planned site maintenance or a partial site outage. Without it, provisioning requests fail when the cluster cannot fully satisfy policy requirements due to reduced site capacity. For information about vSAN stretch cluster deployments, see Overview of Running vSphere Supervisor on vSAN Stretched Cluster.

Managing Thin-Provisioned Storage and Space Reclamation

All virtual disks that back persistent volumes in a Supervisor or VKS cluster are thin-provisioned by default. Because thin-provisioned disks grow without explicit allocation events, you should monitor datastore capacity regularly and configure alarms to catch overprovisioning before it causes provisioning failures.

Maintain 25–30% free space on vSAN datastores to accommodate rebalancing, host rebuild operations, and storage policy changes. For VMFS and NFS datastores, maintain at least 15–20% free space.

For space reclamation configuration, follow the guidance for your datastore type.

VMFS	On VMFS datastores, deleted blocks are returned to the datastore through space reclamation. For information, see Space Reclamation on vSphere VMFS Datastores.
NFS	Space reclamation on NFS requires a VAAI-NAS plugin from the storage vendor that supports UNMAP. Without it, deleted blocks are not reclaimed. For information, see vSphere Hardware Acceleration on NAS Devices.
vSAN	vSAN processes TRIM and UNMAP automatically. No additional configuration is required. Enable Reserved Capacity on vSAN datastores to prevent out-of-space conditions during unexpected workload spikes. For details, see About Reserved Capacity in vSAN Cluster.

Deploying Supervisor with VCF Networking with VPC

For networking and security of vSphere Supervisor, you can use Virtual Private Cloud (VPC). VPC is the recommended workload network for deploying the VMware Cloud Foundation (VCF) stack as it is the supported network stack for VCF Automation environments. The Supervisor can be activated as an optional configuration during workload domain creation and by default, it uses VCF Networking with VPC. If you do not activate the Supervisor during the workload domain creation, you can deploy the Supervisor later with VPC from vCenter to gain the full cloud experience with VCF Automation.

The VPC networking framework is natively multi-tenant and is used by VCF Automation where each tenant organization can create a tenant (NSX Project) in NSX. You can deploy vSphere Namespaces in the NSX default space where the VI administrator creates the VPCs, or in a specific tenant (NSX Project). You can associate workload namespaces with VPCs in an NSX Project or in the NSX default space irrespective of where the vSphere Namespaces have been deployed.

For more information about VPC models, see Workload Networking Detailed Designs.

The NSX Container Plugin (NCP) automatically creates the following:

- A VPC for System Namespaces in the NSX System project.
- A Supervisor Service VPC for Supervisor Services
- An individual VPC per vSphere Namespaces that is separate from the above two if an existing VPC is not configured during namespace creation.

The shared network resources that include the VPC, load balancer, and SNAT IP are grouped in a namespace.

VPCs have three subnet types for workloads:

- Private - VPC subnets: This subnet is strictly isolated to its specific VPC using local IP ranges that can overlap with other VPCs. Workloads on these subnets can only communicate internally unless a NAT rule is explicitly configured for external access.
- Private Transit Gateway subnets: This subnet allows communication between different VPCs connected to the same Transit Gateway, but remains inaccessible from outside the physical network without a NAT rule. These subnets are primarily used by vSphere Pods.
- Public subnets: These subnets are accessible from other VPCs and outside the VPC. Workloads that are connected are assigned unique external IPs from a CIDR carved from an external IP block.

VPCs created for vSphere Namespaces are configured with default outbound NAT enabled allowing workloads on private networks to establish connectivity to outside resources. For more information about VPC, see NSX Virtual Private Clouds.

When you create a VPC in VCF Automation, you can choose to not configure a load balancer. Also, when you create a connectivity profile, you can deselect the default outbound SNAT. For more information see, Create a Virtual Private Cloud in VCF Automation.

Users with the Namespaces role in the VPC can create subnet CRs or SubnetSet CRs. The NSX operator creates NSX subnets in the backend for the two CRDs separately.

Default subnetSet custom resources are created automatically in the namespace which is used to connect the vSphere Pods and VMs when no subnet or subnetSet is specified.

VM Services VMs can be placed on any of the subnet types. By default they are placed on the Private subnet. The default VM SubnetSet and VM network interface cannot be changed. VPC and network constructs are exposed as Kubernetes custom resource definitions in the Supervisor API.

NSX Container Plugin (NCP) and the Avi Kubernetes Operator (AKO) provide networking for the vSphere Pods and VMs, subnet provisioning, IPAM, and service types network policy, ClusterIP, LoadBalancer, and Ingress. For more information, see NSX.

Supervisor Architecture with VCF Networking with VPC

VPCs are connected together by a Transit Gateway which provides connectivity to the datacenter through an external connection.

You can deploy a Supervisor in an environment using one of the following connections:

- Distributed Connection. A distributed connection is a simple connection that directly connects to the VLAN of the datacenter without requiring additional configuration on the physical environment. Distributed connections supporting a Supervisor require a Virtual Network Appliance (VNA) cluster.
- Centralized Connection. A centralized connection attaches the Transit Gateway to a Tier-0 Gateway for external connection. The connection uses the Tier-0 networking capabilities and services like BGP and requires an Edge Cluster.

A Transit Gateway can have multiple connections but all the connections must either be distributed connections or centralized connections. A Transit Gateway is called as a Distributed Transit Gateway or a Centralized Transit gateway depending on the type of connections that it uses.

For more information, see External Connections and Transit Gateways.

When you activate the Supervisor during workload domain creation, you can configure the Virtual Network Appliance (VNA) cluster, network connectivity, external transit gateway, and VPC external IP blocks for the Distributed Transit Gateway. For more information, see Create a New Workload Domain Using VCF Operations.

If you require to configure a Centralized Gateway, you must configure NSX Edge nodes.

> **Note:**
>
> The choice of Transit Gateway for the Supervisor does not limit VCF Automation connection options. Transit Gateways are specific to tenant organizations and when you create an organization through VCF Automation, you can select either centralized or distributed connections.

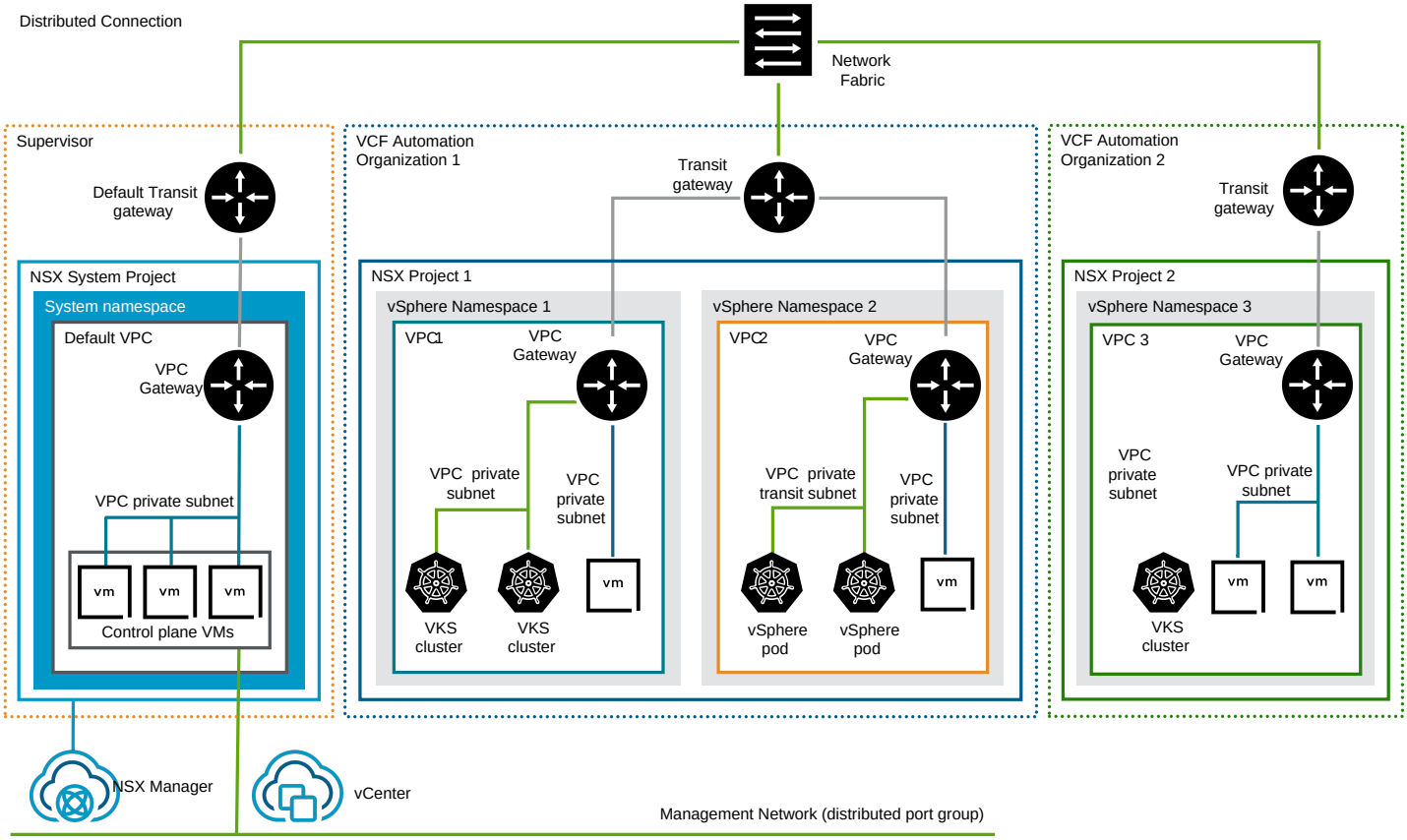
Supervisor networking with VPC with a Distributed Transit Gateway

A VPC with a Distributed Transit Gateway requires a VLAN available on all ESX hosts spanned by the Transit Gateway and connected VPCs. This VLAN and its CIDR block are used directly for IPs allowing outside connectivity.

The Distributed Transit Gateway does not require Edge Nodes, dynamic routing, or any specific configuration on the physical network. It allows a distributed model which scales out as you add hosts since traffic directly goes from the host to the VLAN. Services like DHCP or the ability to expose a workload with an external IP (1:1 NAT) are also distributed. For networking services like Load Balancer or outbound SNAT, the Virtual Network Appliance (VNA) is used. The VNA is multi-tenant and the same VMs can be used for multiple namespaces and multiple VCF Automation Organizations. For more information on VNA, see Transit Gateways.

The diagram shows the architecture of Supervisor with VCF Networking with VPC with a Distributed Transit Gateway.

Figure 512: Supervisor networking with VPC with Distributed Transit Gateway



VCF 9.1.1 introduces support for a VLAN-backed VPC or networking model without TEPs under the Distributed Transit Gateway. In this model, all traffic flows directly on physical VLANs without overlay encapsulation. It does not support Private networks as the IP scheme is limited to the CIDRs backed by physical VLANs.

The region to which the Supervisor belongs to must be homogenous. The network stack must be fully VLAN-backed (model without TEPs). For more information see, Managing Regions in Your VCF Automation.

As an administrator, you can pre-create subnets and assign them to vSphere Namespaces for specific workload types. Subnets are created within VPCs with CIDRs allocated from the VPC external IP blocks. SNAT is not supported in models without TEPs. Because of this restriction, deselect the default outbound SNAT when creating a connectivity profile. For more information about VPC and subnets, see NSX Virtual Private Clouds.

In namespaces with VLAN-backed VPC, the NSX operator does not automatically create a default network for pods or VMs. You must specify a shared subnet explicitly.

In namespaces with the full-stack VPC, if a Private Transit Gateway IPBlock and private CIDR blocks are configured, the NSX operator can automatically create default networks for pods and VMs. You need not set a shared subnet as the default network.

Note:

In the VCF 9.1.1 release, the VLAN-backed VPC model supports only greenfield deployments of Supervisor and VCF Automation.

Note:

vSphere Kubernetes Service 3.7.1 and earlier releases do not support the VLAN-backed VPC model. If you are using vSphere Kubernetes Service 3.7.1 or earlier versions on a 9.1.1 Supervisor, you cannot configure the VLAN-backed VPC for VKS workloads.

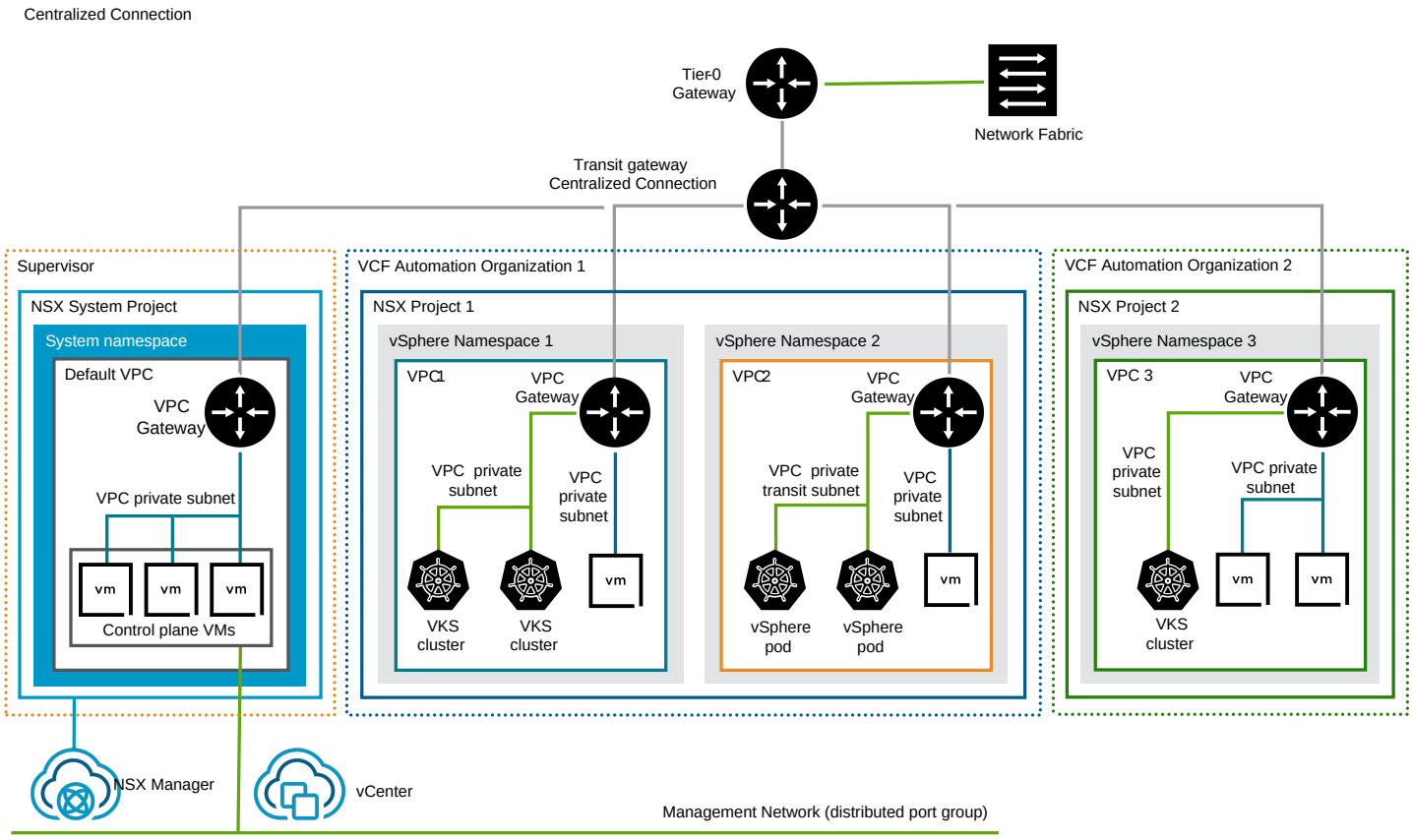
Supervisor networking with VPC with a Centralized Transit Gateway

A VPC with a Centralized Transit Gateway uses centralized connections where traffic is routed through a tier-0 gateway on NSX Edge nodes. Centralized connections allow stateful services and dynamic routing with the physical fabric (BGP) at Tier-0 level.

VCF 9.1 supports a tier-0 gateway in Active/Active configuration and a Centralized Transit Gateway in Active/Standby configuration with the required stateful services such as NAT for the Supervisor.

The diagram shows the architecture of Supervisor with a VPC with a Centralized Transit Gateway.

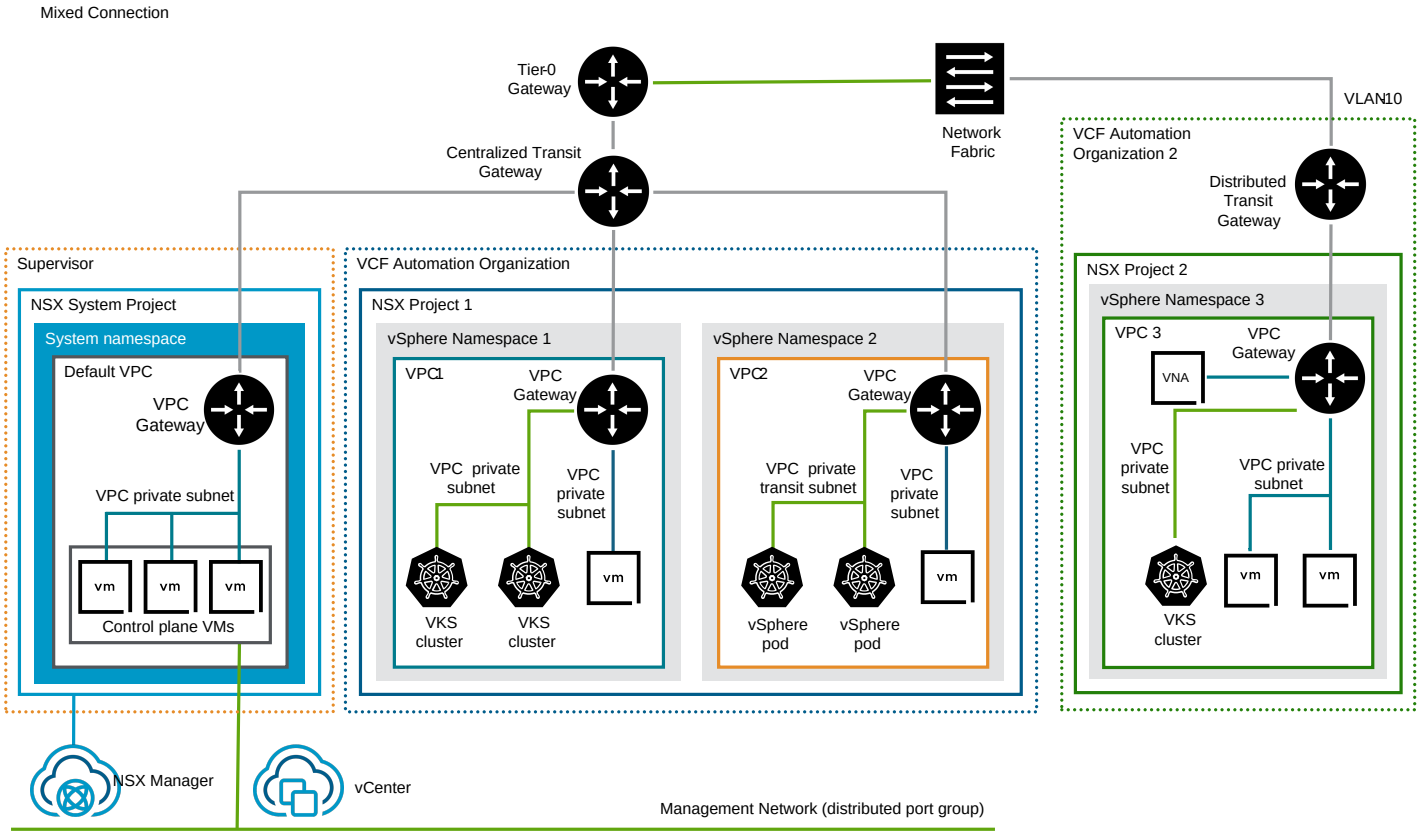
Figure 513: Supervisor networking with VPC with Centralized Transit Gateway



Supervisor networking with VPC with Distributed and Centralized Transit Gateways

The diagram shows the architecture of the Supervisor with VCF Networking with VPC with a mix of Distributed and Centralized Transit Gateways.

Figure 514: Supervisor networking with VPC with Distributed and Centralized Transit Gateways



Load Balancers

You can use either the VCF Native Load Balancer or the Avi Load Balancer. The load balancer that is selected during Supervisor activation depends on what you have configured with NSX.

You can only use one load balancer. If the Supervisor is configured with the VCF Native Load Balancer, a VPC using the Avi Load Balancer cannot be used.

For more information about configuring the Avi Load Balancer, see the Avi documentation.

Workflow to Activate Supervisor with VCF Networking with VPC

At a high level, the workflow to deploy Supervisor with VCF Networking with VPC is as follows:

1. Review the requirements for configuring a Supervisor with VCF Networking with VPC. See Requirements for Supervisor Deployment with VPC Networking.
2. The Supervisor is activated as an optional configuration during workload domain creation and it uses VCF Networking with VPC by default. If you do not activate the Supervisor during the workload domain creation, you can deploy the Supervisor later with VPC from vCenter to gain the full cloud experience with VCF Automation.
3. Configure the distributed port groups or VPC subnets for the Supervisor management network if you use the non-default ESX management network. For more information about subnets see NSX Virtual Private Clouds.
4. Optionally configure a Transit gateway with outside connectivity. When you activate the Supervisor , you have the option of configuring the external connection and gateway. You can select either a distributed connection or a centralized connection.
   a. For a Centralized Transit Gateway, verify that a tier-0 gateway with outside connectivity and an Edge cluster are configured. See Configure the Centralized Transit Gateway.
   b. For a Distributed Transit Gateway, verify that a VLAN-backed network with outside connectivity is available on the required hosts and the Virtual Network Appliance (VNA) with NAT, load balancing is deployed.
5. Optionally, install and configure the Avi Load Balancer.

See the Avi documentation.

6. Activate the Supervisor. See Deploy a Supervisor with VPC Networking.

Requirements for Supervisor Deployment with VPC Networking

Check out the system requirements for configuring a Supervisor with VPC networking.

Minimum Compute Requirements

The table lists the minimum compute requirements for enabling a Supervisor with VPC. You can deploy the Supervisor on one Management Zone or three Management Zones.

System	Minimum Deployment Size	CPU	Memory	Storage
vCenter 9.0	Small	2	21 GB	290 GB
vSphere clusters	• 1 or 3 vSphere clusters. For a Supervisor with three Management Zones, you need three clusters. • vSphere DRS and HA enabled on each vSphere Cluster. vSphere DRS must be in Fully Automate mode. • Independent storage and networking that is configured for each vSphere cluster.	Not applicable	Not applicable	Not applicable
ESX hosts 9.x	For each vSphere cluster: • Without vSAN: 1 ESX host with 1 static IP per host. Recommended number of hosts is 3. • With vSAN: 3 ESX hosts per cluster with at least 2 physical NICs. Recommended number of hosts is 4. **Note:** Make sure that the names of the hosts that join the clusters use lower case letters. Otherwise, the activation of the Supervisor might fail.	8	64 GB per host	Not applicable
NSX Manager	Medium	6	24 GB	300 GB
NSX Edge cluster	Required for the Centralized Connection.	8	32 GB	200 GB

System	Minimum Deployment Size	CPU	Memory	Storage
	Two node cluster of Large form factor.			
Virtual Network Appliance	Required for the Distributed VLAN Connection.	Not applicable	Not applicable	Not applicable
Kubernetes control plane VMs	• 1 for single Management Zone (3 recommended) • 3 for three Management Zones	4	16 GB	48 GB

Supervisor Minimum Networking Requirements

Check the Broadcom Product Interoperability Matrix for the supported NSX versions.

Note:

You cannot create IPv6 clusters with a vSphere 9.0 Supervisor.

VPC Networking Infrastructure Requirements

Table 1782: VPC Networking Requirements

Component	Minimum Quantity	Required Configuration
ESX TEP network	1	IP addresses for the tunnel endpoints (TEPS) and VMkernel on the ESX hosts. Communication between ESX host TEP must exist and support an MTU size greater than 1600. **Note:** This configuration is done (IP allocation, VMkernel creation for the TEP) at the VCF workload domain setup or import.
Centralized Connection Or Distributed Connection	1	Connection that defines the external connectivity of your gateway. If centralized, specify the tier-0 or VRF gateway that is used for outside connectivity.
VPC Connectivity Profile	1	A VPC connectivity profile that specifies the gateway that is connected to the connection. A transit gateway is created by default in each Project. The requirements for the VPC Connectivity Profile are as follows: • Private-Transit Gateway IP Blocks are configured. • Auto SNAT is enabled. • If the VPC Connectivity profile has a gateway connection type set to **Centralized Gateway**, verify that external IP Blocks are configured. Verify that the following conditions are met when you configure the external IP blocks: • The network must not exist as a VLAN or gateway on the physical network nor as a VLAN port group in vCenter. • It must not exist as an overlay in NSX.

Component	Minimum Quantity	Required Configuration
NSX Project	1	Optional as VPCs are available by default in NSX within the Default project The NSX project that you specify during Superviso r enablement is used when you create the vSphere Namespace and the VPC is created in this project. If the default project is not used, the NSX project must be created with the following: • An external connection • An external IP Block • Either an allocated Edge Cluster or a Virtual Network Appliance Cluster. This project is the tenant context that is specified during Supervisor enablement and used when the vSphere Namespace and related VPCs are auto-created. VPCs and namespaces can also be created in an NSX Project that is different from the Supervisor system namespaces.

Table 1783: Centralized Connection

Component	Minimum Quantity	Configuration
Edge TEP network	1 VLAN (optional) 2 IPs per Edge node	Optional network for the Edge VM TEPs that can be shared with the ESX TEP network. The Edge TEP must be able to communicate with the ESX TEP with an MTU size of more than 1600. The minimum requirement is a two Edge node cluster with a recommendation of two TEPs per Edge.
Tier-0 Uplink network and IPs	1 network 1 IP per Edge node for each network An additional IP if using Edge redundancy with HA VIP	A dedicated VLAN network connecting the Tier-0 to the physical infrastructure, that serves as the transit path for traffic entering or exiting the environment. This network must be available on vSphere clusters where the Edge node VMs reside. The requirements for the IP address of the Tier-0 uplink per VLAN network are as follows: • 1 IP per Edge node. • 1 additional IP if using Edge redundancy with HA VIP. • 3 IPs, if you use static routes and Edge redundancy.
Routing between Tier-0 and physical network	1	Ensure that the Tier-0 gateway can route traffic between the physical fabric and internal VPCs either with static routing or dynamic routing like BGP. Static routing: • Tier-0 gateway: On the Tier-0 gateway, configure the physical equipment as the default gateway. • Physical network: Add a static route to the external IP block allocated previously with the next hop as the HA VIP of the Tier-o gateway or an uplink. BGP routing: • Establish peering: Configure BGP neighbors between the Tier-0 gateway and your physical switches. • Ensure proper route advertisement. From the Tier-0 gateway, ensure that the Transit Gateway Static network types are redistributed through BGP.

Table 1784: Distributed Connection

Component	Minimum Quantity	Configuration
ESX TEP network	1 (Optional for VLAN VPC)	IP addresses for the tunnel endpoints (TEPS) and VMkernel on the ESX hosts. **Note:** These IP addresses are configured during VCF setup.
VLAN ID	1	A dedicated VLAN network is required for the external connectivity of the distributed gateway. This network must be available on vSphere clusters with workloads connecting to the VPC.
Gateway CIDR IPv4 Address	1	The gateway CIDR IP address should be the default gateway configured on the external router (TOR).

Workload Network Requirements

Component	Minimum Quantity	Required Configuration
Kubernetes services CIDR range	/16 Private IP addresses	A private CIDR range to assign IP addresses to Kubernetes services. You must specify a unique Kubernetes services CIDR range for each Supervisor. This range must not overlap with other ranges.
Private IP Block - Transit Gateway	/26 CIDR for the IP block	A private CIDR range for inter-VPC connectivity, for instance with Private - Transit Gateway subnets.. This range is not advertised by the tier-0 gateway. Default CIDR: 172.31.0.0/16 for private transit gateway. The CIDR can be changed as needed.
Private VPC CIDR range	/25 Private IP addresses	List of IPv4 CIDRs for IP allocation of Private - VPC subnets not advertised beyond the VPC. The VPC can have overlapping Private ranges. Default CIDR: 172.30.0.0/16 for private VPC. A /26 subnet from the range is reserved for the Avi Service Engine. The private IP blocks must not conflict with resources that exist in the datacenter to allow them to be accessed by workloads deployed in the private networks.

Create vSphere Zones for a Multi Zone Deployment with VPC

To activate a Supervisor, you must create at least one vSphere Zone that will be designated for the management plane of the Supervisor, where the control plane will be placed. To provide zone-level HA to the Supervisor control plane, you deploy the Supervisor on three vSphere Zones designated for management. In Supervisor with three management zones and HA for the control plane, meaning with three control plane VMs, each control plane VM is placed on a different Management Zone.

- Create one or three vSphere clusters to add to the zones:

  - 2 ESX hosts in each zone for NFS or FC storage. For HA, use 4 hosts
  - For storage with vSAN, the cluster must have 3 hosts. For HA, use 4 hosts.
  - For test of POC environments, each cluster must have at least 1 host, and 2 hosts when using vSAN.

    Note:

    Recommendation: Restrict direct access to the ESX host console. Without that restriction, an administrator could override vSphere Zone assignments, undermining the intended isolation and policy enforcement.

- Configure storage with vSAN or other shared storage solution for each cluster.
- Enable vSphere HA and vSphere DRS on Fully Automate or Partially Automate mode.
- All clusters in a zone must be connected to the same vSphere Distributed Switch (VDS). Each zone can use its own management network.

1. In the vSphere Client, navigate to vCenter.
2. Click **Configure**.
3. Select **vSphere Zones**.
4. Click **Add New vSphere Zone**.
5. Name the zone, for example `zone1` and add an optional description.
6. Select a vSphere clusters to add to the zone and click **Finish**.
7. Repeat the steps to create three vSphere Zones.

Create a Distributed Port Group

Create a distributed port groups for the Edge management VLAN.

1. In the vSphere Client, navigate to a data center.
2. In the navigator, right-click the distributed switch and select **Distributed Port Group** > **New Distributed Port Group**.
3. Enter a name for the management VLAN and click **NEXT**.

   For example, `DPortGroup`.
4. Configure **VLAN Type** as **VLAN Trunking**.
5. Enter the VLAN trunk range.

   For example, **100-200, 300-400**.
6. Select **Customize default policies configuration**.
7. Click **NEXT**.
8. Retain the default value in the**Security** and **Traffic shaping** settings.
9. In **Teaming and failover**, configure load balancing and active and standby uplinks.
10. Right-click the distributed switch and from the **Actions** menu, select **Distributed Port Group** > **Manage Distributed Port Groups**.
11. Select **Teaming and failover** and click **NEXT**.
12. Retain the default values in the **Monitoring** and **Miscellaneous** settings.
13. Click **OK** to complete the configuration of the port group.
14. Review the configuration and click **Finish** to create the port group.

Configure the Centralized Transit Gateway

Configure the centralized transit gateway for the NSX VPC.

1. In the vSphere Client, select the vCenter and select the **Networks** tab.
2. Select **Transit Gateways** and click **Setup Network Connectivity**.
3. In **Gateway Type**, select **Centralized Connectivity** and click **Next**.
4. Review and verify that you have completed the networking prerequisites checklist, and click **Continue**.
5. Configure the Edge cluster.
   a) Enter a name for the edge cluster.
   b) The default MTU value is 1700.
   c) Select **Large** form factor.
6. Click ADD to add the Edge nodes. Add a minimum of two Edge nodes to deploy an Edge cluster.
   a) Enter the Edge node name.
   b) In **vSphere Cluster**, select the cluster where you want to deploy the NSX Edge node.
   c) Select the resource pool for the NSX Edge node.
   d) Edge host affinity enables the NSX Edge to be pinned to a given host or group of hosts. Select **Yes** if you want the NSX Edge to be pinned to a host group affinity.
   e) In the Management IP section, configure the management network for the NSX Edge node.

Option	Description
IP Allocation	Select static IP allocation for the NSX Edge node management interface.
Port Group	For management workload, select the VM management port group you created. If a different port group is required, then create the port group before performing this operation.
Management IP CIDR	Enter a CIDR for Management IP if you have selected **Static** option for IP allocation. Each node must have a unique management IP.
Default Gateway	Enter the default gateway address.

   f) Select **Use the host overlay network configuration from the selected vSphere cluster** to get the existing overlay network configuration (such as VLAN ID and TEP IP assignment) is automatically from the hosts in the selected vSphere cluster. This checkbox is automatically selected if all the prerequisites for a prepared vSphere cluster are met for the NSX Edge deployment.
   g) In Edge Node Uplink Mapping, select the active and standby PNICs.
   h) Enter the VLAN ID for Tunnel Endpoints (TEP). This field is automatically populated if the **Use the host overlay network configuration from the selected vSphere cluster** checkbox is selected.
   i) Select the IP allocation method for TEP. Define a vLAN ID for the Host TEP network.

      You can select from the following options. Based on the IP allocation method, provide the other required details.

      - DHCP - DHCP Server provides IP addresses to the hosts. The external DHCP server allocates the TEP IP.
      - IP Pool - Select an IP Pool for allocations.
      - Static IP List - Provide values IPv4 Static List, Static IPv4 Gateway, and IPv4 Subnet Mask.

      This field is automatically populated if the **Use the host overlay network configuration from the selected vSphere cluster** checkbox is selected.
   j) Select the IP pool from the drop-down list.
   k) Enter the static gateway address and subnet mask.
   l) Click **Apply** to create an NSX Edge node.
7. Click **Clone** to clone the NSX Edge node.

   Configure the settings that cannot be cloned.

8. Enter the CLI and Root Administrator passwords to manage the Edge Nodes.
9. Click **Next**.
10. Configure workload domain connectivity.
    a) Enter the name for the gateway.
    b) Select **Active Standby** high availability mode.
11. Configure routing.
    a) Select **BGP** as **Gateway Routing Type**.
    b) Enter the local autonomous system number for use in BGP.
12. Configure the gateway uplinks for the first Edge node.

    Click **Set** and in **Gateway Uplink**, configure the following properties:

Option	Description
Gateway Interface VLAN	Enter the gateway interface vLAN.
Gateway Interface CIDR	Enter the gateway IP address.
BGP Peer IP	Enter the IP for the first uplink peer.
MTU	Enter the MTU for the NSX Edge cluster. The MTU value can be between 1600-9000.
BFD	Click the BFD toggle to activate BFD.
BGP Peer ASN	Enter the ASN for BGP peering.
BGP Peer Password	Enter the BGP password.

13. Configure the **Second Uplink** and click **Apply**.
14. Repeat steps 8 and 9 to configure the gateway uplinks for the second Edge node.
15. In **VPC External IP Blocks**, enter the IPs to allow connectivity to VPC workloads through external IPs or on public subnets.
16. In **Private - Transit Gateway IP Blocks**, enter the private IPs for the VPC configuration.
17. Click **Next**.
18. Review the configuration and click **Deploy** to deploy the gateway.

    The centralized gateway is deployed with two Edges. You can now deploy the Supervisor.

Configure the Distributed Transit Gateway

Configure the distributed transit gateway for the NSX VPC.

1. In the vSphere Client, select the vCenter object and select the **Networks** tab.
2. Select **Transit Gateways** and click **Setup Network Connectivity**.
3. In **Gateway Type**, select **Distributed Connection** and click **Next**.
4. Review and verify that you have completed the networking prerequisites checklist, and click **Continue**.
5. Configure the external network connectivity.

   Configure the following settings:

   - **VLAN ID**. Enter the VLAN ID. Value can be from 0 through 4094.
   - **Gateway CIDR IPv4 Address**. The gateway CIDR must be the same as the VLAN configured for the external router.
   - **External IP Blocks**
   - **Private - Transit Gateway IP Blocks**

   Click **Next**.

Configure VPC services. The Virtual Network Appliance (VNA) cluster is configured in the VPC connectivity profile.

6. Enter a cluster name.
7. Select a form factor.
8. Click **ADD** to add a Virtual Network Appliance (VNA) node.
   a) Enter the VNA Node name.
   b) Select the cluster where you want to deploy the node.
   c) Optionally, select the resource pool and data store for the node.
   d) In the Management IP section, configure the management network for the node.

Option	Description
IP Address Type	Select the type of IP address. For example, **IPv4 Only**.
IP Assignment	Select one of the following options: • DHCP • Static
Port Group	Select the VM management port group that you created. If a different port group is required, then create the port group before performing this step.

9. Click **APPLY**. Repeat step 8 to add additional nodes.
10. Click **NEXT**.
11. Review the configuration and click **DEPLOY** to deploy the gateway.

Deploy a Supervisor with VPC Networking

Learn how to deploy a Supervisor with VPC networking on a vSphere cluster that maps to vSphere Zones.

- Verify that your environment meets the prerequisites for configuring a vSphere cluster as a Supervisor.
- Optionally, configure the Centralized Gateway.
- Optionally install and configure the Avi Load Balancer. If NSX does not detect the Avi Load Balancer, it uses the NSX Edge load balancer.

1. From the home menu, select **Supervisor Management**.
2. On the **Supervisor Management** screen, click **Get Started**.
3. On the **vCenter Server and Network** page, select the vCenter system that is set up for Supervisor deployment and select **VCF Networking with VPC** as the networking stack.
4. Click **Next**.
5. On **Supervisor location** page, configure zones placement of the Supervisor.
   - **vSphere Zones Deployment**
     a. Enter the Supervisor name.
     b. Select the data center where you have created the vSphere Zones that you will use for the deployment.
     c. Enable control plane high availability.
     d. Select the three vSphere Zones that you have previously created for the Supervisor.
   - **Cluster Deployment**

     Deploy the Supervisor on one Management Zone. This option provides host level HA for the Supervisor control plane.

     1. Enter a Supervisor name.
     2. Enable control plane high availability.
     3. Select the data center where you have created the vSphere Zone that you use for the deployment.
     4. Select a compatible vSphere Cluster, and click **Next**.

6. Select the storage policy for placement of the control plane VMs.
7. Click **Next**.
8. On the **Management Network** screen, configure the parameters for the network for Supervisor control plane VMs.

Option	Description
IP Assignment Mode	Select DHCP or Static. If you select DHCP Network, all the IP addresses for the Management Network, such as control plane VM IPs, a floating IP, DNS servers, DNS, search domains, and NTP server are acquired automatically from a DHCP server. To obtain floating IPs, the DHCP server must be configured to support client identifiers. In DHCP mode, all control plane VMs use stable DHCP client identifiers to acquire IP addresses. These client identifiers can be used to set up static IP assignment for the IPs of control plane VMs on the DHCP server to ensure that they do not change. Changing the IPs of control plane VMs and floating IPs is not supported. You can override some of the settings that are inherited from DHCP by entering values in the text fields for these settings.
Network	The network that handles the management traffic for the Supervisor. If you deploy a multi-zone Supervisor, each zone can use its own management network.
Floating IP Addresses	In DHCP mode, this IP address determines the starting point for reserving five consecutive IP addresses for the Supervisor control plane VMs. In Static mode, you enter a block of 5 IP addresses. The Supervisor control plane VM IP addresses are as follows: • An IP address for each of the Supervisor control plane VMs. • A floating IP address for one of the Supervisor control plane VMs to serve as an interface to the Management Network. The control plane VM that has the floating IP address acts as a leading VM for all three Supervisor control plane VMs. The floating IP moves to the control plane VM that is the `etcd` leader in the Supervisor improving availability in a network partition event. • An IP address to serve as a buffer in case a Supervisor control plane VM fails and a new control plane VM is being brought up to replace it.
Subnet Mask	The subnet mask for the management network. For example, `255.255.255.0`

9. In the **Workload Network** pane, configure the following settings:

Option	Description
NSX Project	Select the name of the NSX project that the VPC is part of.
VPC Connectivity Profile	Select the name of the VPC connectivity profile that you want to use.
External IP Blocks	Advertised CIDRs that allow outside connectivity to VPC workloads either on public subnets or through external IPs. The external IP block must be the same as, or a sub-block of, the configured Gateway CIDRs.
Private (Transit Gateway) IP Blocks	Private CIDRs that are available for inter-VPC communication and are not advertised by the Transit Gateway to the outside datacenter.
Private (VPC) CIDRs	CIDR blocks from which private subnets are allocated.
Service CIDR	Specify the CIDR block from which the IP addresses for Kubernetes services are allocated. The IP range must not

Option	Description
	overlap with the ranges of the vSphere Pods, ingress, egress, or other services running in the data center. For the Kubernetes services and the vSphere Pods, you can use the default values which are based on the cluster size that you specify.
DNS Server	Enter the IP addresses of the DNS servers that you use with your environment, if any. For example, `10.142.7.1`. Starting with VCF 9.0, for a new Supervisor, if the DNS servers that you specify are shared between the Management Network and Workload Network, the DNS lookups on the control plane VMs are routed through the management network. For an existing Supervisor, the DNS lookups on the control plane VMs in such a scenario continue to be routed through the workload network.
NTP Server(s)	Enter the NTP Server IP or FQDN that the NSX Manager uses.

10. Click **Next**.
11. In the **Advanced Setting** page, set advanced settings for the Supervisor deployment.

Option	Description
Supervisor Control Plane Size	Select the select the sizing for the control plane VMs. The size of the control plane VMs determines the amount of workloads that you can run on the Supervisor. You can select from: • Tiny - 2 CPUs, 8 GB Memory, 32 GB Storage • Small - 4 CPUs, 16 GB Memory, 32 GB Storage • Medium - 8 CPUs, 16 GB Memory, 32 GB Storage • Large - 16 CPUs, 32 GB Memory, 32 GB Storage **Note:** Once you select a control plane size, you can only scale up. You cannot scale down to smaller size.
API Server DNS Names	Optionally, enter the FQDNs that will be used to access the Supervisor control plane, instead of using the Supervisor control plane IP address. The FQDNs that you enter will be embedded into an automatically-generated certificate. By using FQDNs for the Supervisor, you can omit specifying an IP sand in the load balancer certificate.
Export Configuration	Export a JSON file containing the values of the Supervisor configuration that you have entered. You can later modify and import the file if you want to redeploy the Supervisor or if you want to deploy a new Supervisor with similar configuration. Exporting the Supervisor configuration can save you time from entering all the configuration values in this wizard anew in case of Supervisor redeployment.

12. Click **Finish** when ready with reviewing the settings.

    The activation of the Supervisor initiates the creation and configuration of the control plane VMs and other components.

Deploying vSphere Supervisor with Foundation Load Balancer

Foundation Load Balancer provides basic load balancing to the Supervisor and supports layer 4 services. Foundation Load Balancer is packed with vCenter and provides load balancing support to the Supervisor and the workloads running on it. You can only use Foundation Load Balancer with a Supervisor with VDS networking.

You can deploy an instance of Foundation Load Balancer as part of the Supervisor activation, or add it as the load balancing solution to an existing Simplified Supervisor instance.

Supervisor Architecture with Foundation Load Balancer

Learn what are the supported topologies for Foundation Load Balancer and, it's placement in vSphere Zones, and availability modes.

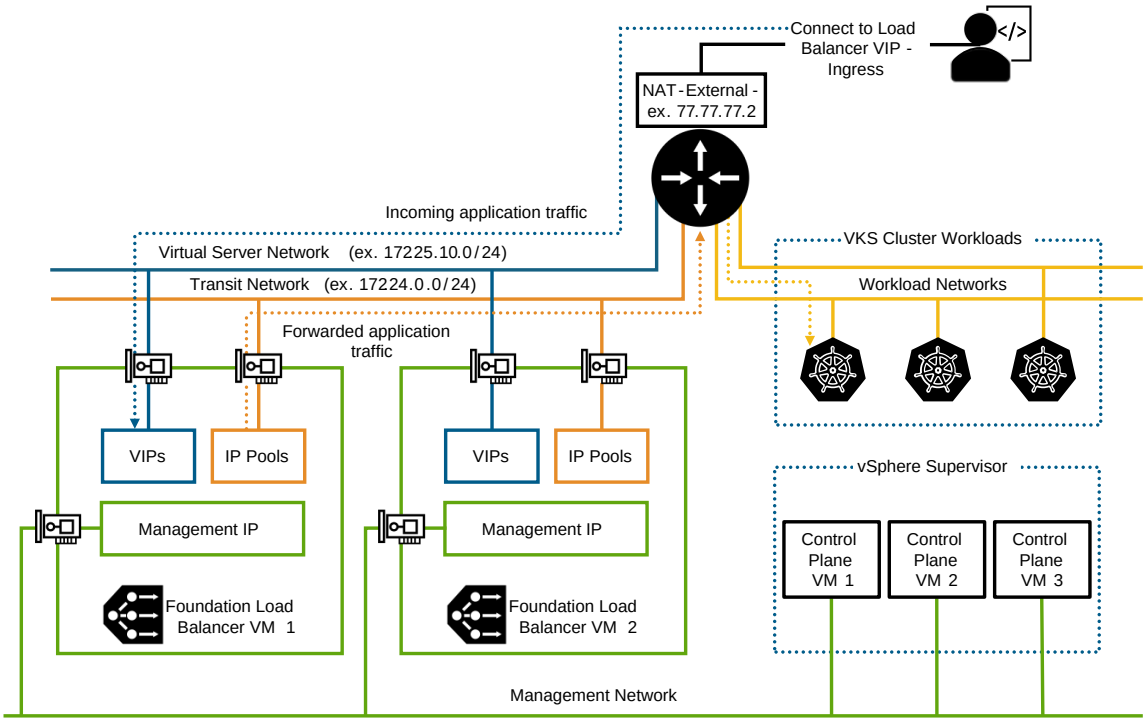
Foundation Load Balancer Networking and Supported Topologies

Foundation Load Balancer handles the following types of network traffic.

- Virtual Server traffic. The incoming traffic from clients to Foundation Load Balancer Virtual IPs (VIPs).
- Traffic to the IP pool members. The outgoing traffic from Foundation Load Balancer that is forwarded to one of the workload IP pool members and then to the Workload Networks.
- Management traffic. The traffic which is used for CRUD operations of Foundation Load Balancer .

Foundation Load Balancer supports three topologies.

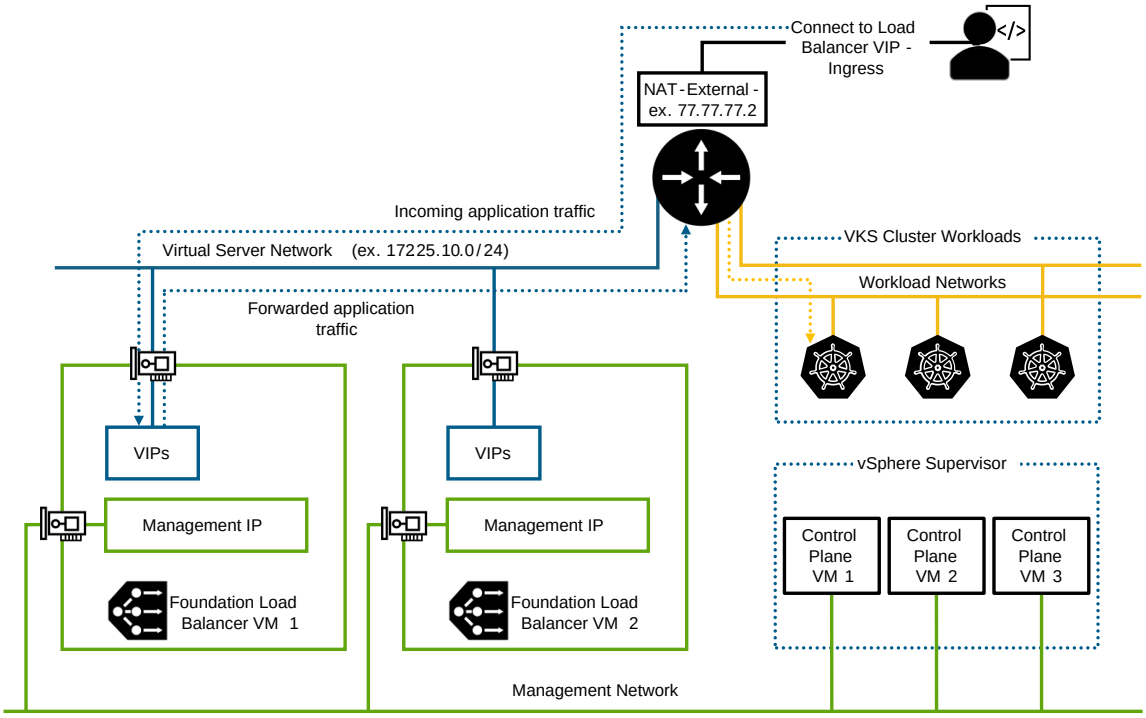
Two-arm configuration



The dataplane has separate Virtual Server and Transit Network interfaces, in addition to a dedicated Management Network interface, for a total of three interfaces per Foundation Load Balancer instance.

In a two-arm Foundation Load Balancer configuration, the incoming application traffic is forwarded to the VIPs and from there it passes through the Transit network to the Workload Networks.

One-arm configuration



The dataplane has a combined Virtual Server and Transit interface, in addition to a dedicated Management interface, which makes a total of two network interfaces per Foundation Load Balancer instance.

In One-Arm configuration, the Virtual Server Network is used both for incoming application traffic and then for forwarded application traffic to the Workload Networks.

One-arm one-nic configuration

The dataplane has a single network interface for all Virtual Server, Transit, and Management traffic. This Foundation Load Balancer topology is only supported in a Simplified Supervisor deployment.

Placement and Availability Modes

Foundation Load Balancer is placed on a Management Zone of the Supervisor instance. In a Supervisor deployed on three Management Zones, one Management Zone is selected for the placement of the Foundation Load Balancer. The Foundation Load Balancer VMs are located in the same folder (`Namespaces > <Supervisor Name>`) and resource pool (`Namespaces`) as the Supervisor control plane VMs.

Foundation Load Balancer supports two availability modes:

- One VM. In this mode, DRS handles the host placement of the Foundation Load Balancer VMs within the vSphere cluster.
- Two VMs. To ensure host-based high availability, the Foundation Load Balancer VMs have soft host anti-affinity rule in the same cluster in the two VM (active-passive) configuration. The active node is determined by leader election performed by both VMs . The elected leader is the active node that serves all of the client traffic coming at the server endpoints. If the active node goes down, for example due to host or VM crash, the passive node takes over leadership and become the new active node. The leader election and failover process is performed without vCenter involvement.

The VM disks of Foundation Load Balancer are place on the datastore determined by the storage policy configured to the Supervisor control plane VMs.

Requirements for Deploying vSphere Supervisor with Foundation Load Balancer

Checkout the requirements for deploying a Supervisor with Foundation Load Balancer. This solution is supported with VVF.

Foundation Load Balancer Requirements

Foundation Load Balancer supports four deployment sizes. When you deploy Foundation Load Balancer with a larger deployment size it has increased performance. Following are the footprint requirements of each Foundation Load Balancer VM. An active-passive deployment requires twice the footprint size.

Use the Small size only for POC environments. Medium, Large, and Extra Large sizes are suitable for production environments depending on the amount of incoming traffic and the number of VIPs.

Table 1785: Foundation Load Balancer Footprint Requirements

Requirements	Small (POC only)	Medium	Large	Extra Large
Compute	2 vCPUs	4 vCPUs	8 vCPUs	16 vCPUs
Memory	4 GB	8 GB	12 GB	16 GB
Storage	8 GB	8 GB	8 GB	8 GB

Table 1786: Foundation Load Balancer Networking Requirements

Requirement	Elements	Quantity	Description
Foundation Load Balancer Management Network	Distributed port group	1	A distributed port group for the Management Network of Foundation Load Balancer. The interface carrying Management traffic must be routable to and from Supervisor Management Floating IP, which is where the runtime control plane endpoint ofFoundation Load Balancer exposed.
	IP range for the Management Network	1	An IP range for the Foundation Load Balancer Management Network. It must reside on the same subnet as the subnet mask for that network. It's only required when the Management Network is configured in static mode.
Foundation Load Balancer Transit Network	Distributed port group	1	A distributed port group for the Foundation Load Balancer Transit Network in a Two-Arm configuration. The interface carrying Transit and Workload traffic must be routable to and from the VIP and Workload endpoints.
	IP range for Transit Network traffic	1	An IP range to be configured to the Transit Network for forwarding outgoing traffic to the Workload Networks.

Requirement	Elements	Quantity	Description
Foundation Load Balancer Virtual Server Network port group	Distributed port group	1	A distributed port group for the subnets of the Virtual IPs. The interface carrying Virtual Server traffic must be routable to and from the Virtual IP clients. An interface carrying combined Virtual Server and Transit traffic must be able to satisfy the routing requirements of both Virtual Server and Transit networks.
	IP range for Virtual IPs	1	An IP range for Virtual IPs of the Virtual Server Network. They must reside on the same subnet as the subnet mask for that network.

Supervisor Minimum Compute Requirements

The table lists the minimum compute requirements for enabling a Supervisor with VDS networking and Foundation Load Balancer . You can deploy the Supervisor on one Management Zone or three Management Zones.

System	Minimum Deployment Size	CPU	Memory	Storage
vCenter 9.0	Small	2	21 GB	290 GB
vSphere clusters	• 1 or 3 vSphere clusters. For a Supervisor with three Management Zones, you need three clusters. • vSphere DRS and HA enabled on each vSphere Cluster. vSphere DRS must be in Fully Automate mode. • Independent storage and networking configured for each vSphere cluster.	Not applicable	Not applicable	Not applicable
ESX hosts 9.0	For each vSphere cluster: • Without vSAN: 3 ESX hosts with 1 static IP per host • With vSAN: 4 ESX hosts per cluster with at least 2 physical NICs. **Note:** Make sure that the names of the hosts that join the clusters use lower case letters. Otherwise, the activation of the Supervisor might fail.	8	64 GB per host	Not applicable
Supervisor control plane VMs	1 to 3	4	16 GB	16 GB

Supervisor Minimum Network Requirements

Note:

You cannot create IPv6 clusters with a vSphere 9 Supervisor, or register IPv6 clusters with Tanzu Mission Control.

Table 1787: Physical Network Requirements

Component	Minimum Quantity	Required Configuration
Physical network setup	1	The Management Network that will handle the traffic of the Supervisor must be connected to the physical network for all clusters part of the Supervisor. The Primary Workload network must also be connected to the same physical network.
Physical Network MTU	1600	The MTU size must be 1600 or greater on any distributed port group.

Table 1788: Supervisor General Networking Requirements

Component	Minimum Quantity	Required Configuration
Latency	100 ms	The maximum recommended latency between each cluster that is part of a vSphere Zone joined together in a Supervis or.
NTP and DNS Server	1	A DNS server and NTP server that can be used with vCenter. **Note:** Configure NTP on all ESX hosts and vCenter .
DHCP Server (optional)	1	Configure a DHCP server to automatically acquire IP addresses for the Management and Workload Networks as well as control plane IPs. The DHCP server must support Client Identifiers and provide compatible DNS servers, DNS search domains, and an NTP server. For the management network, all the IP addresses, such as control plane VM IPs, a Floating IP, DNS servers, DNS, search domains, and NTP server are acquired automatically from the DHCP server. The DHCP configuration is used by the Supervisor. Load balancers may require static IP addresses for Management. DHCP Scopes should not overlap these static IP's. DHCP is not used for virtual IPs. (VIPs)

Table 1789: Supervisor Management Network Requirements

Component	Minimum Quantity	Required Configuration
vSphere Distributed Switch	1	All hosts from all three vSphere clusters that are mapped to the Management Zones where you deploy the Supervisor must be connected to a VDS.
Management traffic network	1	A Management Network that is routable to the ESX hosts, vCenter, the Supervisor and Foundation Load Balancer.

Component	Minimum Quantity	Required Configuration
Management Network Subnet	1	The subnet used for management traffic between ESX hosts and vCenter, and the Supervisor control plane. The size of the subnet must be the following: • One IP address per host VMkernel adapter. • One IP address for the vCenter Appliance. • 3 to 5 IP addresses for the Kubernetes control plane. 1 for each of the 3 nodes, 1 for virtual IP, 1 for rolling cluster upgrade. **Note:** The Management Network and the Workload Network must be on different subnets. Assigning the same subnet to the Management and the Workload Networks is not supported and can lead to system errors and problems.
Management Network VLAN	1	The VLAN ID of the Management Network subnet.

Table 1790: Workload Network Requirements

Component	Minimum Quantity	Required Configuration
Workload Network distributed port group	1	At least one distributed port group must be created on the VDS that you can configure as the Primary Workload Network. Depending on the topology of choice, you can use the same distributed port group as the Workload Network of namespaces or create more port groups and configure them as Workload Networks. Workload Networks must meet the following requirements: • Routability between any Workload Network with the network that Foundation Load Balancer uses for virtual IP allocation. • No overlapping of IP address ranges across all Workload Networks within a S upervisor. **Important:** The workload network must be on a different subnet than the management network.
Kubernetes Services CIDR range	/16 Private IP addresses	A private CIDR range to assign IP addresses to Kubernetes services. You must specify a unique Kubernetes services CIDR range for each Supervisor.

Create vSphere Zones for a Multi-Zone Supervisor Deployment

To activate a Supervisor, you must create at least one vSphere Zone that will become the Management Zone of the Supervisor. To provide cluster-level HA to the Supervisor control plane, you can deploy the Supervisor on three Management Zones. In Supervisor with three Management Zones and HA for the control plane, meaning with three control plane VMs, each control plane VM is placed on a different Management Zone.

- Create one or three vSphere clusters to add to the zones:
  - 2 ESX hosts in each zone for NFS or FC storage. For HA, use 4 hosts
  - For storage with vSAN, the cluster must have 3 hosts. For HA, use 4 hosts.
  - For test of POC environments, each cluster must have at least 1 host, and 2 hosts when using vSAN.

    Note:

    Recommendation: Restrict direct access to the ESX host console. Without that restriction, an administrator could override vSphere Zone assignments, undermining the intended isolation and policy enforcement.
- Configure storage with vSAN or other shared storage solution for each cluster.
- Enable vSphere HA and vSphere DRS on Fully Automate or Partially Automate mode.
- All clusters in a zone must be connected to the same vSphere Distributed Switch (VDS). Each zone can use its own management network.

1. In the vSphere Client, navigate to vCenter.
2. Select **vSphere Zones**.
3. Click **Add New vSphere Zone**.
4. Name the zone, for example `zone1` and add an optional description.
5. Select a vSphere clusters to add to the zone and click **Finish**.
6. Repeat the steps to create three vSphere Zones.

Deploy a Supervisor with Foundation Load Balancer and VDS Networking

Check out how to deploy a Supervisor with Foundation Load Balancer as a load balancing solution. Foundation Load Balancer is a light-weight load balancer that is embedded in vCenter and provides layer 4 services.

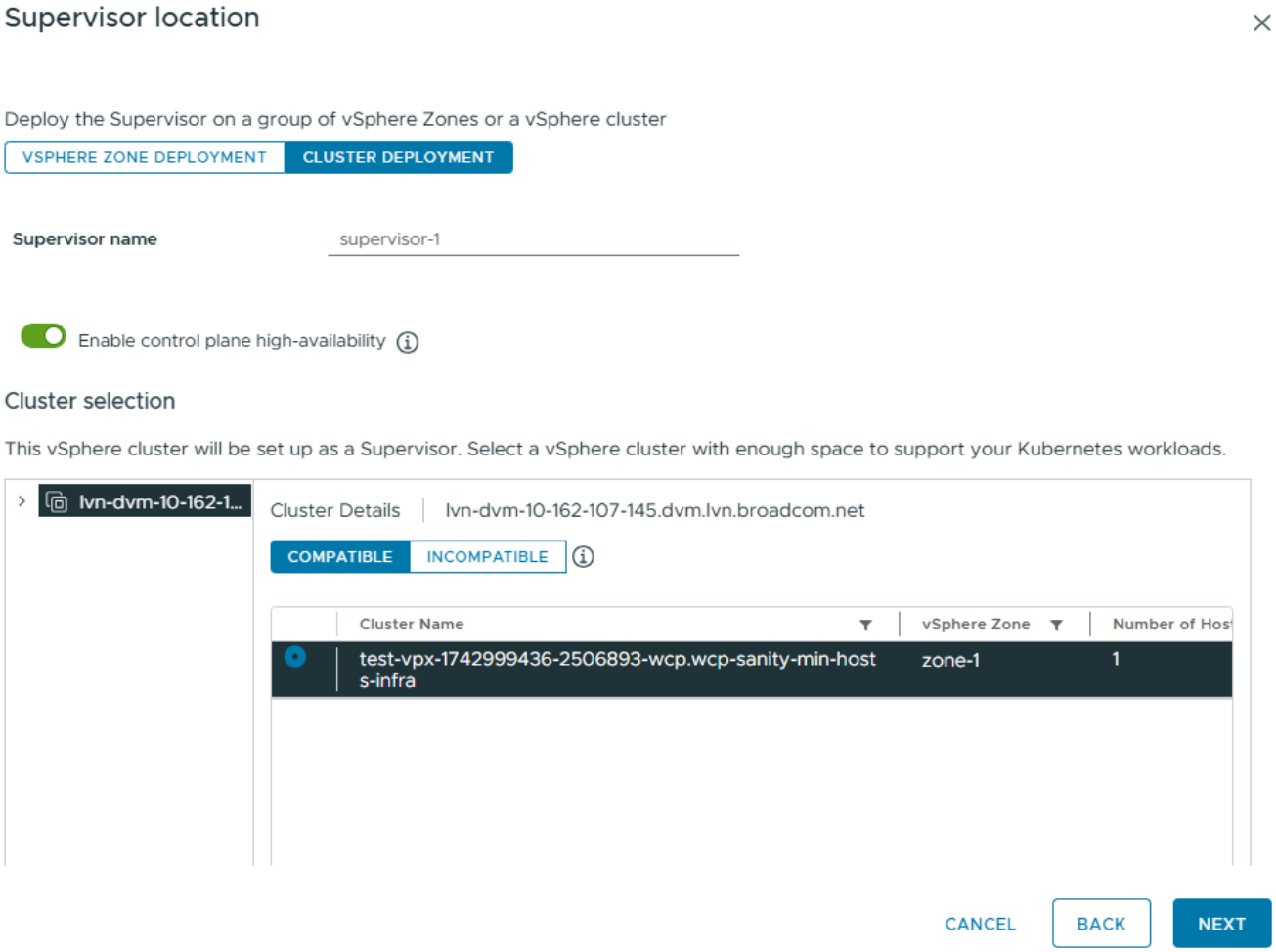
- Verify that your environment meets the requirements for deploying a Supervisor with Foundation Load Balancer. See Requirements for Deploying Supervisor with Foundation Load Balancer.

1. From the vSphere Client home menu, select **Supervisor Management**.
2. Click **Get Started**.
3. On the **vCenter Server and Network** page, configure the following options.
   a) Select the vCenter system where you want to deploy the Supervisor .
   b) Select **vSphere Distributed Switch** and select **Foundation Load Balancer** from the **Load Blanacer Type** menu.
4. To deploy the Supervisor with three control plane nodes for HA, select **Enable control plane high-availability**.

   If you deploy the Supervisor with one control plane VM, you can alter scale to three for HA.
5. On **Supervisor location** page, configure zones placement of the Supervisor.
   - **vSphere Zones Deployment**
     a. Enter the Supervisor name.
     b. Select the data center where you have created the vSphere Zones that you will use for the deployment.
     c. Select the three vSphere Zones that you have previously created for the Supervisor.
   - **Cluster Deployment**

     Deploy the Supervisor on one Management Zone. This option provides host level HA for the Supervisor control plane.

1. Enter a Supervisor name.
2. Select the data center where you have created the vSphere Zones that you will use for the deployment.
3. Select a compatible vSphere Zone.



6. On the **Storage** page, configure storage and file volume support.
   a) Select storage policies for the Supervisor.

      The storage policy you select for each of the following objects ensures that the object is placed on the datastore referenced in the storage policy. You can use the same or different storage policies for the objects.

Control Plane Storage Policy	Select the storage policy for placement of the control plane nodes.
Ephemeral Disks Storage Policy	Select the storage policy for placement of the vSphere Pods.
Image Cache Storage Policy	Select the storage policy for placement of the cache of container images.

7. On the **Management Network** screen, configure the parameters for the network that will be used for Supervisor control plane VMs.

   If you select **DHCP Network**, all the IP addresses for the Management Network, such as control plane VM IPs, a floating IP, DNS servers, DNS, search domains, and NTP server are acquired automatically from a DHCP server. To obtain floating IPs, the DHCP server must be configured to support client identifiers. In DHCP mode, all control plane VMs use stable DHCP client identifiers to acquire IP addresses. These client identifiers can be used to setup static IP assignment for the IPs of control plane VMs on the DHCP server to ensure they do not change. Changing the IPs of control plane VMs and floating IPs is not supported.

You can override some of the settings inherited from DHCP by entering values in the text fields for these settings.

DHCP	Static	Description
Network	**Network**	The network that will handle the management traffic for the Supervisor. If you deploy a multi-zone Supervisor, each zone can use its own management network.
Floating IP	**IP Addresses**	In DHCP mode, this is the IP address that determines the starting point for reserving five consecutive IP addresses for the Sup ervisor control plane VMs. In Static mode, you enter a block of 5 IP addresses. The Su pervisor control plane VM IP addresses are as follows: • An IP address for each of the Superviso r control plane VMs. • A floating IP address for one of the S upervisor control plane VMs to serve as an interface to the Management Network. The control plane VM that has the floating IP address assigned acts as a leading VM for all three Supervis or control plane VMs. The floating IP moves to the control plane VM that is the etcd leader in the Supervisor. This improves availability in the case of a network partition event. • An IP address to serve as a buffer in case a Supervisor control plane VM fails and a new control plane VM is being brought up to replace it.
N/A	**Subnet Mask**	The subnet mask for the management network. For example, `255.255.255.0`
N/A	**Gateway**	A gateway for the management network.
DNS Servers	**DNS Servers**	The addresses of the DNS servers that you use in your environment. If the vCenter system is registered with an FQDN, you must enter the IP addresses of the DNS servers that you use with the vSphere environment so that the FQDN is resolvable in the Supervisor.
DNS Search Domains	**DNS Search Domains**	Domain names that DNS searches inside the Kubernetes control plane nodes, such as `corp.local`, so that the DNS server can resolve them.
NTP Servers	**NTP Servers**	The addresses of the NTP servers that you use in your environment, if any.

8. In the **Workload Network** page, enter the settings for the network that will handle the networking traffic for workloads running on the Supervisor.

   Note:

   If you select using a DHCP server to provide the networking settings for Workload Networks, you will not be able to create any new Workload Networks once you complete the Supervisor configuration.

In **DHCP Network** network mode, all networking settings for the Workload Networks are acquired through DHCP. You can also override some of the settings inherited from DHCP by entering values in the text fields for these settings: for these settings:

DHCP	Static	Description
Network Name	**Network Name**	The network name.
Port Group	**Port Group**	The port group that will serve as the primary workload network to the Supervisor The primary network handles the traffic for the Kubernetes control plane VMs and Kubernetes workload traffic. Depending on your networking topology, you can later assign a different port group to serve as the Workload Network to each vSphere Namespace. VKS clusters use only the network that is assigned to the namespace where they are deployed or they use the Primary network if there is no explicit network assigned to that vSphere Namespace.
Internal Network for Kubernetes Services	**IP Address Range(s)**	Enter a CIDR notation that determines the range of IP addresses for VKS clusters and services that run inside the clusters.
N/A	**Subnet Mask**	The subnet mask IP address.
N/A	**Gateway**	The gateway for the primary network.
DNS Servers	**DNS Servers**	The IP addresses of the DNS servers that you use with your environment, if any. For example, `10.142.7.1`. When you enter IP address of the DNS server, a static route is added on each control plane VM. This indicates that the traffic to the DNS servers goes through the workload network. Starting with VCF 9.0, for a new Supervis or, if the DNS servers that you specify are shared between the Management Network and Workload Network, the DNS lookups on the control plane VMs are routed through the management network. For an existing Supervisor, the DNS lookups on the control plane VMs in such a scenario continue to be routed through the workload network.
NTP Servers	**NTP Servers**	The address of the NTP server that you use with your environment if any.

9. On the **Load Balancer** page, configure the settings for Foundation Load Balancer.
   a) Enter a name for Foundation Load Balancer.
   b) From **Network Topology**, select the topology for deploying Foundation Load Balancer.
      - **Two Arm** topology with a separated Virtual and Transit interfaces in addition to the Management interfaces, which makes three interfaces in total.
      - **One Arm** topology with a dataplane that has a combined Virtual Server and Transit interface, in addition to a dedicated Management interface, which makes a total two network interfaces.

c) From **Size**, select the deployment footprint for Foundation Load Balancer.

Requirements	Small (POC only)	Medium	Large	Extra Large
Compute	2 vCPUs	4 vCPUs	8 vCPUs	16 vCPUs
Memory	4 GB	8 GB	12 GB	16 GB
Storage	8 GB	8 GB	8 GB	8 GB

d) From **High-Availability**, select whether to deploy Foundation Load Balancer with one or two VMs for HA.

If you select the deploy Foundation Load Balancer with two VMs, the deployment footprint that you selected in the previous step will double in size.

e) In **Management Network**, click **Assign** and configure the options for the Foundation Load Balancer Management Network.

Static	DHCP	Description
Port Group	**Port Group**	The distributed port group that you have created to serve as the Management Network for Foundation Load Balancer.
Network Name	**Network Name**	The name of the Management Network. You can keep the same name as the port group, or enter a different one.
IP Address Range(s)	N/A	The IP address for the Foundation Load Balancer management. A minimum IP address are required.
Subnet Mask	N/A	The **Subnet Mask** for the Management Network subnet.
Gateway	N/A	The Management Network gateway.
DNS Server(s)	**DNS Server(s)**	The IP addresses of the DNS servers.
NTP Server(s)	**NTP Server(s)**	The IP addresses of the NTP servers that you have configured in your environment.
Syslog	**Syslog**	The IP address of the syslog server for forwarding logs from Foundation Load Balancer.
Syslog Certificate	**Syslog Certificate**	The certificate for the syslog server.

f) In **Virtual Server Network**, click **Assign** and configure the following settings.

1. Click **Assign** and configure the following settings.

Option	Description
Port Group	The distributed port group that you have created to serve as the Virtual Server Network for Foundation Load Balancer.
Network Name	The name of the Virtual Server Network. You can keep the same name as the port group, or enter a different one.
IP Address Range(s)	The IP address ranges for the virtual IPs where client applications connect. Must be on the same subnet as the gateway.
Subnet Mask	The subnet mask for the Virtual Server Network subnet.
Gateway	The gateway for the Virtual Server Network.

2. Once you click OK, configure **Virtual IP Address Range(s)** for the service type LB**.**

g) In **Transit Network**, click **Assign** and configure the following settings.

Option	Description
Port Group	The distributed port group that you have created to serve as the Transit Network for Foundation Load Balancer.
Network Name	The name of the VTransit Network. You can keep the same name as the port group, or enter a different one.
IP Address Range(s)	IP address ranges where forwarded application traffic is sent. Must be on the same subnet as the gateway.
Subnet Mask	The subnet mask for the Transit Network subnet.
Gateway	The gateway for the Transit Network.

10. In the **Advanced Setting** page, set advanced settings for the Supervisor deployment.

Option	Description
Supervisor Control Plane Size	Select the select the sizing for the control plane VMs. The size of the control plane VMs determines the amount of workloads that you can run on the Supervisor. You can select from: • Tiny - 2 CPUs, 8 GB Memory, 32 GB Storage • Small - 4 CPUs, 16 GB Memory, 32 GB Storage • Medium - 8 CPUs, 16 GB Memory, 32 GB Storage • Large - 16 CPUs, 32 GB Memory, 32 GB Storage **Note:** Once you select a control plane size, you can only scale up. You cannot scale down to smaller size.
API Server DNS Names	Optionally, enter the FQDNs that will be used to access the Supervisor control plane, instead of using the Supervisor control plane IP address. The FQDNs that you enter will be embedded into an automatically-generated certificate. By using FQDNs for the Supervisor, you can omit specifying an IP sand in the load balancer certificate.
Export Configuration	Export a JSON file containing the values of the Supervisor configuration that you have entered. You can later modify and import the file if you want to redeploy the Supervisor or if you want to deploy a new Supervisor with similar configuration. Exporting the Supervisor configuration can save you time from entering all the configuration values in this wizard anew in case of Supervisor redeployment.

11. On the **Ready to complete** page, click **Finish** when ready with reviewing the settings.

    The activation of the Supervisor initiates the creation and configuration of the control plane VMs and other components.

Maintaining Foundation Load Balancer

Learn what are the high availability options that are available for the Foundation Load Balancer , as well as monitoring and observability, failover and recovere, backup and restore, and more.

High Availability and Failover Performance

In an event of a ESX host or VM crash affecting the health of the active Foundation Load Balancer VM, the recovery time for the Foundation Load Balancer active VM is 5 seconds. During these 5 seconds, the standby Foundation Load Balancer VM takes over as the new active VM. VIPs on the new Foundation Load Balancer active VM start working when the standby VM takes over the active VM.

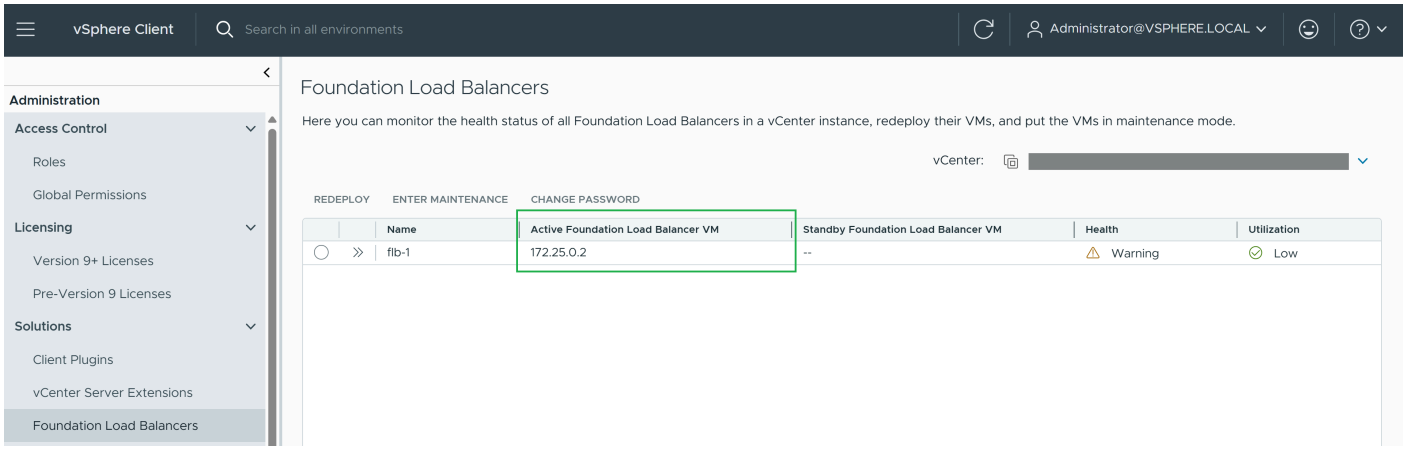
Impact on active TCP sessions	Connection states are not synchronized between the active and standby Foundation Load Balancer VMs. Existing active TCP connections will break during the 5 seconds downtime while the active Foundation Load Balancer is being recovered. If a client or server application does not send any TCP packets during the recovery of the active Foundation Load Balancer VM, they will not notice that the connection is broken. When a client or server sends TCP packets after the recovery of the active Foundation Load Balancer VM: • When a client sends TCP packets to the new active Foundation Load Balancer VM, the Foundation Load Balancer VM does not have the connection information. The Foundation Load Balancer VM responds to the client with a TCP RST (reset) packet. • When a server sends TCP packets to the new active Foundation Load Balancer VM, the Foundation Load Balancer VM responds to the server with a TCP RST (reset) packet. Depending on the application, it might attempt to reestablish the TCP session.
Restoring Foundation Load Balancer when both active and standby Foundation Load Balancer VMs are down	Information about the Foundation Load Balancer is stored in the vCenter database. In case both the Foundation Load Balancer VMs are down, vCenter automatically restores the VMs.
Redeployment of the Foundation Load Balancer VMs	You can redeploy any of the Foundation Load Balancer VMs from the vSphere Client **Administration** > **Solutions** > **Foundation Load Balancers** view. • If you redeploy the standby Foundation Load Balancer VM, workloads are not impacted. • If you redeploy the active Foundation Load Balancer VM, an automatic active-standby failover is triggered before the redeployment takes place. The current active Foundation Load Balancer VM becomes standby and the standby VM becomes the active one. If you want to redeploy two Foundation Load Balancer VMs in a single Foundation Load Balancer instance, you must wait for the first Foundation Load Balancer VM redeployment to complete before you start redeploying the second VM

Establish an SSH Session with the Foundation Load Balancer VMs

To acquire Foundation Load Balancer logs and statistics from Envoy, you establish an SSH sessions with the active Foundation Load Balancer VM.

Before you begin, make sure that you are logged in as an administrator in vCenter.

1. In vCenter, from **Administration** > **Solutions** > **Foundation Load Blanacer**, view the Foundation Load Balancer instances and check the IP address of the active VMs:



2. Select the active Foundation Load Balancer VM, click **Change Password**, and enter a password for the active Foundation Load Balancer VM.
3. Establish an SSH session at the IP address of the active Foundation Load Balancer VM by using the following credentials:
   Username: `vmware-system-user`
   Password: the password that you have set earlier

Collect the Foundation Load Balancer Support Bundle

To collect a support bundle for Foundation Load Balancer:

1. Establish an SSH session with either of the active or standby Foundation Load Balancer VMs.
2. Run the following command:
   ```
   flbctl supportbundle
   ```
3. Collect the generated bundle file.

Monitoring and Observability

You can view and monitor all the Foundation Load Balancer instances available in vCenter from **Administration** > **Solutions** > **Foundation Load Blanacer**. See Monitor Foundation Load Balancer for more information. VCF Operations dashboards are not available for Foundation Load Balancer.

Foundation Load Balancer Health Errors

You view the errors that occur in Foundation Load Balancer instances from **Administration** > **Solutions** > **Foundation Load Blanacer** option in the vSphere Client. See Foundation Load Balancer Heath Errors and Troubleshooting.

Backup and Restore

The backup of Foundation Load Balancer VIPs configurations and endpoints are included in the Supervisor control plane backup. Once the Supervisor is restored from a backup, Foundation Load Balancer resynchronizes the load balancer configuration with the restored Supervisor control plane. See Backup and Restore the Supervisor Control Plane for more information.

In the case when the Foundation Load Balancer instance is still active, the restored state of the Supervisor includes a reference to the existing Foundation Load Balancer instance. Supervisor reconnects the Foundation Load Balancer service to vCenter to refresh the Foundation Load Balancer authentication tokens. This allows the Foundation Load Balancer instance to re-establish connection with its XDS runtime control plane in the Supervisor and to obtain the up-to-date load balancer virtual server configurations for the Supervisor.

See High Availability and Failover Performance for the cases when both Foundation Load Balancer VMs are down and redeployed.

Health Checks and Statistics

To provide health checks on VIPs and endpoints of Foundation Load Balancer , you can configure liveness and readiness probes on the Foundation Load Balancer endpoints. Foundation Load Balancer only consumes healthy endpoints from Supervisor Kubernetes API.

You can get real-time statistics from Envoy. Establish an SSH session with the active Foundation Load Balancer VM and use the following command:

```
curl http://<active FLB VM IP>:19000/stats
```

For information on the Envoy statistics, check the listener and server statistics in the Envoy documentation.

Get Foundation Load Balancer Logs

You can get the Foundation Load Balancer logs from the following locations in case the Supervisor Kubernetes Services are not realized:

- Envoy logs in the active Foundation Load Balancer VM. Establish an SSH session with the active Foundation Load Balancer VM and use the following command:
  ```
  sudo journalctl -u envoy
  ```
- Logs generated by the `flb-agent` component in the active Foundation Load Balancer VM. Establish an SSH session with the active Foundation Load Balancer VM and find the `flb-agent` logs in the following location:
  ```
  /var/log/flb-agent/
  ```

- Logs from the Foundation Load Balancer support bundle. Establish an SSH session with any of the Foundation Load Balancer VMs and use the following command:

  ```
  flbctl supportbundle
  ```
- Logs that are generated from the Foundation Load Balancer system pod, which is named, for example `flb-controller-5689c6bf74-t5rk2` and is located in the `vmware-system-flb` system namespace. You can check the exact name of the pod from the **Pods** view on the Supervisor in the vSphere Client. Then log into the Supervisor as an administrator by using VCF CLI and use the following command:

  ```
  kubectl logs flb-controller-5689c6bf74-t5rk2 -n vmware-system-flb
  ```
- Perform a tcpdump on the active Foundation Load Balancer VM front-end interface and workload network interface. This captures live client traffic and indentify potential issues on the physical network, and provide additional useful information.

Configure a Syslog Server to Foundation Load Balancer

As a VI admin, you can also configure a remote syslog server to Foundation Load Balancer in case one has not been configured during the initial. Foundation Load Balancer deployment.

Before you begin, make sure that you are logged in as an administrator.

1. In **Supervisor Management**, select a Supervisor.
2. Select **Configure** > **Network** > **Load Balancer** and click **Edit**.
3. Under **Management Network**, select **Show advanced settings**.
4. Enter the IP address or FQDN of the remote syslog server and a valid certificate.

Check the Live Envoy Configuration

Establish an SSH session is the active Foundation Load Balancer VM and run the following command:

```
curl http://127.0.0.1:19000/config_dump?include_eds > config_dump.json
```

Troubleshooting VIP Traffic Connectivity Loss

In case the VIP traffic is down, perform the following actions:

- First, check the `flb-controller`, `flb-agent`, and Envoy logs like shown from above.
- Establish an SSH session is the active Foundation Load Balancer VM and check the following VIP configurations:
  - `/var/lib/flb-agent/listeners/`
  - `/var/lib/flb-agent/clusters/`
  - `/var/lib/flb-agent/endpoints`
- Check if a VIP is configured to the active Foundation Load Balancer VM. Establish an SSH session with the activeFoundation Load Balancer VM and use the following commands:
  - `sudo ip a show vsphere-lb0`
  - `sudo ip route show table all`
  - `sudo ip rule show`
  - `sudo iptables-save`
- Raise the Envoy log level and check the Envoy logs in the active Foundation Load Balancer VM.Establish an SSH session with the activeFoundation Load Balancer VM and use the following command to raise the Envoy logs level:
  - `curl -X POST http://127.0.0.1:19000/logging?level=trace`
  - Restore the log level to info: `curl -X POST http://127.0.0.1:19000/logging?level=info`

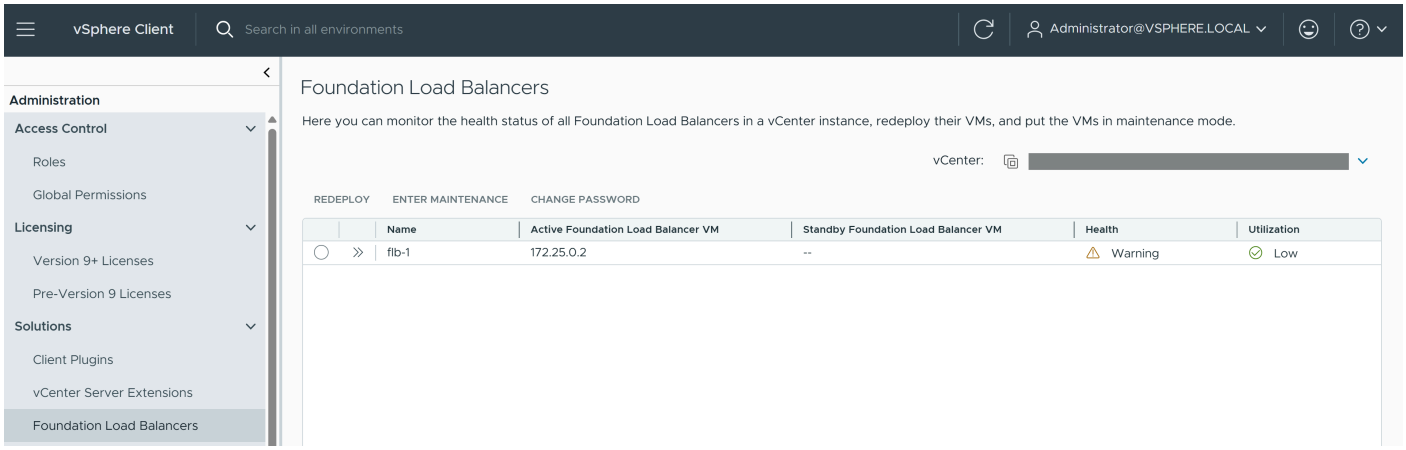
Lastly, perform `tcpdump` on Foundation Load Balancer VM front-end interface and workload network interface.

Monitor Foundation Load Balancer

Learn how to monitor the state of the Foundation Load Balancer instances in vCenter, put load balancer VMs in maintenance mode, track their resource utilization, or redeploy any of the Foundation Load Balancer VMs.

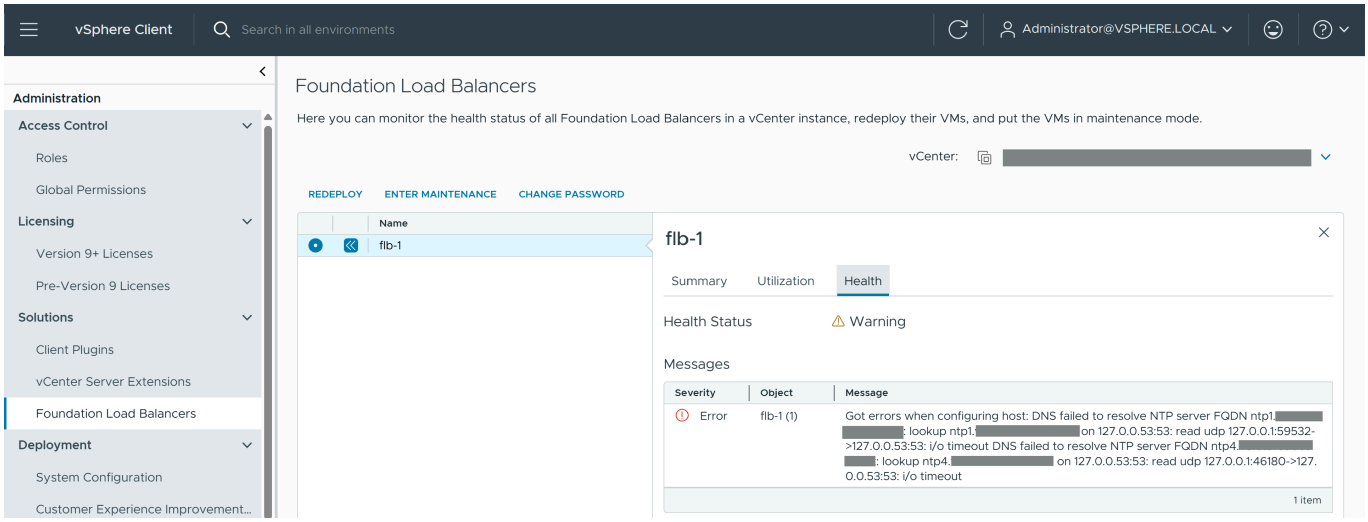
1. From the vSphere Client Home menu select **Administration** > **Solutions** > **Foundation Load Balancers**.

   The available Foundation Load Balancer instances are displayed:

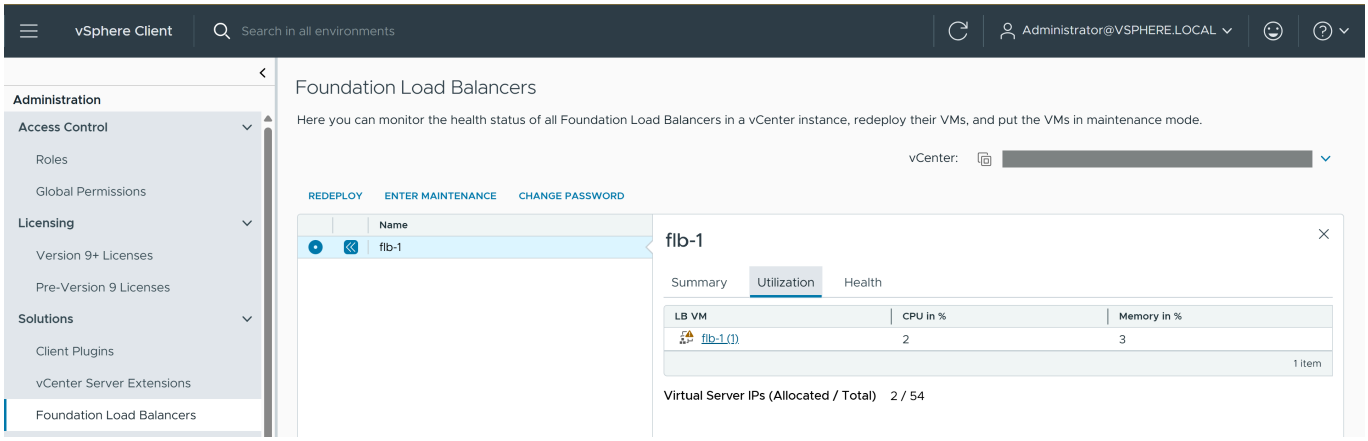


2. Track the status of the Foundation Load Balancer instances in vCenter.

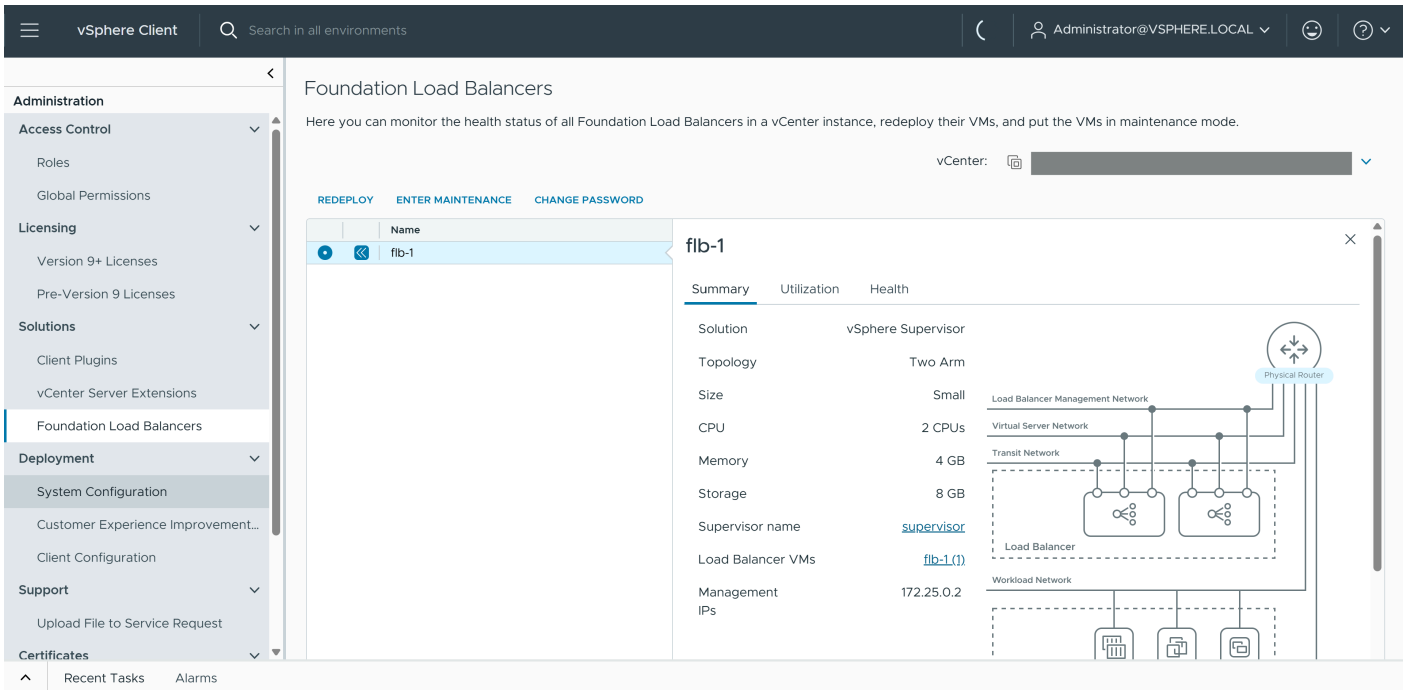
   - **Name** - The Foundation Load Balancer instance name.
   - **Active Foundation Load Balancer VM** - The VM that is active in the active-passive Foundation Load Balancer configuration.
   - **Standby Foundation Load Balancer VM** - The VM that is standby in an active-passiveFoundation Load Balancer configuration.
   - **Health** - The health status of the Foundation Load Balancer VM. The possible status are **Healthy** and **Unhealthy**. In case a VM is in **Unhealthy** state, the possible reason could be a resource over-utilization. In such case, you can increase the deployment footprint of the Foundation Load Balancer VMs to provide additional resources. See Change Load Balancer Settings on a Supervisor Configured with VDS Networking. See Troubleshooting Foundation Load Balancer Heath Errors for other errors and how to troubleshoot them.



   - **Utilization** - The resource utilization of the Foundation Load Balancer VM.



3. View the topology of a Foundation Load Balancer instance.



Troubleshooting Foundation Load Balancer Health Errors

Under **Administration** > **Solutions** > **Foundation Load Balancers** you can view the available Foundation Load Balancer instances and troubleshoot any errors that are displayed under the Health option for each Foundation Load Balancer instance.

Error	Error Message	Condition	Recommended Solution
`Envoy (DP in FLB VM) crashed`	`Envoy status is unexpected. Current status: inactive`	When the Envoy daemon is down, systemd restarts Envoy. If Envoy cannot be restarted, there might be un unexpected issue in the Foundation Load Balancer VM.	• Log in to the Foundation Load Balancer VM and check Envoy logs. • Collect the Foundation Load Balancer support bundle.
`xDS connection is not ready`	`xDS connection is not ready. Current state: STREAM_NOT_READY. Foundation LB instance is unable to connect to Supervisor Control Plane. Please check network connectivity between the Foundation LB management interface and the Supervisor cluster management floating IP. Run "flbctl supportbundle" on the Foundation LB VM to collect support bundle.`	The Foundation Load Balancer VM fails to connect to the Foundation Load Balancer controller in the Supervisor for fetching VIPs and endpoint configurations.	• Check the connectivity of the management network that is configured to the Foundation Load Balancer VM. • Collect the Supervisor and Foundation Load Balancer support bundles.

Error	Error Message	Condition	Recommended Solution

LB VIP capacity	Consumption >= 80% error message:
	`Listener count exceeds 80% of the supported capacity of the current LB VM size. Expect performance degradation when all capacity are consumed. Capacity: CAPACITY_NUM. Listener count: LISTENER_COUNT`
	Consumption >= 100% error message:
	`Listener count exceeds the supported capacity of the current LB VM size. Expect performance degradation. Capacity: CAPACITY_NUM. Listener count: LISTENER_COUNT`
		The size that is configured to the Foundation Load Balancer VM can support a maximum number of Kubernetes services. The Foundation Load Balancer VM raises an alarm when approximately 80% of its capacity is reached.	Configuring more Kubernetes services than the capacity of the Foundation Load Balancer VM can handle might cause performance degradation. In such case, use a larger Foundation Load Balancer VM size.

Load Balancer Endpoint capacity	Consumption >= 80% error message:
	`Endpoint count exceeds 80% of the supported capacity of the current LB VM size. Expect performance degradation when all capacity are consumed. Capacity: CAPACITY_NUM. Endpoint count: ENDPOINT_COUNT`
	Consumption >= 100% error message:
	`Endpoint count exceeds the supported capacity of the current LB VM size. Expect performance degradation. Capacity: CAPACITY_NUM. Endpoint count: ENDPOINT_COUNT`
		The size that is configured to the Foundation Load Balancer VM can support a maximum number of endpoints.	Increase the size of the Foundation Load Balancer VMs.

Error	Error Message	Condition	Recommended Solution
`Peer node down`	`One of the LB FLB_NODE_ID (IP: MGMT_IP_ADDRESS) from the LB cluster FLB_CLUSTER_ID has unexpected state ONLINE_STATE` `ONLINE_STATE` can be *`ONLINE_STATE_UNSPECIFIED`*, *`ONLINE_STATE_LEFT`*. Example: FLB_NODE_ID: flb-1-0, example FLB_CLUSTER_ID: foundation-loadbalancer-0. You can click the Foundation Load Balancer VM and check its DNS name, such as `flb-1-0.foundation-loadbalancer-0`.	If two Foundation Load Balancer VMs are deployed in active-standby mode, the VMs probe each other on the management network. If one Foundation Load Balancer VM is down, the remaining VM raises an alarm to indicate the peer Foundation Load Balancer VM is down.	• Check the status of the Foundation Load Balancer VM which is reported as being down. • Check management network connectivity between the Foundation Load Balancer VMs.

`Fail to configure network interfaces with static or DHCP IPs`	`failed to wait for static networks to be ready: REASON`
	Or
	`failed to wait for DHCP networks to be ready: REASON`
		Incorrect static IP configuration due to an unexpected issue or the DHCP server is not responding.	• Check whether any conflict IPs are already configured on the network.
			• Check whether the DHCP server is working.
			• If the management network down, log in to the Foundation Load Balancer VM through the ESX UI console and manually configure an IP address.
			• Collect a Foundation Load Balancer support bundle.

Error	Error Message	Condition	Recommended Solution

`Fail to config NTP servers`	`DNS failed to resolve NTP servers: DNS failed to resolve NTP server FQDN NTP_SERVER1: REASON`
	
	`DNS failed to resolve NTP server FQDN NTP_SERVER2: REASON`
	
	`DNS failed to resolve NTP server FQDN NTP_SERVER3: REASON`
	`----`
	Where REASON is the error message reported by the system DNS resolver library.
	
	`failed to query NTP server NTP_SERVER1 (IP_ADDRESS1, IP_ADDRESS2, ...)`
	
	`failed to query NTP server NTP_SERVER2 (IP_ADDRESS1, IP_ADDRESS2, ...)`
	
	`failed to query NTP server NTP_SERVER3 (IP_ADDRESS1, IP_ADDRESS2, ...)`
	
	`...`
		• The DNS server cannot resolve the NTP server names.
		• The Foundation Load Balancer VM cannot connect to NTP server IPs.
			• Check whether Foundation Load Balancer can connect to the DNS servers through the management network.
			• Check whether Foundation Load Balancer can connect to the NTP servers through the management network.

`Fail to configure remote syslog server`	`DNS failed to resolve syslog server SYSLOG_SERVER: REASON`
	Where REASON is the error message reported by the system DNS resolver library.
	
	`failed to probe Syslog server PROTOCOL:// SYSLOG_SERVER (IP_ADDRESS1, IP_ADDRES2, ...)`
	
	Where PROTOCOL is TLS or TCP.
		• The DNS server cannot resolve the syslog server names.
		• The Foundation Load Balancer VM cannot connect to the rsyslog server through TCP or TLS.
		• The UDP syslog server is not probed in the current implementation.
			• Check whether Foundation Load Balancer can connect to the DNS servers.
			• Check whether Foundation Load Balancer can connect to syslog servers through the management network.

Deploying Supervisor with a Simplified Deployment Flow

VMware Cloud Foundation 9.0 allows you to create a simplified Supervisor deployment requiring fewer resources and configuration steps than a typical deployment.

This simplified deployment allows access to a limited subset of Supervisor capabilities. You can then expand the deployment over time to build out to full Supervisor functionality.

The initial deployment consists of the following:

- A single control-plane VM.
- A single network used for both management and workload traffic.
- Support for the VM Service only. vSphere Pods and Supervisor Services are not supported with this configuration.

The next step in expanding this configuration is to add a load balancer. Currently Avi and Foundation load balancers are supported. After the load balancer has been added, support for vSphere Pods and Supervisor Services is enabled.

After the load balancer is added, you have two additional options for modifying the topology of the Supervisor, which can be done in any order:

- Scaling out control-plane VMs: Scaling from 1 control-plane VM to 3 control-plane VMs enables high-availability for the Supervisor.
- Adding a NIC: Adding a NIC allows you to set a different network to be the primary workload network, that is, the network that the control-plane VMs dedicate for their workload traffic.

Requirements for Simplified Supervisor Deployment

Ensure that you meet the following requirements for deploying Supervisor in a simplified deployment.

Network Routing Requirements

Ensure that network routing is configured between the following components.

Components	Purpose
Supervisor Management Network and vCenter	Management traffic between vCenter and Supervisor.
Supervisor Management Network and Spherelet vmkernel management NIC.	Needed for ESX hosts to be managed as Kubernetes worker nodes.
Workload Network and virtual IP (when adding a load balancer)	Needed for load balancer DNS lookups.
Load balancer backend interface and control plane VM workload network	The workload network may either be the same as the management network (single homed) or not (dual homed). The initial simplified deployment is single homed.

Requirement for Secure Access Using FQDN

Use of a fully-qualified domain name (FQDN is required for login to the Supervisor in order to prevent certificate issues. Server Name Indication (SNI) is used to provide the VIP certificate to incoming requests when the request is made by FQDN.

Configure the DNS record for the FQDN to point to whatever address is indicated for the control plane in the vCenter UI.

- If there is no load balancer, this is the floating IP address.
- If there is a load balancer, this is the virtual IP address for the service type load balancer fronting the Supervisor.

Limitations of Simplified Supervisor Configurations

Simplified Supervisor configurations have certain limitations when compared to a full Supervisor deployment.

Make sure you understand the limitations before deciding to use a particular type of configuration.

Limitations of Configurations Without a Load Balancer

Limitations of configurations without a load balancer are:

- vSphere Pods and Supervisor Services are not supported with this configuration. Only the VM Service is available.
- Simplified Supervisor does not support provisioning of persistent volumes.
- Kubernetes Service type load balancers do not receive an external IP address.
- Supervisor Control Plane scale out is not possible.

Add a load balancer to the simplified deployment to allow for scale out and additional workloads.

Limitations of a Single Control Plane VM

Limitations of a single control plane VM configuration are:

- The single control plane VM is a single point of failure. Any downtime of the control plane VM affects control plane availability. Any form of `etcd` data corruption might require a Supervisor restore operation to recover the system.
- A single control plane configuration experiences brief downtime during the following operations:
  - Upgrade of Supervisor Services that run on the control plane
  - Upgrade of the Supervisor itself.
- The single control plane VM configuration might result in memory and/or CPU contention among system controllers due to the inability to distribute work across multiple control plane nodes. Control plane VMs run a number of vSphere controllers to provide various services. A three control plane VM configuration runs two instances of most of these controllers, allowing for another instance to take over if a single node goes down. On a single control plane configuration, all controllers must run on a single node.

Limitations of a Single Network

Limitations of a single network are:

- Workload and management traffic travel on the same network, which might affect how you configure routing in your environment.
- In a dual-homed (two network) configuration, management traffic such as that for vCenter, etcd, backup-restore, and so on uses the management network (eth0), while intra-cluster traffic for operators and services uses the workload network (eth1). Using a single network means potentially more contention and less isolation for management traffic at the Linux VM level.

Activate Supervisor Using the Simplified Deployment Flow

1. Log in to the vCenter using the vSphere Client and select the cluster where you want to activate the Supervisor.
2. Select **Actions** > **Activate Supervisor**.
3. Select a network mode to use for the management and workload network.

   In the initial deployment, workloads run on the management network.

   - DHCP - In this networking mode, all of the IP addresses and network settings for the management and workload network are acquired through DHCP.

Option	Description
Port Group	Select the port group that will serve as the Primary Workload Network to the Supervisor. The primary network handles the traffic for the Kubernetes control plane VMs and Kubernetes workload traffic. Depending on your networking topology, you can later assign a different port group to serve as the network to each namespace. This way, you can provide layer 2 isolation between the namespaces in the Supervisor. Namespaces that do not have a different port group assigned as their network use the primary network.
Service CIDR	Enter a CIDR notation that determines the range of IP addresses for Service type ClusterIP addresses in the Supervisor.
FQDN	Enter the FQDN that will be used to access the Supervisor control plane instead of using the Supervisor control plane IP address. Because SNI (Server Name Indication) is used to select certificates when connecting to a Supervisor with a single NIC, use of the FQDN is required for secure connection to this configuration. The DNS record for the FQDN should point to the Floating IP address.

- Static - Manually enter the settings for the management and workload network.

Option	Description
Port Group	Select the port group that will serve as the Primary Workload Network to the Supervisor. The primary network handles the traffic for the Kubernetes control plane VMs and Kubernetes workload traffic. Depending on your networking topology, you can later assign a different port group to serve as the network to each namespace. This way, you can provide layer 2 isolation between the namespaces in the Supervisor. Namespaces that do not have a different port group assigned as their network use the primary network. VKS clusters use only the network that is assigned to the namespace where they are deployed or they use the primary network if there is no explicit network assigned to that namespace.
IP Address Ranges	Enter an IP range for allocating IP address of Kubernetes control plane VMs and workloads. This address range is used by the Supervisor nodes and, in the case of a single Workload Network, also connects the cluster nodes. This IP range must not overlap with the load balancer VIP range.
Subnet Mask	Enter the subnet mask IP address.
Gateway	Enter the gateway for the primary network.
NTP Servers	Enter the address of the NTP server that you use with your environment if any.
DNS Servers	Enter the IP addresses of the DNS servers that you use with your environment, if any.
Service CIDR	Enter a CIDR notation that determines the range of IP addresses for Service type ClusterIP addresses in the Supervisor.
FQDN	Enter the FQDN that will be used to access the Supervisor control plane instead of using the Supervisor control plane IP address. Because SNI (Server Name Indication) is used to select certificates when connecting to a Supervisor with a single NIC, use of the FQDN is required for secure connection to this configuration. The DNS record for the FQDN should point to the Floating IP address.

4. Select a Storage Policy for the Control Plane VM.
5. Click **Activate**.

Supervisor is activated in a simplified deployment with a single control-plane VM, a single NIC, and no load balancer.

Add a Load Balancer to a Simplified Supervisor Deployment

After the initial simplified Supervisor deployment, the next step you can take to scale out your deployment is to add a load balancer.

Back up both the Supervisor and vCenter before making this or any other topology change to your simplified Supervisor deployment. If you need to restore the Supervisor to the state it was in prior to the topology change, you must also restore vCenter in order to ensure that vCenter recognizes the correct configuration of the Supervisor.

If you are using the Avi Load Balancer, deploy the load balancer before you begin this task as described in the Avi documentation.

If you are using the Foundation Load Balancer, you do not need to deploy it in advance. It is deployed as part of this workflow.

Adding a load balancer enables use of the vSphere Pods and Supervisor Services in addition to the VM Service.

Currently, the following load balancers are supported:

- Foundation Load Balancer
- Avi Load Balancer

After you add a load balancer to your Supervisor deployment, it cannot be removed.

1. From the vSphere Client home menu, select **Supervisor Management**.
2. Select the **Supervisors** tab, and click the Supervisor you want to add the load balancer to.
3. Select **Network** and expand the **Load Balancer** section.
4. Select the load balancer to deploy and click **Add Load Balancer**.
   - Avi Load Balancer
   - Foundation Load Balancer
5. Complete the required parameters for the selected load balancer.
   - Avi Load Balancer Load Balancer

Option	Description
Name	Enter a name for the load balancer deployment.
Endpoint	The IP address of the load balancer controller. The default port is443.
Username	The user name that is configured for theAvi Load Balancer. You use this user name to access the Controller.
Password	The password for the user name.
Server Certificate	The certificate used by the Controller. Provide the certificate that you assigned during the load balancer deployment and configuration. For more information see the Avi documentation.

   - Foundation Load Balancer

Option	Description
Name	Enter a name for the load balancer deployment.
Networks Topology	Select the topology for the load balancer network. For more information on Foundation Load Balancertopologies, see Architecture with Foundation Load Balancer.
Size	Select the size of the Foundation Load Balancer appliance. For more information on compute, storage, and memory requirements for Foundation Load Balancer, see Requirements for Deploying with Foundation Load Balancer.
High Availability	Select the number of load balancer VMs to deploy.
Networks	Configure network parameters for the **Management Network**and **Virtual Server Network**. Select an existing network or click **Create** to create a new network.

After you add the load balancer, reconfigure the DNS record for the Supervisor FQDN so that it points to the virtual IP address for the service type load balancer fronting the Supervisor.

Add a Second NIC to a Simplified Supervisor Deployment

After you have added a load balancer to your simplified deployment, you can add a second NIC to allow for separating Workload Network traffi from Management Network traffic.

- Ensure that you have added a load balancer to your simplified deployment.
- Review the network routing requirements listed in Requirements for Simplified Supervisor Deployment to ensure that your new primary network meets these requirements.

Back up both the Supervisor and vCenter before making this or any other topology change to your simplified Supervisor deployment. If you need to restore the Supervisor to the state it was in prior to the topology change, you must also restore vCenter in order to ensure that vCenter recognizes the correct configuration of the Supervisor.

Adding a second NIC and adding additional control plane VMs can be done in either order. After you have added a second NIC, you cannot revert to a single NIC configuration.

1. Add a workload network to the Supervisor by following the instructions in Add Workload Networks to a Configured with VDS Networking.

   Select **primary** as the port group and set the network name to `primary`.
2. Set the new workload network to be the primary workload network.
   a) In the vSphere Client, select **Supervisor Management**.
   b) Under **Supervisors**, select the Supervisor, and select **Configure**.
   c) Select **Network** and expand **Workload Network**.
   d) Select the **primary** network, and click **Set Primary**.
   e) Click **OK** to confirm that you want to set the new primary network.

Scale Out Control Plane VMs in a Simplified Supervisor Deployment

After you have added a load balancer, you can scale out the number of control plane VMs from 1 to 3 to allow for high-availability.

Ensure that you have added a load balancer to your simplified deployment.

Back up both the Supervisor and vCenter before making this or any other topology change to your simplified Supervisor deployment. If you need to restore the Supervisor to the state it was in prior to the topology change, you must also restore vCenter in order to ensure that vCenter recognizes the correct configuration of the Supervisor.

Adding a second NIC and adding additional control plane VMs can be done in either order. After you have scaled out to 3 control plane VMs, this operation cannot be reversed.

1. From the vSphere Client home menu, select **Management**.
2. Under **Supervisors**, select the Supervisor, and select **Configure**.
3. Select **General** and expand the **Control Plane**section.
4. Next to the text **Control Plane HA is not activated**, click **Activate**.
5. Click **Activate** to confirm that you want to scale out the control plane VMs.

Additional control plane VMs are deployed. You can monitor the status of the reconfiguration on the **Supervisors** tab of the **Supervisor Management** page.

Deploying vSphere Supervisor with NSX and Avi Load Balancer

A Supervisor that is configured with NSX networking uses the software-based networks of the solution and Avi Load Balancer to provide connectivity to external services and DevOps users.

For information about how to deploy the Supervisor with NSX networking and Avi Load Balancer, see the Avi documentation.

Deploying vSphere Supervisor with vDS and Avi Load Balancer

In a Supervisor environment that uses vSphere Distributed Switch (vDS) networking, you can use the Avi Load Balancer for load balancing services.

For information about how to deploy the Supervisor with vDS and Avi Load Balancer, see the Avi documentation.

Deploying vSphere Supervisor from an Exported Configuration

Export a Supervisor Configuration

Checkout how to export the configuration of an existing Supervisor, which you can later import in the Supervisor activation wizard to deploy a new Supervisor instance with a similar configuration. The Supervisor exports in a JSON configuration file, which you can modify as needed and use to deploy a new Supervisor instance.

Deploy a Supervisor.

Exporting a Supervisor configuration allows you to:

- Persist Supervisor configurations. You can export all previous Supervisor configurations and reuse them when needed.
- More efficient troubleshooting. If a Supervisor activation fails, you can adjust the Supervisor configuration directly into the JSON file and restart the process. This allows for quick troubleshooting, as you can modify the settings directly in the JSON file before importing it.
- Streamlined administration. You can share the exported Supervisor configuration with other administrators, to setup new Supervisors with similar settings.
- Consistent format. The exported Supervisor configurations adhere to a standardized format that is applicable to the supported deployment types.

You can also export the Supervisor configuration during the Supervisor activation workflow.

1. Go to **Supervisor Management** > **Supervisor** > **Supervisors**.
2. Select a Supervisor and select **Export Config**.

The configuration is exported and saved in a ZIP file named `wcp-config.zip` that is stored locally in the browser's default download folder. Within the `wcp-config.zip` file you can find:

- A JSON file containing the Supervisor configuration named `wcp-config.jso`. Each configuration setting has a corresponding name and location in the JSON file. This JSON file adheres to a hierarchical data structure.
- A valid JSON schema file named `wcp-config-schema.json`. This file outlines all exportable settings for the Supervisor, encompassing their type, location within the JSON file, and whether they are required. You can use the schema file to generating a sample configuration JSON file , which you can populate manually and import to a fresh activation workflow.

Edit the JSON configuration as needed and use it to deploy new Supervisors.

Deploy a Supervisor by Cloning an Existing Configuration

Learn how to deploy a Supervisor by cloning the configuration of an existing Supervisor instance. Clone a Supervisor in case you want to deploy a new Supervisor instance with similar settings as an already deployed Supervisor.

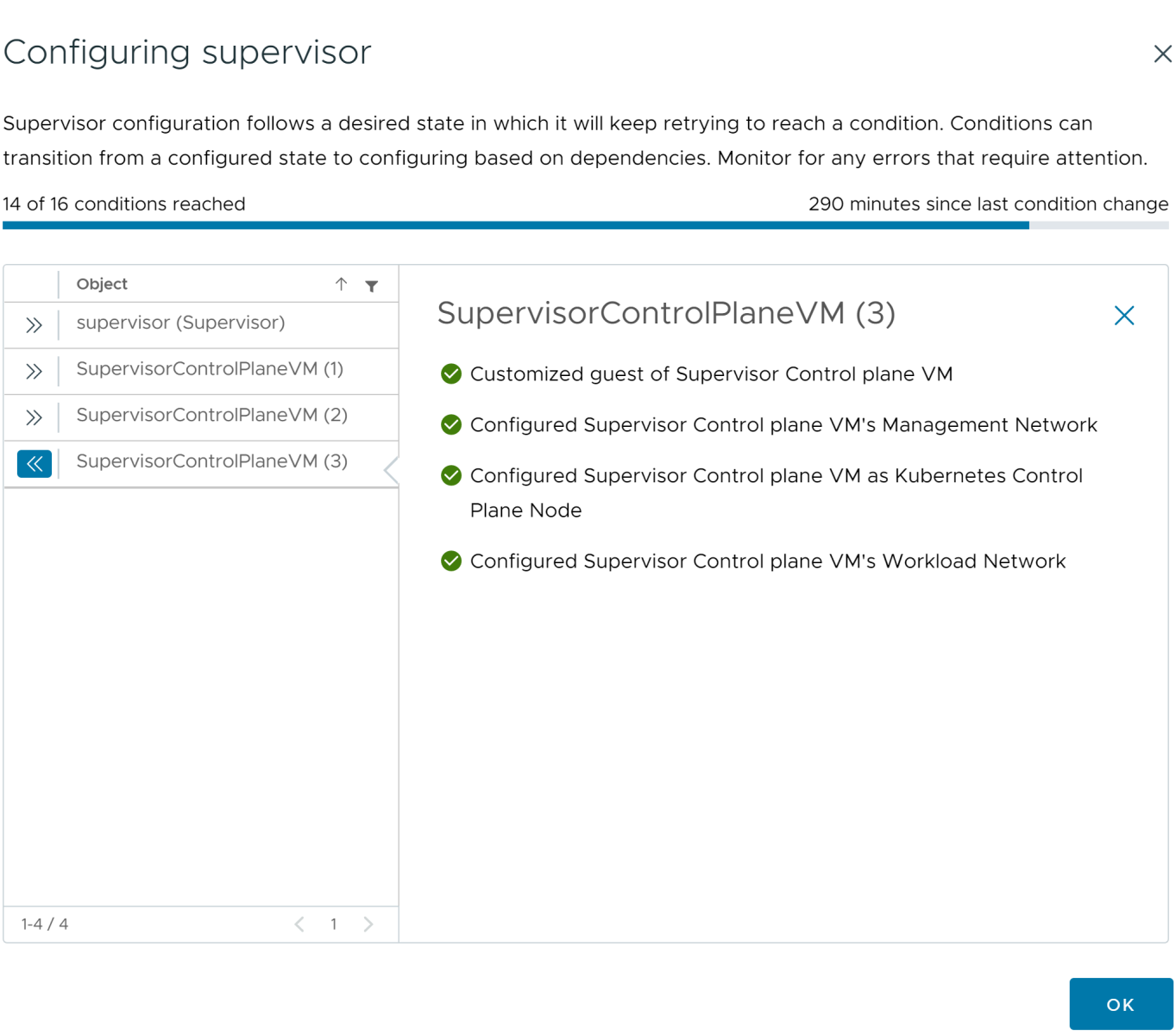
- Complete the prerequisites for configuring vSphere clusters as a Supervisor. See the requirements for each deployment type in the relevant chapters.

1. Go to **Supervisor Management** > **Supervisor** > **Supervisors**.
2. Select the Supervisor, which you want to clone and select **Clone Config**.

   The Supervisor activation wizard opens with values pre-filled from the Supervisor that you selected.
3. Go through the wizard by modifying the values as needed.

Once you complete the wizard for enabling a Supervisor, you can track the activation process and observe potential issues that require troubleshooting. In the **Config Status** column, click **view** next to the status of the Supervisor.

Figure 515: Supervisor activation view



For the deployment process to complete, the Supervisor must reach the desired state, which means that all 16 conditions are reached. When a Supervisor is successfully enabled, its status changes from Configuring to Running. While the Supervisor is in Configuring state, reaching each of the 16 conditions is retried continuously. If a condition is not reached, the operation is retried until success. Because of this reason, the number of conditions that are reached could change back and forth, for example *10 out of 16 conditions reached* and then *4 out of 16 conditions reached* and so on. In very rare cases the status could change to Error, if there are errors that prevent reaching the desired state.

For more information on the deployment errors and how to troubleshoot them, see Resolving Errors Health Statuses on Supervisor Cluster During Initial Configuration Or Upgrade.

Deploy a Supervisor by Importing a JSON Configuration File

Checkout how to automatically populate all the configuration values in the Supervisor activation wizard by importing a JSON configuration file that you have exported from previous Supervisor deployments. When troubleshooting an unsuccessful Supervisor deployment or deploying a new Supervisor with a similar configuration, you can change the configuration values directly in the JSON file before you import it in the wizard. This way you save time from filling in all the values in the activation wizard manually and can just focus on the areas that need change.

- Complete the prerequisites for configuring vSphere clusters as a Supervisor. See the requirements for deploying a Supervisor int the relevant chapters.
- Verify that you have a JSON configuration file that you have exported from an existing Supervisor deployment. The default name of the file is `wcp-config.json`.

You can export the configuration of a Supervisor in two ways:

- During the Supervisor deployment on the **Ready to complete** page of the wizard.
- Export the configuration of an already deployed Supervisor.

1. Initiate a Supervisor deployment in one of the following ways:
   - If you haven't successfully deployed Supervisor yet, click **Get Started** on the Supervisor **Management** page.
   - If you want to deploy an additional Supervisor in your environment, select **Supervisor Management** > **Supervisor** > **Supervisors** > **Add Supervisor**.
2. From the top right corner, select **Import Config**.

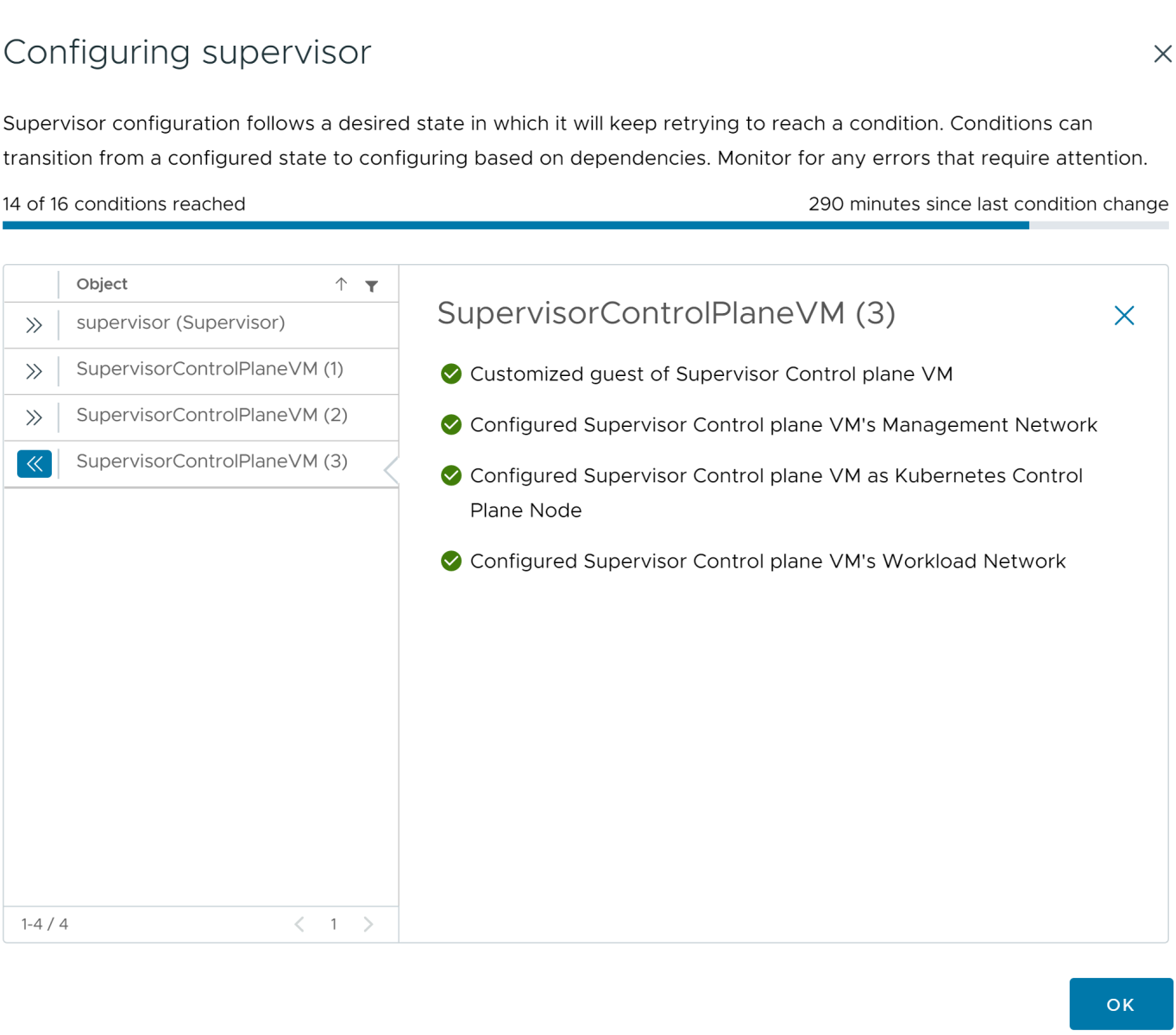
   The vSphere Client verifies the values in the JSON file. Errors might be displayed if the uploaded file is invalid or corrupted JSON. Similarly, errors will appear if the JSON file lacks the spec version or if the spec version exceeds what is currently supported by the client. Therefore, you should only edit the setting that you needed before importing the configuration file. If the file is corrupt, you can use the JSON schema to generate a an empty Supervisor configuration, that you can fill in with the values that you need.
3. In the **Upload Supervisor Configuration** dialog, click **Upload** and select a JSON configuration file you have exported previously.
4. Click **Import**.

   The values recorded in the JSON configuration file are populated in the Supervisor activation wizard. You still might need to enter certain settings manually, such as load balancer password.
5. Click **Next** through the wizard and fill in any values where required.
6. In the **Review and Confirm** page, scroll up and review all the settings that you configured thus far and make final changes if needed.
7. Click **Finish** when ready with reviewing the settings.

   The activation of the Supervisor initiates the creation and configuration of the control plane VMs and other components.

Once you complete the wizard for enabling a Supervisor, you can track the activation process and observe potential issues that require troubleshooting. In the **Config Status** column, click **view** next to the status of the Supervisor.

Figure 516: Supervisor activation view



For the deployment process to complete, the Supervisor must reach the desired state, which means that all conditions are reached. When a Supervisor is successfully enabled, its status changes from Configuring to Running. While the Supervisor is in Configuring state, reaching each of the conditions is retried continuously. If a condition is not reached, the operation is retried until success. Because of this reason, the number of conditions that are reached could change back and forth, for example *10 out of 16 conditions reached* and then *4 out of 16 conditions reached* and so on. In very rare cases the status could change to Error, if there are errors that prevent reaching the desired state.

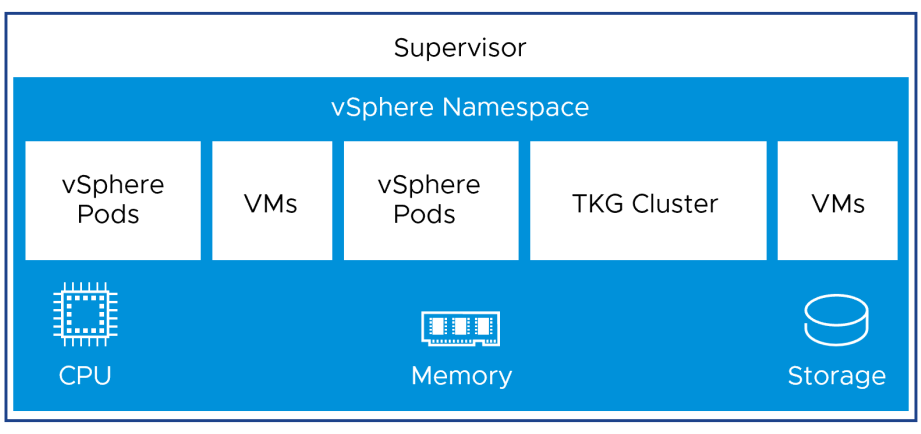
For more information on the deployment errors and how to troubleshoot them, see Resolving Errors Health Statuses on Supervisor Cluster During Initial Configuration Or Upgrade.

Configuring and Managing vSphere Namespaces

vSphere Supervisor workloads, including vSphere Pods, VMs, and VKS clusters, are deployed to a vSphere Namespace. A VI administrator defines the namespace on a Supervisor and configures it with resource quota and user permissions. Depending on the needs of DevOps engineers and workloads they plan to run, the VI administrator might also assign storage policies, VM classes, and content libraries for fetching VM images.

When initially created, the namespace has unlimited resources within the Supervisor. As a VI administrator, you can set limits for CPU, memory, storage, and the number of Kubernetes objects that can run within the namespace. Storage limitations are represented as storage quotas in Kubernetes. A resource pool is created in vSphere per each namespace on the Supervisor

In a Supervisor activated on vSphere Zones, a namespace resource pool is created on the vSphere clusters that are mapped to the zones. For vSphere Zones that have more than one cluster associated with them, a resource pool for the vSphere Namespace is create to only one of the clusteres that are part of the zone. The resources that are utilized to the namespace that spread to more than one vSphere Zone are taken from all three underlying vSphere clusters where the namespaces resouroce pool resides in equal parts. For example, if you dedicate 300 MHz of CPU, 100 MHz are taken from each vSphere cluster.



To provide access to namespaces to DevOps engineer, a VI administrator assigns permission to users or user groups available within an identity source that is associated with vCenter Single Sign-On or from an ODIC provider that is registered with the Supervisor. For more information, see Understanding Authentication and Authorization in vSphere Supervisor.

After a vSphere Namespace is created and configured with resource, object limits, permissions, and storage policies, as a DevOps engineer, you can access the namespace to run the following workloads:

- For information about running Supervisor Services, see https://techdocs.broadcom.com/bin/gethidpage?ux-context-string=svservices-1&appid=sv-async&language=&format=rendered.
- For information about running vSphere Pods, see https://techdocs.broadcom.com/bin/gethidpage?ux-context-string=vsphere-pods-1&appid=vcf-9-1&language=&format=rendered.
- For information about deploying VMs, see Deploying and Managing Virtual Machines in vSphere Supervisor.

Create and Configure a vSphere Namespace on the Supervisor

Check out how to create and configure a vSphere Namespace on the Supervisor. As a vSphere administrator, after you create a vSphere Namespace, you set resources limits to the namespace and permissions so that DevOps engineers can access it. You provide the URL of the Kubernetes control plane to DevOps engineers where they can run workloads on the namespaces for which they have permissions.

- Deploy a Supervisor.
- Create users and groups for DevOps engineers and developers, who need access to the vSphere Namespace. Create the users or groups in identity sources that are connected to vCenter Single Sign-On or in an OIDC provider configured with the Supervisor.
- Create storage policies for persistent storage. If the vSphere Namespace is in a multi-zone Supervisor, use topology aware policies. You cannot assign storage policies that are not topology aware to the multi-zone vSphere Namespace .
- Create VM classes and content libraries for stand-alone VMs.
- Create a vSphere Zone to associate with the vSphere Namespace. For isolation of between the workload and management plane, you can assign a vSphere Zone that is different from the zone where the Supervisor is running. See Add a Cluster as a vSphere Zone to a Supervisor.

- Required privileges: `Namespaces.view`, `Namespace.edit`, and `Namespaces.owner`. See How Permissions Work in Supervisorfor more information.

You can also set resources limits to the vSphere Namespace, assign permissions, and provision or activate the vSphere Namespace service on a cluster as a template. As a result, DevOps engineers can create a vSphere Namespace in a self-service manner and can deploy workloads within it. For more information, see Provision a Self-Service Namespace Template in vSphere Supervisor.

If you use NSX for your Supervisors, you choose to override the networking settings on vSphere Namespace level. Have in mind the following considerations if you select that option:

Table 1791: vSphere Namespace Network Planning Considerations

Consideration	Description
NSX Installation	To override Supervisor network settings for a particular vSphere Namespace, the NSX must include an Edge Cluster dedicated for Tier-0 Gateways (routers) and another Edge Cluster dedicated for Tier-1 Gateways. Refer to the NSX installation instructions provided in the guide *Installing and Configuring vSphere Supervisor*.
IPAM Required	If you override Supervisor network settings for a particular vSphere Namespace, the new vSphere Namespace network must specify Ingress, Egress, and Namespace Network subnets that are unique for the Supervisor and from any other vSphere Namespace network. You will need to manage IP address allocation accordingly.
Supervisor Routing	The Supervisor must be able to route directly to the VKS cluster nodes and ingress subnets. When selecting a Tier-0 Gateway for the vSphere Namespace, you have two options for configuring the required routing: • Use a Virtual Routing and Forwarding (VRF) Gateway to inherit the configuration from the Supervisor Tier-0 Gateway • Use the Border Gateway Protocol (BGP) to configure routes between the Supervisor Tier-0 Gateway and the dedicated Tier-0 Gateway Refer to the NSX Tier-0 Gateways documentation for details on these options.

1. From the vSphere Client home menu, select **Supervisor Management**.
2. Select the **Namespaces** tab.
3. Click **New Namespace**.
4. Select the Supervisor where you want to place the vSphere Namespace.
5. Enter a name for the vSphere Namespace.
   The name must be in a DNS-compliant format.
6. From the **Default Network** drop-down menu, select a Workload Network for the vSphere Namespace.
7. From the **Additional Networks**, select the additional network for the vSphere Namespace.
8. If you have configured the NSX networking stack for your Supervisor, you can select **Override cluster network settings** to override the Supervisor network settings and configure network settings for the namespace.
   Configure the following network settings for the namespace:

Option	Description
Tier-0 Gateway	Select the Tier-0 gateway to associate with the namespace Tier-1 gateway. Selecting a Tier-0 gateway overrides the tier-0 gateway you configured while enabling the cluster, so you must configure the CIDR ranges again. **Note:** The Supervisor must be able to route directly to the VKS cluster nodes and ingress subnets. • If you select a VRF gateway that is linked to the Tier-0 gateway, the network and subnets are automatically configured.

Option	Description
	• If you have selected the NAT mode, you must configure the subnet, ingress, and egress CIDRs. • If you deselect the NAT mode, you must only configure the subnet and ingress CIDRs. **Note:** Once you select a Tier-0 gateway, you cannot change it.
NAT Mode	The NAT mode is selected by default. If you deselect this option, all the workloads such as the vSphere Pods, VMs, and VKS clusters node IP addresses are directly accessible from outside the tier-0 gateway and you do not have to configure the egress CIDRs. **Note:** Once you enable a namespace mode, you cannot change it.
Load balancer Size	Select the size of the load balancer instance on the tier-1 gateway for the namespace.
Namespace Network	Enter one or more IP CIDRs to create subnets/segments and assign IP addresses for workloads that are connected to namespaces. **Note:** Enter the CIDR range if you did not configure it for the cluster. You can configure additional CIDRs after you create the namespace, by editing the namespace network settings.
Namespace Subnet Prefix	Enter the subnet prefix that specifies the size of the subnet reserved for namespaces segments. The default is 28. **Note:** Once you specify the subnet prefix, you cannot change it.
Ingress	Enter a CIDR annotation that determines the ingress IP range for the virtual IP addresses published by the load balancer service for vSphere Pods or VKS clusters. You can configure additional CIDRs after you create the namespace, by editing the namespace network settings.
Egress	Enter a CIDR annotation that determines the egress IP range for the SNAT IP addresses. You can configure additional CIDRs after you create the namespace, by editing the namespace network settings.

9. Select a zone to associate with the vSphere Namespace.
10. Enter a description, and click **Finish**.

    The namespace is created on the Supervisor.
11. Set permissions for users who can access the namespace.

    See How Permissions Work in Supervisor for more information on setting up permissions for users and groups on the vSphere Namespace.

    a) From the **Permissions** pane, select **Add Permissions**.
12. Assign storage to the namespace.

    Storage policies that you assign to the namespace make persistent storage available to the DevOps team and determine the storage placement of persistent volumes.

    a) From the **Storage** pane, select **Add Storage**.
    b) Select a storage policy to control datastore placement of persistent volumes and click **OK**.

    After you assign the storage policy, the system creates a matching Kubernetes storage class in the vSphere Namespace. The storage class is automatically replicated from the namespace to the VKS cluster. When you

assign multiple storage policies to the namespace, a separate storage class is created for each storage policy. For information about verifying storage classes, see Display Storage Classes in a Namespace.

13. From the Capacity and Usage pane, select **Edit Limits** and configure resource limitations to the namespace.

Option	Description
CPU	The amount of CPU resources to reserve for the namespace.
Memory	The amount of memory to reserve for the namespace.
Storage	The total amount of storage space to reserve for the namespace.
Storage policies limits	Set the amount of storage dedicated individually to each of the storage policies that you associated with the namespace.

    A resource pool for the namespace is created on vCenter.

14. Set up VM Service for stand-alone VMs.

    For information, see Deploying and Managing Virtual Machines in vSphere Supervisor.

Share the Kubernetes Control Plane URL with DevOps engineers as well as the user name they can use to log in to the Supervisor through the VCF Consumption CLI. You can grant access to more than one namespace to a DevOps engineer. See Connecting to Supervisor and VKS Clusters.

Managing vSphere Namespaces on a Supervisor with NSX VPC

As a vSphere administrator, you can create vSphere Namespaces on a Supervisor configured with NSX VPC. After you create the namespaces, you can manage the namespaces and resources.

You can create and manage namespaces on Supervisors configured with either the full-stack VPC or VLAN-backed (TEP-less) VPC.

Create and Configure a vSphere Namespace on a Supervisor with NSX VPC

Check out how to create and configure a vSphere Namespace on a Supervisor that uses NSX VPC.

- Deploy a Supervisor.
- Configure NSX VPC for the Supervisor.
- Create users and groups for DevOps engineers and developers, who require access to the vSphere Namespace . Create the users or groups in identity sources that are connected to vCenter Single Sign-On or in an OIDC provider that is configured with the Supervisor.
- Create storage policies for persistent storage. If the namespace is in a three-zone Supervisor, use topology aware policies. You cannot assign storage policies that are not topology aware to the three-zone namespace.
- Create VM classes and content libraries for stand-alone VMs.
- Required privileges: `Namespaces.view`, `Namespace.edit`, and `Namespaces.owner`. See How Permissions Work in Supervisorfor more information.

If NSX VPC is configured on the Supervisor, a Namespace in the Supervisor must use the VPC networking. The Namespace can either inherit the NSX Project of the Supervisor or select another existing NSX Project.

1. From the vSphere Client home menu, select **Supervisor Management**.
2. Select the **Namespaces** tab.
3. Click **New Namespace**.
4. Select the Supervisor where you want to place the vSphere Namespace.
5. Enter a name for the namespace.

    The name must be in a DNS-compliant format.

6. Select the option to override the networking settings.
7. You can use the existing VPC or can create one.
   a) If you select an existing VPC, select the NSX project from the list of compatible projects, the VPC that you want to use, and default subnet size.

      The default Subnet size is 32.
   b) If you create a VPC, the namespace uses the default NSX Project and VPC Connectivity Profile that is configured in the Supervisor.

      Configure the following settings:

Option	Description
Default Subnet Size	The default is 32.
Private VPC CIDRs	List of IPv4 CIDRs. This list is used for IP allocation of private segments. A /26 subnet from the range is reserved for the Avi Service Engine. This list also serves as the default value of private CIDRs for workload namespaces. Separate IP pools are created for each VPC matching the private CIDR ranges. The pools are not shared across namespaces.

8. Review the settings and click **Finish**.

   The namespace is created on the Supervisor.

Create vSphere Namespaces with Auto-created VPC Subnets

As a VI administrator or Tenant administrator, you can create vSphere Namespaces with auto-created full-stack VPC subnets or can assign the VPC subnets to the namespaces.

You can create namespaces in VCF Automation with auto-created VPC subnets. For more information, see Create a Namespace Class in VCF Automation.

Use the steps in this topic to create or assign vSphere Namespaces with auto-created VPC subnets using CLI commands.

1. Log in to the Supervisor.

   See Connecting to Supervisor and VKS Clusters.
2. Create a vSphere Namespaces with the auto-created VPC subnet.

```
com vmware vcenter namespaces instances createv2 --namespace test-vpc --network-spec-network-provider
 NSX_VPC  --network-spec-vpc-network-default-subnet-size 64 --network-spec-vpc-network-vpc-config-private-
cidrs '[{"address": "<IP_ADDRESS>", "prefix":25}]' --supervisor supervisor_name
```

3. Update a vSphere Namespaces with the auto-created VPC subnet.

```
com vmware vcenter namespaces instances update --namespace test-vpc --network-spec-network-provider
 NSX_VPC --network-spec-vpc-config-private-cidrs '[{"address": "<IP_ADDRESS>", "prefix"
:25}, {"address": "<IP_ADDRESS>", "prefix":25}]'
```

Create vSphere Namespaces with Shared VPC Subnets

As a VI administrator or Tenant administrator, you can create vSphere Namespaces with shared VPC subnets or assign shared VPC subnets to the namespaces.

1. Log in to the Supervisor.

   See Connecting to Supervisor and VKS Clusters.
2. Verify if the Supervisor is configured with the Distributed Transit Gateway or the Centralized Transit Gateway. Run the following `kubectl` command to check the status conditions in the VPCNetworkConfiguration CR:

```
kubectl get vpcnetworkworkconfiguration system -o yaml
```

If the `GatewayConnectionReady` is `True`, it indicates that a Centralized Transit Gateway is configured.

For example:

```
Status:
       Conditions:
       - lastTransitionTime: "2025-09-24T21:37:32Z"
         Status: "True"
         Type: GatewayConnectionReady
```

If `ServiceClusterReady` is **True**, it indicates that a Distributed Gateway is configured.

For example:

```
Status:
       Conditions:
       - lastTransitionTime: "2025-09-24 T21:37:32Z"
         Status: "True"
         Type: ServiceClusterReady
```

3. Check if `shared subnets` is activated on the vSphere Namespaces

```
kubectl get capabilities supervisor-capabilities -o
 jsonpath='{.status.supervisor.supports_shared_subnets_with_namespace}'
```

If `shared subnets` is activated, the command returns `true`:

```
 {"activated":true}
```

4. Create a vSphere Namespaces with shared VPC subnet.

Example DCLI command:

```
com vmware vcenter namespaces instances createv2 --namespace test-ns --supervisor supervisor_name --
network-spec-network-provider NSX_VPC --network-spec-vpc-network-shared-subnets /orgs/default/projects/
project-quality/vpcs/ns-test/subnets/vlan-subnet
```

- Instances.CreateSpecV2.NetworkConfigCreateSpecV2.network_provider

Create vSphere Namespaces on VPCs without SNAT and Load Balancer

As a VI administrator or Tenant administrator, you can create vSphere Namespaces on VPCs that have do not have the SNAT and load balancer configured.

vSphere Namespaces usually require load balancing and SNAT as most workloads require ingress/egress. However, internal-only services, batch processing jobs, development, test environments, and deployments where each VM sits on a public Subnet do not require access to the external network. In such environments, you can create vSphere Namespaces on VPCs without SNAT and Load Balancer.

If you provision VKS clusters on vSphere Namespaces without SNAT and load balancers, the clusters might not deploy. To prevent deployments from failing, a validation webhook runs a check on the network information in the NetworkInfo custom resource.

Keep the following considerations in mind when SNAT or load balancers are not enabled:

- VMs. If SNAT is not enabled, network paths might be affected if VMS use Private or Private Gateway access modes. Different VPCs or project can have the same Private or PrivateTGW CIDRs.
- If the load balancer is not activated, then the LoadBalancer-type services, including `VirtualMachineService` resources, cannot be deployed. For more information on load balancers, see NSX Native Load Balancers.

Clusters are created only if the following requirements are met:

- If NAT is disabled, the VKS cluster can only be created with `accessmode: public`.
- If load balancer is disabled, the VKS cluster can only be created with a Custom ClusterClass.

Create vSphere Namespaces with Shared Subnets on a Supervisor with VLAN-backed VPC

As a VI administrator or Tenant administrator, you can create vSphere Namespaces with shared subnets or assign shared subnets to the namespaces on a VLAN-backed VPC.

1. Log in to the Supervisor.

   See Connecting to Supervisor and VKS Clusters.

2. Verify that the Supervisor is configured with the Distributed Transit Gateway. Run the following `kubectl` command to check the status conditions in the VPCNetworkConfiguration CR:

   ```
   kubectl get vpcnetworkworkconfiguration system -o yaml
   ```

   If `ServiceClusterReady` is **True**, it indicates that a Distributed Gateway is configured.

   For example:

   ```
   Status:
          Conditions:
          - lastTransitionTime: "2025-09-24 T21:37:32Z"
            Status: "True"
            Type: ServiceClusterReady
   ```

3. Create a vSphere Namespaces with shared VPC subnet.

   Example DCLI command:

   ```
   com vmware vcenter namespaces instances createv2 --namespace test-ns --supervisor a9e3438f-
   f958-4f95-87f8-2fe013ea5714 --network-spec-network-provider NSX_VPC --network-spec-vpc-network-
   shared-subnets '[{"path":"/orgs/default/projects/project-quality/vpcs/test1_8bfon/subnets/subnet1",
    "pod_default": true, "name":"subnet1", "vm_default": true}]' --namespace testns1  --network-spec-vpc-
   network-vpc /orgs/default/projects/project-quality/vpcs/vpc1
   ```

Managing Existing Namespaces on a Supervisor with NSX VPC

As an administrator you can manage existing vSphere Namespaces that are configured on Supervisors by configuring network settings and changing subnet assignments.

You can manage existing namespaces on Supervisors configured with either the full-stack VPC or VLAN-backed (TEP-less) VPC.

Change Network Settings for a vSphere Namespace on a Supervisor with NSX VPC

As a VI administrator, you can configure network settings when you configure a vSphere Namespace. If you must change the initial network settings, perform this task.

1. In the vSphere Client, navigate to the namespace.
   a) From the vSphere Client home menu, select **Supervisor Management**.
   b) Click the **Namespaces** tab and click the namespace.
2. Click the **Configure** tab and click **General**.
3. In the **Network Settings** section, you can change the following network settings:
   - Default Subnet Type
   - Default Subnet Size
   - Default Private Subnet CIDRs and ranges.

Change Workload Network Settings on a Supervisor configured with VPC Networking

You can assign additional networks to your Namespaces on the Supervisor.

Verify that you have configured the distributed port group. See Create and Configure a vSphere Distributed Switch.

1. In the vSphere Client, select **Supervisor Management**.
2. Under **Supervisors**, select the **Supervisor** and select **Configure**.
3. Select **Network** and expand **Workload Network**.
4. Click **ADD** and configure the workload network settings.

IP Assignment Mode	Select one of the following options: • DHCP • Static • None. Select **None** if you do not want to attach static routes or DHCP client. **Note:** If you select **None**, you cannot use the workload network as the only network, the default network, or with the Foundational load balancer.
Network Role	Select **Workload**.
Port Group	Select the port group.
Network Name	Enter a name for the network.
IP Address Ranges	Enter the IP ranges to be used with workloads on that network. The IP ranges must be in the subnet that is indicated by the subnet mask.
Subnet Mask	Enter the subnet mask.
Gateway	Enter the gateway IP.

5. Click **ADD**.

   The network appears in the list of networks.

Update a vSphere Namespace with a Shared VPC subnet

As a VI administrator or Tenant administrator, you can update existing vSphere Namespaces with shared subnets.

1. Log in to the Supervisor.

   See Connecting to Supervisor and VKS Clusters.
2. To update a namespace with a shared subnet, run the following DCLI command:

   Example DCLI command:

```
com vmware vcenter namespaces instances update --namespace test-ns --network-spec-network-provider NSX_VPC
 --network-spec-vpc-network-shared-subnets '[{"path":"/orgs/default/projects/project-quality/vpcs/vpc1/
subnets/subnet1", "pod_default": true, "name":"subnet1"},{"path":"/orgs/default/projects/project-quality/
vpcs/vpc1/subnets/subnet2", "vm_default": true, "name":"subnet2"}]' --namespace testns1
```

Remove Shared VPC Subnets from a vSphere Namespace

As a VI administrator or Tenant administrator, you can remove shared VPC subnets from a vSphere Namespace.

1. Log in to the Supervisor.

   See Connecting to Supervisor and VKS Clusters.
2. To remove a shared subnet from a namespace, run the following DCLI command:

   Example DCLI command:

```
com vmware vcenter namespaces instances update --namespace test-ns --network-spec-network-provider NSX_VPC
 --network-spec-vpc-config-shared-subnets /orgs/default/projects/project-quality/vpcs/ns-test/subnets/
test-subnet
```

3. Remove all shared subnets from a namespace.

   Example DCLI command:

   ```
   com vmware vcenter namespaces instances set --namespace test-ns --network-spec-network-provider NSX_VPC --
   network-spec-vpc-config-default-subnet-size 32
   ```

   Note:

   You cannot remove a shared subnet that is in use. The update command fails with the following error:

   ```
   Server error: com.vmare.vapi.std.errors.InvalidArgument
   Error message: The removed subnet /orgs/default/projects/project-quality/vpcs/ns-test/subnets/test-
   subnet is still in use.
   ```

Change Network Settings on a Supervisor configured with VDS Networking

You can assign additional networks to your Namespaces on the Supervisor that is configured with VDS networking.

1. In the vSphere Client, select **Supervisor Management**.
2. Under **Supervisors**, select the **Supervisor** and select **Configure**.
3. Select **Network** and expand **Workload Network**.
4. Click **ADD** and configure the workload network settings.

IP Assignment Mode	Select one of the following options: • DHCP • Static • None. **None** indicates that IP address is not assigned and that the DHCP client must not be configured. This mode indicates that the network is intended for L2 connectivity purposes. **Note:** If you select **None**, you cannot use the workload network as the only network, the default network, or with the Foundational load balancer.
Network Role	Select **Workload**.
Port Group	Select the port group.
Network Name	Enter a name for the network.

5. Click **ADD**.

   The network appears in the list of networks.

Change Storage Settings on a vSphere Namespace

Storage policies assigned to a namespace in a Supervisor make persistent storage available to the DevOps team. These storage policies control how persistent volumes and VKS cluster nodes are placed within vSphere datastores. As a VI administrator, you typically assign storage policies when configuring the namespace. If you need to make any changes in the initial storage policy assignments, perform this task.

- Before deleting a storage policy from vCenter or a vSphere Namespace, or changing the storage policy assignment, make sure that no persistent volume claim with the corresponding storage class runs in the namespace. Also, ensure that no VKS cluster is using the storage class.
- If the namespace is in a three-zone Supervisor, use topology aware policies. You cannot assign storage policies that are not topology aware to the three-zone namespace.

1. In the vSphere Client, navigate to the namespace.
   a) From the vSphere Client home menu, select **Supervisor Management**.
   b) Click the **Namespaces** tab and click the namespace.
2. Click the **Storage** tab and click **Storage Policies**.
3. Click the **Edit** icon to change storage policy assignments.

Remove a Workload Network from a Supervisor

You can remove a secondary workload network from the Namespaces on the Supervisor.

1. In the vSphere Client, select **Supervisor Management**.
2. Under **Supervisors**, select the **Supervisor** and select **Configure**.
3. Select **General** to see a list of configured networks.
4. Select the network that you want to remove and click **REMOVE**.

   You cannot remove the default network. When you remove a network, it is marked for deletion. You cannot deploy new workloads on the removed network. Workloads running on the removed network continue to use it until they are migrated to a different network. Once the workloads are migrated, the network is automatically removed.

Remove a vSphere Namespace from Supervisor

You can remove a vSphere Namespace from Supervisor.

- Remove any deployed workloads, including VMs, vSphere Pods, and VKS clusters.
- Required privileges:
  - **Namespaces** > **Modify cluster-wide configuration**
  - **Namespaces** > **Modify namespace configuration**

1. From the vSphere Client home menu, select **Supervisor Management**.
2. Click the **Namespaces** tab.
3. From the list of namespaces available on the Supervisor, select the vSphere Namespace you want to remove.
4. Click **Remove**.

   The system removes the vSphere Namespace. It might take time for the process to complete.

Set Resource Limits to a vSphere Namespace

As a VI administrator, you can set resource limits and container defaults on a vSphere Namespace. DevOps engineers can later override the container defaults in pod specifications but without exceeding the total resource limits set to the namespace by the VI administrator. Container requests translate to resource reservations in pods.

- Verify that you have the **Modify namespace configuration** privilege on the Supervisor.

1. From the vSphere Client home menu, select **Supervisor Management**.
2. Select a vSphere Namespace, select **Configure**, and click **Resource Limits**.
3. Click **Edit**.

   Setting resource limits on a vSphere Namespace where a VKS cluster runs affects the cluster differently depending on the VM class type used for the nodes. Before applying limits, understand the difference between best effort and guaranteed VM classes. For details, see https://techdocs.broadcom.com/bin/gethidpage?ux-context-string=vks-16&appid=sv-async&language=&format=rendered.

Option	Description
CPU	Set a limit to the CPU consumption on the vSphere Namespace.
Memory	Set a limit to the memory consumption on the vSphere Namespace.
Storage	Set a limit on the storage consumption on the vSphere Namespace per storage policy that is used.

Option	Description
Container Defaults	Set the defaults for CPU limits, CPU requests, memory requests, and memory limits for containers in the vSphere Namespace.

Configure Object Limitations on a vSphere Namespace

You can configure limitations for object running in the vSphere Namespace, such as the number of deployments, jobs, daemon sets, stateful sets, and others The limitations that you configure for an object depend on the specifics of your applications and the way you want them to consume resources within avSphere Namespace.

- Verify that you have the **Modify namespace configuration** privilege on the Supervisor.

1. From the vSphere Client home menu, select **Supervisor Management**.
2. Select thevSphere Namespace to which you want to apply object or container restrictions.
3. Select **Configure** and select **Object Limits**.
4. Click **Edit**.

Option	Description
vSphere Pods	The number of vSphere Pods that can run in thevSphere Namespace.
Deployments	The number of deployments that can run in thevSphere Namespace.
Jobs	The number of jobs that can run in thevSphere Namespace.
Daemon Sets	The number of daemon sets that can run in thevSphere Namespace.
Replica Sets	The number of replica sets in thevSphere Namespace.
Replication Controllers	The number of replication controllers that can run in thevSphere Namespace.
Stateful Sets	The number of StatefulSets that can run in thevSphere Namespace.
Config Maps	The number of ConfigMaps that can run in thevSphere Namespace.
Secrets	The number of secrets that can run in thevSphere Namespace.
Persistent Volume Claims	The persistent volume claims that can exists in thevSphere Namespace.
Services	The services that can exists in thevSphere Namespace.

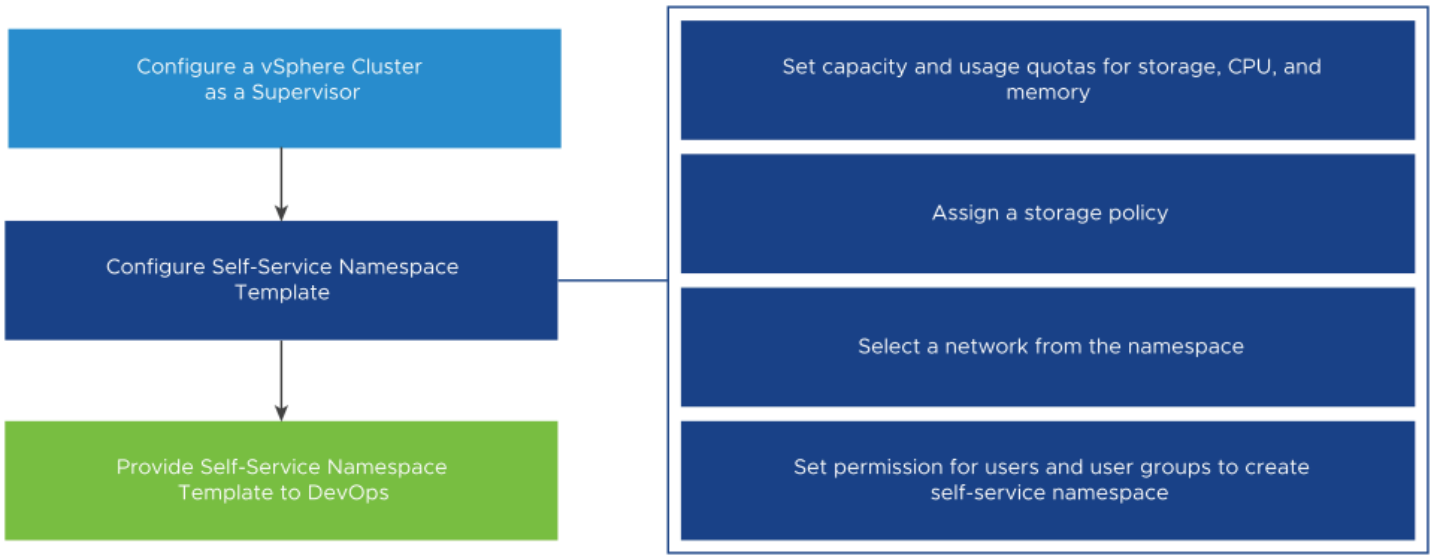
Provision a Self-Service Namespace Template in vSphere Supervisor

As a VI administrator, you can create a vSphere Namespace, set CPU, memory, and storage limits to the namespace, assign permissions, and activate the namespace service on a cluster as a template. As a result, DevOps engineers can create a vSphere Namespace in a self-service manner and deploy workloads within it.

Self-Service Namespace Creation and Configuration Workflow

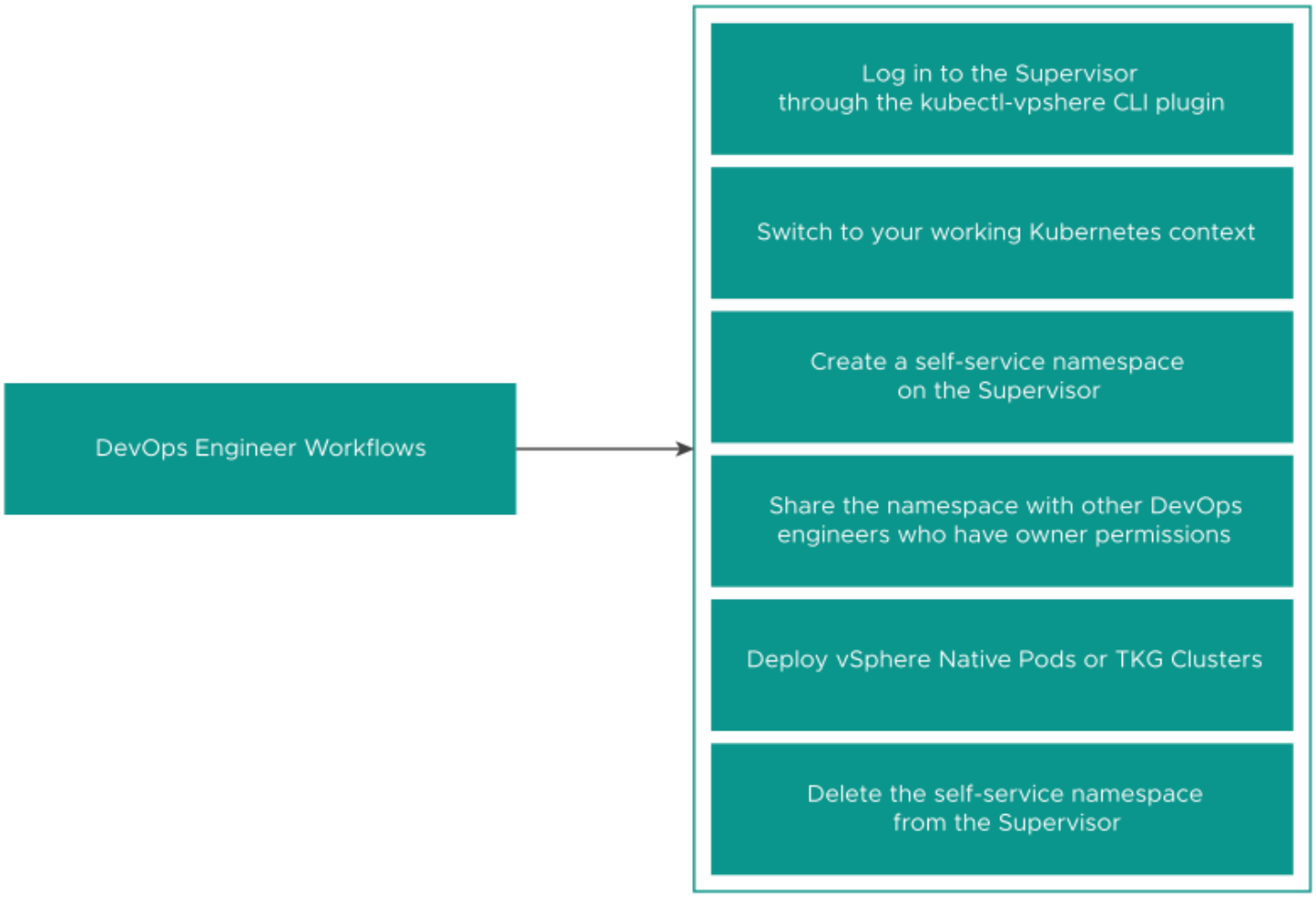
As a VI administrator, you can create a vSphere Namespace, set CPU, memory, and storage limits to the namespace, assign permissions, and provision or activate the namespace service on a cluster as a template.

Figure 517: Self-service Namespace Template Provisioning Workflow



As a DevOps engineer, you can create a vSphere Namespace in a self-service manner and deploy workloads within it. You can share it with other DevOps engineers or delete it when it is no longer required. To share the namespace with other DevOps engineers, contact the VI administrator.

Figure 518: Self-service Namespace Creation Workflow



Create and Configure a Self-Service Namespace Template

As a VI administrator, you can create and configure a Supervisor Namespace as a self-service namespace template. DevOps engineers can then create and delete Supervisor Namespaces using the `kubectl` command line.

Enable a Supervisor.

1. In the vSphere Client, navigate to the Supervisor.
2. Click the **Configure** tab, and select **General** under **Supervisor**.
3. Select **Namespace Service**.
4. Toggle the **Status** switch to enable the feature.
   The **Create Namespace Template** page appears.
5. In the **Configuration** pane, configure resources for the namespace.

Option	Description
CPU	The amount of CPU resources to reserve for the namespace.
Memory	The amount of memory to reserve for the namespace.
Storage	The total amount of storage space to reserve for the namespace.
Storage Policy	Storage policies to use with workloads that require persistent storage.
Network	From the **Network** drop-down menu, select a network for the namespace.
VM Classes	VM classes to deploy stand-alone VMs.
Content Libraries	Content libraries with VM images to use for VM deployments.

6. Click **Next**.
7. In the **Permissions** pane, add DevOps engineers and groups to empower them to use the template to create namespaces.
   Select an identity source and a user or a group and click **Next**.
8. In the **Review and Confirm** pane, the properties you configure are displayed.
   Review the properties and click **Done**.

A namespace template is configured and is in Active state. As a VI administrator, you can edit the template. DevOps engineers can use the template to create namespaces.

Deactivate a Self-Service Namespace

As a VI administrator, you can deactivate a self-service namespace on the cluster.

When you deactivate a self-service namespace template, DevOps engineers cannot use the template to create new namespaces on the cluster. They can delete the namespaces that they have already created.

1. In the vSphere Client, navigate to the Supervisor.
2. Click the **Configure** tab, and select **General** under **Supervisor**.
3. In the **Namespace Service** pane, toggle the **Status** switch to deactivate the template.
4. To activate the template again, toggle the **Status** switch.
   You can either create another self-service namespace or use the existing one.

Create a Self-Service Namespace

As a DevOps Engineer, you can create a self-service namespace and run workloads within it. Once you create the namespace, you can share it with other DevOps Engineers or delete it when it is no longer required.

- Verify that a VI administrator has created and activated a self-service namespace template on the cluster. See Create and Configure a Self-Service Namespace Template .
- Verify that you are added to the permissions list in the self-service namespace template either individually or as a member of a group .
- Get the IP address of the Supervisor control plane.

1. Using the VCF CLI, authenticate with the Supervisor. See Connect to the Supervisor as a vCenter Single Sign-On User.

   ```
   kubectl vsphere login --server=IP-ADDRESS--vsphere-usernameUSERNAME
   ```

2. Switch context to the Supervisor.

   ```
   kubectl config use-contextSUPERVISOR-CLUSTER-IP
   ```

3. Create a self-service namespace on the cluster.

   ```
   kubectl create namespaceNAMESPACE NAME
   ```

   For example

   ```
   kubectl create namespace test-ns
   ```

   Note:

   Owner permissions are available to DevOps Engineers after you enable the Supervisor and upgrade the cluster. If you have only upgraded vCenter and not the cluster, the DevOps engineers will only have edit permissions on the namespaces.

   The namespace that you create appears in the cluster. To share the namespace with other DevOps engineers, contact the VI administrator.

Create a Self-Service Namespace with Annotations and Labels

DevOps engineers can create self-service namespaces with annotations and labels using the kubectl command line.

DevOps engineers can use a YAML manifest with user-defined annotations and labels.

1. Log in to the Supervisor.

   ```
   kubectl vsphere login --server IP-ADDRESS-SUPERVISOR-CLUSTER --vsphere-username VCENTER-SSO-USERNAME
   ```

2. Create a namespace YAML manifest file with annotations and labels.

   ```
   kubectl create -f ns-create.yaml
   ```

   For example, create the following `ns-create.yaml` file:

   ```
   apiVersion: v1
   kind: Namespace
   metadata:
     name: test-ns-yaml
     labels:
       my-label: "my-label-val-yaml"
     annotations:
       my-ann-yaml: "my-ann-val-yaml"
   ```

3. Apply the YAML manifest.

   ```
   kubectl create -f ns-create.yaml
   ```

   Or

   ```
   kubectl apply -f ns-create.yaml
   ```

4. Describe the namespace that you created to see the changes.

```
root@localhost [ /tmp ]# kubectl describe ns test-ns-yaml
Name:         test-ns-yaml
Labels:       my-label=my-label-val-yaml
              vSphereClusterID=domain-c50
Annotations:  my-ann-yaml: my-ann-val-yaml
              vmware-system-namespace-owner-count: 1
              vmware-system-resource-pool: resgroup-171
              vmware-system-resource-pool-cpu-limit: 0.4770
              vmware-system-resource-pool-memory-limit: 2000Mi
              vmware-system-self-service-namespace: true
              vmware-system-vm-folder: group-v172
Status:       Active

Resource Quotas
 Name:             test-ns-yaml
 Resource          Used  Hard
 --------          ---   ---
 requests.storage  0     5000Mi

 Name:                                                                          test-ns-yaml-storagequota
 Resource                                                                       Used  Hard
 --------                                                                       ---   ---
 namespace-service-storage-profile.storageclass.storage.k8s.io/requests.storage  0     9223372036854775807

No LimitRange resource.
```

Update a Self-Service Namespace Using kubectl annotate and kubectl label

As a DevOps engineer you can update or delete self-service namespace annotations and labels using the `kubectl annotate` and `kubectl label` commands.

Verify that you have owner permissions on the namespace that you want to update.

1. Log in to the Supervisor.

   ```
   kubectl vsphere login --server IP-ADDRESS-SUPERVISOR-CLUSTER --vsphere-username VCENTER-SSO-USERNAME
   ```

2. Describe the namespace that you want to update.

```
root@localhost [ /tmp ]# kubectl describe ns testns
Name:         testns
Labels:       my-label=test-label-2
              vSphereClusterID=domain-c50
Annotations:  my-ann: test-ann-2
              vmware-system-namespace-owner-count: 2
              vmware-system-resource-pool: resgroup-153
              vmware-system-resource-pool-cpu-limit: 0.4770
              vmware-system-resource-pool-memory-limit: 2000Mi
              vmware-system-self-service-namespace: true
              vmware-system-vm-folder: group-v154
Status:       Active


Resource Quotas
 Name:             testns
 Resource          Used  Hard
 --------          ---   ---
 requests.storage  0     5000Mi
```

```
 Name:                                                                                testns-storagequota
 Resource                                                                             Used  Hard
 --------                                                                             ---   ---
 namespace-service-storage-profile.storageclass.storage.k8s.io/requests.storage  0     9223372036854775807
```

3. Update annotations using the `kubectl annotate` command.
   For example, `kubectl annotate --overwrite ns testns my-ann="test-ann-3"`
   To delete an annotation, run the command `kubectl annotate --overwrite ns testns my-ann-`
4. Update labels using the `kubectl label` command.
   For example, `kubectl label --overwrite ns testns my-label="test-label-3"`
   To delete a label, run the command `kubectl label --overwrite ns testns my-label-`
5. Describe the namespace to see the updates.

```
root@localhost [ /tmp ]# kubectl describe ns testns
Name:         testns
Labels:       my-label=test-label-3
              vSphereClusterID=domain-c50
Annotations:  my-ann: test-ann-3
              vmware-system-namespace-owner-count: 2
              vmware-system-resource-pool: resgroup-153
              vmware-system-resource-pool-cpu-limit: 0.4770
              vmware-system-resource-pool-memory-limit: 2000Mi
              vmware-system-self-service-namespace: true
              vmware-system-vm-folder: group-v154
Status:       Active


Resource Quotas
 Name:             testns
 Resource          Used  Hard
 --------          ---   ---
 requests.storage  0     5000Mi


 Name:                                                                                testns-storagequota
 Resource                                                                             Used  Hard
 --------                                                                             ---   ---
 namespace-service-storage-profile.storageclass.storage.k8s.io/requests.storage  0     9223372036854775807


No LimitRange resource.
```

Update a Self-Service Namespace Using kubectl edit

As a DevOps engineer you can update self-service namespaces using the `kubectl edit` command.

Verify that you have owner permissions on the namespace that you want to update.

1. Log in to the Supervisor.

   ```
   kubectl vsphere login --server IP-ADDRESS-SUPERVISOR-CLUSTER --vsphere-username VCENTER-SSO-USERNAME
   ```

2. Describe the namespace that you want to update.

```
 kubectl describe ns testns-1
 Name:         testns
```

```
Labels:       vSphereClusterID=domain-c50
Annotations:  my-ann: test-ann-2
              vmware-system-namespace-owner-count: 2
              vmware-system-resource-pool: resgroup-153
              vmware-system-resource-pool-cpu-limit: 0.4770
              vmware-system-resource-pool-memory-limit: 2000Mi
              vmware-system-self-service-namespace: true
              vmware-system-vm-folder: group-v154
Status:       Active


Resource Quotas
 Name:             testns-1
 Resource          Used  Hard
 --------          ---   ---
 requests.storage  0     5000Mi


 Name:                                                                          testns-1-storagequota
 Resource                                                                       Used  Hard
 --------                                                                       ---   ---
 namespace-service-storage-profile.storageclass.storage.k8s.io/requests.storage  0     9223372036854775807
```

3. Edit the namespace using the `kubectl edit` command.

   For example, `kubectl edit ns testns-1`

   The `kubectl edit` command opens the namespace manifest in the text editor defined by your KUBE_EDITOR or the EDITOR environment variable.

4. Update the labels.

   For example, `my-label=test-label`

5. Update the annotations.

   For example, `my-ann: test-ann`

6. Describe the namespace to see the updates.

```
root@localhost [ /tmp ]# kubectl describe ns testns-1
Name:         testns-1
Labels:       my-label=test-label
              vSphereClusterID=domain-c50
Annotations:  my-ann: test-ann
              vmware-system-namespace-owner-count: 1
              vmware-system-resource-pool: resgroup-173
              vmware-system-resource-pool-cpu-limit: 0.4770
              vmware-system-resource-pool-memory-limit: 2000Mi
              vmware-system-self-service-namespace: true
              vmware-system-vm-folder: group-v174
Status:       Active


Resource Quotas
 Name:             testns-1
 Resource          Used  Hard
 --------          ---   ---
 requests.storage  0     5000Mi


 Name:                                                                          testns-1-storagequota
 Resource                                                                       Used  Hard
```

```
--------                                                                    ---   ---
namespace-service-storage-profile.storageclass.storage.k8s.io/requests.storage  0     9223372036854775807


No LimitRange resource.
```

Delete a Self-Service Namespace

As a DevOps Engineer, you can delete a self-service namespace that you create.

Verify that you have created a self-service namespace using the VCF CLI.

1. Using the VCF CLI, authenticate with the Supervisor. See Connect to the Supervisor as a vCenter Single Sign-On User.
2. Delete the self-service namespace from the cluster.

   ```
   kubectl delete namespace NAMESPACE NAME
   ```

   For example:

   ```
   kubectl delete namespace test-ns
   ```

Add Security Policies to an NSX vSphere Namespace

A Supervisor that uses NSX networking supports network security policies configured through a security policy CRD.

Create a Security Policy

As a DevOps, you can configure the security policy CRD to apply an NSX based security policy to a Supervisor namespace. The security policy protects traffic for vSphere Pods and VMs. VMs include VKS cluster nodes and other VMs deployed in the Supervisor.

Use NSX version 3.2 or later.

1. Create a security policy CRD.

   For the fields to use and CRD examples, see NSX Operator Security Policy CRD documentation on GitHub.
2. Access your namespace in the Kubernetes environment.

   See Connecting to Supervisor and VKS Clusters.
3. Apply the security policy to the namespace.

   ```
   kubectl apply -f policy-name.yaml
   ```
4. View your security policy.
   a) View details for the security policy.

      ```
      kubectl get securitypolicy policy-name
      ```

   b) View a description of your security policy.

      ```
      kubectl describe securitypolicy policy-name
      ```

You can also use the NSX UI to view details of your policy. For information, see the *VMware NSX Documentation* page.

Configuring and Managing vSphere Zones

Check out how to manage vSphere Zones in vCenter. After the initial deployment, you can then add more vSphere Zones to the Supervisor and assign them to vSphere Namespaces. You can also increase the capacity of a vSphere Zone by adding new clusters to it. To provide separation between the workload and management planes in your environment, you can dedicate the vSphere Zones where the Supervisor runs to the management plane only and assign different vSphere Zones to vSphere Namespaces for workloads only. You can also remove vSphere Zones from vSphere Namespaces and the Supervisor.

Add a Cluster as a vSphere Zone to a Supervisor

You can add a vSphere Cluster as a vSphere Zone to a Supervisor and assign it as a workload zone to vSphere Namespaces.

- Configure vSphere DRS in fully-automated mode to the cluster.
- Configure vSphere HA to the cluster.
- Configure shared storage with vSAN to a different solution to the cluster.
- Add the cluster to the same VDS where the Supervisor is connected.
- Required privileges:
  - **Namespaces** > **Modify cluster-wide configuration**
  - **Namespaces** > **Modify namespace configuration**
- In NSX-based setup, the cluster must have connectivity to the same NSX overlay network as the Supervisor.
- In a VDS networking setup, verify that the cluster is connected to distributed port group that is configured as Workload Network to the Supervisor.

1. In the vSphere Client, navigate to the cluster that you want to add as a vSphere Zone to a Supervisor.
2. Select **Actions** > **Add as a Zone to a Supervisor**
3. Select **New Zone** and click **Next**.
4. Enter a vSphere Zone name and click **Next**.
5. Select the Supervisor where you want to add the zone and click **Next**.
6. Review your settings and click **Finish**.

The new zone is added to the Supervisor and you can add to vSphere Namespaces.

Add a New Cluster to an Existing vSphere Zone

You can expand the capacity of the existing vSphere Zones by adding a new vSphere cluster to them. vSphere Namespaces that are already associated with the vSphere Zone will not use the newly-added cluster. Only when you associate new vSphere Namespaces with the vSphere Zone, they can use the newly-added cluster for resource utilization.

- Configure vSphere DRS in fully automated mode to the cluster.
- Configure vSphere HA to the cluster.
- Configure shared storage with vSAN to a different solution to the cluster.
- Add the cluster to the same VDS where the Supervisor is connected.
- Required privileges:
  - **Namespaces** > **Modify cluster-wide configuration**
  - **Namespaces** > **Modify namespace configuration**
- In NSX based setup, the cluster must have connectivity to the same NSX overlay network as the Supervisor.
- In a VDS networking setup, verify that the cluster is connected to distributed port group that is configured as workload network to the Supervisor.
- If you add the cluster to a vSphere Zone that is used for workloads.
  - The cluster must have access to at least 1 datastore that is compatible with the storage policy that is configured with the vSphere Namespaces the vSphere Zone is associated with.
- If you want to add the cluster to a vSphere Zone used for management.
  - The cluster must be configured with the Supervisor management network.
  - The cluster must have acces to at least one datastore that is compatible with the storage policy used for the Supervisor control plane.

1. In the vSphere Client, navigate to the vCenter instance.
2. Select **vSphere Zones**.
3. Right-click the zone to which you want to add a new cluster and select **Associate cluster**.
4. Select from clusters that are not associated with any vSphere zone.
5. Click **Run Compatibility Check** to determine which clusters do not meet the compatibility requirements for that zone.

   If a cluster is not connected to the workload or management networks that are associated with the zone, an error is displayed.
6. Click **Associate cluster**.

Add a vSphere Zone to a vSphere Namespace

You can configure up to three vSphere Zones to a vSphere Namespace in a Supervisor to increase the capacity for running workarounds on the vSphere Namespace.

- Add a vSphere cluster as a zone to the Supervisor.
- Required privileges:
  - **Namespaces** > **Modify cluster-wide configuration**
  - **Namespaces** > **Modify namespace configuration**

1. In the vSphere Client, navigate to **Supervisor Management**.
2. Select **Namespaces**, and select the vSphere Namespace where you want to add the zone.
3. Select **Zones** and select **Add Zone**.
4. Select a zone that you want to add as a vSphere Zone to the vSphere Namespace and click **Add Zone**.

Remove a vSphere Zone from a vSphere Namespace

Check out how to remove a vSphere Zone from a vSphere Namespace in case you want to move it another vSphere Namespace or stop using the zone for running containerized workloads.

- Migrate all the workloads currently running on the zone to a different zone or vSphere Namespace.
- Verity that there's at least one vSphere Zone that is assigned to the vSphere Namespace. You cannot remove a Zone if it's the only zone that is assigned to the vSphere Namespace.

1. In the vSphere Client, navigate to **Supervisor Management**.
2. Select **Namespaces**, and select the vSphere Namespace that is associated with the Workload Zone that you want to remove.
3. Select **Zones**.
4. Select the vSphere Zone that you want to remove and click **Mark for Removal**.

The vSphere Zone is marked for removal and you are unable to run any new workloads on it. The zone will be automatically removed once it's empty of workloads, or after a few seconds in case it's emptied of workloads already.

Remove a vSphere Zone from a Supervisor

Remove a vSphere Zone from the Supervisor and respectively from all the vSphere Namespaces that are part of the Supervisor. You cannot remove the vSphere Zone marked as Management where the Supervisor is running.

- Migrate all the workloads currently running on the zone to a different zone.

  Note:

  Migration of Persistent Volume Claims (PVCs) and Persistent Volumes (PVs) to a different zone is not supported. If you remove a zone, the PVC or PV from the removed zone cannot be used by a pod in a different zone.
- Verify that all vSphere Namespaces with which the vSphere Zone is associated have at least one different zone configured.

1. In the vSphere Client, navigate to **Supervisor Management**.
2. Select **Supervisors**, and select the Supervisor from which you want to remove the vSphere Zone.
3. Select **Zones**.
4. Select the vSphere Zone that you want to remove and click **Mark for Removal**.

The vSphere Zone is marked for removal, and you are unable to run new workloads on it. The zone will be automatically removed once it is removed from any other vSphere Namespaces it is associated with, or after a few seconds if there are no vSphere Namespaces using it.

Remove a vSphere Cluster from a vSphere Zone

Remove a cluster from a vSphere Zone when you want to decommission the zone or to use the cluster resources in a different zone. A vSphere Zone must have at least one cluster that is associated with it. You cannot remove a cluster from a vSphere Zone if the cluster is the last remaining in that zone.

Workloads that are running on the custer that you remove will be migrated to the remaining clusters in the vSphere Zone. The vSphere Zone must have enough resources to accommodate the workloads running on the cluster that you want to remove.

1. In the vSphere Client, navigate to the vCenter instance.
2. Select **vSphere Zones**.
3. Select the zone from which you want to remove a cluster.
4. Select the cluster that you want to remove and click **Remove Cluster**.

Once all the workloads are migrated from the cluster to be removed to the remaining clusters in the vSphere Zone, the cluster is removed from the zone.n

Connecting to Supervisor and VKS Clusters

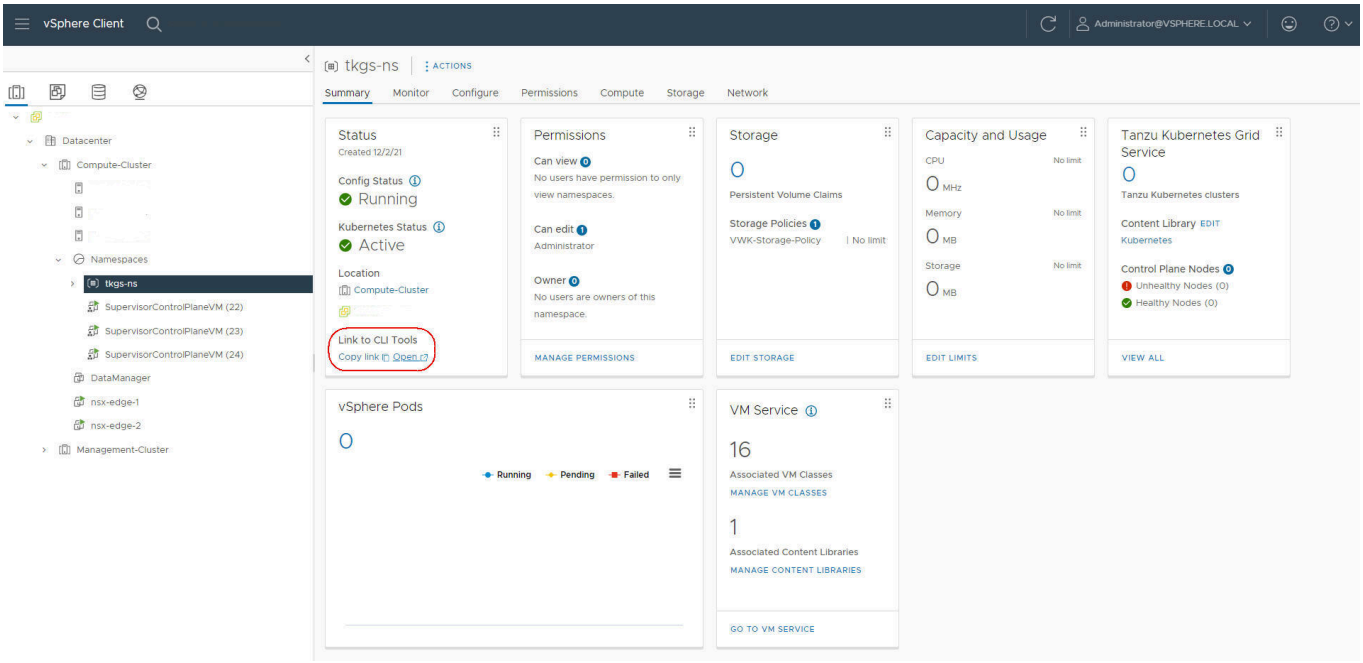
You connect to the Supervisor to provision VKS clusters, vSphere Pods, and VMs. Once provisioned, you can connect to VKS clusters by using various methods and authenticate based on your role and objective.

Download and Install the VCF Consumption CLI

You can use the VCF Consumption CLI to log in to the Supervisor, access the vSphere Namespaces for which you have permissions, and deploy and manage vSphere Pods, VKS clusters, and VMs.

The VCF Consumption CLI download package includes an executable binary, called `vcf`. You can use it to connect to the Supervisor and VKS clusters with vCenter Single Sign-On credentials.

1. In the vSphere Client, get the URL for the VCF Consumption CLI download instructions.
   a) Navigate to **Supervisor Management** > **Namespaces**, and select a vSphere Namespace.
   b) Select the **Summary** tab and locate the **Status** pane.
   c) Under **Link to CLI Tools**, click **Open** or **Copy Link**.



2. In a browser, open the VCF Consumption CLI download URL.



3. Select the operating system.
4. Download the `vcf-cli.tar.gz` (or `vcf-cli.zip`) file.
5. Extract the contents of the compressed file to a working directory.

   The `vcf-cli.tar.gz (.zip)` package contains an executable file: `vcf-cli-{os}_{arch}`, for example, `vcf-cli-darwin_amd64`. This binary helps you authenticate with the Supervisor and VKS clusters using your vCenter Single Sign-On credentials.

6. Rename the executable file to `vcf` and add the location of the executable to your system's PATH variable.
7. To verify the installation of the VCF CLI, start a shell, terminal, or command prompt session and run the `vcf` command.

   You see the VCF CLI banner message, and the list of command-line options for the CLI.

Configure Secure Login for vSphere Supervisor.

Configure Secure Login for vSphere Supervisor

To log in securely to the Supervisor and VKS clusters, configure VCF CLI with the appropriate TLS certificate and ensure that you are running the latest edition of VCF CLI .

Supervisor CA Certificate

vSphere Supervisor supports vCenter Single Sign-On for cluster access by using the VCF CLI command `vcf context create`…. To install and use this utility, see Download and Install the VCF CLI.

The VCF CLI defaults to secure login and requires a trusted certificate, the default being the certificate signed by the vCenter root CA. Although the CLI supports the `--insecure-skip-tls-verify` flag, for security reasons this is not recommended.

To securely log in to the Supervisor and VKS clusters by using the VCF CLI, you have two options:

Option	Instructions
Download and install the vCenter root CA certificate on each client machine.	Refer to the Broadcom knowledge base article How to download and install vCenter root certificates.
Replace the VIP certificate used for the Supervisor with a certificate signed by a CA each client machine trusts.	See *Replace the VIP Certificate to Securely Connect to the API Endpoint* in the Supervisor documentation.

Note:

For additional information on vSphere authentication, including vCenter Single Sign-On, managing and rotating vCenter certificates, and troubleshooting authentication, see the *vSphere Authentication* documentation. For more information about vSphere Supervisor certificates, see Broadcom Knowledge Base article 323421.

VKS Cluster CA Certificate

To connect securely with the VKS cluster using the VCF CLI, you download the VKS cluster CA certificate. Although the CLI supports the `--insecure-skip-tls-verify` flag, for security reasons this is not recommended.

For details on connecting to the VKS cluster, see Connect to a VKS Cluster with VCF CLI.

Connect to the Supervisor as a vCenter Single Sign-On User

To provision vSphere Pods, VKS clusters, or VMs, you connect to the Supervisor using the VCF CLI and authenticate with your vCenter Single Sign-On credentials.

- Get your vCenter Single Sign-On credentials from the VI administrator.
- Get the IP address of the Supervisor control plane from the VI administrator.
- To log in by using an FQDN instead of the control plane IP address, get an FQDN configured during the Supervisor enablement.
- Get the name of the vSphere Namespace for which you have permissions.
- Get confirmation that you have **Edit** permissions on the vSphere Namespace to be able to perform such actions as create resources in the namespace.
- Download and Install the VCF CLI.

- Verify that the certificate served by the Supervisor is trusted on your system, either by having the signing CA installed as a Trust Root or by adding the certificate as a Trust Root directly. See Configure Secure Login for Supervisor Clusters.

After you log in to the Supervisor, the VCF CLI generates the context for the Supervisor. Configuration context contains a Supervisor, a vSphere Namespace, and a user. You can view the cluster context in the file `.kube/config` in the user's home directory. This file is commonly called the `kubeconfig` file.

> **Note:**
> If you have an existing `kubeconfig` file, it is appended with each Supervisor context. The VCF CLI respects the KUBECONFIG environment variable that kubectl itself uses. Although not required, it can be useful to set this variable before running `vcf context create...`, so that the information is written to a new file, instead of being added to your current `kubeconfig` file.

1. To view the command syntax and options for logging in, run the following command.

   ```
   vcf context create --help
   ```

2. To connect to the Supervisor, run the following command, using the Supervisor IP address or FQDN as the endpoint.

   ```
   vcf context create --endpoint <SUPERVISOR-ADDRESS> --username <VCENTER-SSO-USER> --ca-certificate <PATH-TO-CERTIFICATE-FILE>
   ```

   For example:

   ```
   vcf context create --endpoint 10.92.42.13 --username administrator@example.com --ca-certificate ~/ca_root.cert
   vcf context create --endpoint wonderland.example.com --username administrator@example.com --ca-certificate ~/ca_root.cert
   ```

   This action creates a configuration file with the JSON Web Token (JWT) to authenticate to the Kubernetes API.

3. To authenticate, enter the password for the user.

   You can also set an environment variable `VCF_CLI_VSPHERE_PASSWORD` with the password value.

   After you connect to the Supervisor, you can view the configuration contexts that are accessible to you.

4. To view details of your configuration contexts, run the following command:

   ```
   vcf context list
   ```

   The CLI displays the details for each available context.

5. To switch between contexts, use the following command:

   ```
   vcf context use <example-context-name>
   ```

You can now connect to a VKS cluster as a vCenter Single Sign-On user.

Connect to a VKS Cluster with VCF CLI

Connect to a VKS cluster using the VCF Consumption CLI.

Obtain the following information from your VI administrator:

- Get your vCenter Single Sign-On credentials.
- Get the IP address of the Supervisor control plane.
- Get the name of the vSphere Namespace.
- Download and Install the VCF Consumption CLI.

After you log in to the VKS cluster, the VCF CLI generates the context for the cluster. In Kubernetes, a configuration context contains a cluster, a namespace, and a user. You can view the cluster context in the file `.kube/config`. This file is commonly called the `kubeconfig` file.

> **Note:**
> If you have an existing `kubeconfig` file, it is appended with each cluster context. The VCF CLI respects the KUBECONFIG environment variable that kubectl itself uses. Although not required, it can be useful to set this variable before running `vcf context create...` so that the information is written to a new file, instead of being added to your current `kubeconfig` file.

Log in to a VKS Cluster Using vCenter Single Sign-On Credentials

Follow these steps to log in to the VKS cluster using vCenter Single Sign-On credentials and create CLI context. The context allows you to use other VCF CLI commands to manage the cluster.

1. To view the command syntax and options for logging in, run the following command.

   ```
   vcf context create --help
   ```

2. To connect to the VKS cluster, run the following command.

   ```
   vcf context create --endpoint <SUPERVISOR-ADDRESS> --username <VCENTER-SSO-USER> --ca-certificate <PATH-TO-CERTIFICATE-FILE> --workload-cluster-name <VKS-CLUSTER-NAME> --workload-cluster-namespace <VSPHERE-NAMESPACE>
   ```

   For example,

   ```
   vcf context create --endpoint 10.92.42.137 --username operator@example.com --ca-certificate ~/ca_root.cert
    --workload-cluster-name vks-cluster-01 --workload-cluster-namespace vks-cluster-ns
   ```

   Or, if the Supervisor has been enabled with a fully qualified domain name (FQDN):

   ```
   vcf context create --endpoint wonderland.example.com --username operator@example.com --ca-certificate ~/ca_root.cert --workload-cluster-name vks-cluster-01 --workload-cluster-namespace vks-cluster-ns
   ```

   This action creates a configuration file with the JSON Web Token (JWT) to authenticate to the Kubernetes API.

3. To authenticate, enter your vCenter Single Sign-On password.

   You can also set an environment variable `VCF_CLI_VSPHERE_PASSWORD` with the password value.

   If the operation is successful, you see the message `Logged in successfully`, and you can run kubectl commands against the cluster. If the command returns `Error from server (Forbidden)`, it usually means you do not have the required permissions.

4. To get a list of contexts available to you, run the following command.

   ```
   vcf context list
   ```

   This command lists the configuration contexts you have access to. You see a configuration context for the target cluster, such as `vcf-cluster-01`.

5. To use the context for the target cluster, run the following command.

   ```
   vcf context use <example-context-name>
   ```

Log in Using the VKS Cluster Plugin

Alternatively, you can log in using the cluster plugin of VCF CLI. This method works with the Identity Provider (IdP) configured on the Supervisor cluster.

1. To view the command syntax and options for logging in, run the following command.

   ```
   vcf context create --help
   ```

2. Connect to the Supervisor and activate the namespace context that has the VKS cluster.

   For information, see Connect to the Supervisor as a vCenter Single Sign-On User.

3. Check if the cluster plugin is installed.

   ```
   vcf plugin list
   ```

   You see all plugins installed on your system. Look for the cluster plugin in the list.

4. If the cluster plugin is not installed, run the following command.

   ```
   vcf plugin install cluster
   ```

5. Use the cluster plugin to create a kubecontext for the VKS cluster.

   ```
   vcf cluster kubeconfig get <VKS-CLUSTER-NAME> --export-file <PATH-TO-VKS-KUBECONFIG-FILE>
   ```

For example,

```
vcf cluster kubeconfig get vks-cluster-01 --export-file ~/vks_kube_config
```

6. After successful context creation, to list cluster nodes, run the following command.

   ```
   kubectl get nodes --kubeconfig <PATH-TO-VKS-KUBECONFIG-FILE>
   ```

   You see the control plane and worker nodes in this cluster.

7. To list all the cluster pods, run the following command.

   ```
   kubectl get pods -A --kubeconfig <PATH-TO-VKS-KUBECONFIG-FILE>
   ```

   You see all the pods in this cluster across all Kubernetes namespaces that you have access to. If you have not deployed any workloads, you do not see any pods in the default namespace.

Grant Developer Access to VKS Clusters

Developers are the target users of Kubernetes. Once a VKS cluster is provisioned, you can grant developer access using vCenter Single Sign-On authentication.

Authentication for Developers

A cluster administrator can grant cluster access to other users, such as developers. Developers can deploy pods to clusters directly using their user accounts, or indirectly using service accounts.

- For user account authentication, VKS clusters support vCenter Single Sign-On users and groups. The user or group can be local to the vCenter, or synchronized from a supported directory server.
- For service account authentication, you can use service tokens. For more information, see the Kubernetes documentation.

Adding Developer Users to a Cluster

To grant cluster access to developers:

1. Define a `Role` or `ClusterRole` for the user or group and apply it to the cluster. For more information, see the Kubernetes documentation at
   Using RBAC Authorization.
2. Create a `RoleBinding` or `ClusterRoleBinding` for the user or group and apply it to the cluster. See the following example.

Example `RoleBinding`

To grant access to a vCenter Single Sign-On user or group, the subject in the `RoleBinding` must contain either of the following values for the `name` parameter.

Table 1792: Supported User and Group Fields

Field	Description
`sso:`*`USER-NAME@DOMAIN`*	For example, a local user name, such as `sso:joe@vsphere.local`.
`sso:`*`GROUP-NAME@DOMAIN`*	For example, a group name from a directory server integrated with the vCenter, such as `sso:devs@ldap.example.com`.

The following example RoleBinding binds the vCenter Single Sign-On local user named Joe to the default ClusterRole named `edit`. This role permits read/write access to most objects in a namespace, in this case the `default` namespace.

```
kind: RoleBinding
apiVersion: rbac.authorization.k8s.io/v1
metadata:
```

```
  name: rolebinding-cluster-user-joe
  namespace: default
roleRef:
  kind: ClusterRole
  name: edit                          #Default ClusterRole
  apiGroup: rbac.authorization.k8s.io
subjects:
- kind: User
  name: sso:joe@vsphere.local         #sso:<username>@<domain>
  apiGroup: rbac.authorization.k8s.io
```

Configuring and Managing a Supervisor

As a VI administrator, you enable a vSphere cluster as a Supervisor. You can select between creating the Supervisor with the vSphere networking stack or with VMware NSX (NSX) as the networking solution. A cluster configured with NSX supports running a vSphere Pod and a VKS cluster created through the vSphere Kubernetes Service. A Supervisor that is configured with the vSphere networking stack only supports VKS clusters.

After you enable the Supervisor, you can use the vSphere Client to manage and monitor the cluster.

Replace the VIP Certificate to Securely Connect to the Supervisor API Endpoint

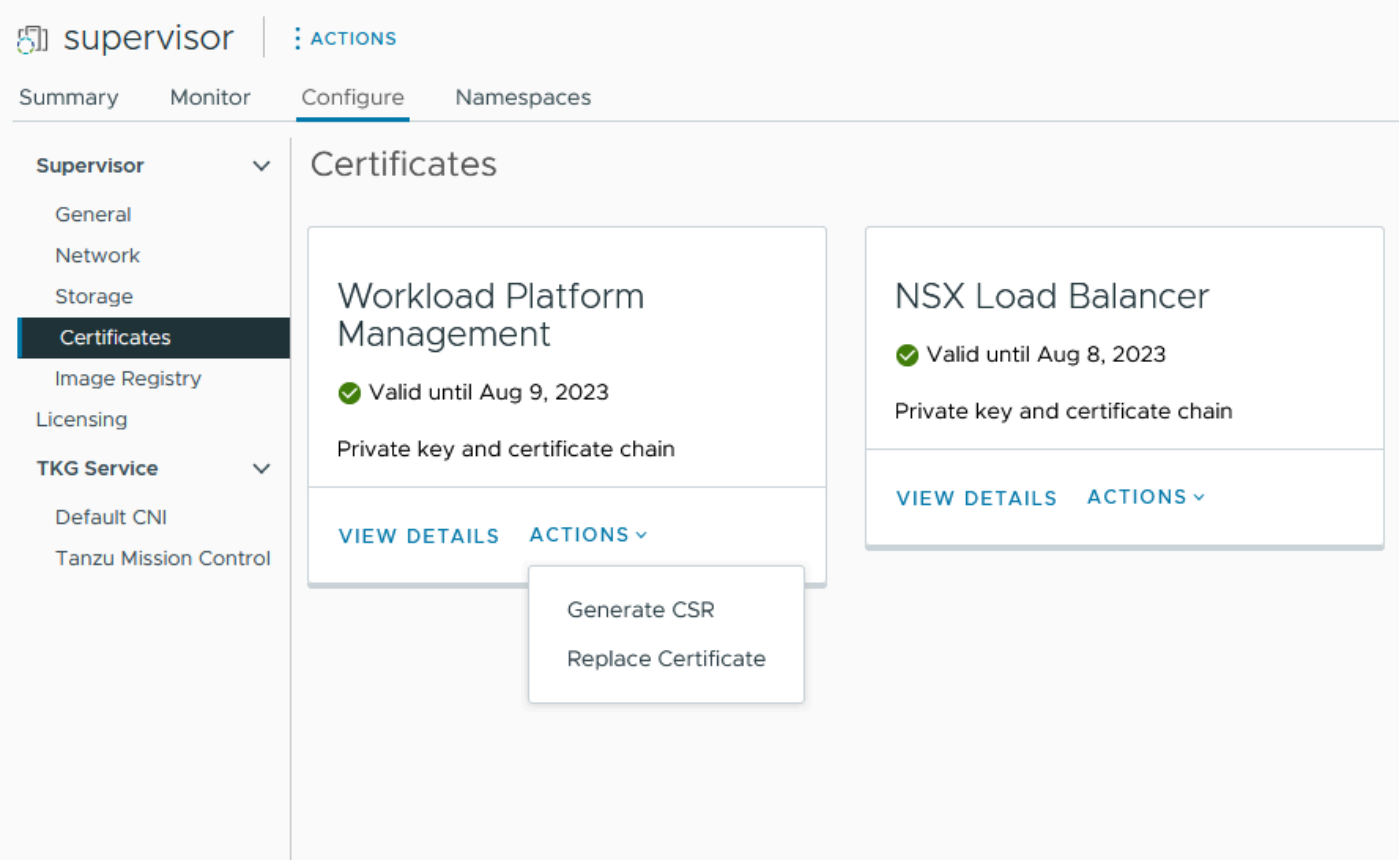
As a VI administrator, you can replace the certificate for the virtual IP address (VIP) to securely connect to the Supervisor API endpoint with a certificate signed by a CA that your hosts already trust. The certificate authenticates the Kubernetes control plane to DevOps engineers, both during login and subsequent interactions with the Supervisor.

Verify that you have access to a CA that can sign CSRs. For DevOps engineers, the CA must be installed on their system as a trusted root.

For more information about the Supervisor certificate, see Supervisor CA Certificate.

1. In the vSphere Client, navigate to **Supervisor Management**.
2. Select **Supervisors** and the select the Supervisor from the list.
3. Click **Configure** and select **Certificates**.
4. In the **Supervisor Management Platform** pane, select **Actions** > **Generate CSR**.

   Figure 519: Replacing Supervisor default certificate



5. Provide the details for the certificate.

   Note:
   If you use an identity provider service, you must also include the entire certificate chain. The chain is not required for standard HTTPS traffic, however.
6. Once the CSR is generated, click **Copy**.
7. Sign the certificate with a CA.
8. From the **Workload Platform Management** pane, select **Actions** > **Replace Certificate**.
9. Upload the signed certificate file and click **Replace Certificate**.
10. Validate the certificate on the IP address of the Kubernetes control plane.
    For example, you can open the VCF Consumption CLI download page and confirm that the certificate is replaced successfully by using the browser. On a Linux or Unix system you can also use `echo | openssl s_client -connect <SUPERVISOR_VIP_IP>:443 2>/dev/null | openssl x509 -text -noout`.

Set the Default CNI for VKS Clusters

As a VI administrator, you can set the default container network interface (CNI) for VKS clusters.

Default CNI

VKS supports two CNI options for VKS clusters: Antrea and Calico.

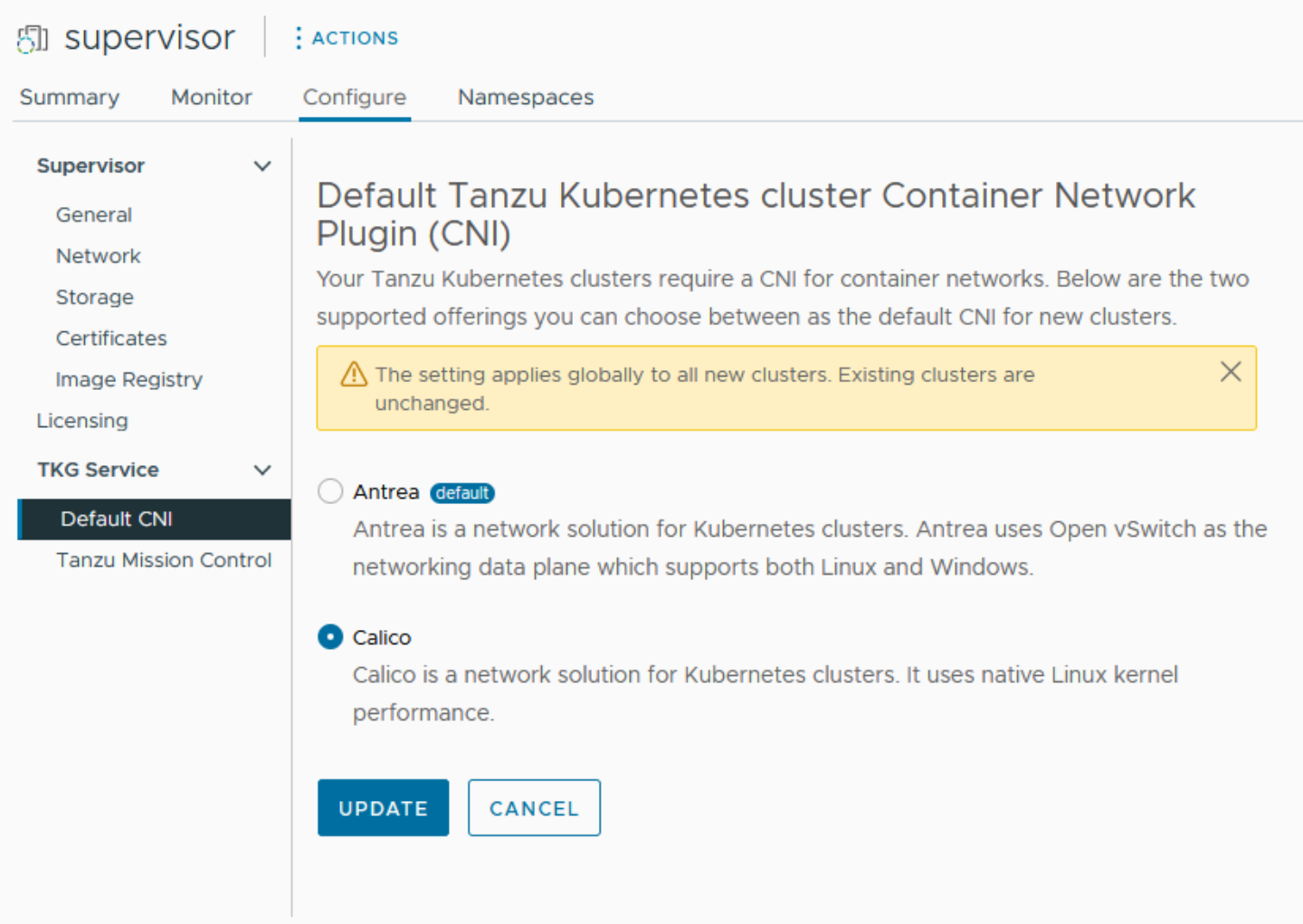
The system-defined default CNI is Antrea. For more information on the default CNI setting, see *Using TKG Service with vSphere Supervisor*.

You can change the default CNI by using the vSphere Client. To set the default CNI, complete the following procedure.

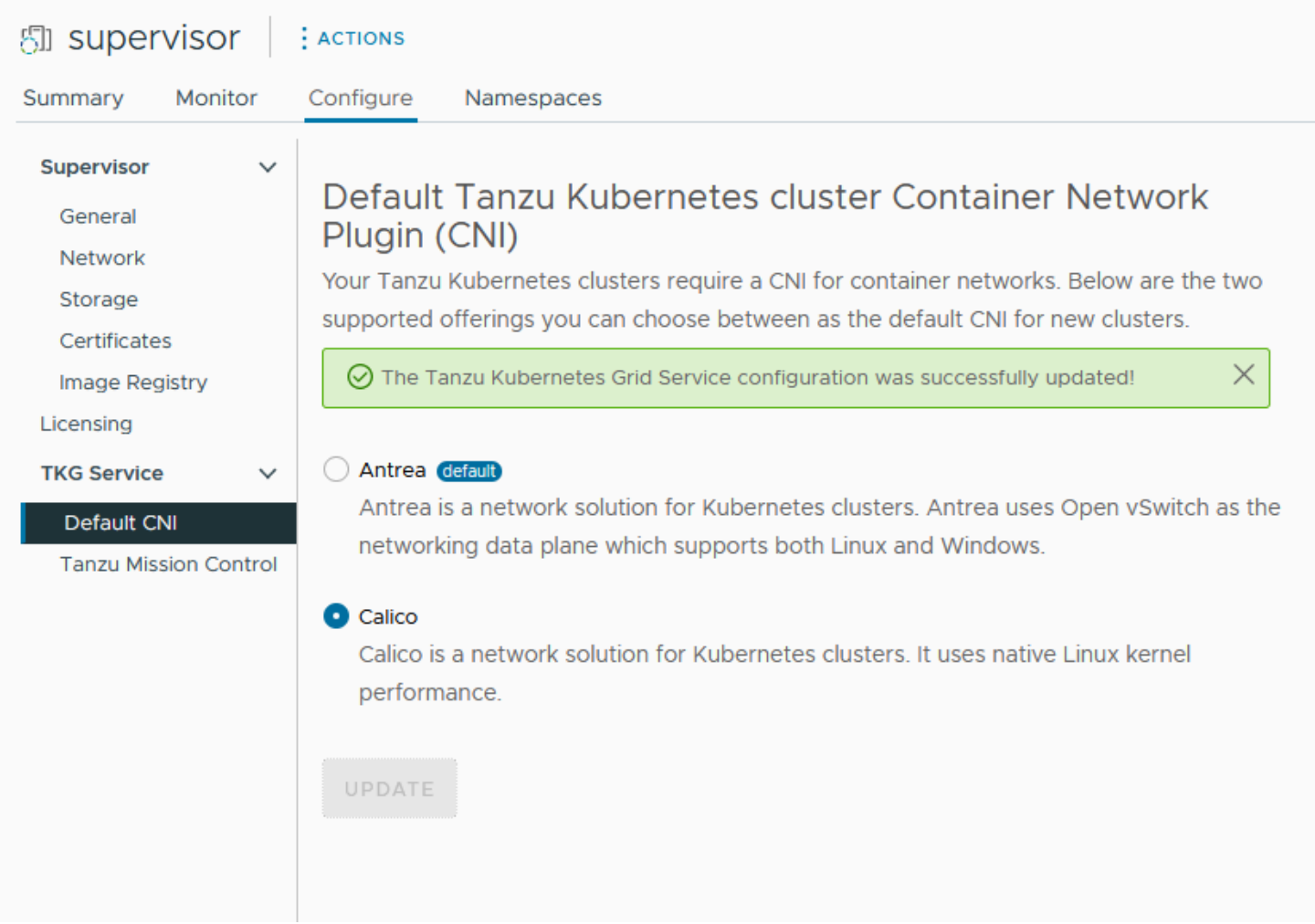
> **CAUTION:**
> Changing the default CNI is a global operation. The newly set default applies to all new clusters created by the service. Existing clusters are unchanged.

1. Log in to your vSphere Supervisor environment using the vSphere Client.
2. Select **Supervisor Management** and select **Supervisors** .
3. Select the Supervisor instance from the list.
4. Select **Configure** and select **Kubernetes Service** > **Default CNI**
5. Choose the default CNI for new clusters.
6. Click **Update**.

The following image shows the default CNI selection.



The following image shows changing the CNI selection from Antrea to Calico.



Change the Control Plane Size of a Supervisor

Check out how to change the size of the Kubernetes control plane VMs of a Supervisor in your vSphere Supervisor environment.

- Verify that you have the **Modify cluster-wide configuration** privilege on the cluster.

1. In the vSphere Client, navigate to the **Supervisor Management**.
2. Under **Supervisors** select the Supervisor.
3. Select **Configure** and select **General** .
4. Expand **Control Plane Size**.
5. Click **Edit** and select new control plane size from the drop-down menu.

Option	Description
Tiny	2 CPUs, 8-GB Memory, 32-GB Storage
Small	4 CPUs, 16-GB Memory, 32-GB Storage
Medium	8 CPUs, 24-GB Memory, 32-GB Storage
Large	16 CPUs, 32-GB Memory, 32-GB Storage

Note:
Once you select a control plane size, you cannot scale down. For example, if you have already set the Tiny option during Supervisor activation, you can only scale up from it.

6. Click **Save**.

   You can only scale up the control plane size.

Change Load Balancer Settings on a Supervisor Configured with VDS Networking

Checkout how to change the settings of the load balancer configured with the VDS networking stack on your Supervisor. You can change settings such as user name and password, add new IP ranges, and update the certificate used with the load balancer.

- Verify that you have the **Modify cluster-wide configuration** privilege on the cluster.

1. In the vSphere Client, navigate to the **Supervisor Management**.
2. Under **Supervisors**, select the Supervisor, and select **Configure**.
3. Select **Network** and expand **Workload Network**.
   - For Foundation Load Balancer, click Edit and change any of the settings.

Option	Description
Network Topology	Change the network topology from one-arm to two-arm. The opposite is not supported.
Size	Change the deployment footprint of Foundation Load Balancer in case there's resource over-utilization which leads to unhealthy status of the Foundation Load Balancer VMs. See Monitor Foundation Load Balancer.
High Availability	Change the HA mode Foundation Load Balancer.
Networks	Click **Create** and add new networks to Foundation Load Balancer.
Management Network	Change the Management Network of Foundation Load Balancer.
Virtual Server Network	Change the Virtual Server Network of Foundation Load Balancer.
Transit Network	Change the Transit Network of Foundation Load Balancer.

   - HA Proxy Load Balancer.

Option	Description
User name	Edit the user name that the Supervisor uses to authenticate with the load balancer end point.
Password	Change the password the Supervisor uses to authenticate with the load balancer endpoin
Virtual IP Ranges	Add IP ranges that are a subset of the virtual IP CIDR range that you have initially configured with the load balancer. **Note:** You can only add new IP ranges. You cannot remove or change existing IP ranges.
TLS Certificate	Change the TLS certificate that is use to ensure secure connection between the Supervisor and the load balancer.

Add Workload Networks to a Supervisor Configured with vSphere VLAN Port Groups

For a Supervisor that is configured with the vSphere networking stack, you can provide Layer 2 isolation for your Kubernetes workloads by creating Workload Networks and assigning them to namespaces. Workload Networks provide connectivity to VKS clusters in the namespace and are backed by distributed port groups on the switch that is connected to the hosts in the Supervisor.

- Create a distributed port group that will back the Workload Network.

- Verify that the IP range that you will assign to the Workload Network is unique within all Supervisors available in your environment.

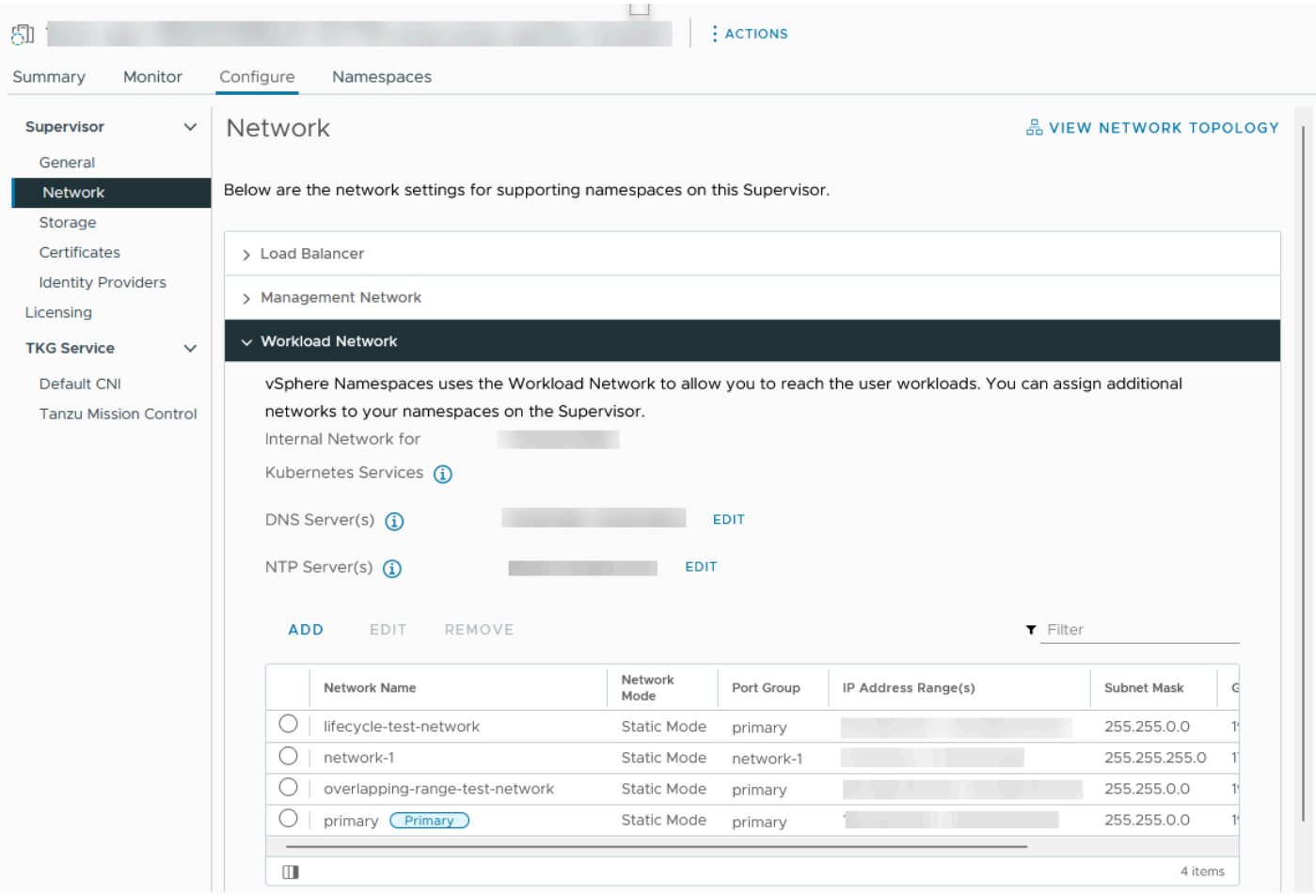
For more information on the topologies that you can implement for the Supervisor, see Supervisor Architecture and Deployment Options.

Note:

If you have configured the Supervisor with a DHCP server providing networking settings for Workload Networks, you cannot create new Workload Networks post Supervisor configuration.

1. In the vSphere Client, navigate to the **Supervisor Management**.
2. Under **Supervisors** and select the Supervisor.
3. Select **Configure** and select **Network**.

Figure 520: Adding a Supervisor workload network



4. Select **Workload Network** and click **Add**.

Option	Description
Port Group	Select the distributed port group to be associated with this Workload Network. The vSphere Distributed Switch (VDS) that is configured for the Supervisor networking contains the port groups from which you can select.
Network Name	The network name that identifies the Workload Network when assigned to namespaces. This value is automatically populated from the name of the port group that you select, but you can change it as appropriate.

Option	Description
IP Address Ranges	Enter an IP range for allocating IP addresses of cluster nodes. . The IP range must be in the subnet indicated by the subnet mask. **Note:** You must use a unique IP address ranges for each Workload Network. Do not configure the same IP address ranges for multiple networks.
Subnet Mask	Enter the IP address of the subnet mask for the network on the port group.
Gateway	Enter the default gateway for the network on the port group. The gateway must be in the subnet indicated by the subnet mask. **Note:** Do not use the gateway that is assigned to the HAProxy loadbalancer.

5. Click **Add**.

Assign the newly-created Workload Network to vSphere Namespaces.

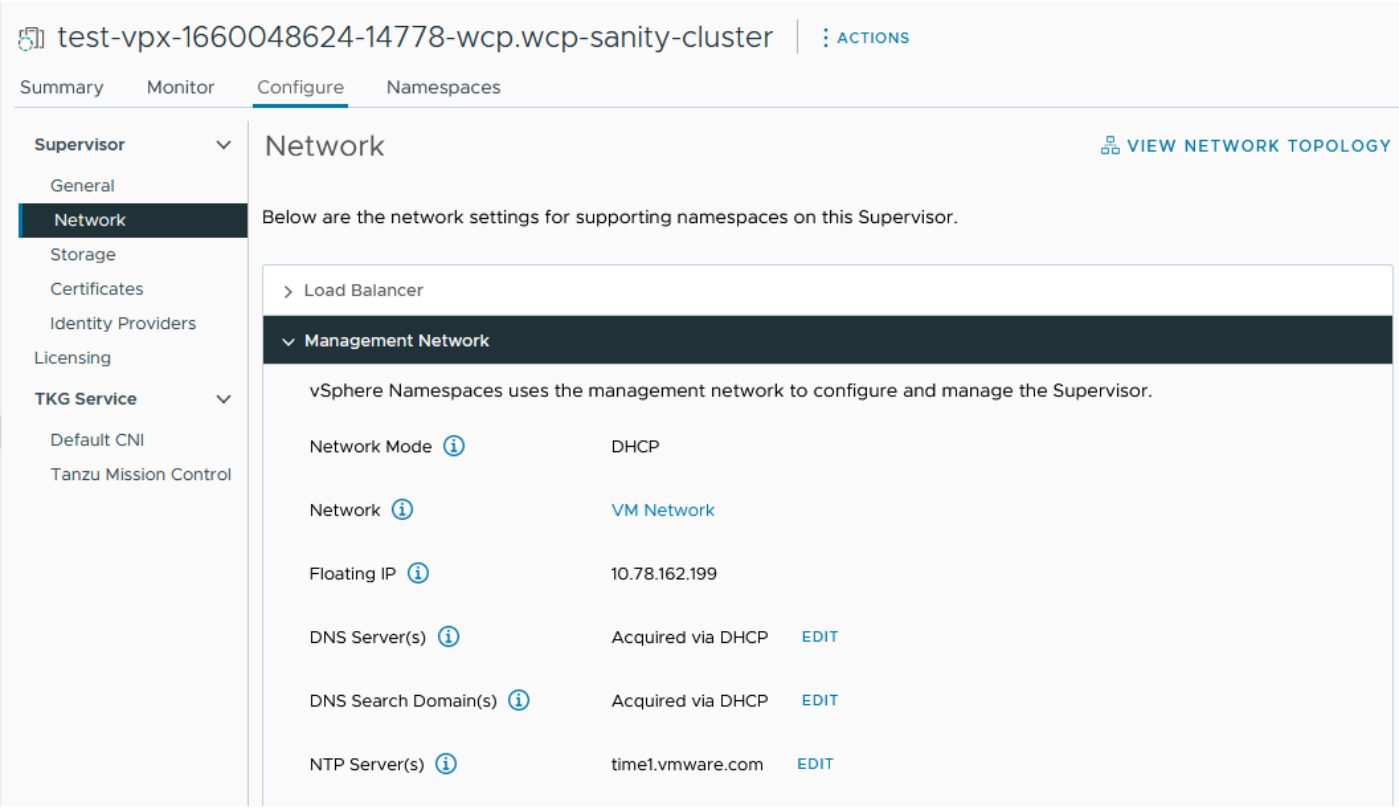
Change the Management Network Settings on a Supervisor

Learn how to update the DNS and NTP settings on the Supervisor management network on your vSphere Supervisor environment.

- Verify that you have the **Modify cluster-wide configuration** privilege on the cluster.

1. In the vSphere Client, select **Supervisor Management**.
2. Under **Supervisors**, select the Supervisor, and select **Configure**.
3. Select **Network** and expand **Management Network**.

Figure 521: Updating Supervisor management network settings



4. Edit the DNS and NTP settings.

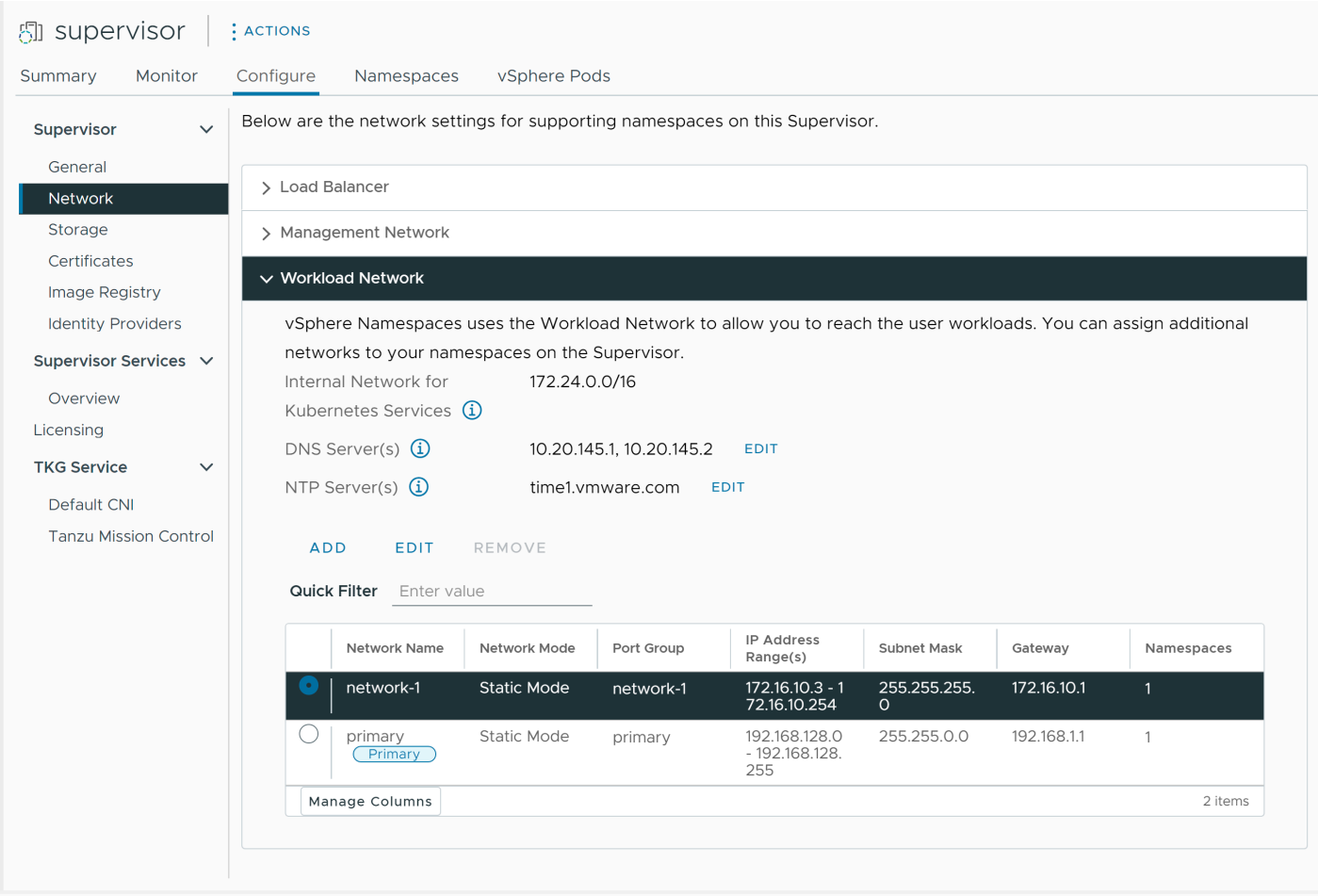
Option	Description
DNS Server(s)	Enter the addresses of the DNS servers that you use in your environment. If the vCenter system is registered with an FQDN, you must enter the IP addresses of the DNS servers that you use with the vSphere environment so that the FQDN is resolvable in the Supervisor.
DNS Search Domain(s)	Enter domain names that DNS searches inside the Kubernetes control plane nodes, such as `corp.local`, so that the DNS server can resolve them.
NTP Server(s)	Enter the addresses of the NTP servers that you use in your environment, if any.

Change the Workload Network Settings on a Supervisor Configured with VDS Networking

Checkout how to change the NTP and DNS server settings for the Workload Networks of a Supervisor configured with the VDS networking stack. The DNS servers that you configure for Workload Networks are external DNS servers exposed to Kubernetes workloads and they resolve default domain names that are hosted outside of the Supervisor.

- Verify that you have the **Modify cluster-wide configuration** privilege on the cluster.

1. In the vSphere Client, select **Supervisor Management**.
2. Under **Supervisors**, select the Supervisor, and select **Configure**.
3. Select **Network** and expand **Workload Network**.



Note:
You cannot remove a workload network that is already assigned to a vSphere Namespace. If you need to remove a workload network, you must delete all vSphere Namespaces attached to that network. Also you cannot edit or remove the primary workload network.

4. Edit the DNS server settings.

   Enter the addresses of DNS Servers that can resolve the domain names of the vSphere management components, such as vCenter .

   For example, `10.142.7.1`.

   When you enter the IP address of the DNS server, a static route is added on each control plane VM. This indicates that the traffic to the DNS servers goes through the workload network.

   If the DNS servers that you specify are shared between the management network and workload network, the DNS lookups on the control plane VMs are routed through the Workload Network after initial setup.

5. Edit the NTP settings as needed.
6. Edit workload network settings.
   a) Select a workload network and click **Edit**.
   b) Click **Add** next to **IP Address Range(s)** to add new IP ranges to be used with workloads on that network.
      The IP ranges must to be in the subnet indicated by the subnet mask.

Note:
The IP ranges that you add must not overlap with the Virtual IPs of the Frontend Network configuration of the load balancer.

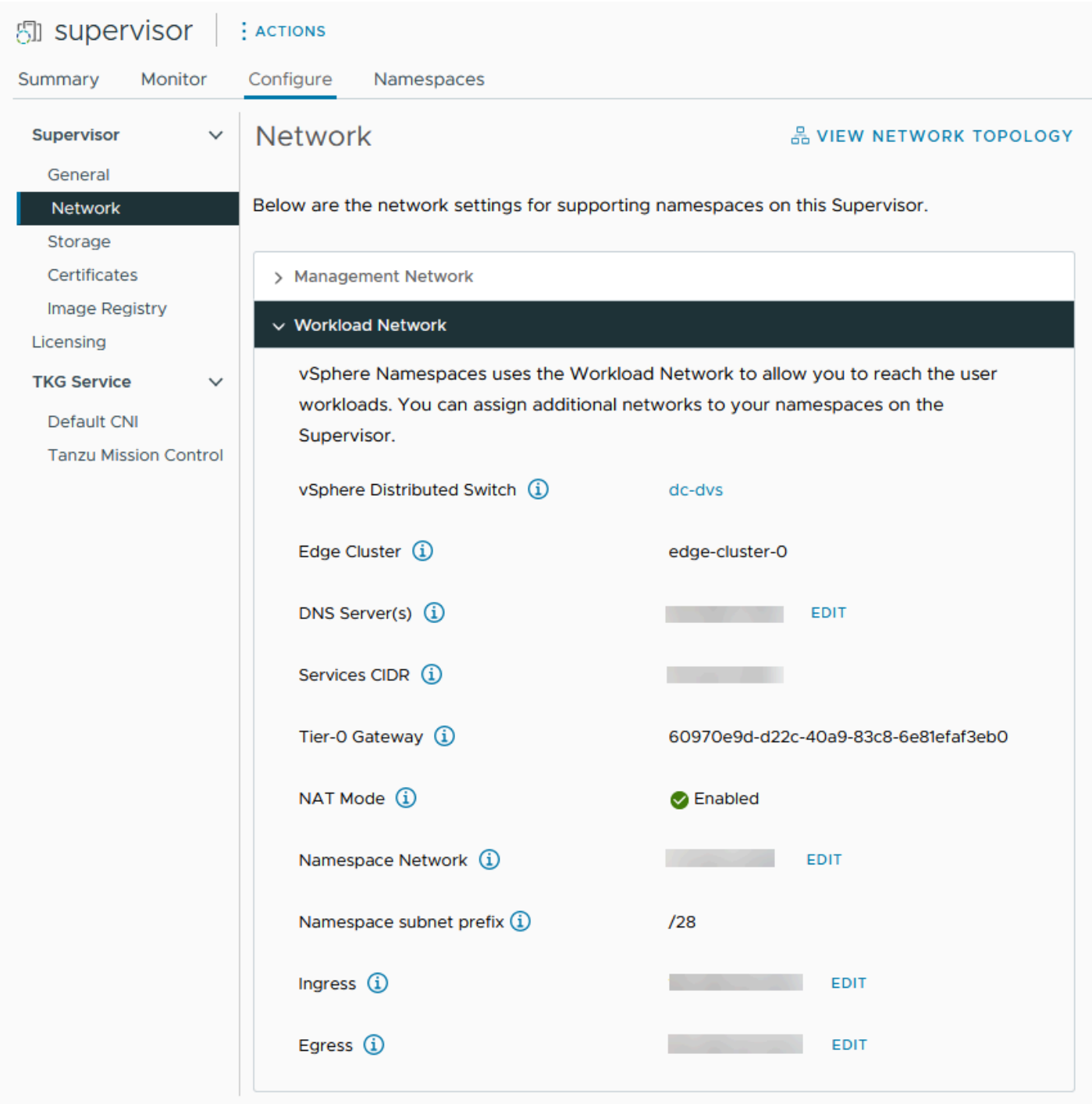
Change Workload Network Settings on a Supervisor Configured with NSX

Learn how to change the networking settings for DNS server, namespace networks, ingress and egress of a Supervisor configured for NSX as the networking stack.

- Verify that you have the **Modify cluster-wide configuration** privilege on the cluster.

1. In the vSphere Client, navigate to **Supervisor Management**.
2. Under **Supervisors**, select the Supervisor, and select **Configure**.
3. Select **Network** and expand **Workload Network**.

Figure 522: Updating Supervisor workload network settings



4. Change networking settings as needed.

Option	Description
DNS Server(s)	Enter the addresses of DNS Servers that can resolve the domain names of the vSphere management components, such as vCenter . For example, `10.142.7.1`.

Option	Description
	When you enter IP address of the DNS server, a static route is added on each control plane VM. This indicates that the traffic to the DNS servers go through the workload network. If the DNS servers that you specify are shared between the management network and workload network, the DNS lookups on the control plane VMs are routed through the workload network after initial setup.
Namespace Network	Enter a CIDR annotation to change the IP range for Kubernetes workloads that are attached to the namespace segments of the Supervisor. If NAT Mode is not configured, then this IP CIDR range must be a routable IP Address.
Ingress	Enter a CIDR annotation to change the ingress IP range for the Kubernetes services. This range is used for services of type load balancer and ingress. For clusters, publishing services through ServiceType loadbalancer will also get the IP addresses from this IP CIDR block. **Note:** You can only add CIDRs to ingress and workload network fields, but you cannot edit or remove existing ones.
Egress	Enter a CIDR annotation for allocating IP addresses for SNAT (Source Network Address Translation) for traffic exiting the Supervisor to access external services. Only one egress IP address is assigned for each namespace in the Supervisor. The egress IP is the IP address that the vSphere Pods in the particular namespace use t o communicate outside of NSX.

Configuring HTTP Proxy Settings in vSphere Supervisor

Checkout how to configure HTTP proxy settings to the Supervisor and VKS clusters and what is the workflow for configuring a proxy when you register the Supervisor and VKS clusters with Tanzu Misson control.

You can configure a proxy to the Supervisor through the vSphere Client, the Cluster Management API, or DCLI commands. You can use a proxy if you need to handle container traffic or image pulling from networks external to the Supervisor. For on-premises Supervisors that you register as management clusters in Tanzu Mission Control, you use an HTTP proxy for image pulling and container traffic .

Configuring Proxy Settings on Newly-Created vSphere 7.0 Update 3 and Later Supervisors

For newly-created Supervisors on a vSphere 7.0 Update 3 and later environment, HTTP proxy settings are inherited from vCenter. No matter if you create the Supervisors before or after you configure HTTP proxy settings on vCenter, the settings are inherited by the clusters.

See *Configure the DNS, IP Address, and Proxy Settings* in the vSphere documentation to learn how to configure the HTTP proxy settings on vCenter.

You can also override the inherited HTTP proxy configuration on individual Supervisors through the vSphere Client, Cluster Management API, or DCLI.

Since inheriting the vCenter proxy settings is the default configuration for newly-created vSphere 7.0.3 Supervisors, you can also use the Cluster Management API or DCLI to not inherit any HTTP proxy settings in case the Supervisors don't require a proxy, but vCenter still does.

Configuring Proxy Settings on Supervisors Upgraded to vSphere 7.0 Update 3 and Later

If you have upgraded your Supervisors to vSphere 7.0 Update 3 and later, the HTTP proxy settings of vCenter are not automatically inherited. In that case, you configure proxy settings Supervisors by using the vSphere Client, `vcenter/namespace-management/clusters` API, or DCLI command line.

Configuring HTTP Proxy to VKS Clusters in vSphere Supervisor

Use one of the following methods to configure a proxy to your VKS clusters in vSphere Supervisor:

- Configure proxy settings to individual VKS clusters.
- Create a global proxy configuration that will be applied to all VKS clusters.

> **Note:**
>
> If you use Tanzu Mission Control to manage your VKS clusters, you do not have to configure proxy settings through the cluster YAML file in vSphere Supervisor. You can configure proxy settings when you add the VKS clusters as workload clusters to Tanzu Mission Control.

Configure HTTP Proxy Setting on the Supervisor by Using the vSphere Client

Checkout how to configure HTTP proxy settings to the Supervisor through the vSphere Client. You can override the proxy settings inherited from vCenter on individual Supervisors or select not to use any proxy settings at all.

- Verify that you have the **Modify cluster-wide configuration** privilege on the cluster.

1. In the vSphere Client, navigate to **Supervisor Management**.
2. Under **Supervisors**, select the Supervisor, and select **Configure**.
3. Select **Network**, expand **Proxy Configuration**, and click **Edit**
4. Select Configure proxy settings on Supervisor and enter the proxy settings.

Option	Description
TLS Certificate	The proxy TLS root CA bundle which is used to verify the proxy's certificates. Enter the bundle in plain text.
Hosts and IP addresses excluded from proxy	A comma separated list of IPv4 addresses, FQDNs, or domain names that do not require the proxy server and can be accessed directly.
HTTPS Configuration	HTTPS settings such as URL, port, user name, and password.
HTTP Configuration	HTTP settings such as URL, port, user name, and password.

5. Click **OK**.

The proxy settings you configured on this Supervisor override the settings inherited from vCenter.

Using the Cluster Management API and DCLI to Configure HTTP Proxy to Supervisors

You can configure the Supervisor proxy settings through the `vcenter/namespace-management/clusters` API or DCLI.

The API provides three options for proxy configuration on the Supervisor:

API Setting	Newly-Created vSphere 7.0.3 and Later Supervisors	Supervisors Upgraded to vSphere 7.0.3 and Later
VC_INHERITED	This is the default setting for new Supervisors and you don't have to use the API to configure the Supervisors proxy settings. You can just configure proxy settings on vCenter through its management interface.	Use this setting to push the HTTP proxy configuration to Supervisors upgraded to vSphere 7.0.3 and later.
CLUSTER_CONFIGURED	Use this setting to override the HTTP proxy configuration inherited from vCenter in one of the following cases: • A Supervisor resides on a different subnet than vCenter and a different proxy server is required. • The proxy server uses custom CA bundles.	Use this setting to configure HTTP proxy to individual Supervisors upgraded to vSphere 7.0.3 and later in one of the following cases: • You cannot use the vCenter proxy because the Supervisor resides on a different subnet thanvCenter and a different proxy server is required. • The proxy server uses custom CA bundles.
NONE	Use this setting when the Supervisor has direct connectivity to the internet while vCenter requires a proxy. The NONE settings prevents the proxy settings of vCenter to be inherited by Supervisors.	

To set an HTTP proxy to a Supervisor or modify the existing settings, use the following commands in an SSH session with vCenter :

```
vc_address=<IP address>
cluster_id=domain-c<number>
session_id=$(curl -ksX POST --user '<SSO user name>:<password>' https://$vc_address/api/session | xargs -t)
curl -k -X PATCH -H "vmware-api-session-id: $session_id" -H "Content-Type: application/
json" -d '{ "cluster_proxy_config": { "proxy_settings_source": "CLUSTER_CONFIGURED",
 "http_proxy_config":"<proxy_url>" } }' https://$vc_address/api/vcenter/namespace-management/clusters/
$cluster_id
```

You only need to pass the `domain_c<number>` from the full cluster ID which. For example, take `domain-c50` from the following cluster ID: `ClusterComputeResource:domain-c50:5bbb510f-759f-4e43-96bd-97fd703b4edb`.

When using the `VC_INHERITED` or `NONE` settings, omit `"http_proxy_config:<proxy_url>"` from the command.

To use a custom CA bundle, add a `"tlsRootCaBundle": "<TLS_certificate>"` to the command by providing the TSL CA certificate in plain text.

For HTTPS proxy settings use the following command:

```
curl -k -X PATCH -H "vmware-api-session-id: $session_id"
-H "Content-Type: application/json" -d '{ "cluster_proxy_config":
{ "proxy_settings_source": "CLUSTER_CONFIGURED", "https_proxy_config":"<proxy_url>" } }'
https://$vc_address/api/vcenter/namespace-management/clusters/$cluster_id
```

Using DCLI to Configure HTTP Proxy Settings on Supervisors

You can use the following DCLI command to configure HTTP proxy settings to Supervisors by using the `CLUSTER_CONFIGURED` setting.

```
<dcli> namespacemanagement clusters update --cluster domain-c57 --cluster-proxy-config-http-proxy-config
 <proxy URL> --cluster-proxy-config-https-proxy-config <proxy URL> --cluster-proxy-config-proxy-settings-
source CLUSTER_CONFIGURED
```

Configuring HTTP Proxy Settings on the Supervisor and VKS Clusters for Tanzu Mission Control

To configure an HTTP proxy on Supervisors that you want to register as management clusters with Tanzu Mission Control, follow the steps:

1. In vSphere, configure HTTP proxy on Supervisors by either inheriting the HTTP proxy settings from vCenter, or configuring proxy settings on individual Supervisors through the vSphere Client, Namespace Management Clusters APIs, or DCLI command line.
2. In Tanzu Mission Control, create proxy configuration object by using the proxy settings you configured to the Supervisors in vSphere Supervisor. See Create a Proxy Configuration Object for a Tanzu Kubernetes Grid Service Cluster in the *VMware Tanzu Mission Control Product Documentation*.
3. In Tanzu Mission Control, use this proxy configuration object when you register the Supervisors as a Management Cluster. See Register a Management Cluster with Tanzu Mission Control and Complete the Registration of a Supervisor Cluster in the in the *VMware Tanzu Mission Control Product Documentation*.

To configure an HTTP proxy to VKS clusters that you provision or add as workload clusters in Tanzu Mission Control:

1. Create a proxy configuration object with the proxy settings that you want to use with VKS clusters. See https://techdocs.broadcom.com/bin/gethidpage?ux-context-string=vsphclient_001&appid=vsphere-9-1&language=&format=rendered.
2. Use that proxy configuration object when you provision or add VKS clusters as workload clusters. See Provision a Cluster and Add a Workload Cluster into Tanzu Mission Control Management in the *VMware Tanzu Mission Control Product Documentation*

Configure an External Identity Provider for VKS Clusters

You can configure Supervisor with any OIDC-compliant identity provider (IDP), such as Okta. To complete the integration, you configure the IDP with the callback URL for Supervisor.

Supported External Identity Provider

You can configure Supervisor with any OIDC-compliant identity provider. The table lists common ones and includes links for configuration instructions.

External IDP	Configuration
Okta	Configure Okta as an OIDC provider for Pinniped
Workspace ONE	Configure Workspace ONE Access as an OIDC provider for Pinniped
Dex	Configure Dex as an OIDC provider for Pinniped
GitLab	Configure GitLab as an OIDC provider for Pinniped
Google OAuth	Using Google OAuth 2

Configuring the IDP with the Callback URL for Supervisor

Supervisor acts as an OAuth 2.0 client to the external identity provider. The Supervisor callback URL is the redirect URL used to configure the external IDP. The callback URL takes the form *https://SUPERVISOR-VIP/wcp/pinniped/callback*.

If the IP portion of the callback URL is represented as a fully qualified domain name (FQDN), ensure that the DNS server used by both the Supervisor and the VKS can resolve the FQDN to the correct virtual IP (VIP).

> **Note:**
> When performing the IDP registration, the callback URL might be called the "redirect URL" in the OIDC provider you are configuring.

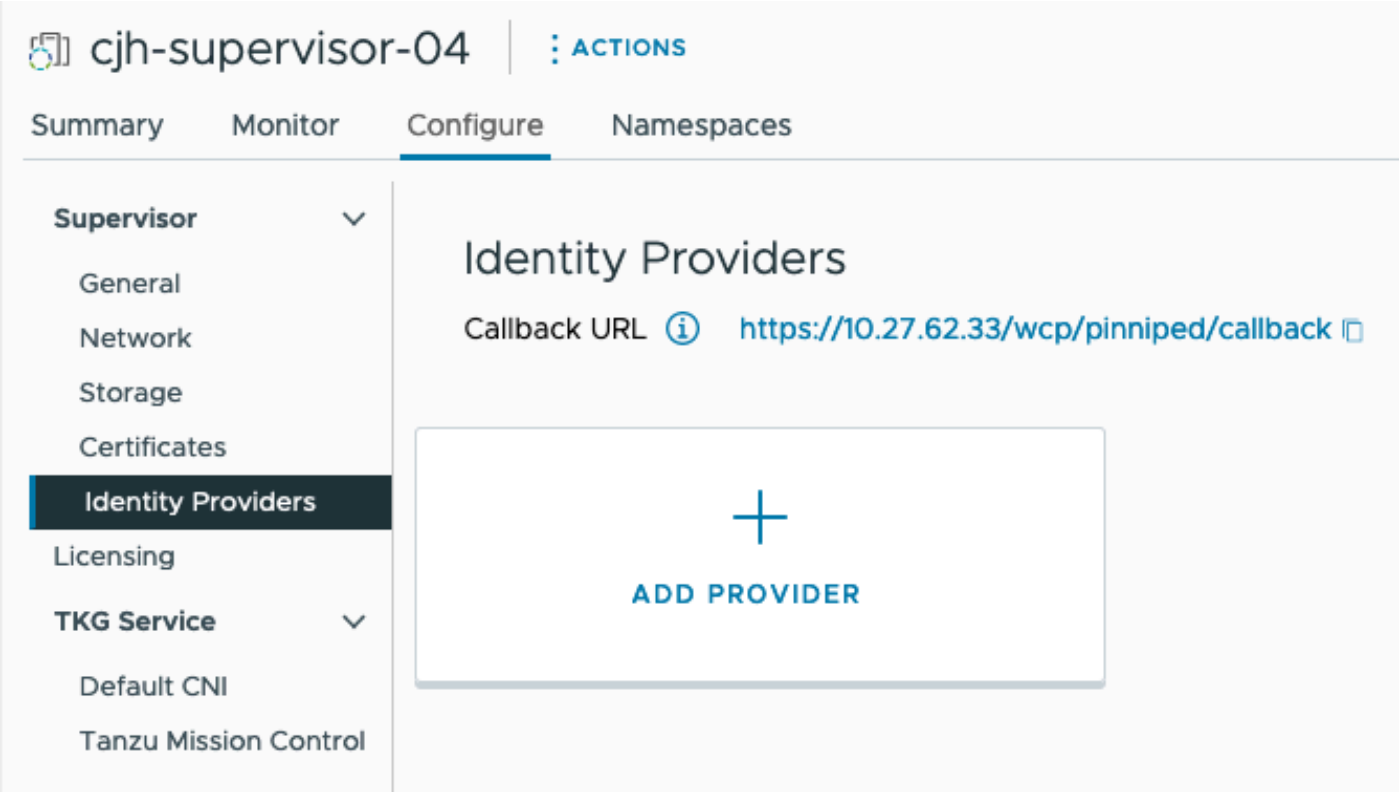
When configuring the external identity provider for use with VKS on Supervisor, you supply the external identity provider with the **Callback URL** that is available in vCenter at **Supervisor Management** > **Supervisors** > **Configure** > **Identity Providers** screen.

Example Identity Provider Configuration Using Okta

Okta lets users sign-in to applications using the OpenID Connect protocol. When you configure Okta as an external identity provider for VKS on Supervisor, the Pinniped pods on Supervisor and on VKS clusters control user access to both vSphere Namespaces and workload clusters.

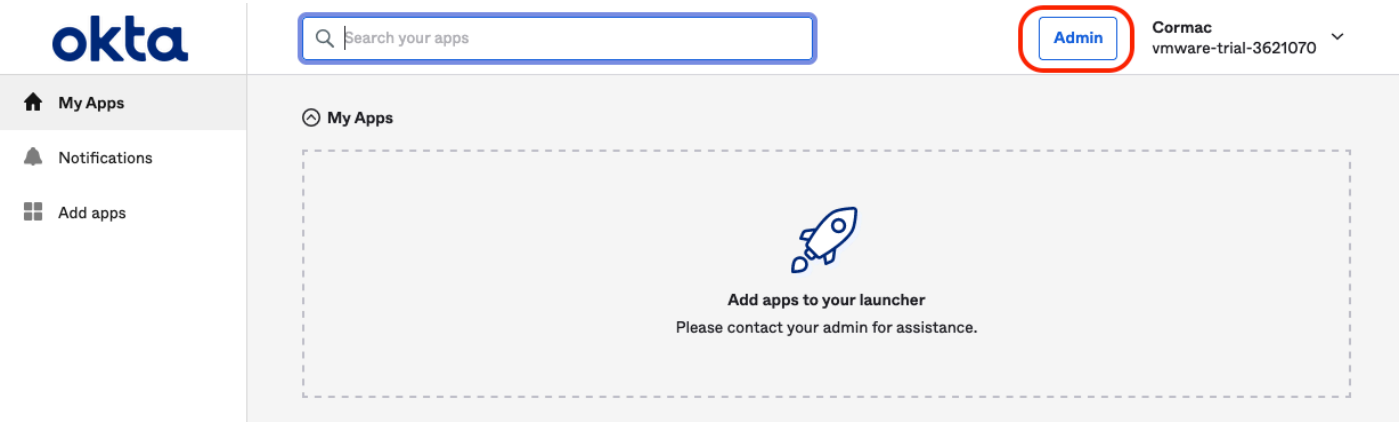
1. Copy the Identity Provider Callback URL which you need to create an OIDC connection between Okta and vCenter. Using the vSphere Client, get the Identity Provider Callback URL at **Supervisor Management** > **Supervisors** > **Configure** > **Identity Providers**. Copy this URL to a temporary location.

   Figure 523: IDP Callback URL



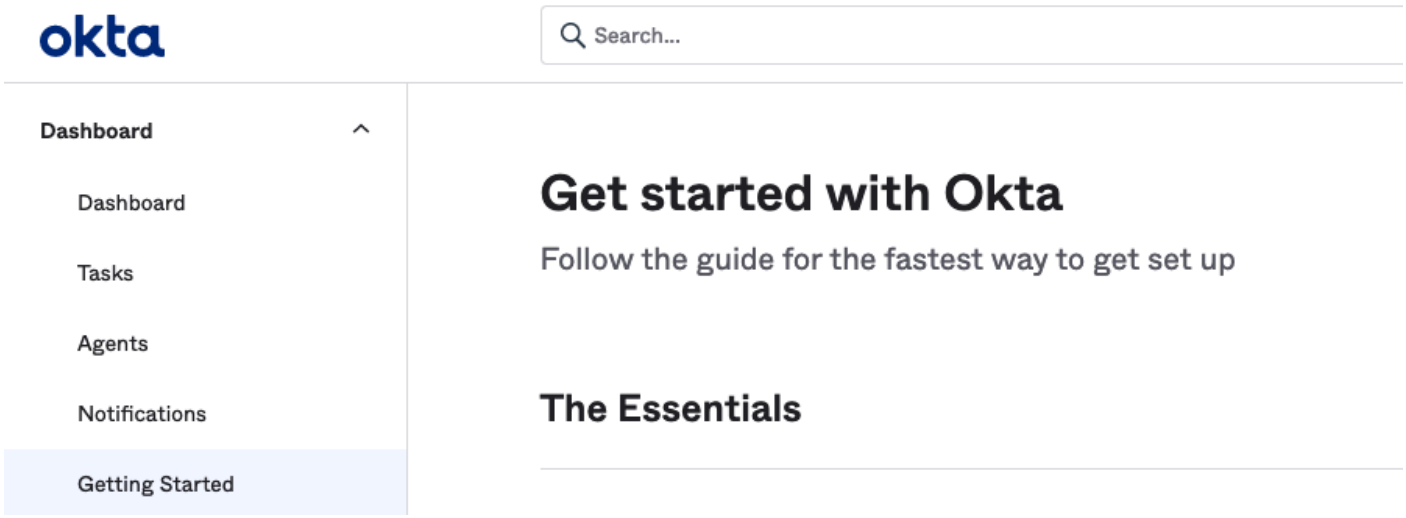
2. Log in to the Okta account for your organization, or create a trial account at https://www.okta.com/. Click the **Admin** button to open the Okta admin console.

   Figure 524: Okta Admin Console



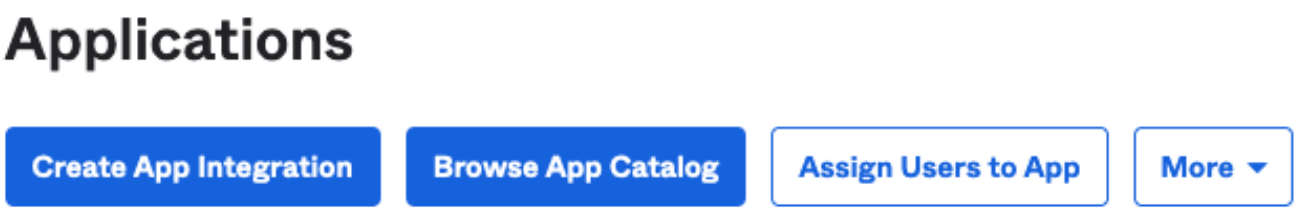
3. From the Getting Started page in the admin console, navigate to **Applications** > **Applications**.

Figure 525: Okta Getting Started



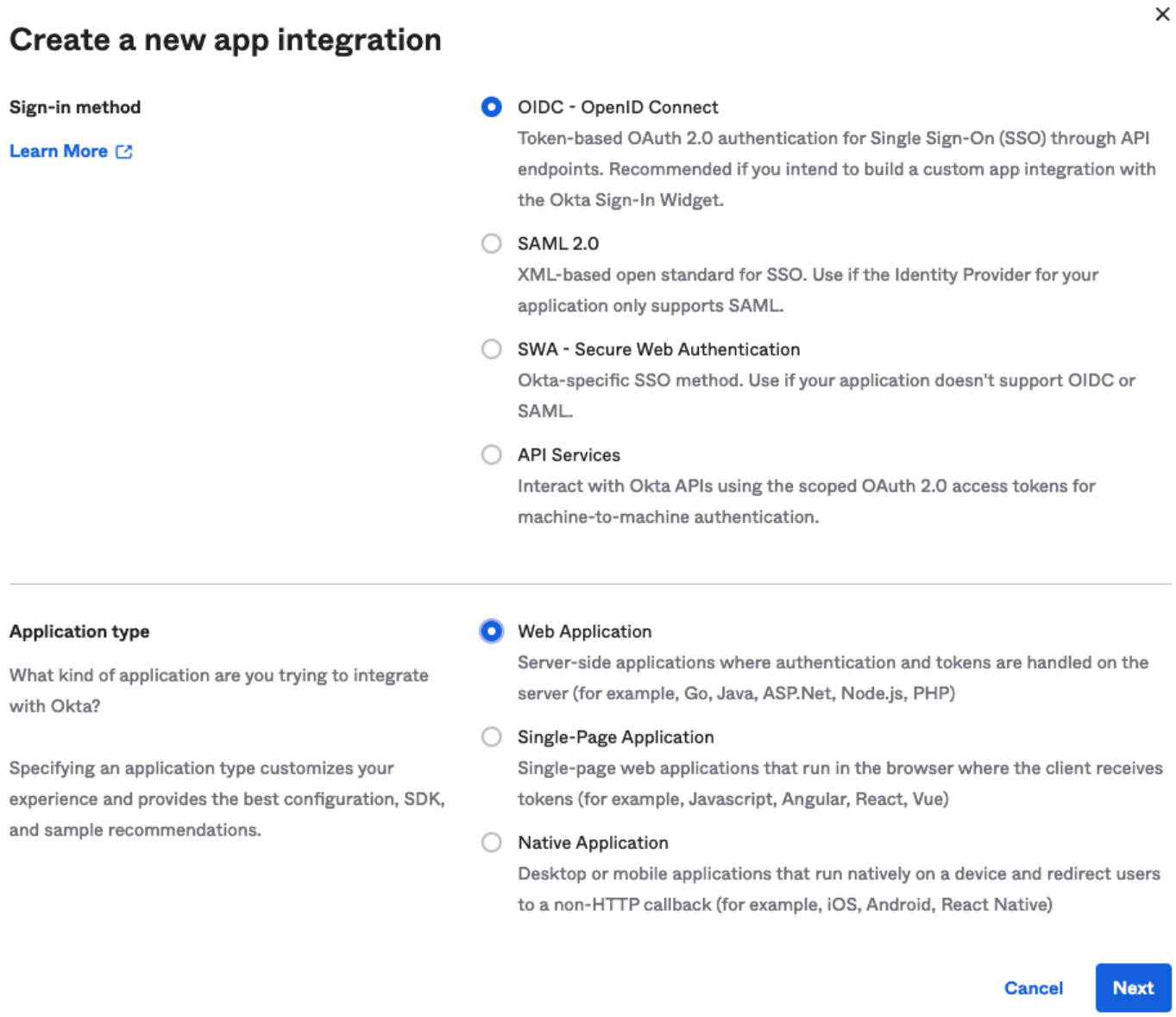
4. Select the option to **Create App Integration**.

Figure 526: Okta Create App Integration



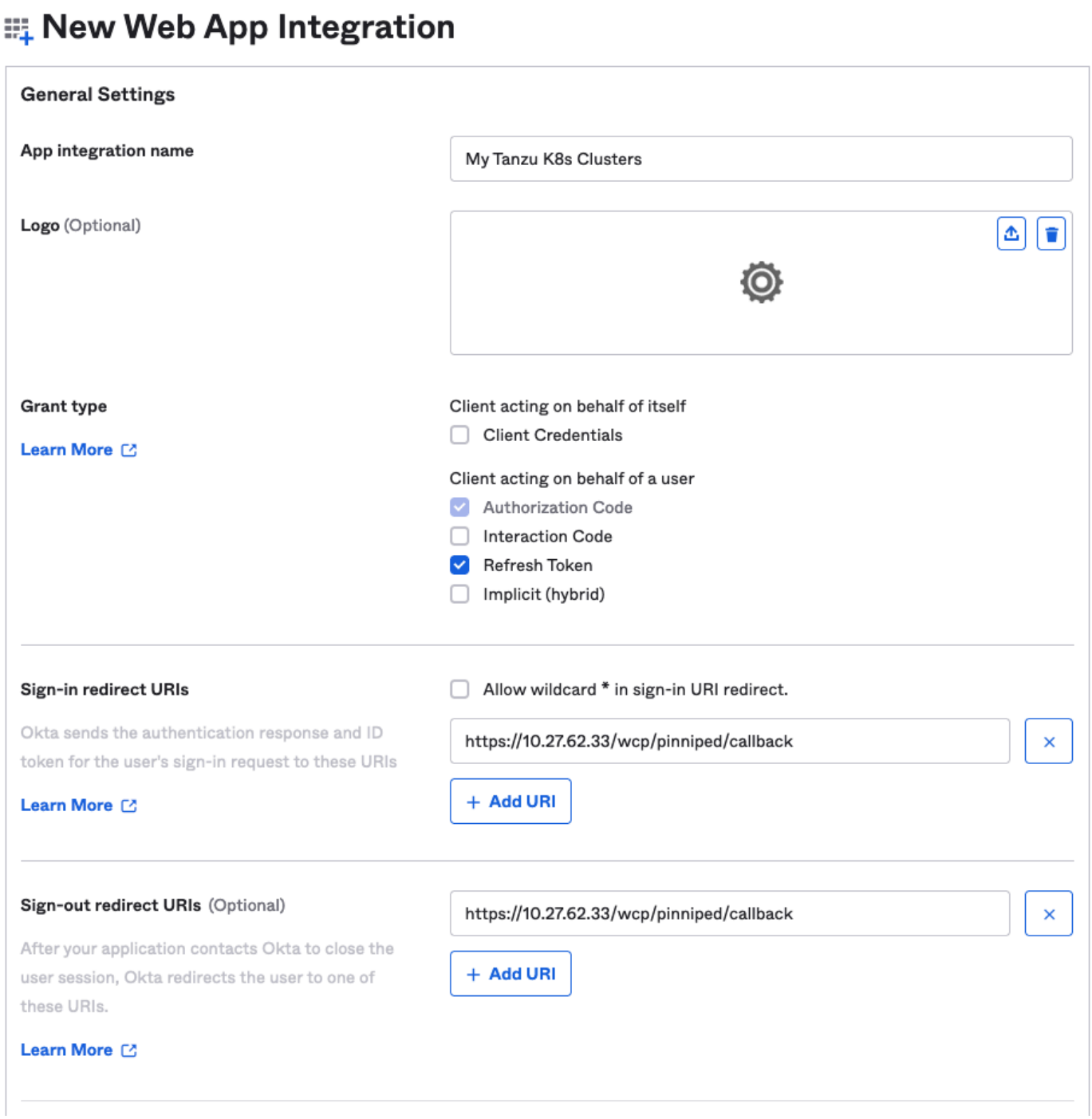
5. Create the new app integration.
   - Set the Sign-in method to **OIDC - OpenID Connect**
   - Set the Application type to **Web Application**

Figure 527: Okta Sign-On Method and Application Type



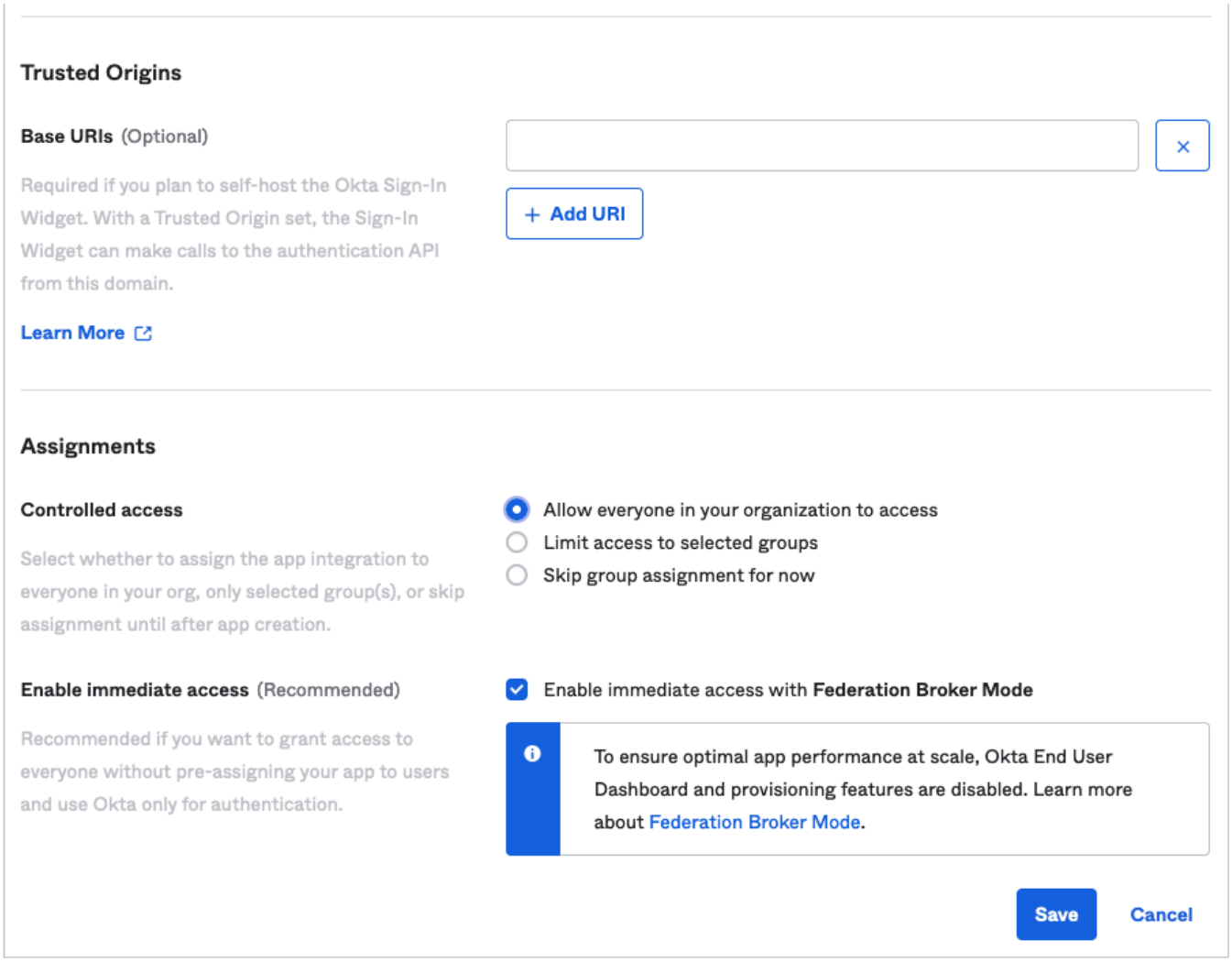
6. Configure the Okta web application integration details.
   - Provide an **App integration name**, which is a user-defined string.
   - Specify the **Grant type**: Select **Authorization Code** and also select **Refresh Token**.
   - Sign-in redirect URIs: Enter the Identity Provider Callback URL that you copied from Supervisor (see Step 1), such as https://10.27.62.33/wcp/pinnipend/callback.
   - Sign-out redirect URIs: Enter the Identity Provider Callback URL that you copied from Supervisor (see Step 1), such as https://10.27.62.33/wcp/pinnipend/callback.

Figure 528: Okta Web Application Integration Details



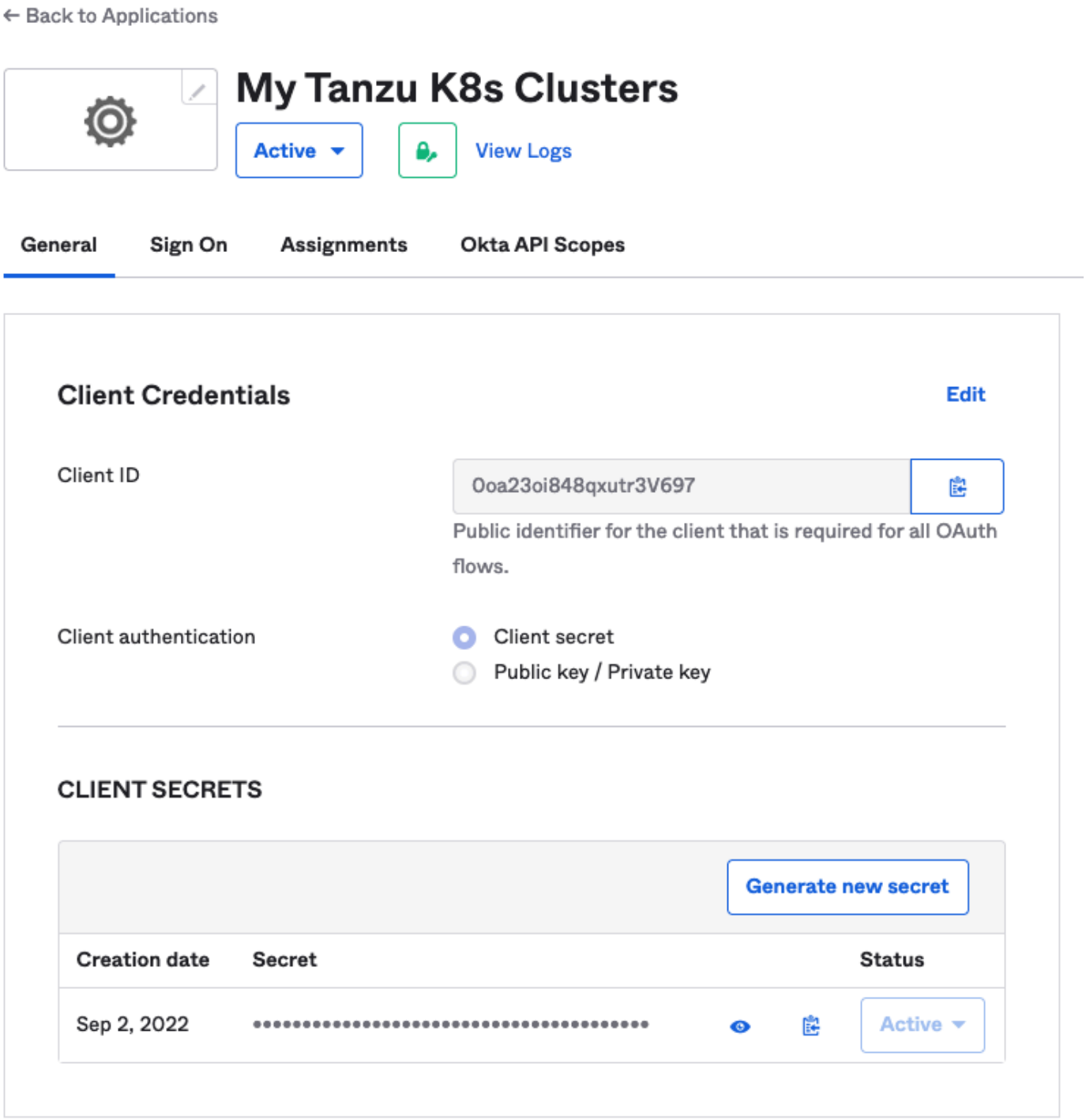
7. Configure user access control.
   In the **Assignments** > **Controlled access** section, you can optionally control which of the Okta users that exist in your organization can access VKS clusters. In the example, allow everyone defined in the organization has access.

Figure 529: Okta Access Control



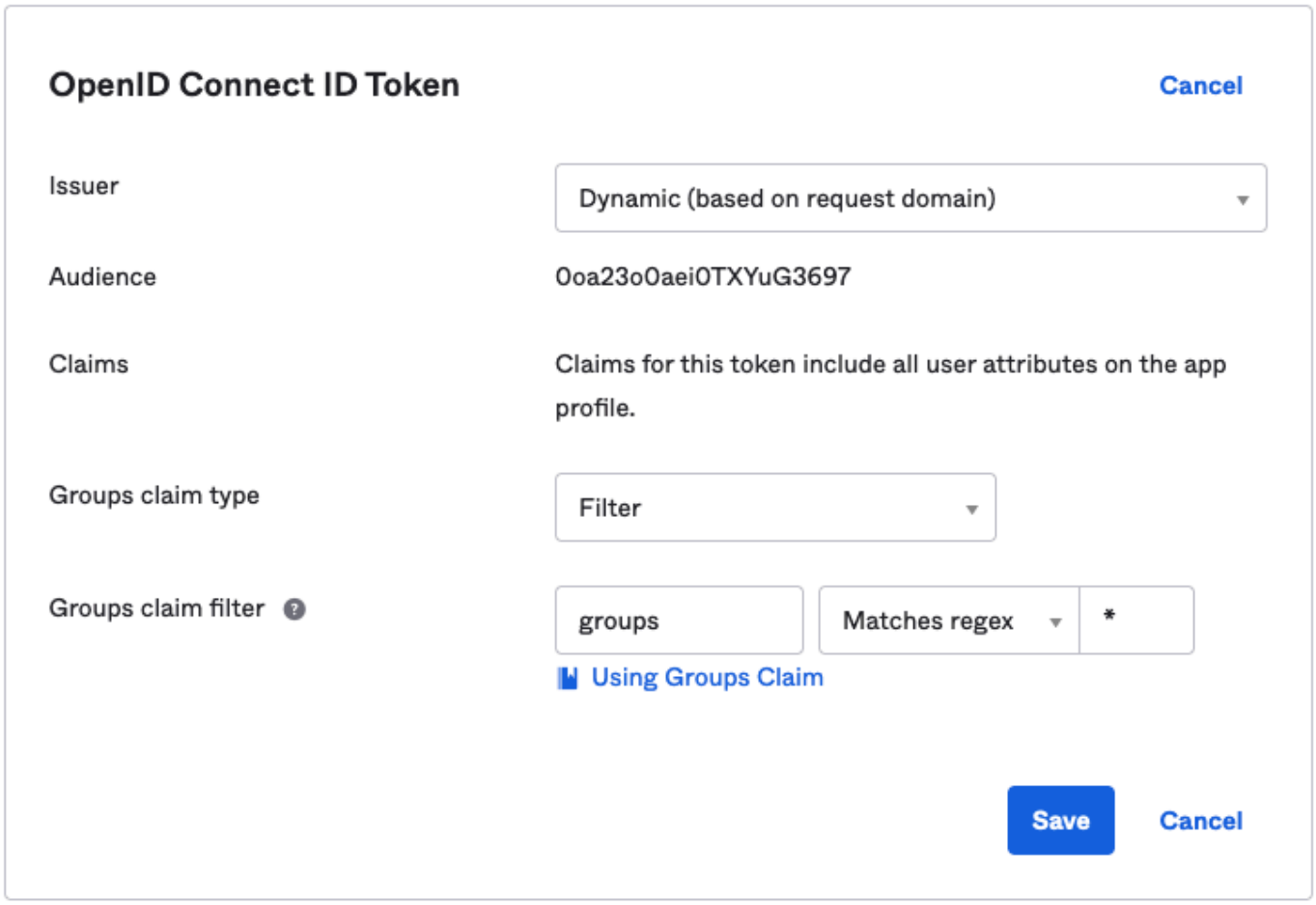
8. Click **Save** and copy the **Client ID** and **Client Secret** that are returned.
   When you save the OKTA configuration, the admin console provides you with a **Client ID** and **Client Secret**. Copy both pieces of data because you need them to configure Supervisor with an external identity provider.

Figure 530: OIDC Client ID and Secret



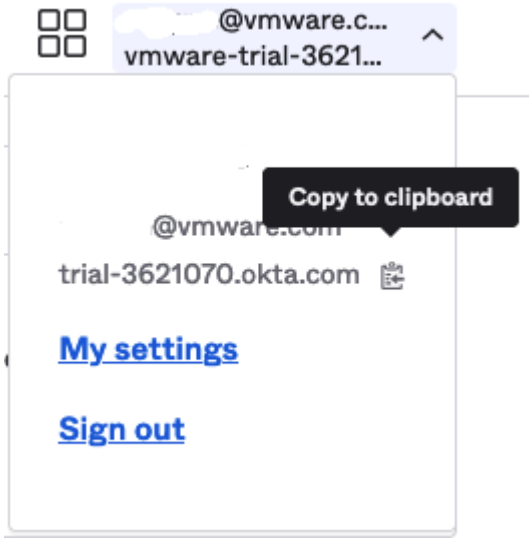
9. Configure the OpenID Connect ID Token.
   Click the **Sign On** tab. In the **OpenID Connect ID Token** section, click the Edit link, fill in the **Groups claim type** filter, and **Save** the settings.
   For example, if you want the claim name "groups" to match all groups, select **groups** > **Matches regex** > ***.

Figure 531: OpenID Connect ID Token



10. Copy the **Issuer URL**.
    To configure Supervisor, you need **Issuer URL**, in addition to the **Client ID** and **Client Secret**.
    Copy the **Issuer URL** from the Okta admin console.

    Figure 532: Okta Issuer URL



Register an External Identity Provider with Supervisor

To connect to VKS clusters on Supervisor using an external identity provider, register your identity provider with Supervisor.

Prerequisites

Before you register an external identity provider with Supervisor, complete the following prerequisites:

- Enable Supervisor Management and deploy a Supervisor instance.
- Configure an external OpenID Connect identity provider with the Supervisor callback URL. See Configure an External Identity Provider for VKS Clusters.
- Get the Client ID, Client Secret, and Issuer URL from the external IDP. See Configure an External Identity Provider for VKS Clusters.

Register an External Identity Provider with Supervisor

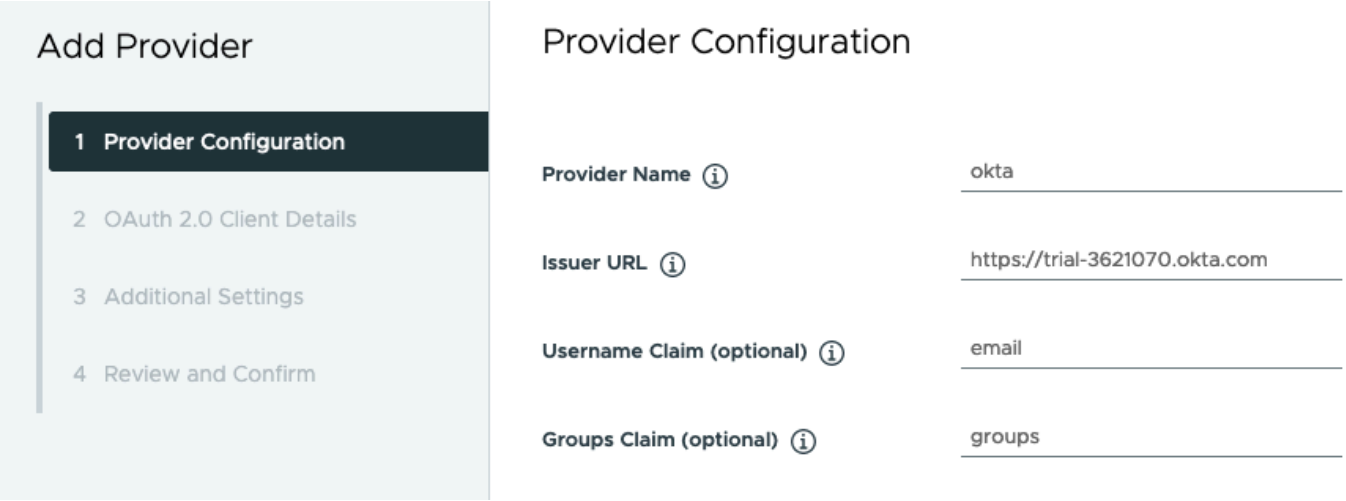
Supervisor runs as pods the Pinniped Supervisor and the Pinniped Concierge components. VKS clusters run as pods the Pinniped Concierge component only. For more information about these components and how they interact, refer to the Pinniped authentication service documentation.

Once you register an external identity provider with Supervisor, the system updates the Pinniped Supervisor and Pinniped Concierge pods on Supervisor and the Pinniped Concierge pods on VKS clusters. All VKS clusters running on that Supervisor instance are automatically configured with that same external identity provider.

To register an external ODIC provider with Supervisor, complete the following procedure:

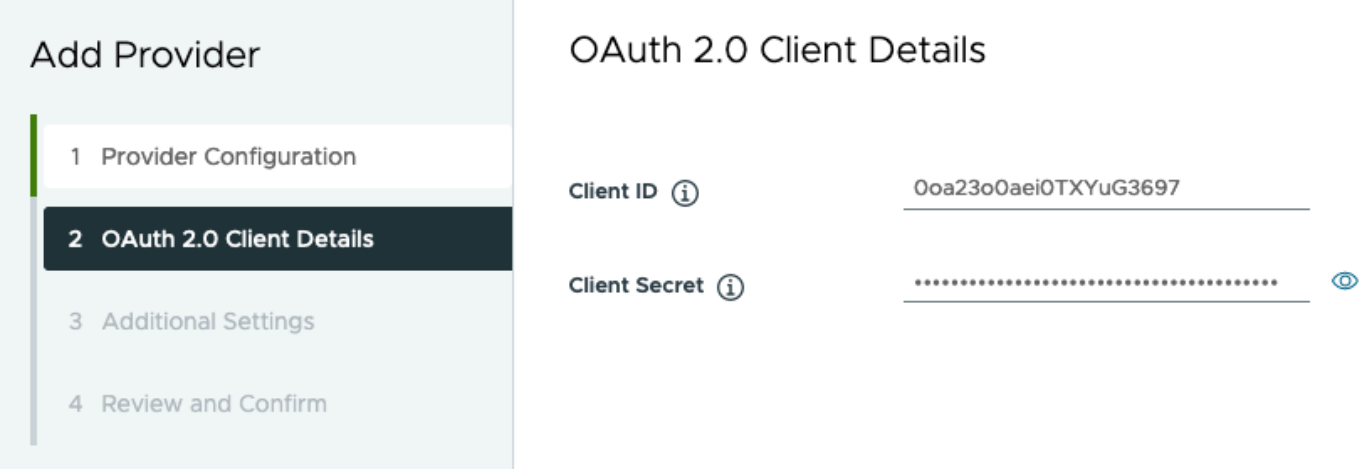
1. Log in to the vCenter using the vSphere Client.
2. Select **Supervisor Management** > **Supervisors** > **Configure** > **Identity Providers**.
3. Click the plus sign to begin the registration process.
4. Configure the provider. See Identity Provider Configuration.

   Figure 533: OIDC Provider Configuration



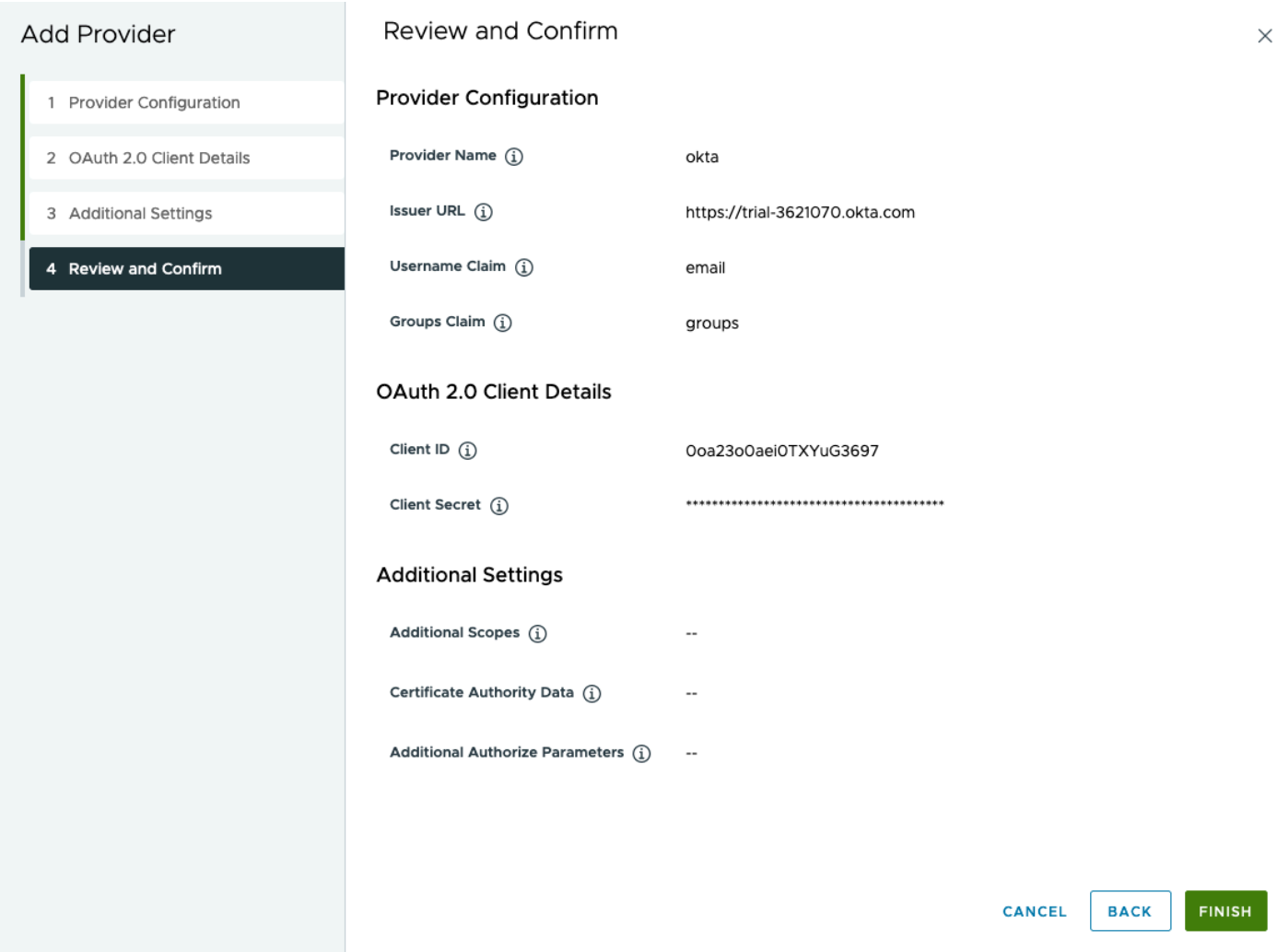
5. Configure OAuth 2.0 client details. See OAuth 2.0 Client Details.

Figure 534: OAuth 2.0 Client Details



6. Configure additional settings. See Additional Settings.
7. Confirm provider settings.

Figure 535: Confirm Provider Settings



8. Click **Finish** to complete the OIDC provider registration.

Identity Provider Configuration

Refer to the following provider configuration details when registering an external provider with Supervisor.

Table 1793: Identity Provider Configuration

Field	Importance	Description
Provider Name	Required	User-defined name for the external identity provider.
Issuer URL	Required	The URL to the identity provider issuing tokens. The OIDC discovery URL is derived from the issuer URL. For example, the Okta Issuer URL may look like the following and can be obtained from the Admin console: *https://trial-4359939-admin.okta.com*.
Username Claim	Optional	The claim from the upstream identity provider ID token or user info endpoint to inspect to obtain the user name for the given user. If you leave this field empty, the upstream issuer URL is concatenated with the "sub" claim to generate the user name to be used with Kubernetes. This field specifies what Pinniped should look at from the upstream ID token to determine to authenticate. If not provided, the user identity will be formatted as *https://IDP-ISSUER?sub=UUID*.
Groups Claim	Optional	The claim from the upstream identity provider ID token or user info endpoint to inspect to obtain the groups for the given user. If you leave this field empty, no groups are used from the upstream identity provider. The Group Claims field tells Pinniped what to look at from the upstream ID token to authenticate user identity.

OAuth 2.0 Client Details

Refer to the following provider OAuth 2.0 client details when registering an external identity provider with Supervisor.

Table 1794: OAuth 2.0 Client Details

OAuth 2.0 Client Details	Importance	Description
Client ID	Required	Client ID from the external IDP
Client Secret	Required	Client Secret from the external IDP

Additional Settings

Refer to the following additional settings when registering an external identity provider with Supervisor.

Table 1795: Additional Settings

Setting	Importance	Description
Additional Scopes	Optional	Additional scopes to be requested in tokens
Certificate Authority Data	Optional	TLS certificate authority data for secure external IDP connection
Additional Authorize Parameters	Optional	Additional parameters during the OAuth2 Authorization request

Change Storage Settings on the Supervisor

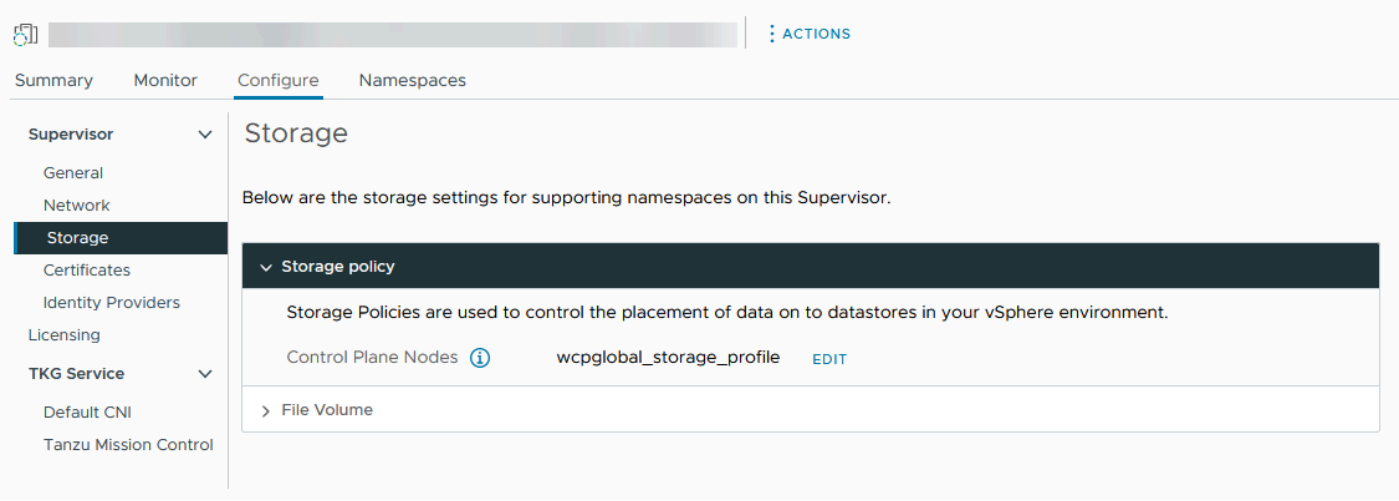
Storage policies assigned to the Supervisor manage how such objects as a control plane VM, vSphere Pod ephemeral disk, and container image cache are placed within datastores in the vSphere storage environment. As a VI administrator, you typically configure storage policies when enabling the Supervisor. If you need to make any changes in the storage policy assignments after initial Supervisor configuration, perform this task. You can also use this task to activate or deactivate file volume support for ReadWriteMany persistent volumes in VKS clusters.

If you plan to activate file volume support on VKS clusters for persistent volumes in ReadWriteMany mode, make sure to activate vSAN File Services must be enabled on the vSAN cluster. For additional information, follow the guidelines in *Using vSAN File Service to Create ReadWriteMany Volumes in Supervisor Environment* in the *vSphere Supervisor Services and Workloads* documentation.

Generally, the changes you make to the storage settings apply only to new objects in Supervisor. If you use this procedure to activate file volume support in VKS clusters, you can do it for the existing clusters.

1. In the vSphere Client, navigate to **Supervisor Management**.
2. Click the **Supervisors** tab and select the Supervisor to edit from the list.
3. Click the **Configure** tab and click **Storage**.

   Figure 536: Updating Supervisor storage settings



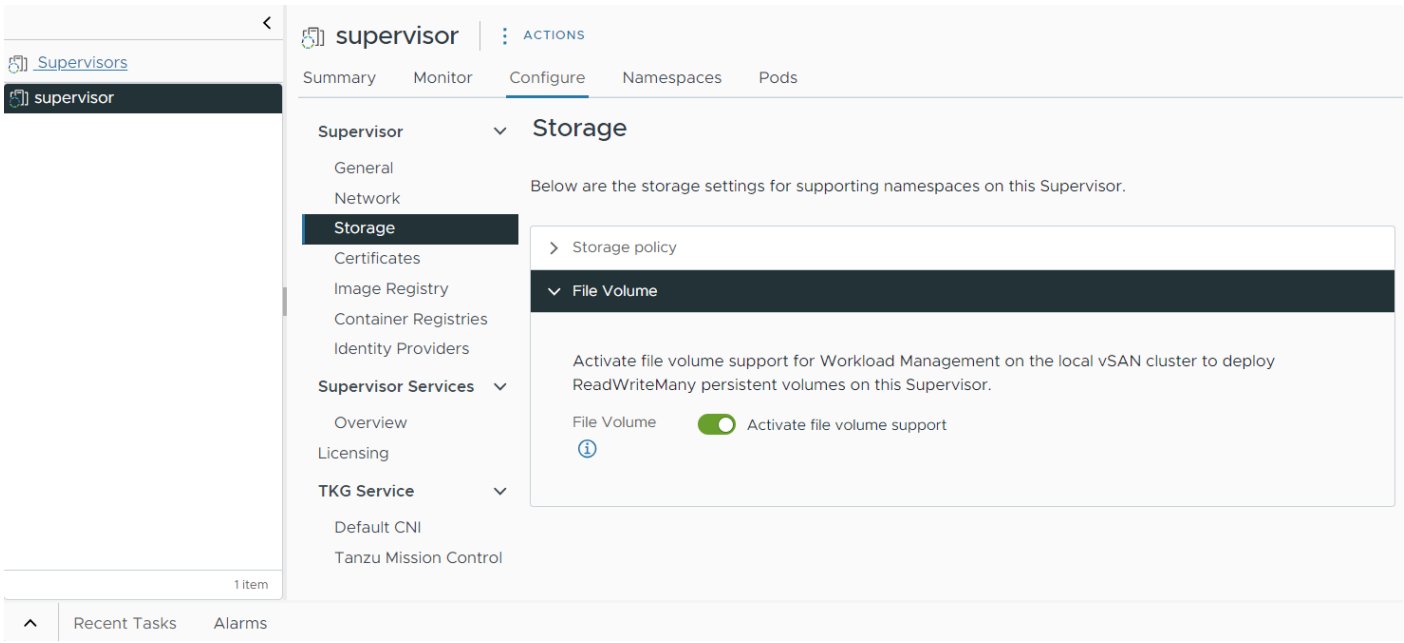
4. Change storage policy assignments for the control plane VMs.

   If your environment supports a vSphere Pod, you can also change storage policies for an ephemeral virtual disk and container image cache.

Option	Description
Control Plane Node	Select the storage policy for placement of the control plane VMs.
Pod Ephemeral Disks	Select the storage policy for placement of the vSphere Pods.
Container Image Cache	Select the storage policy for placement of the cache of container images.

5. Enable file volume support to deploy ReadWriteMany persistent volumes.

   This option is available only if your environment has been configured with vSAN File Service. See *Enable vSAN File Service* in the *vSAN Administration* guide.



Modify the List of Supervisor Control Plane DNS Names

Checkout how to modify the list of FQDNs for accessing the Supervisor control plane. You can provide a list of Supervisor FQDNs during the Supervisor enablement and later update that list. You can also set a list Supervisor FQDNs if you did not provide one during the Supervisor enablement.

Use the following DCLI command to update the list of FQDNs for the Supervisor control plane:

```
dcli com vmware vcenter namespacemanagement clusters update --cluster <cluster_ID> --master-dns-name <FQDN 1>
 --master-dns-name <FQDN_2>
```

- To add a new FQDN to the list you pass the already existing names as arguments as well as add the new FQDN.
- To remove an FQDN from the list, you call the `update` command by omitting the FQDN that you want to remove and passing the rest of the FQDNs that you want to keep.

For a three-zone Supervisor you can pass the ID of any of the clusters that are part of the Supervisor.

In the following example, the Supervisor running on cluster `domain-c50` is already configured with an FQDN `supervisor.acme.com`. You are adding a new `supervisor.vmware.com` FQDN to the list of DNS names for the Supervisor:

```
dcli com vmware vcenter namespacemanagement clusters update --cluster domain-c50 --master-dns-name
 supervisor.acme.com --master-dns-name supervisor.vmware.com
```

- The VIP certificate to securely connect to the Supervisor is not automatically updated with the new FQDNs. Therefore, you need to do that manually, see Replace the VIP Certificate to Securely Connect to the Supervisor API Endpoint.
- Once you update the VIP certificate to connect to the Supervisor, log in to the Supervisor control plane by using the newly-added FQDNs. See Connect to the Supervisor as a vCenter Single Sign-On User

Change the Default Key Size of Supervisor Certificates

You can change the default key size of the Supervisor internal certificates. The Supervisor generates certificates that arae used for securing internal communication between its components. For example, `authproxy`, `apiserver`, scheduler, `etcd`, and others. All of these components use their own certificates with a default algorithm and key size of RSA-2048, which you can change. The default certificate size is 3072 bits for a greenfield VCF deployment and Supervisor Kubernetes version 1.31 and later.

- Upgrade vCenter to VCF 9.1.

- Upgrade the Supervisor Kubernetes version to 1.31.

1. In the vSphere Client, navigate to **Supervisor Management**.
2. Click the **Supervisors** tab and select the Supervisor to edit from the list.
3. Click the **Configure** tab and **Certificates**.
4. In the Advanced section, select **Edit**.
5. Select a new size for the Supervisor generated certificates and click **Update**.

It takes few minutes for the certificates to rotate. Check for an event with ID *com.vmware.wcp.supervisor.internal.certificates. keySizeUpdate* in the Events tab of the vSphere Clusters where the Supervisor control plane VMs reside.

Configuring a Supervisor Management Proxy

The Supervisor Management Proxy allows workloads to access components that are exposed in the VCF management domain. Currently, components in VCF that run workloads requiring access on the VCF management domain include the software depot component, AVI controller, VCF Operations for Networks, and VCF Operations for gathering VKS cluster metrics. Workloads that can require access to the Supervisor management network are VKS cluster nodes, vSphere Pods, and VMs created by using the VM Service.

Components that run workloads requiring access to the VCF management domain are called management services. Every management service has access grants assigned for the workloads it runs. An access grant is the combination from the vSphere Namespace and workload types that are given access to the VCF management domain.

Create Management Services on the Supervisor

You create management services to provide access of workloads running on the Supervisor to the VCF management domain.

1. In the vSphere Client, navigate to **Supervisor Management**.
2. Click the **Supervisors** tab and select the Supervisor to edit from the list.
3. Click the **Configure** tab select `Network`.
4. Click **Add Management Service**.
5. Enter a name and a description for the management service.
6. Enter the IP address of the endpoint of the management service.
7. Configure ports and certificates for the management service.
8. Click **Add**.

Create access grants for the new management service.

Create Access Grants for Management Services on the Supervisor

You add access grants to management services to provide access to the VCF management domain to the workloads running in the management service.

1. In the vSphere Client, navigate to **Supervisor Management**.
2. Click the **Supervisors** tab and select the Supervisor to edit from the list.
3. Click the **Configure** tab select `Network`.
4. Select from the available network services and click **Add Access Grant**.
5. Select the vSphere Namespace to which you want to configure access grants to management services.
6. Select the scope for access grant.

   You select virtual machine and vSphere Pods. Virtual machines include nodes of VKS clusters and VMs created by using the VM Service.

Edit a Management Service on the Supervisor

You can edit the properties of a management service on the Supervisor .

1. In the vSphere Client, navigate to **Supervisor Management**.
2. Click the **Supervisors** tab and select the Supervisor to edit from the list.
3. Click the **Configure** tab select `Network`.
4. Click the arrows (>>) next to the service that you want to edit and click **Edit**.
5. Edit the properties of the management service.
6. Click **Save Changes**.

Edit Access Grants to Management Services on the Supervisor

You can edit the access grants of a management service to suspend workload types from accessing the VCF management domain.

1. In the vSphere Client, navigate to **Supervisor Management**.
2. Click the **Supervisors** tab and select the Supervisor to edit from the list.
3. Click the **Configure** tab select `Network`.
4. Click the management service where you want to edit access grants.
5. Click the three dots next to a namespace from the list and click **Edit**.
6. Edit the scope for the namespace.

Delete a Management Service on the Supervisor

Delete a management service in case that you want to suspend the access of that service to the VCF management domain.

> **Note:**
>
> Do not delete the pre-configured management services such as the software depot component, AVI controller, VCF Operations for Networks, and VCF Operations for gathering VKS cluster metrics.

1. In the vSphere Client, navigate to **Supervisor Management**.
2. Click the **Supervisors** tab and select the Supervisor to edit from the list.
3. Click the **Configure** tab select `Network`.
4. Click the three dots next to the service that you want to delete.
5. Click **Delete**.

Deactivate an Access Grant to a Management Service on the Supervisor

Deactivate an access grant if you want to temporarily suspend the access of certain management service workloads to the VCF management domain.

1. In the vSphere Client, navigate to **Supervisor Management**.
2. Click the **Supervisors** tab and select the Supervisor to edit from the list.
3. Click the **Configure** tab select `Network`.
4. Click a management service.
5. Click the three dots next to the access grant that you want to deactivate and click **Deactivate**.

Preserve Client IP Addresses on vSphere Supervisor

Configure the preservation of the source IP addresses of authentication and Kubernetes APIServer requests to the Supervisor. You can later inspect these IPs in the Supervisor Authenticating Proxy logs (Authproxy) and Kubernetes APIServer audit logs. This functionality is only available when the Supervisor is configured with the Avi Load Balancer and VDS as the networking stack.

- Verify that the Supervisor is deployed with dual NIC control plane VMs.
- Verify that Avi Load Balancer is the load balancer configured with the Supervisor.

1. In the vSphere Client, navigate to **Supervisor Management**.
2. Click the **Supervisors** tab and select the Supervisor to edit from the list.
3. Select **Configure** and under **General**, select **Control Plane**.
4. Enable the **Preserve Client IP Address** option.

Maintenance Mode for a Datastore Used by Supervisor

Starting with vSphere 9.0, you can decommission a datastore that is used for different Supervisor workloads. To do this, you need to place the datastore in maintenance mode.

Requirements

- vSphere 9.0 or later.
- Supervisor is enabled.
- Storage DRS is enabled. It is enabled only to support the datastore decommissioning and maintenance mode migration flow, and is not required for standard stable-state operations.
- The datastore is not part of a vSAN cluster or shared across multiple vCenter instances.

Prerequisites

Before putting the datastore in maintenance mode, make sure to follow these guidelines.

- Wait for ongoing operations on the datastore to complete, or cancel them manually.
- Do not create new content libraries on the datastore.
- Stop adding new items to any content library that resides on the datastore.
- Identify any applications that create First Class Disks (FCDs) independently, for example, OpenStack or direct API use. Prepare to migrate, back up and restore, or delete these FCDs as needed.
- For vSphere Pods not managed by deployments or stateful sets, prepare to recreate them manually from spec or pipeline.
- Unmount all ISO or shared images attached to VMs, VM Service VMs, vSphere Pods, and content libraries on the datastore.
- For vSAN clusters, verify that sufficient capacity and the required number of vSAN nodes meet the storage policy in use before initiating maintenance mode. Use the vSAN maintenance mode pre-check to confirm the cluster can satisfy all existing storage policies after the host is removed.

Migrating Storage Components

A Supervisor or VKS deployment involves four distinct storage components. Before putting the datastore in maintenance mode, account for each component and its migration method.

Before starting a migration, verify storage policy assignments, namespace configuration, and available capacity on the destination datastore. See Migrate Persistent Volumes.

Note:

If only virtual disks backing persistent volumes reside on the datastore, but no VMs, the system does not warn you before entering maintenance mode. Migrate those volumes before putting the datastore in maintenance mode to avoid accidentally deleting a datastore that still contains active volumes.

Component	Description	Migration method
Supervisor cluster VM VMDKs	Virtual disks for the Supervisor control plane VMs	Automatic
Supervisor PVCs	PVCs used by Supervisor services, including vSphere Pod VMs	Manual, backup and restore
VKS cluster node VMDKs	Virtual disks for VKS controller and worker node VMs	Automatic
VKS cluster workload PVCs	Application PVCs in VKS clusters	Manual, backup and restore

For PVCs, use a backup and restore solution to migrate workloads to the destination datastore. For additional information, see KB 319402.

Automatic Migrations

When the datastore enters maintenance mode, the system automatically migrates the following items. No administrator actions are required.

- VMs with attached VMDKs.
- Supervisor VM Service VMs.
- VKS clusters.

Manual Migration

The following items require manual handling by an administrator. After the datastore enters the maintenance mode, you must perform a number of tasks to migrate several of these items manually.

- Content library images.
- Standalone FCDs not attached to any VMs.
- vSphere Pods
- Orphaned or unused items in the datastore:
  - Unregistered or orphaned VMs.
  - Orphaned VMDKs.
  - Stale VM snapshots.
  - Orphaned Supervisor images and snapshots.
  - Orphaned vSphere Pods image disks.
  - ISO images.

Place a Datastore Used by Supervisor in Maintenance Mode

Begin the process by placing the Supervisor datastore into maintenance mode. This action triggers automatic migration for supported workloads.

1. Browse to the datastore in the vSphere Client.
2. Right-click the datastore and select **Maintenance Mode** > **Enter Maintenance Mode**.

   A list of recommendations appears for datastore maintenance mode migration.

3. Optional: On the **Recommendations** tab, review the recommendations and deselect any recommendations you do not want to apply.

   Note:

   The datastore cannot enter maintenance mode without evacuating all disks. If you deselect the recommendations, you must manually move all disks before entering maintenance mode.

4. Check the **Faults** tab for any VM artifacts that cannot be migrated.
5. Click **Continue** to initiate migration using Storage vMotion.

   After you place the datastore in maintenance mode, perform additional manual migration tasks.

Migrate vSphere Pods

Manually delete and prepare to recreate vSphere Pods that are not automatically handled during datastore maintenance.

1. Identify the datastore MoID.
2. Go to the following URL to list all vSphere Pods using the datastore.

   `https://<VC>/mob/?moid=<datastore-moid>`

3. Use the `kubectl delete` command to remove the vSphere Pods.

   ```
   kubectl delete pods <pod-name> --namespace <namespace>
   ```

4. Delete dangling image disks after vSphere Pods deletion.

   ```
   kubectl delete imagedisk <imagedisk>
   ```

   vSphere Pods that are part of a deployment or stateful set will restart automatically. vSphere Pods that are deployed outside of deployments or stateful sets are not recreated. You must manually recreate them.

Migrate Content Library Images

Move all content library data from the datastore and ensure no ISO images remain mounted to dependent VMs.

1. In the vSphere Client, select the datastore and click the **Files** tab.
2. Filter by folder name: `contentlib-`.
3. Identify all content libraries present on the datastore.
4. For each library, perform the following steps.

   Ensure the correct source and destination datastores are selected. Unmount any ISOs from VMs or VM Service VMs before migration.

   a) Click **Actions** > **Migrate**.
   b) Ensure the correct source and destination datastores are selected.
   c) Unmount any ISOs from VMs or VM Service VMs before migration.

Migrate Detached FCDs

Manually migrate First Class Disks (FCDs) that are not attached to VMs, using different methods depending on whether they are managed by CNS or not.

To access and migrate CNS-managed FCDs, make sure that you have the ServiceProviderUsers role.

Note:

FCDs that are attached to VM Service VMs or vSphere Pods are migrated automatically with a corresponding VM and placed in the VM folder.

1. Migrate FCDs managed by CNS.

   For more information, see Monitoring and Migrating Persistent Volumes in Supervisor Environment.

a) In the vSphere Client, navigate to the datastore cluster, and click **Monitor** > **Container Volumes**.
b) Select the volumes and click **Migrate**.
c) Select the destination datastore, and confirm.

2. Migrate FCDs not managed by CNS.
   a) Go to the following URL:

      `https://<VC>/mob/?moid=VStorageObjectManager`
   b) Use the `ListVStorageObject` method with the datastore MoID to get a list of FCD IDs.
   c) Use the `RelocateVStorageObject_Task` method to move each FCD.

      Enter the FCD ID, source, and destination datastore MoIDs.

Decommissioning a Supervisor Cluster

VCF does not support decommissioning a vSphere cluster while a Supervisor is active on the cluster or zone. You must disable the Supervisor before decommissioning the cluster or zone.

Disable the Supervisor only after you have completed this workflow:

1. Delete all workloads running in VKS clusters in the affected zone.
   For information about workloads, see Deploying Workloads on VKS Clusters.
2. Delete all VKS clusters in the affected namespaces.
   See Delete a VKS Cluster Using Kubectl or Using VCF CLI.
3. Remove all Supervisor namespaces.
   See Remove a vSphere Namespace from Supervisor.
4. Disable the Supervisor on the cluster or zone.
   See Disable a Supervisor.

Monitoring vSphere Supervisor

You can use different tools to monitor the vSphere Supervisor state and workloads.

You can use VCF Operations to monitor the state and the resources of vSphere Supervisor. See vSphere Supervisor Monitoring.

Monitor and Manage Resources in a vSphere Namespace

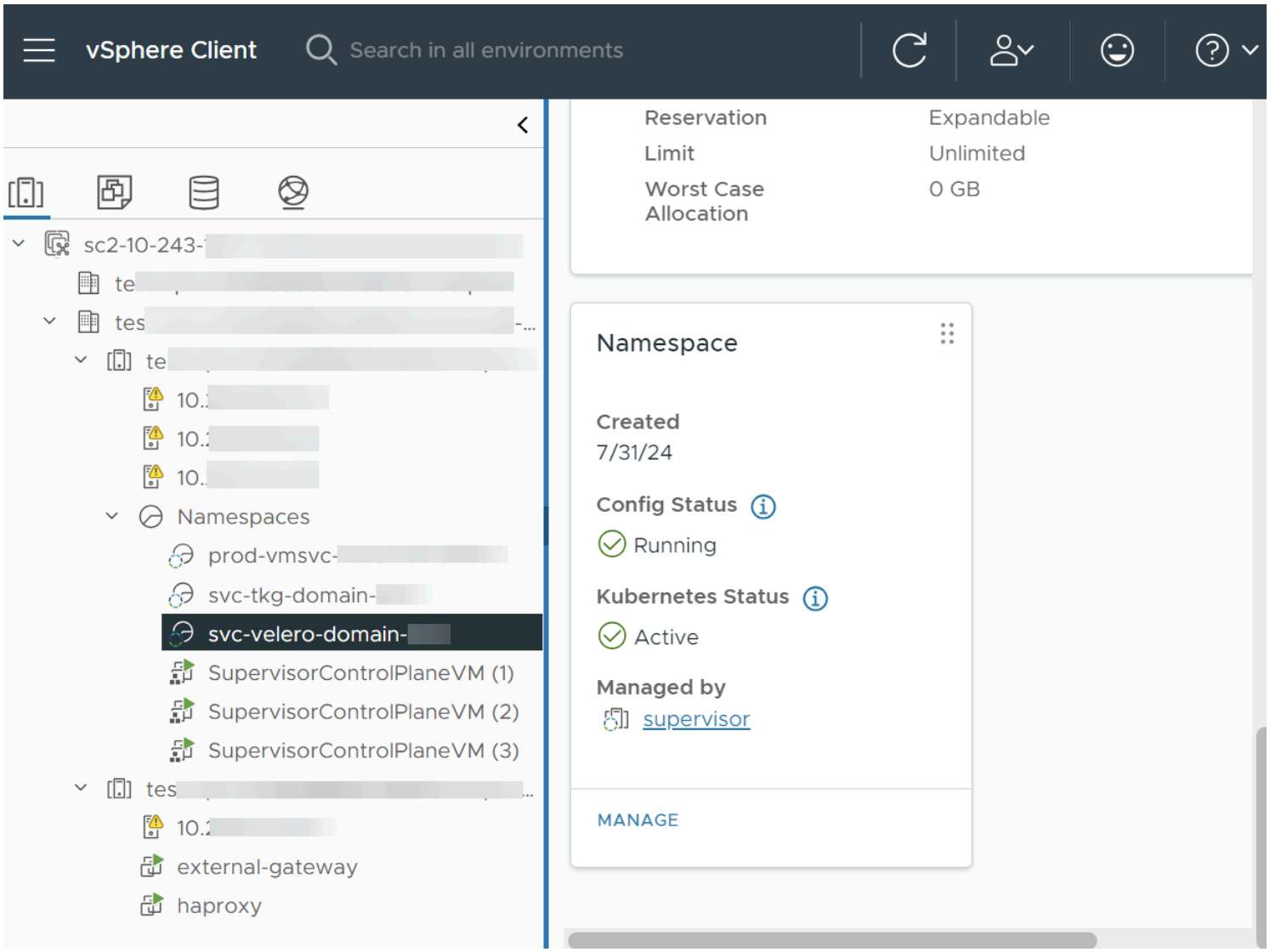
You can monitor and manage different aspects of a vSphere Namespace, such as resource consumption for the namespace as well as the number of different objects that exist in a namespace and their states.

Create and Configure a vSphere Namespace.

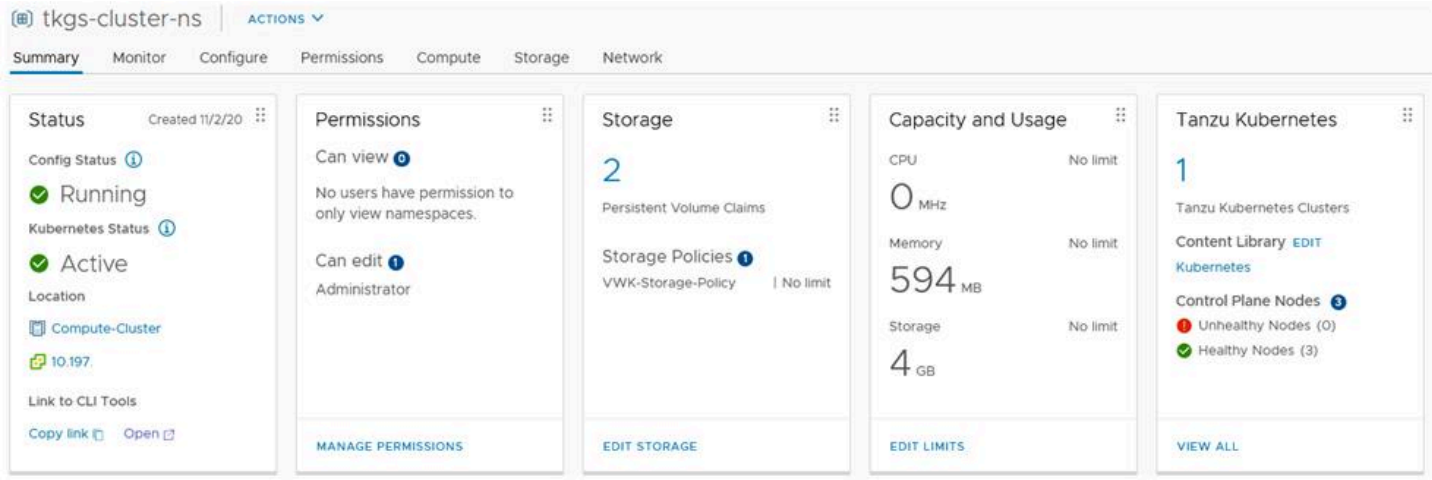
1. Log in to the vCenter using the vSphere Client.
2. In the inventory, navigate to the **Menu** > **Hosts and Clusters** view.
3. Select the host cluster where you have enabled **Supervisor Management**.
4. Expand the **Namespaces** resource pool.

   The Supervisor control plane nodes are located in the Namespaces resource pool. In addition, you can see individual resource pools created for each namespace on this Supervisor.

5. Select the namespace resource pool you want to review, click the **Summary** tab, and click **Manage** on the **Namespace** pane.



In the **Summary** tab for this namespace you see the various configuration sections for the vSphere Namespace, including **Status**, **Permissions**, **Storage**, **Capacity and Usage**, and so on. From this screen, you can manage any of these settings.



Monitoring and Migrating Persistent Volumes in Supervisor Environment

As a vSphere administrator, you can use the vSphere Client to view details and monitor key metrics related to persistent volumes, such as PV health status, storage compliance status, and binding information. You can identify potential issues like PV detachment or storage non-compliance and, if needed, remediate the issues by reapplying the storage policy or migrating the volume to another datastore.

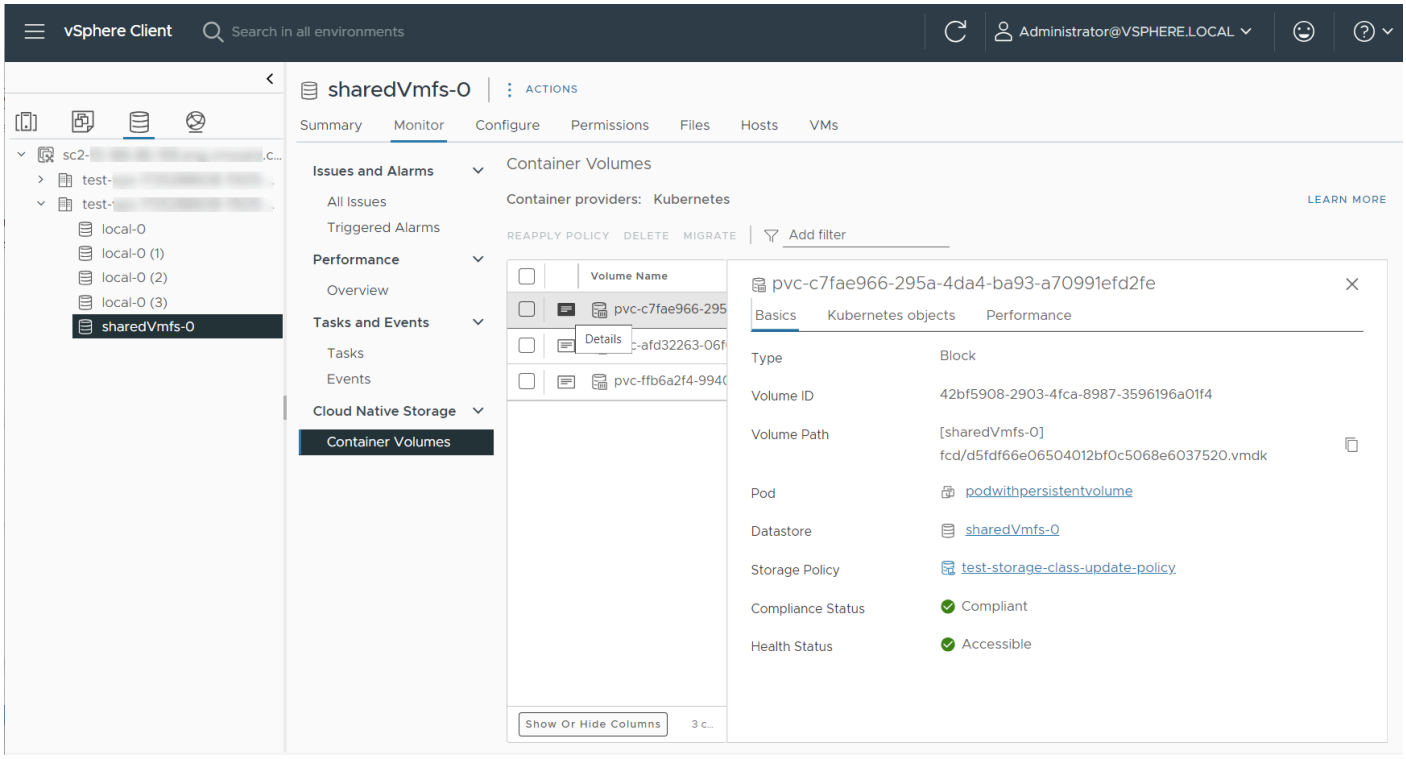
Monitor and Manage Persistent Volumes

Use the vSphere Client to display persistent volumes and review their details.

For information about using `govc` to identify orphaned volumes, see Identify and Remove Orphaned Volumes.

To monitor datastores, see Monitor Datastore Capacity.

1. In the vSphere Client, navigate to the namespace that has persistent volumes.
   a) From the vSphere Client home menu, select **Supervisor Management**.
   b) Click the **Namespaces** tab, and select the namespace from the list.
2. Click the **Storage** tab and click **Persistent Volume Claims**.
   The vSphere Client lists all persistent volume claim objects and corresponding volumes available in the namespace.
3. To view details of a selected persistent volume, click the name of the volume in the **Name** column.
   The **Container Volumes** panel opens.
4. Click the **Details** icon, and switch between the tabs.



The tabs display the following information related to the Kubernetes persistent volume.

Item	Description
Type	Type of storage that backs the persistent volume, such as block.
Volume ID	An ID of the volume.

Item	Description
Volume Path	Path to the virtual disk that backs the volume. The path includes the datastore name, the folder on the datastore, and the name of the disk. For example, `[sharedVmfs-0] fcd/1b325557eb7f4a0991fb47ab1f32f74d.vmdk`.
Pod or VM	Name of the pod or VM that uses the volume. You can click the link to display the item in the vSphere inventory.
Datastore	Datastore where the virtual disk that backs the volume resides. You can click the link to display the datasotre in the inventory. Use the datastore browser to navigate to the virtual disk. See *Using Datastore Browser in vSphere Environment* in *vSphere Storage*.
Storage Policy	The storage policy that the volume uses.
Compliance Status	The storage compliance status of the volume.
Health Status	The health status of the volume.
Namespace	The namespace where the volume resides.
Persistent Volume Claim	Information related to the persistent volume claim bound to the volume.
Physical Placement	The device information, such as name, identifier or UUID, number of devices used for a persistent volume, and how it is mirrored across hosts.
Performance	You can use vSAN performance service to monitor the performance of your vSAN environment, and investigate potential problems. For more information, see the *Monitoring vSAN Performance* documentation.

For more details, see https://knowledge.broadcom.com/external/article?legacyId=90607.

Check Health and Storage Compliance Status of a Persistent Volume

On the **Basics** tab, you can check the health status of the persistent volume. You can also check its storage compliance status and reapply the storage policy if the status is out of date.

1. Click the **Basics** tab and verify the health status of the volume.

Health Status	Description
Accessible	The persistent volume is accessible and available for use.
Inaccessible	The persistent volume is inaccessible and cannot be used. For example, the volume becomes inaccessible if the datastore that stores the volume cannot be reached by the hosts that connect to the datastore.

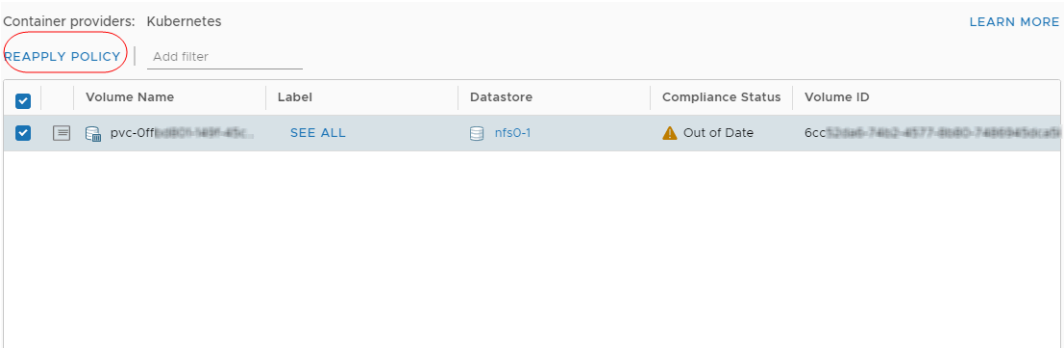
2. Verify the storage compliance status.

   You can see one of the following statuses in the **Compliance Status** column.

Compliance Status	Description
Compliant	The datastore where the virtual disk backing the volume resides has the storage capabilities that the policy requires.
Out of Date	This status indicates that the policy has been edited, but the new storage requirements have not been communicated to the datastore. To communicate the changes, reapply the policy to the volume that is out of date.

Compliance Status	Description
Noncompliant	The datastore supports specified storage requirements, but cannot currently satisfy the storage policy. For example, the status might become Noncompliant when physical resources of the datastore are unavailable. You can bring the datastore into compliance by making changes in the physical configuration of your host cluster, for example by adding hosts or disks to the cluster. If additional resources satisfy the storage policy, the status changes to Compliant. If you cannot bring the datastore in the compliance, you can migrate the volume to a compliant datastore.
Not Applicable	The storage policy references datastore capabilities that are not supported by the datastore.

3. If the compliance status is Out of Date, select the volume and click **Reapply Policy**.



The status changes to Compliant.

Migrate Persistent Volumes

In the Supervisor environment, you can use the vSphere Client to migrate persistent volumes to a different datastore. For example, you might need to migrate a volume when its current datastore can no longer satisfy the volume storage requirements.

- Make sure that the original and target datastore is accessible to all Supervisor nodes. This applies to one-zone and three-zone deployments.
  For example, in a single-zone deployment, the target datastore must be shared across the nodes. The system doesn't prevent you from moving a volume from shared storage to non-shared storage, but the volume becomes inaccessible, which can affect the apps that use the volume.
- Ensure that a storage policy covering the target datastore is assigned to the namespace. If the target datastore uses a different storage policy than the source, assign both policies to the namespace before starting the migration.
- If a volume is attached to a VM, the VM remains locked during volume relocation. If the volume you relocate is attached to a node VM, such as the VKS node, the system does not allow any other control operations on the VM. The operations include attaching or detaching other volumes to the VM, migrating the VM to a different datastore, changing any configuration of the VM, and so on.
- Because PMEM datastores are not supported, attempts to relocate a volume to a PMEM datastore fail.
- The target datastore must have sufficient free capacity to accommodate the migrated volume. If the free space of a datastore is less than the volume size, you cannot select the datastore for migration.
- The system does not support storage policy changes during the migration operation. However, the system doesn’t prevent you from selecting a target datastore that is not compatible with the volume policy.

For additional details and known issues, see the following Broadcom Knowledge Base articles:

- 313416

- 313308
- 319402

Note:
When VMs reside on a datastore, the system alerts you and prevents the task from completion if you attempt to put the datastore into maintenance mode. However, if virtual disks backing persistent volumes are on the datastore, it can enter maintenance mode without warning. This means no safeguards prevent you from deleting a datastore in maintenance mode that still contains active volumes. To avoid accidentally deleting such a datastore, consider migrating the volumes to a different datastore before putting the original datastore into maintenance.

1. In the **Container Volumes** panel, select the volume or several volumes and click the **Migrate** button.
2. In the **Migrate volume** dialog box, select the target datastore.
   Make sure that the target datastore is compatible and accessible to the volume.
3. Acknowledge that you accept the potential consequences of the migration.
   For example, the volume might become inaccessible or the target datastore might be incompatible with the storage policy the volume uses.
4. Click **Migrate**.

Forwarding Supervisor Logs to External Monitoring Systems

Checkout how to configure forwarding of Supervisor control plane logs to external monitoring systems, such as Grafana Loki, or Elastic Search, by using Fluent Bit.

Supervisor control plane logs are automatically forwarded to the syslog server configured to the vCenter appliance by using Fluent Bit. Fluent Bit is an open-source lightweight logging and metrics processor and forwarder that provides configurations to support various log data types, filtering and log tag enhancements.

During Supervisor activation or upgrade, bootstrap logs are still forwarded by rsyslog to the syslog servers that are configured to the vCenter appliance. Once the Supervisor control plane VMs are up and running, Fluent Bit becomes the default log forwarder for the Supervisor control plane logs.

As a VI administrator, you can use Fluent Bit to:

- Forward Supervisor control plane logs and system journal logs to major external log monitoring platforms such as Loki, Elastic Search, Grafana, and other platforms that are supported by Fluent Bit.
- Update or reset the log forwarding configuration for the Supervisor control plane by using the k8s API.

Fluent Bit runs as a DeamonSet on Supervisor control plane nodes. It exposes the `fluentbit-config-custom` ConfigMap under the `vmware-system-logging` namespace that VI administrators can edit to configure log forwarding to external platforms by defining log servers.

```
inputs-custom.conf: |
[INPUT]
   Name              tail
   Alias             audit_apiserver_tail
   Tag               audit.apiserver.*
   Path              /var/log/vmware/audit/kube-apiserver.log
   DB                /var/log/vmware/fluentbit/flb_audit_apiserver.db
   Buffer_Max_Size   12MBb
   Mem_Buf_Limit     32MB
   Skip_Long_Lines   On
   Refresh_Interval  10

filters-custom.conf: |
[FILTER]
   Name            record_modifier
   Alias           audit_apiserver_modifier
   Match           audit.apiserver.*
   Record          hostname ${NODE_NAME}
   Record          appname audit-kube-apiserver
```

```
    Record         filename kube-apiserver.log

outputs-custom.conf: |
[OUTPUT]
  Name                 syslog
  Alias                audit_apiserver_output_syslog
  Match                audit.apiserver.*
  Host                 <syslog-server-host>
  Port                 <syslog-server-port>
  Mode                 tcp
  Syslog_Format        rfc5424
  Syslog_Message_key   log
  Syslog_Hostname_key  hostname
  Syslog_Appname_key   appname
  Syslog_Msgid_key     filename
```

Customize Fluent Bit Log Forwarding

Follow the steps to customize the Fluent Bit log forwarding configuration:

1. Log in to the Supervisor.
   For information, see Connecting to Supervisor and VKS Clusters.
2. Update or add a syslog output in the `outputs-custom.conf` section in the `fluentbit-config-custom` ConfigMap, which will forward all control plane VM system logs to an external server.

   ```
   [OUTPUT]
     Name                 syslog
     Alias                syslog_system
     Match                system*
     Host                 <syslog-server-host>
     Port                 <syslog-server-port>
     Mode                 tcp
     Syslog_Format        rfc5424
     Syslog_Message_key   log
     Syslog_Hostname_key  hostname
     Syslog_Appname_key   appname
     Syslog_Msgid_key     filename
     # add the following if the mode is TLS
     Tls                  on
     Tls.verify           off
     Tls.ca_file          /etc/ssl/certs/vmca.pem
   ```

3. Apply the changes to the `fluentbit-config-custom` ConfigMap.

   ```
   > kubectl -n vmware-system-logging edit cm fluentbit-config-custom

   # use the below command if the change is stored in outputs-custom.conf file
   > kubectl -n vmware-system-logging create configmap fluentbit-config-custom --from-file=filters-
   custom.conf --from-file=inputs-custom.conf --from-file=outputs-custom.conf -o yaml --from-file=parsers-
   custom.conf --dry-run | kubectl replace -f -
   ```

4. Monitor the Fluent Bit pod to apply the configuration changes automatically, and query Supervisor logs on the syslog server. If the Fluentbit DaemonSet runs to an error after updated configuration isreloaded, repair or reset the configuration in the `fluentbit-config-custom` ConfigMap to ensure that the Fluentbit DaemonSet healthy.

   ```
   > kubectl -n vmware-system-logging get pod
   > kubectl -n vmware-system-logging logs <fluentbit-pod-name>
   ```

Forward Kubernetes API Server Audit Logs to a Grafana Loki Server

Follow the steps to configure log forwarding to an external Grafana Loki server:

1. Log in to the Supervisor.
   For information, see Connecting to Supervisor and VKS Clusters.
2. Update or add a Loki output in the `outputs-custom.conf` section in the `fluentbit-config-custom` ConfigMap, which will forward all control plane VM system logs to the Loki log server.
   ```
   [OUTPUT]
        Name loki
        Alias system_output_loki
        Match system*
        Host <loki-server-host>
        Port <loki-server-port>
        Labels $hostname,$appname,$filename,$procid,$labels
   ```
3. Apply the changes to the `fluentbit-config-custom` ConfigMap.
   ```
   > kubectl -n vmware-system-logging edit cm fluentbit-config-custom

   # use the below command if the change is stored in outputs-custom.conf file
   > kubectl -n vmware-system-logging create configmap fluentbit-config-custom --from-file=filters-
   custom.conf --from-file=inputs-custom.conf --from-file=outputs-custom.conf -o yaml --from-file=parsers-
   custom.conf --dry-run | kubectl replace -f -
   ```
4. Monitor the Fluent Bit pod to apply the configuration changes automatically, and query Supervisor logs on the syslog server. If the Fluentbit DaemonSet runs to an error after updated configuration isreloaded, repair or reset the configuration in the `fluentbit-config-custom` ConfigMap to ensure that the Fluentbit DaemonSet healthy.
   ```
   > kubectl -n vmware-system-logging get pod
   > kubectl -n vmware-system-logging logs <fluentbit-pod-name>
   ```

Forward Logs to Elastic Search

Follow the steps to configure log forwarding to an external Elastic Search server:

1. Log in to the Supervisor.
   For information, see Connecting to Supervisor and VKS Clusters.
2. Update or add a Elastic Search output in the `outputs-custom.conf` section in the `fluentbit-config-custom` ConfigMap, which will forward all control plane VM system logs to the ES log server.
   ```
   [OUTPUT]
       Name es
       Alias system_output_es
       Match system*
       Host <es-server-host>
       Port <es-server-port>
       Index supervisor
       Type controlplanevm
   ```
3. Apply the changes to the `fluentbit-config-custom` ConfigMap.
   ```
   > kubectl -n vmware-system-logging edit cm fluentbit-config-custom

   # use the below command if the change is stored in outputs-custom.conf file
   > kubectl -n vmware-system-logging create configmap fluentbit-config-custom --from-file=filters-
   custom.conf --from-file=inputs-custom.conf --from-file=outputs-custom.conf -o yaml --from-file=parsers-
   custom.conf --dry-run | kubectl replace -f -
   ```
4. Monitor the Fluent Bit pod to apply the configuration changes automatically, and query Supervisor logs on the syslog server.
   ```
   > kubectl -n vmware-system-logging get pod
   > kubectl -n vmware-system-logging logs <fluentbit-pod-name>
   ```

Forward Supervisor Auth Logs to a Syslog Server

Follow the steps to configure forwarding of Supervisor Auth logs to an external syslog server:

1. Add `kubectl-plugin-vsphere` and `authproxy` input to `fluentbit-config-custom` ConfigMap:

```
[INPUT]
 Name              tail
 Tag               auth.kubectl-plugin.*
 Path              /var/log/containers/kubectl-plugin-vsphere*.log
 DB                /var/log/vmware/fluentbit/flb_auth_kubectl-plugin.db
 Skip_Long_Lines   Off
 Refresh_Interval  10

[INPUT]
 Name              tail
 Tag               auth.authproxy.*
 Path              /var/log/containers/wcp-authproxy*.log
 DB                /var/log/vmware/fluentbit/flb_auth_authproxy.db
 Skip_Long_Lines   Off
 Refresh_Interval  10
```

2. Add `kubectl-plugin-vsphere` and `authproxy` filter to the `fluentbit-config-custom` ConfigMap:

```
[FILTER]om
 Name           kubernetes
 Match          auth.*
 Kube_URL       https://localhost:6443
 Tls.verify     Off
 K8S-Logging.Parser On
 K8S-Logging.Exclude On

[FILTER]
 Name           record_modifier
 Match          auth.*
 Operation      lift
 Nested_under   kubernetes

[FILTER]
 Name           modify
 Match          auth.*
 Rename         container_name appname
 Rename         host hostname
 Rename         pod_name procid
```

3. Add `kubectl-plugin-vsphere` and `authproxy` output to syslog server to the `fluentbit-config-custom` ConfigMap:

```
[OUTPUT]
 Name                syslog
 Match               auth.*
 Host                <syslog-server-host>
 Port                <syslog-server-port>
 Mode                tcp
 Syslog_Format       rfc5424
 Syslog_Message_key  log
 Syslog_Hostname_key hostname
 Syslog_Appname_key  appname
 Syslog_Msgid_key    filename
```

4. Monitor the Fluent Bit pod to apply the configuration changes automatically, and query the Supervisor auth logs on the syslog server.

```
> kubectl -n vmware-system-logging get pod
> kubectl -n vmware-system-logging logs <fluentbit-pod-name>
```

Streaming Supervisor Metrics to a Custom Observability Platform

Learn how to stream Supervisor metrics gathered by Telegraf to a custom observability platform. Telegraf is enabled by default on the Supervisor and it collects metrics in Prometheus format from Supervisor components, such as the Kubernetes API server, VM service, VKS, and others. As a VI administrator, you can configure an observability platform such as VCF Operations, Grafana, and others to view and analyse the collected Supervisor metrics.

Telegraf is a server-based agent for collecting and sending metrics from different systems, data bases, and IoT. Each Supervisor component exposes an endpoint where Telegraf connects. Telegraf then sends the collected metrics to an observability platform of your choice. You can configure any of the output plug-ins that Telegraf supports as an observability platform for aggregating and analyzing Supervisor metrics. See the Telegraf documentation for the supported output plug-ins.

The following components expose endpoints where Telegraf connects and gathers metrics: Kubernetes API server, etcd, kubelet, Kubernetes controller manager, Kubernetes scheduler, VKS, VM service, VM image service, NSX Container Plug-in (NCP), Container Storage Interface (CSI), certificate manager, NSX, and various host metrics such as CPU, memory, and storage.

View the Telegraf Pods and Configuration

Telegraf runs under the `vmware-system-monitoring` system namespace on the Supervisor. To view the Telegraf pods and ConfigMaps:

1. Log in to the Supervisor.
   For information, see Connecting to Supervisor and VKS Clusters.
2. Use the following command to view the Telegraf pods:
   ```
   kubectl -n vmware-system-monitoring get pods
   ```
   The resulting pods are the following:
   ```
   telegraf-csqsl
   telegraf-dkwtk
   telegraf-l4nxk
   ```
3. Use the following command to view the Telegraf ConfigMaps:
   ```
   kubectl -n vmware-system-monitoring get cm
   ```
   The resulting ConfigMaps are as follows:
   ```
   default-telegraf-config
   kube-rbac-proxy-config
   kube-root-ca.crt
   telegraf-config
   ```
   The `default-telegraf-config` ConfigMap holds the default Telegraf configuration and it is read-only. You can use it as fallback option to restore the configuration in `telegraf-config` in case the file is corrupt or you want to just restore to the defaults. The only ConfigMap that you can edit is `telegraf-config`, it defines which components are sending metrics to the Telegraf agents, and to which platforms.
4. View the `telegraf-config` ConfigMap:
   ```
   kubectl -n vmware-system-monitoring get cm telegraf-config -o yaml
   ```

The `inputs` section of the `telegraf-config` ConfigMap defines all the endpoints of Supervisor components from which Telegraf gathers metrics as well as the types of metrics themselves. For example the following input defines the Kubernetes API server as an endpoint :

```
[[inputs.prometheus]]
     # APIserver
     ## An array of urls to scrape metrics from.
     alias = "kube_apiserver_metrics"
```

```
      urls = ["https://127.0.0.1:6443/metrics"]
      bearer_token = "/run/secrets/kubernetes.io/serviceaccount/token"
      # Dropping metrics as a part of short term solution to vStats integration 1MB metrics payload limit
      # Dropped Metrics:
      # apiserver_request_duration_seconds
      namepass = ["apiserver_request_total", "apiserver_current_inflight_requests",
"apiserver_current_inqueue_requests", "etcd_object_counts",
"apiserver_admission_webhook_admission_duration_seconds", "etcd_request_duration_seconds"]
      # "apiserver_request_duration_seconds" has _massive_ cardinality, temporarily turned off. If histogram,
maybe filter the highest ones?
      # Similarly, maybe filters to _only_ allow error code related metrics through?
      ## Optional TLS Config
      tls_ca = "/run/secrets/kubernetes.io/serviceaccount/ca.crt"
```

The `alias` property indicates the component from where metrics are gathered. The `namepass` property specifies which component metrics are exposed and respectively collected by the Telegraf agents.

Although the `telegraf-config` ConfigMap already contains a broad range of metrics, you can still define additional ones. See Metrics For Kubernetes System Components and Kubernetes Metrics Reference.

Configure Observability Platform to Telegraf

In the `outps` section of `telegraf-config` you configure where Telegraf streams the metrics that it gathers. Several options exist such as `outputs.file`, `outputs.wavefront`, `outputs.prometheus_client`, and `outps-https`. The `outps-https` section is where you can configure the observability platforms that you want to use for aggregation and monitoring of the Supervisor metrics. You can configure Telegraf to send metrics to more than one platform. To edit the `telegraf-config` ConfigMap and configure an observability platform for viewing Supervisor metrics, follow the steps:

1. Log in to the Supervisor.
   For information, see Connecting to Supervisor and VKS Clusters.
2. Save the `telegraf-config` ConfigMap to the local kubectl folder:
   ```
   kubectl get cm telegraf-config -n vmware-system-monitoring -o jsonpath="{.data['telegraf
   \.conf']}">telegraf.conf
   ```
   Make sure to store the `telegraf-config` ConfigMap in a version control system before you make any changes to it in case you want to restore to a previous version of the file. In case you want to restore to the default configuration, you can use the values from the `default-telegraf-config` ConfigMap.
3. Add `outputs.http` sections with the connection settings of the observability platforms of your choice by using a text editor, such as VIM:
   ```
   vim telegraf.config
   ```
   You can directly uncomment the following section and edit the values accordingly, or add a new `outputs.http` sections as needed.
   ```
      #[[outputs.http]]
      #   alias = "prometheus_http_output"
      #   url = "<PROMETHEUS_ENDPOINT>"
      #   insecure_skip_verify = <PROMETHEUS_SKIP_INSECURE_VERIFY>
      #   data_format = "prometheusremotewrite"
      #   username = "<PROMETHEUS_USERNAME>"
      #   password = "<PROMETHEUS_PASSWORD>"
      #   <DEFAULT_HEADERS>
   ```
   For example, here's how an `outputs.http` configuration for Grafana looks like:
   ```
      [[outputs.http]]
     url = "http://<grafana-host>:<grafana-metrics-port>/<prom-metrics-push-path>"
     data_format = "influx"
     [outputs.http.headers]
       Authorization = "Bearer <grafana-bearer-token>"
   ```

See Stream metrics from Telegraf to Grafana for more information on configuring dashboards and consuming metrics from Telegraf.

And here's an example with VMware Aria Operations for Applications (former Wavefront):

```
[[outputs.wavefront]]
  url = "http://<wavefront-proxy-host>:<wavefront-proxy-port>"
```

The recommended way of ingesting metrics to Aria Operations for Applications is through a proxy. See Wavefront Proxies for more information.

4. Replace the existing `telegraf-config` file on the Supervisor with the one that you edited on your local folder:

   ```
   kubectl create cm --from-file telegraf.conf -n vmware-system-monitoring telegraf-config --dry-run=client -o yaml | kubectl replace -f -
   ```

5. Check if the new configuration is saved successfully:
   - View the new `telegraf-config` ConfigMap:

     ```
     kubectl -n vmware-system-monitoring get cm telegraf-config -o yaml
     ```

   - Check if all the Telegraf pods are up and running:

     ```
     kubectl -n vmware-system-monitoring get pods
     ```

   - In case some of the Telegraf pods is not running, check the Telegraf logs for that pod to troubleshoot:

     ```
     kubectl -n vmware-system-monitoring logs <telegraf-pod>
     ```

Migrate Telegraf Configuration after Upgrading to VCF 9

Learn how to migrate the Telegraf configuration values after upgrading your environment to VCF 9 so that you can keep collecting metrics with the same configuration values prior the upgrade.

When you upgrade the Supervisor from version 8.x to 9.0, the `telegraf-user-config` configmap is created to maintain user managed configuration. The data from the `telegraf-config` configmap is automatically backed up to the `telegraf-config-backup` configmap and you need to copy your configuration data from the backup to the `telegraf-user-config`.

1. Log in to the Supervisor as a vCenter Single Sign-On user.

   ```
   vcf context create --endpoint <SUPERVISOR-ADDRESS> --username <VCENTER-SSO-USER> --ca-certificate <PATH-TO-CERTIFICATE-FILE>
   ```

2. Switch to the context of the Supervisor.

   ```
   kubectl config use-context <SUPERVISOR-ADDRESS>
   ```

3. Verify if your vCenter Single Sign-On user has edit permission on the `telegraf-user-config` configmap.

   ```
   kubectl auth can-i update cm/telegraf-user-config -n vmware-system-monitoring
   ```

   The command returns `yes` if the role assigned to the user has `Namespaces.Observe` privilege.

4. Copy the required configuration from the `telegraf-config-backup` configmap.

   ```
   kubectl get cm/telegraf-config-backup -n vmware-system-monitoring -o yaml
   ```

5. Update the `telegraf-user-config` configmap with the data copied from the `telegraf-config-backup` configmap.

   ```
   kubectl edit cm/telegraf-user-config -n vmware-system-monitoring
   ```

   You have to copy the `telegraf-config-backup` information in an appropriate section in the `telegraf-user-config` configmap. Following are the different configuration sections that are available in the `telegraf-user-config` configmap that are mounted as separate files on the Telegraf pods.

   - `telegraf.conf` - The base configuration that is mounted on both deamonset and deployment instances. It provides the common configurations to both the instances like agent config, output plug-ins, and others.
   - `telegraf-deployment.conf` - This configuration is mounted on the Telegraf deployment pods. It provides input plug-in configuration for collecting metrics from IaaS controllers running in the Supervisor.
   - `telegraf-daemonset.conf` - This configuration file is mounted on Telegraf daemonset instance pods. It provides input plug-in configuration for collecting metrics from kube-system and node metrics.

```
kubectl get cm/telegraf-user-config -n vmware-system-monitoring -o yaml
apiVersion: v1
data:
  telegraf-daemonset.conf: ""
  telegraf-deployment.conf: ""
  telegraf.conf: ""
```

Once the `telegraf-user-config` configmap is updated, the Telegraf daemonset and deployment pods will sync the new changes and start collecting and pushing metrics.

Updating vSphere Supervisor

Learn how to update the vSphere Supervisor environment, including the version of vSphere Supervisor instances and the underlying infrastructure.

Updating the vSphere Supervisor Environment

You can update to the latest version of vSphere Supervisor, including the vSphere infrastructure supporting Supervisors, VKS clusters, and the VCF Consumption CLI.

> **Important:**
>
> When upgrading vCenter from 8.0 Update 3 and later, 9.0.0, 9.0.1, or 9.0.2 to 9.1.0 or 9.1.1, Supervisor data can potentially be lost if the vLCM database was previously reset. To avoid data loss, you must determine if your environment if affected and take mitigation steps. See Broadcom KB article 455737 for more information.

> **Note:**
>
> Before you update vCenter, make sure that the Kubernetes version of all Supervisors is of the minimum supported version, and that the vSphere Kubernetes release version of the VKS clusters is of the minimum supported version. See the compatibility matrix for more information.

In VCF 9.1 online deployments, Supervisor artifacts required for upgrade are automatically synchronized to the local Software Depot. No manual repository or registry configuration is required before initiating the upgrade.

How vSphere Supervisor Updates Work

Check out how to update the Supervisors and VKS clusters in your vSphere Supervisor environment. vSphere Supervisor supports rolling updates for Supervisors and VKS clusters, and for the infrastructure supporting these clusters.

When you update your vSphere Supervisor environment, you update the Kubernetes version of your Supervisors and VKS clusters along with the underlying infrastructure components and services. Therefore, the term *update* is used to describe this process instead of the term *upgrade*, which is a limited form of update that increments the software version.

Supervisors and VKS clusters are built by using a common Kubernetes distribution core. Kubernetes versions for the Supervisor and VKS clusters are delivered differently:

- Starting from VCF 9, some Supervisor Kubernetes releases, such as patches, can be delivered independently from vCenter. For these releases, you can update the Supervisor version by deploying a Supervisor release version from the Supervisor Content Library. For other Supervisor releases, in which more significant architecture changes are involved, the Supervisor update is still delivered through vCenter, and requires a vCenter upgrade.
- VKS clusters are built by using vSphere Kubernetes releases (VKrs). A VKr provides upstream aligned Kubernetes software distribution, signed, tested, and supported by VMware. VKrs are intended for VKS clusters. VKrs are released independently from vCenter. A VKr is a combination of an operating system such as Photon or Ubuntu combined with VKS core components necessary to provide pod functionality. A VKr includes components such as Antrea and Calico that you can deploy in VKS clusters. For more information, see the *VMware vSphere Kubernetes releases Release Notes*.

You can only update the Kubernetes version of Supervisors and VKS clusters sequentially due to the update policy of upstream Kubernetes. You cannot skip a minor version. For example, if a Supervisor is running 1.28, you cannot update straight to 1.30. The correct update path is, 1.28, 1.29, 1.30.

Support Policy for the Supervisor and VKS Cluster Kubernetes Versions

Both Supervisor Kubernetes versions and VKrs implement an N-2 support policy for Kubernetes releases, similarly to the upstream Kubernetes support policy. This means that each Kubernetes release in vSphere Supervisor (Supervisor or VKr) is supported for a minimum of 12 months from the time it is released.

Note however, that the Supervisor and VKS clusters must run compatible Kubernetes versions, see the Compatibility Matrix for the compatible VKrs releases and Verify VKS Cluster Compatibiity for Updates.

Rolling Updates of Supervisors and VKS Clusters

vSphere Supervisor uses a rolling update model for Supervisors and VKS clusters. The rolling update model ensures that there is minimal downtime for cluster workloads during the update process. Rolling updates include upgrading the Kubernetes software versions and the infrastructure and services supporting the VKS clusters, such as virtual machine configurations and resources, vSphere Namespaces, and custom resources. For more information, see Understanding the Rolling Update Model for VKS Clusters for more details.

> **Important:**
>
> Upgrading to version 9.1 in environments with regional Harbor triggers an automatic rolling update of all VKS clusters. The rolling update occurs so that nodes can adopt new certificates and DNS entries. To avoid an unplanned rollout, before starting the upgrade, upgrade the VKS to 3.6.1 or later, retire legacy (v1alpha3) clusters, and manually upgrade your VKS clusters to ClusterClass 3.6.0 or later.

For the update to succeed, your environment must meet several compatibility requirements. The system enforces precheck conditions to ensure that clusters are ready for updates, and supports rollback if cluster upgrade is not successful.

Updating the Supervisor

You can update the Kubernetes version of the Supervisor, and the infrastructure supporting the Supervisor and VKS clusters. You might need to update the Supervisor Kubernetes versions 3 times per year to maintain pace with the Kubernetes release cadence. The following is the Supervisor Kubernetes version update sequence that is still applicable for VCF 9.

1. Create a Supervisor content library.
2. Update the Supervisor Kubernetes version and infrastructure components. See Update the Supervisor

When you initiate an update of the Supervisor Kubernetes version the update is performed in the following order of operations:

1. The system creates a new control plane VM and joins it to the existing Supervisor control plane. During this phase of the update, the vSphere inventory shows four control plane VMs as the system adds a new updated VM and then removes the older out-of-date VM.
2. Objects are migrated from one of the old control plane VMs to the new one, and the old control plane VM is then removed. This process repeats one-by-one until all control plane VMs are updated.
3. Once all the control plane VMs are updated, the worker nodes are updated in a similar rolling update fashion. The worker nodes are the ESXi hosts, and each spherelet process on each ESXi host is updated one-by-one.

You can choose between the following updates:

- Update the Supervisor Kubernetes version only.
- Update everything, including VMware versions and Kubernetes versions.

Updating VKS Clusters

Updating your VKS clusters depends on the VKS service version that is running on the Supervisor as well as the VKr versions the clusters are running. For more information, see Installing and Updating VKS and Updating VKS Clusters.

Configure a Subscribed Content Library for Supervisor Images

You can use a subscribed content library for Supervisor enablement and upgrade. Starting from VCF 9, the Supervisor releases can be delivered separately from vCenter.

Ensure that the Software Depot is installed and configured in your environment and that you have the Software Depot subscription endpoint URL for your environment.

To find your subscription URL, log in to the vCenter Developer Center, open the API Explorer, and execute a GET request on the lcm/depot/services API. Use the returned `VcenterLcmDepotServicesAddress` and `base_url` values to format your URL as: https://{VcenterLcmDepotServicesAddress}{base_url}/PROD/COMP/SUPERVISOR/lib.json. See Configure a Software Depot Connection Mode.

To enable or upgrade vSphere Supervisor, you can create a subscribed content library and synchronize the Supervisor release images. Starting with VCF 9.1, you can create a content library that subscribes to Supervisor image publications directly from the Software Depot.

The subscribed content library can be in immediate or on-demand synchronization mode. Immediate synchronization on the public content delivery network might take more time and disk space.

Important:

Do not attempt to delete or edit the content library that is assigned with the Supervisor release images. You must unassign the Supervisor release images before you attempt to delete or edit the content library. Use the following DCLI command:

```
> dcli> com vmware vcenter namespacemanagement lifecycle content libraries unassign --library
 <libraryId>
```

1. Log in to the vCenter as a VI administrator.
2. Create the subscribed content library for Supervisor release images.
   a) From the Home menu, select **Content Libraries**.
   b) Click **Create**.
   c) Name the content library, for example, **Supervisor Releases Content Library**, and click **Next**.
   d) On the **Configure content library** page, select **Subscribed Content Library**.
   e) Set the subscription URL to the Software Depot address you constructed in prerequisites.
   f) In the **Download content** section, select the synchronization mode of the content library and click **Next**.

Option	Description
Immediately	The subscription process synchronizes both the library metadata and images. If items are deleted from the published library, their contents remain in the subscribed library storage and you have to manually delete them.
When needed	Use this option to save storage space. The subscription process synchronizes only the library metadata. The system downloads the images when published. If you no longer need the item, you can delete the item contents to free storage space.

   g) When prompted, accept the SSL certificate thumbprint.

      The SSL certificate thumbprint is stored on your system until you delete the subscribed content library from the inventory.

h) On the **Add storage** page, select a datastore as a storage location for the content library contents and click **Next**.
i) Review the details and click **Finish**.

3. Assign the content library to the vSphere Supervisor platform.
   a) From the Home menu, select **Supervisor Management**.
   b) Select **Content Distribution**.
   c) On the **Supervisor Images Library** card, click **Assign**.
   d) Select the Content Library that you created and click **Assign**.

The new content library starts synchronizing which might take some time. Once the synchronization completes, the new Supervisor Kubernetes versions contained in the images appear in the available versions on the **Updates** tab.

Configuring a Subscribed Content Library for Supervisor Images in Air-Gapped Environment

Learn how to create a subscribed content library in an air-gapped environment. You can then use this subscribed content library to enable and upgrade the Supervisor. Starting from VCF 9, the Supervisor Kubernetes releases can be delivered separately from vCenter. To enable or upgrade the Supervisor in an air-gapped environment, you can create a subscribed content library and synchronize vSphere Supervisor release images.

Broadcom publishes vSphere Supervisor images to a content delivery network. In an air-gapped environment, where direct internet access is restricted, you must first download the vSphere Supervisor release artifacts on a separate host that is connected to the internet.

Once the artifacts are downloaded, depending on the specific environment properties, you can apply the following approaches to create a subscribed content library:

- For VCF 9.1.1 and later. Use the VCF Download Tool to download the Supervisor release images and upload them to the Software Depot. Then create a subscribed content library subscribing to the Software Depot directly. For the subscribed library creation steps, see Configure a Subscribed Content Library for Supervisor Images.
  To download the Supervisor release images:
  ```
  ./bin/vcf-download-tool artifacts download \
      --vcf-version=9.1 \
      --component=SUPERVISOR \
      --depot-download-activation-code-file=activation-code.txt
  ```
  To upload the Supervisor release images to the Software Depot:
  ```
  ./bin/vcf-download-tool depot artifacts upload \
      --vcf-version=9.1 \
      --component=SUPERVISOR \
      --depot-store=./depot-store \
      --depot-fqdn=<DEPOT_FQDN> \
      --ops-fqdn=<OPS_FQDN> \
      --ops-user=admin \
      --ops-user-password-file=vcfops.txt
  ```
- For VCF earlier than 9.1.1, or for environments that require legacy Supervisor versions or a customized publisher. Use one of the following approaches to set up a local publisher, then create a subscribed content library that subscribes to it:
  – Remote content library publisher in your local environment.
  – vCenter publisher.

  Note:

  New Supervisor release images might not be published to https://wp-content.broadcom.com/supervisor/v1/latest. All new Supervisor release images are published to the Broadcom online depot and are accessible through the Software Depot.

For all approaches, you can select between the immediate and on-demand synchronization modes. Immediate synchronization can take substantial time and disk space.

Important:

Do not attempt to delete or edit the content library that is assigned with the Supervisor release images. These attempts might cause Supervisor node deployment failures.

You must unassign the Supervisor release images before you attempt to delete or edit the content library. Do not unassign a content library if its Supervisor release images are actively used by running Supervisor.

To unassign, use the following DCLI command:

```
> dcli> com vmware vcenter namespacemanagement lifecycle content libraries unassign --library
 <libraryId>
```

Create a Remote Content Library Publisher for vSphere Supervisor Releases in an Air-Gapped Environment

Learn how to create a subscribed content library for vSphere Supervisor release images by setting up a remote content publisher in your local network. This approach is for pre-9.1.1 deployments, or for 9.1.1+ deployments where you need legacy Supervisor versions or want a customized publisher.

Note:

For VCF 9.1.1 and later, the recommended approach is to create a subscribed content library subscribing to the Software Depot directly. See Configure a Subscribed Content Library for Supervisor Images.

The following steps apply to pre-9.1.1 deployments, or to 9.1.1 and later deployments where you need legacy Supervisor versions or want a customized publisher that includes both legacy and newer versions.

You set up a remote content publisher in your local network by using a staging host that downloads release artifacts and acts as a publisher for internal consumption. You can then create a subscribed content library to update the Supervisor releases in an air-gapped environment where internet access is prohibited.

1. Download the Supervisor release bundle.

   The download method depends on the Supervisor version you need.

   - For legacy Supervisor versions.

     Using a machine with internet access, download the full release bundle including the lib.json file from https://wp-content.broadcom.com/supervisor/v1/latest (version 9.0) or https://wp-content.broadcom.com/ (version 9.1).
   - For newer Supervisor versions (9.1.1 and later).

     Use the VCF Download Tool to download the Supervisor release images and upload them to the Software Depot:

     ```
     ./bin/vcf-download-tool artifacts download \
         --vcf-version=9.1 \
         --component=SUPERVISOR \
         --depot-store=./depot-store \
         --depot-download-activation-code-file=activation-code.txt
     ./bin/vcf-download-tool depot artifacts upload \
         --vcf-version=9.1 \
         --component=SUPERVISOR \
         --depot-store=./depot-store \
         --depot-fqdn=<DEPOT_FQDN> \
         --ops-fqdn=<OPS_FQDN> \
         --ops-user=admin \
         --ops-user-password-file=vcfops.txt
     ```

     Note:

     New Supervisor release images might not be published to https://wp-content.broadcom.com/supervisor/v1/latest. All new Supervisor release images are published to the Broadcom online depot and are accessible through the Software Depot.

The resulting directory structure must look in the following way. Use this as an example as the exact artifact release numbers depend on the vSphere Supervisor release number:

```
spherelet-v1.28
    item.json
    spherelet-depot-9.0.1.28.2.0-24522473.zip
    spherelet-solution-9.0.1.28.2.0-24522473.json
spherelet-v1.29
    item.json
    spherelet-depot-9.0.1.29.3.0-24522471.zip
    spherelet-solution-9.0.1.29.3.0-24522471.json
spherelet-v1.30
    item.json
    spherelet-depot-9.0.1.30.5.0-24526496.zip
    spherelet-solution-9.0.1.30.5.0-24526496.json
supervisor-9.0.0.0100-24845085
    item.json
    photon-ova.cert
    photon-ova.mf
    photon-ova.ovf
    photon-ova.vmdk
items.json
lib.json
```

2. Host the release artifacts on a local Web server.
   a) Copy the contents of the downloaded bundle to a directory served by a local HTTP server.
   b) Ensure the `lib.json` file is accessible through a stable internal HTTP URL, for example, `http://<host-ip>/supervisor/lib.json`.
3. Log in to the vCenter as a VI administrator.
4. Navigate to **Menu** > **Content Libraries**.
5. Click **Create**.
6. Enter a name of the content library and click **Next**.

   For example, you can enter a descriptive name, such as `Supervisor-CL` and include a description, such as `Subscribed library for Supervisor Image`.
7. On the **Configure content library** page, select the **Subscribed content library** option.
8. Enter the URL of the `lib.json` hosted in step 2.
9. For the **Download content** option, select one of the following.

Option	Description
Immediately	The subscription process synchronizes both the library metadata and images. If items are deleted from the published library, their contents remain in the subscribed library storage, and you have to manually delete them.
When needed	The subscription process synchronizes only the library metadata. The system downloads the images when published. When you no longer need the item, you can delete the item contents to free storage space. To save storage this option is recommended.

10. When prompted, accept the SSL certificate thumbprint.

    The SSL certificate thumbprint is stored on your system until you delete the subscribed content library from the inventory.

11. On the **Apply security policy** page, configure the OVF security policy and click **Next**.

    When you select the **Apply security policy** option, the system verifies the OVF signing certificate during the synchronization process. An OVF template that does not pass certificate validation is marked with the Verification Failed tag. The template metadata is kept, but the OVF files cannot be synchronized.

    Note:

    Currently, the OVF default policy is the only supported security policy.

12. On the **Add storage** page, select a datastore as a storage location for the content library contents and click Next.
13. On the **Ready to complete** page, review the details and click **Finish**.
14. Assign Supervisor content library in the **Supervisor Management** Page.
    a) From the **Home** menu, select **Supervisor Management** > **Supervisor Content Distribution**.
    b) On the **Supervisor Images Library** card, click **Assign**.
    c) Select the content library that you created and click **Assign**.

Create a vCenter Publisher for vSphere Supervisor Releases Content Library

Create a vCenter publisher for vSphere Supervisor release artifacts by uploading downloaded artifacts to a local content library on a vCenter system and enabling publishing. This approach is for pre-9.1.1 deployments, or for 9.1.1 and later deployments where you need legacy Supervisor versions or want a customized publisher.

Note:

For VCF 9.1.1 and later, the recommended approach is to create a subscribed content library subscribing to the Software Depot directly. See Configure a Subscribed Content Library for Supervisor Images.

The following steps apply to pre-9.1.1 deployments, or to 9.1.1 and later deployments where you need legacy Supervisor versions or want a customized publisher that includes both legacy and newer versions.

You upload the downloaded vSphere Supervisor release artifacts to a local content library on a vCenter system. The content library is marked as published, exposing a subscription URL for downstream consumption. The same or a separate vCenter system can then host a subscribed content library that pulls content from the published library.

A critical configuration step in this workflow is to update the library item type for Spherelet. Manually uploaded Spherelet artifacts are marked as type `file`, which causes the Supervisor assign operation to fail. To fix this, the item type must be changed to type `other` by using the DCLI tool.

Important:

If the content library item type remains `file`, the content library assign operation will fail.

1. Download the vSphere Supervisor release bundle.

   The download method depends on the Supervisor version you need.

   - For legacy Supervisor versions.

     Using a machine with internet access, download the full release bundle including the lib.json file from https://wp-content.broadcom.com/supervisor/v1/latest.
   - For newer Supervisor versions (9.1.1 and later).

     Use the VCF Download Tool to download the Supervisor release images. For the prerequisites and download procedure, see Download Binaries to Software Depot in Disconnected Mode by Using the VCF Download Tool.

     Note:

     New Supervisor release images might not be published to https://wp-content.broadcom.com/supervisor/v1/latest. All new Supervisor release images are published to the Broadcom online depot and are accessible through the Software Depot.

2. Log in to the vCenter and navigate to **Menu** > **Content Libraries**.
3. Click **Create**.
4. Enter a name for the content library and click **Next**.

   For example, you can enter a descriptive name, such as `Supervisor_Local-CL` and include a description, such as `Local Content library for Supervisor Image.`
5. On the **Configure content library** page, select the **Local content library** option and make sure to check the **Enable Publishing** option.
6. On the **Apply security policy** page, configure the **OVF security policy** and click **Next**.

   When you select the **Apply security policy** option, the system verifies the OVF signing certificate during the synchronization process. An OVF template that does not pass certificate validation is marked with the **Verification Failed** tag. The template metadata is kept, but the OVF files cannot be synchronized.

   Note:

   Currently the OVF default policy is the only supported security policy.
7. On the **Add storage** page, select a datastore as a storage location for the content library contents and click **Next**.
8. On the **Ready to complete** page, review the details and click **Finish**.
9. On the **Content Libraries** page, click the content library that you created (for example, **Supervisor_Local-CL**).
10. Next to the newly created content library (**Supervisor_Local-CL**), click **Actions**, and click on **Import Item**.
11. Upload vSphere Supervisor release artifacts.

    In the downloaded vSphere Supervisor release artifacts, you can see three spherelet folders, such as v1.28, v1.29, v1.30. These versions change in subsequent releases.

    Repeat the steps that follow for each spherelet folder, starting with the lowest number (1.28, for example).

    a) Click on **Local file** and then click on **UPLOAD FILES**.
    b) Upload the following files that are located in the spherelet folder.

    `spherelet-depot-9.x.y.28.2.0-2xxxxxxx.zip`

    `spherelet-solution-9.x.y.28.2.0-2xxxxxxx.json`

    The naming convention for item name must strictly follow the versioning in the folder structure, for example, `spherelet-v1.28.`
    c) Click **Import**.
    d) Repeat the above steps for creating a separate library item for each spherelet version in the release artifacts.
    e) Upload the vSphere Supervisor OVA files.

    Name the library item exactly as it appears in the folder structure, for example, `supervisor-9.x.y.0100-24737045`.

    Upload the following files:

    `.mf`, `.ovf`, `.cert`, and `.vmdk`

    Wait for the upload to complete by monitoring **Recent Tasks**.
12. On the page for the content library (**Supervisor_Local-CL** for example), select the **Publication** card and then copy the **Subscription URL** and keep it handy.

At this stage your local content library where publishing is enabled is created. Next you need to update the library item type for Spherelet**.**

Update the Library Item Type for Spherelet

Once you have created a local content library with enabled publishing for the vSphere Supervisor release artifacts, you need to update the library item type for Spherelet through DCLI.

1. SSH to vCenter and launch DCLI.

```
ssh root@<vc-ip>
dcli +i
```

2. Use the following command to update the spherelet's type.
   a) List the available content libraries (the output will be similar to the below one).

```
dcli> com vmware content library list
- 4f253700-88ef-4109-808a-92cd4c0682de
```

   b) Get the library details.

```
dcli> com vmware content library get --library-id 4f253700-88ef-4109-808a-92cd4c0682de
creation_time: 2025-06-11T08:36:13.445Z
storage_backings:
   - datastore_id: datastore-51
     type: DATASTORE

last_modified_time: 2025-06-16T10:02:30.150Z
optimization_info:
   optimize_remote_publishing: False
server_guid: 587d6a6e-6f3b-44e5-8ab5-cd443251c5be
last_sync_time:
description: moving datastore to another
security_policy_id:
type: LOCAL
version: 3
subscription_info:
state_info:
   state: ACTIVE
name: async-cl-hosted-locally
unset_security_policy_id:
id: 4f253700-88ef-4109-808a-92cd4c0682de
```

   c) List the items inside the library.

```
dcli> com vmware content library item list --library-id 4f253700-88ef-4109-808a-92cd4c0682de
- aae3c54d-a678-4470-ad34-a0bb98e4e62e
- d67523ae-9ac0-44d4-b1d8-b4b95ce7bd50
- b84b5871-b676-4bd4-a6b2-702e1fe8fff0
- d4ce9cbb-7f4b-47a5-af80-41c48e31f83b
```

   d) Check the type of a specific item (e.g., spherelet-v1.28).

   For each library item, first check if it is a spherelet library item. Here in the name of the library item, you can see it starts with a `spherelet`, hence, you will update the library item type from `file` to `other`.

```
dcli> com vmware content library item get --library-item-id b84b5871-b676-4bd4-a6b2-702e1fe8fff0
creation_time: 2025-06-11T08:38:38.280Z
last_modified_time: 2025-06-11T08:42:07.307Z
last_sync_time:
certificate_verification_info:
description:
type: file
version: 1
content_version: 2
library_id: 4f253700-88ef-4109-808a-92cd4c0682de
size: 93973412
cached: True
name: spherelet-v1.28
security_compliance: True
id: b84b5871-b676-4bd4-a6b2-702e1fe8fff0
```

```
source_id:
metadata_version: 1
```

e) Update the item type to **other**.

```
com vmware content library item update --type other --library-item-id b84b5871-b676-4bd4-
a6b2-702e1fe8fff0
```

f) Repeat for each Spherelet library item.

By completing these steps, now the local content library that is enabled for publishing is ready for subscription.

Create a Subscribed Supervisor Content Library for Supervisor Images by Using a Local Content Library
You can now create a subscribed content library for the local content library containing the Supervisor release artifacts that you created in the previous tasks.

1. To create the subscribed content library on a different vCenter system, log in to that vCenter system through the vCenter.

   Otherwise, remain logged in to the same vCenter system.

2. Navigate to **Menu** > **Content Libraries** and click **Create**.
3. Enter a name for the content library and click **Next**.

   For example, you can enter a descriptive name, such as `Supervisor_Subscribed-CL` and include a description, such as `Subscribed library for Supervisor Image`.

4. On the **Configure content** library page, select the **Subscribed content library** option.
5. Enter the **Subscription URL** of the `lib.json` that you obtained in Create a vCenter Publisher for vSphere Supervisor Releases Content Library in step 12.
6. For the **Download content** option, select one of the following.

Option	Description
Immediately	The subscription process synchronizes both the library metadata and images. If items are deleted from the published library, their contents remain in the subscribed library storage, and you have to manually delete them.
When needed	The subscription process synchronizes only the library metadata. The system downloads the images when published. When you no longer need the item, you can delete the item contents to free storage space. To save storage this option is recommended.

7. When prompted, accept the SSL certificate thumbprint.

   The SSL certificate thumbprint is stored on your system until you delete the subscribed content library from the inventory.

8. On the **Apply security policy** page, configure the **OVF security policy** and click **Next.**

   When you select the **Apply security policy** option, the system verifies the OVF signing certificate during the synchronization process. An OVF template that does not pass certificate validation is marked with the **Verification Failed** tag. The template metadata is kept, but the OVF files cannot be synchronized.

   Note:

   Currently, the OVF default policy is the only supported security policy.

9. On the **Add storage** page, select a datastore as a storage location for the content library contents and click **Next**.
10. On the **Ready to complete** page, review the details and click **Finish**.
11. Assign the content library in the Supervisor Management Page.
    a) From the **Home** menu, select **Supervisor Management** > **Supervisor Content Distribution**.
    b) On the **Supervisor Images Library** card, click **Assign**.
    c) Select the content library that you created and click **Assign**.

Update the Supervisor

Learn how to update your Supervisors, including the Kubernetes version that the Supervisors are running. The Supervisors in your environment must always run a supported Kubernetes version.

- Check the supported Kubernetes versions for the Supervisor in the vSphere Supervisor Release Notes.
- Create a content library for the Supervisor releases.
- Create a content library containing the most recent Supervisor build.
- For additional details and critical information regarding VKS rolling updates in version 9.1, see How vSphere Supervisor Updates Work.

  Note:

  Updating a Supervisor can trigger a rolling update of the VKS clusters deployed there. See Understanding the Rolling Udate Model for VKS Clusters.

There is a version entity for the Supervisor. The version entity is a semantic version string in the form `v1.30.10+vmware.1-fips-vsc9.0.0.0100`, where the prefix is the Kubernetes version (`v1.30.10`) and the suffix is the Supervisor Kubernetes release version release number (`vsc9.0.0.0100`).

1. Log in to the vCenter as a VI administrator.
2. Select **Menu** > **Supervisor Management**.
3. Select the **Namespaces** > **Updates** tab.
4. Select the **Available Version** that you want to update to.

   For example, select the version `v1.30.10+vmware.1-fips-vsc9.0.0.0100`.

   Note:
   You must update incrementally. You cannot skip updates, such as from 1.28 to 1.30. The path should be 1.28, 1.29, 1.30.
5. Select a Supervisor to update.
6. Click **Apply Updates**.

   The system runs a series of pre-checks to verify the compatibility of the different components against the Supervisor Kubernetes version to which you want to update. When the pre-checks are completed successfully, you can update the Supervisor.

What are Supervisor Update Pre-checks and How to Troubleshoot Them

Learn which are the pre-checks that are run before updating the Supervisor and how to troubleshoot in case of errors resulting from failed pre-checks.

The Supervisor and the workloads that it runs interact with multiple vSphere resources (storage, memory, CPU, and network) running on different layers of vSphere. As a VI administrator, when you initiate a Supervisor update, the system runs pre-checks to verify that workload compatibility and resource requirements are met. These pre-checks include health checks, resource checks, and dependency checks on different software components to ensure that a successful Supervisor update is possible.

Supervisor Update Pre-checks and Troubleshooting Errors

Pre-check	Description	Error	Troubleshooting
Datastore requirement verified	Checks if the datastore has enough resource to create the Supervisor control plane VMs.	Not enough free space on chosen datastore to deploy one more Supervisor CPVM. Need at least *<value>*.	Reclaim the unused space from the configured datastores for Supervisor control plane VMs or add additional datastores to the control plane storage policy.
Supervisor Control Plane VMs health verified	Checks if the health of the target Supervisor control plane VM is good enough.	Cluster *<Supervisor>* is unhealthy: *<error message>*	Depending on the reported error, follow the resolutions at Resolving Errors Health Statuses on a Supervisor Control Plane VMs During Activation Or Update
Etcd cluster health verified	Check if the etcd cluster of the current Supervisor has a good health status	Etcd members are not healthy during cluster upgrade pre-check.	Follow the steps in Tanzu Kubernetes Cluster Upgrade Stuck - EtcdMemberHealthy condition is unknown
Network compatibility verified	Checks if the NSX version is compatible with NSX Container Plug-in (NCP) version.	NSX version *<version>* is not compatible with NCP version *<version>*. Please choose NSX version from compatible list *<list>* but not in incompatible list *<list>*.	Make sure that the NSX and NCP versions are compatible.
Workload cluster(s) compatibility verified	Checks if running VKS clusters are going to be incompatible after upgrade	Proposed upgrade to Namespaces cluster < Supervisor> is incompatible with one or more Tanzu Kubernetes cluster(s) **Note:** The error message for VKS clusters differ in format based on the source Supervisor from which the upgrade is performed. BD	Update the VKr version of the incompatible VKS clusters to a version that is compatible with the new Kubernetes version of the Supervisor. See Verify VKS Cluster Compatibility for Updates. This pre-check only occurs only during the first Supervisor update in vSphere 8 Update 3.
Supervisor Service compatibility verified	Checks if the versions of the installed Supervisor Services are compatible with the target Supervisor Kubernetes version.	The incoming Supervisor requires the Supervisor Service*<service name>* version *<number>* but the installed Service is on version *<number>*. Upgrading the Supervisor will make the installed service incompatible. The Supervisor Service*<service name>* version *<number>* requires Supervisor release version *<number>* but the Supervisor will be on version *<number>*. Upgrading the Supervisor will make the installed service incompatible.	Update the versions of the incompatible Supervisor Services.

Resolving Errors Health Statuses on a Supervisor Control Plane VMs During Activation Or Update

After you activate a Supervisor, update the Supervisor Kubernetes version, or edit the settings of an existing Supervisor, all the settings that you have specified are validated and applied to the Supervisor until the configuration completes. Health checks are performed on the entered parameters that might detect errors in the configuration resulting in an error health status of the Supervisor. You must resolve these error health statuses so that the configuration or update of the Supervisor is possible.

Table 1796: vCenter Connection Errors

Error Message	Cause	Solution
Unable to resolve the vCenter Primary Network Identifier <*FQDN*> with the configured management DNS server(s) on control plane VM <*VM name*>. Validate that the management DNS servers <server name> can resolve <*network name*>.	• At least one management DNS server is reachable. • At least one management DNS is statically supplied. • The management DNS servers do not have any host name lookup for the vCenter PNID. • The vCenter PNID is a domain name, not a static IP address.	• Add a host entry for the vCenter PNID to the management DNS servers. • Verify the configured DNS servers are correct.
Unable to resolve the vCenter Primary Network Identifier <*network name*> with the DNS server(s) acquired via DHCP on the management network of the control plane VM <*VM name*>. Validate that the management DNS servers can resolve <*network name*>.	• The management DNS servers supplied by the DHCP server (at least one) are reachable. • The management DNS servers are statically supplied. • The management DNS servers do not have any host name lookup for the vCenter PNID. • The management DNS servers do not have any host name lookup for the vCenter PNID. • The vCenter PNID is a domain name, not a static IP address.	• Add a host entry for the vCenter PNID to the management DNS servers supplied by the configured DHCP server. • Verify the DNS servers supplied by the DHCP server are correct.
Unable to resolve the host <*host name*> on control plane VM <*VM name*> , as there are no configured management DNS servers.	• The vCenter PNID is a domain name, not a static IP address. • There are no DNS servers configured.	Configure a management DNS server.
Unable to resolve the host <*host name*> on control plane VM <*VM name*>. The hostname ends with the '.local' top level domain, which requires 'local' to be included in the management DNS search domains.	The vCenter PNID contains `.local` as a top-level domain (TLD), but the configured search domains do not include`local.`	Add `local` to the management DNS search domains.

Error Message	Cause	Solution
Unable to connect to the management DNS servers <*server name*> from control plane VM <*VM name*>. The connection was attempted over the workload network.	• The management DNS servers are unable to be connected to vCenter. • The provided `worker_dns` values wholly contain the provided management DNS values. This means that traffic is routed via the workload network, as the Supervisor must pick one network interface to direct static traffic to these IPs.	• Check the Workload Network to verify that it can route to the configured management DNS servers. • Verify there are no conflicting IP addresses that might trigger alternate routing between the DNS servers and some other servers on the Workload Network. • Verify the configured DNS server is, in fact, a DNS server, and is hosting its DNS port on port 53. • Verify the workload DNS servers are configured to allow connections from the IPs of the control plane VMs (the Workload Network provided IPs). • Verify that there are no typos in the management DNS servers' addresses. • Verify search domains don't include an unnecessary '~' that could be resolving the host name incorrectly.
Unable to connect to the management DNS servers <*server name*> from the control plane VM <*VM name*>.	Unable to connect to the DNS servers.	• Check the management network to verify that routes to the management DNS servers exist. • Verify there are no conflicting IP addresses that may trigger alternate routing between the DNS servers and other servers. • Verify the configured DNS server is, in fact, a DNS server, and is hosting its DNS port on port 53. • Verify the management DNS servers are configured to allow connections from the IPs of the control plane VMs. • Verify that there are no typos in the management DNS servers' addresses. • Verify that search domains do not include an unnecessary '~' that could be resolving the host name incorrectly.
Unable to connect to <*component name*> <*component address*> from control plane VM <vm name>. Error: *error message text*	• A generic network failure occurred. • Error occurred while connecting to actual connecting to vCenter.	• Validate that the host name or IP address of the configured components, such as vCenter Server, HAProxy, NSX Manager, or Avi Load Balancer Avi Load Balancer are correct. • Validate any external network settings such as conflicting IPs, firewall rules, and others, on the management network.
The control plane VM <*VM name*> was unable to validate the vCenter <*vCenter Server name*> certificate. The vCenter server certificate is invalid.	The certificate provided by vCenter is in invalid format, and therefore is untrusted.	• Restart `wcpsvc` to verify that the Trusted Roots bundle in the control plane VMs are up-to-date with the latest vCenter root certificates. • Verify that the vCentercertificate is actually a valid certificate.

Error Message	Cause	Solution
The control plane VM <*VM name*> does not trust the vCenter <*vCenter Server name*>certificate.	• The `vmca.pem` certificate presented by vCenter is different from what is configured to the control plane VMs. • The trusted root certificates were replaced in the vCenter appliance, but `wcpsvc` wasn't restarted.	• Restart `wcpsvc` to verify that the Trusted Roots bundle in the control plane VMs are up-to-date with the latest vCenter certificate roots.

Table 1797: NSX Manager Connection Errors

The control plane VM <*VM name*> was unable to validate the NSX Server<*NSX server name*> certificate. The thumbprint returned by the server <*NSX-T address*> doesn't match the expected client certificate thumbprint registered in vCenter <*vCenter Server name*>	The SSL thumbprints registered to the Sup ervisor don't match the SHA-1 hash of the certificate presented by the NSX manager.	• Re-enable trust on the NSX manager between NSX and the vCenter instance. • Restart `wcpsvc` on vCenter.
Unable to connect to <*component name*> <*component address*> from control plane VM <*vm name*>. Error: *error message text*	A generic network failure occurred.	• Validate any external network settings,conflicting IPs, firewall rules, and others, on the management network for the NSX manager. • Verify the NSX manager IP in the NSX extension is correct. • Verify that the NSX manager is running.

Table 1798: Load Balancer Errors

The control plane VM <*vm name*> does not trust the load balancer's (<*load balancer*>- <*load balancer endpoint*>) certificate.	The certificate the load balancer presents is different from the certificate that is configured to the control plane VMs.	Verify that you have configured the correct Management TLS certificate to the load balancer.
The control plane VM <*vm name*> was unable to validate the load balancer's (<*load balancer*>- <*load balancer endpoint*> certificate. The certificate is invalid.	The certificate the load balancer presents is in an invalid format, or expired.	Correct the server certificate of the configured load balancer.
The control plane VM <*vm name*> was unable to authenticate to the load balancer (<*load balancer*>- <*load balancer endpoint*> with the username <*user name*> and the supplied password.	The user name or password of the load balancer are incorrect.	Verify the if the user name and password configured to the load balancer are correct.
An HTTP error occurred when attempting to connect to the load balancer (<*load balancer*>- <*load balancer endpoint*> from the control plane VM <*vm name*>.	The control plane VMs can connect to the load balancer endpoint, but the endpoint does not return a successful (200) http response	Verify that the load balancer is healthy and accepting requests.
Unable to connect to <*load balancer*> (<*load balancer endpoint*>) from control plane VM <*vm name*>. Error: <*error text*>	• A generic network failure occurred. • Typically, it means the load balancer is not working, or some firewall blocks the connection.	• Validate that the load balancer endpoint is accessible • Validate no firewalls are blocking the connection to the load balancer.

Update the VCF CLI

Once you have performed a vSphere Namespace update and upgraded the Supervisor, update the VCF CLI, if it is recommended.

If a newer version of the VCF CLI is available, you will be prompted with the message in the terminal when you use the VCF CLI.

To download and install the the VCF CLI, see Download and Install the VCF CLI.

Working with vSphere Lifecycle Manager

As a VI administrator, you can enable vSphere Supervisor on vSphere clusters that you manage with a single VMware vSphere Lifecycle Manager image. You can then use the Supervisor while it is managed by vSphere Lifecycle Manager.

vSphere Lifecycle Manager allows you to manage the ESXi hosts and clusters in your environment. You can upgrade a Supervisor to the latest version of vSphere Supervisor. You can also upgrade the ESXi version of the hosts in the Supervisor.

vSphere Lifecycle Manager is a service that runs in vCenter. When you deploy vCenter, the vSphere Lifecycle Manager user interface is enabled in the HML5-based vSphere Client.

For more information about the vSphere Lifecycle Manager, see the *Managing Host and Cluster Lifecycle* documentation.

Requirements

To configure vSphere Supervisor on a vSphere cluster that is managed by vSphere Lifecycle Manager, your environment must meet specific requirements.

System Requirements

To enable vSphere Supervisor, verify that the components on the vSphere cluster meet the following requirements:

- Verify that vCenter and ESX are of version 7.0 Update 2 and later if you use NSX.
- Verify that vCenter and ESX are of at least version 7.0 Update 1 and later if you use vSphere Networking.
- Verify that HA and DRS are enabled on the vSphere cluster.
- Verify that vSphere Distributed Switch version 7.0 Update 2 and later is configured.
- Verify that vSphere Networking or NSX 3.1 or a later version is configured on the cluster. You cannot use vSphere Lifecycle Manager image to manage a cluster that is configured with earlier versions of NSX.

Enable vSphere Supervisor on a Cluster Managed by vSphere Lifecycle Manager

To run Kubernetes workloads, you can enable vSphere Supervisor on a cluster that you manage with a single vSphere Lifecycle Manager image. After it is enabled, you can manage the Supervisor using vSphere Lifecycle Manager.

When you enable the cluster with vSphere Supervisor that uses NSX, the vSphere Lifecycle Manager installs the Spherelet vSphere Installation Bundle (VIB) on every ESX host in the cluster. Enabling the cluster assigns the version of the Kubernetes version that ships with vCenter. After the installation is complete, the WCP service performs the post installation tasks such as, starting and configuring the Spherelet.

For the steps to enable the cluster, see Deploying Supervisor with VCF Networking with VPC.

Upgrade a Supervisor

You can update to the latest version of vSphere Supervisor, including the vSphere infrastructure supporting vSphere Supervisor clusters, the Kubernetes versions, and the VCF CLI on clusters that use a single vSphere Lifecycle Manager image.

You upgrade the ESX version of the hosts in the Supervisor. During the upgrade the Spherelet VIB on every ESX host is upgraded.

vSphere Lifecycle Manager uses DRS and puts the hosts in maintenance mode before remediation. DRS first attempts to migrate the virtual machine running vCenter Server to another host, such as VMs that are affinitized to the host or running on the local storage of the host, and workloads, including vSphere Pods to other hosts, so that the remediation succeeds.

Note:
You can use vSphere Lifecycle Manager to upgrade a Supervisor only on clusters that use a single vSphere Lifecycle Manager image.

1. From the vSphere Client menu, select **Workload Management**.
2. Select the **Updates** tab.
3. Select the **Available Version** that you want to update to.
   For example, select the version `v1.17.4-vsc0.0.2-16293900`.
4. Select the Supervisor to apply the update to.
5. To initiate the update, click **Apply Updates**.
6. Use the **Recent Tasks** pane to monitor the status of the update.

Add Hosts to a Supervisor

As a VI administrator, you might need to scale out the Supervisor to run more workloads. To add capacity to a cluster, you can add ESXi hosts to the cluster thats uses a single vSphere Lifecycle Manager image.

- Obtain the user name and password of the root user account for the host.
- Verify that hosts behind a firewall might communicate with the vCenter.

When you add a host to the Supervisor that is configured with NSX, vSphere Lifecycle Manager installs the Spherelet VIB and the image on the host. After it is installed, vSphere Supervisor configures the Spherelet process on the newly added host, which allows running containers natively on ESXi.

1. From the vSphere Client menu, select **Workload Management**.
2. Right-click the data center, cluster, or folder and select **Add Host**.
3. Enter the IP address or the name of the host and click **Next**.
4. Enter the administrator credentials and click **Next**.
5. Review the host summary and click **Next**.
6. Assign a license to the host and click **Finish**.
7. In the **Add Host** wizard, click **Next**.
8. Review the summary and click **Finish**.

Note:
If a host is part of the same data center, you can move it into the Supervisor. To move the host, put it in maintenance mode and drag it into the cluster.

Remove Hosts from a Supervisor

As a VI administrator, you might need to scale in the Supervisor to save cost. To reduce the capacity of a Supervisor, you can remove ESXi hosts from a cluster that uses a single vSphere Lifecycle Manager image.

Before you can remove a host from a cluster, you must power off all virtual machines that are running on the host or migrate the virtual machines to a new host.

When you remove a host from the Supervisor that is configured with NSX, vSphere Supervisor clears the configuration of the Spherelet and stops the Spherelet process on the ESXi host. vSphere Lifecycle Manager then uninstalls the Spherelet VIB and image from the host and vSphere Supervisor removes the host metadata from the cluster control plane.

1. In the vSphere Client, navigate to the cluster from which you want to remove the host.
2. Right-click the host and select **Enter Maintenance Mode** from the pop-up menu.
3. In the confirmation dialog box that appears, click **Yes**.

   The confirmation dialog box also asks if you want to move virtual machines that are powered off to other hosts. Select this option if you want those virtual machines to remain registered to a host within the cluster.

   The host icon changes and the term “maintenance mode” is added to the name in parentheses.
4. Select the host icon from the inventory, and drag it to the new location.

   The host can be moved to another cluster or another data center.

   vCenter Server moves the host to the new location.
5. Right-click the host, and select **Exit Maintenance Mode** from the pop-up menu.
6. Optional: Restart any virtual machines, if needed.

Disable a Supervisor

You can disable vSphere Supervisor from a vSphere cluster that uses a single vSphere Lifecycle Manager image, to make it available for traditional workloads.

When you disable a vSphere Supervisor on a cluster, vSphere Lifecycle Manager uninstalls the Spherelet VIB and image from each ESXi host and the `WCP` service stops and deletes all workloads from the cluster.

1. From the vSphere Client menu, select **Workload Management**.
2. Select the **Clusters** tab.
3. Select the cluster on which you want to disable vSphere Supervisor.
4. Click **Disable**.

   The **Disable Cluster** dialog box appears with a message that all Kubernetes workloads and NSX configuration will be disabled on the cluster.
5. Click **Disable**.

Backing Up and Restoring vSphere Supervisor

Considerations for Backing Up and Restoring vSphere Supervisor

Learn what is the backup and restore process for the Supervisor environment, and get familiar with high-level considerations for implementing your backup and restore strategy for Supervisors and workloads.

Scenario	Tools	Comments
Backup and restore the Supervisor control plane	vCenter file-based backup and restore from the Workload Management UI	Configure backing up the state of the Supervisors in vCenter as part of the scheduled file-based backups in vCenter. Later, you can restore the state of the Supervisors in vCenter through the Workload Management interface in the vSphere Client. **Note:** Restoring the state of the Supervisors in vCenter and restoring the state of vCenter are two different workflows. Restoring vCenter does not lead to restoring the Supervisors is hosts. See Backup and Restore the Supervisor Control Plane.
Backup and restore vSphere Pods	Velero Plugin for vSphere	Install and configure the plug-in on the Supervisor. See Installing and Configuring the Velero Plugin for vSphere on a Supervisor. See Backup and Restore vSphere Pods Using the Velero Plugin for vSphere.
Backup and restore VKS cluster and workload.	vCenter Server file-based backup and restore for VKS cluster. Velero for VKS cluster workloads.	Backup and restore VKS cluster with vCenter Server file-based backup and restore. Backup and restore VKS cluster workloads with Velero. See Backing Up and Restoring VKS Clusters and Workloads.
vCenter Configuration	vCenter	If vCenter is lost, use vCenter to back up and restore vCenter objects. See *File-Based Backup and Restore of vCenter Server* in *vCenter Deployment and Setup*.
NSX	NSX Manager	Load balancer and ingress services depend on NSX backup. NSX provides in-product backup and recovery that supports backup and restore of the NSX Manager Nodes and objects. For more information, see Backing Up and Restoring NSX Manager in the NSX documentation.

When to Back Up and Restore the vSphere Supervisor Control Plane

This topic outlines when it is safe to back up and restore the Supervisor. It also highlights scenarios where backups or restores should not be performed.

CAUTION:

You must exercise caution when performing a Supervisor restore. Use the restore only for specific recovery scenarios. You should not use it as a general solution for addressing issues that might occur during lifecycle

operations, such as a Supervisor upgrade. Restoring might result in existing workloads to become orphaned or the Supervisor control plane not fully return to a healthy running state.

When to Take a Supervisor Backup

This section lists scenarios in which it is recommended to take a backup of the Supervisor. It also identifies cases when you should avoid taking a backup. Backing up the Supervisor in unsupported states might lead to inconsistent or unusable backups and unpredictable behavior during restore.

vSphere Supervisor State	Backup Recommended	Details
Supervisor is enabled and status is `Running`	Yes	The Supervisor is in a stable state and backup is safe.
Supervisor is upgraded and status is `Running`	Yes	The upgrade is complete and the Supervisor is stable.
Supervisor is in an error state	No	The error state might be due to network issues, pod failures, or etcd issues. Backing up in this condition can result in an unusable restore.
Supervisor is being upgraded	No	A backup taken during an upgrade might be inconsistent and can cause problems when restoring.
Supervisor upgrade failed	No	A failed upgrade leaves the cluster in an unstable state. Backups taken in this state can lead to inconsistent backup data and cause restore issues.
Supervisor Service is being upgraded	No	Do not take a backup when a Supervisor Service, such as VKS, is being upgraded. Backups taken in this state might record inconsistent data. For example, etcd and Supervisor Service metadata might be out of sync.

When Not to Perform a Supervisor Restore

Restoring the Supervisor is not supported in certain scenarios. Avoid restore operations in the following cases as they might cause orphaned resources, version mismatches, an unstable cluster state, or other unpredictable behavior.

Restore Scenarios to Avoid	Details
Supervisor Kubernetes Version Mismatch	Do not restore if the Supervisor Kubernetes version in the backup differs from the version in your current environment. Example: You backed up Supervisor running Kubernetes 1.28.3 in a vCenter 8.x environment. Later, you upgraded to vCenter 9.0, where the default Supervisor version is 1.28.14. If you attempt a restore, the version mismatch blocks the operation.
Restoring vCenter After Upgrading and Restoring the Supervisor	If you restore vCenter from a backup taken before a Supervisor upgrade or restore, Supervisor might become out of sync with vCenter. You can encounter several problems. For example, existing control plane VMs become orphaned and are placed to the Discovered Virtual Machines folder. Or you might see duplicate control plane VMs that cannot be powered on. If those problems occur, and you want to recover the Supervisor, perform these steps: 1. Delete discovered and orphaned control plane VMs before you start the Supervisor restore. 2. Restore the Supervisor using a backup taken when vCenter backup was taken.

Restore Scenarios to Avoid	Details
Restoring Supervisor After New VKS Cluster Creation or VKS Upgrade	Avoid restoring the Supervisor if either of these applies: • You created new VKS clusters after the backup. Any VMs from those clusters might become orphaned during the restore, even if the VKS service was not upgraded. See Cleaning Up Orphaned Objects After Supervisor Restore from a Backup. • You upgraded the VKS after the backup. Restoring at this point causes a service downgrade. The downgrade can make existing clusters incompatible and unmanageable by VKS.
Restoring Supervisor After New vSphere Pods Were Created	Do not restore the Supervisor if you created new vSphere Pods after the backup was taken. If new vSphere Pods were created after the backup was taken, during restore, the system might attempt to downgrade internal components on the ESX hosts to match the backup version. The downgrade might fail or get blocked if ESX hosts fail to enter maintenance mode, with an error such as `Waiting for all VMs to be powered off or suspended or migrated.` If this situation occurs and you want to recover the Supervisor, cleanup orphaned VMs associated with vSphere Pod resources and wait for a couple of minutes for the restore operation to succeed. See Cleaning Up Orphaned Objects After Supervisor Restore from a Backup.

Backup and Restore the Supervisor Control Plane

You can include the option for recording the state of the Supervisors in vCenter as part of the vCenter file-based backups. Later on you can restore the Supervisor control plane from the created backup files.

Back up the Supervisor State

Learn how to back up the state of the Supervisors in your environment. You can include the backup of the Supervisors available in vCenter as part of the vCenter file-based backups.

- You must have an FTP, FTPS, HTTP, HTTPS, SFTP, NFS, or SMB server up and running with sufficient disk space to store the backup.
- The lowest supported version for the Supervisor control plane backup is vSphere 8 Update 3. If the Supervisor control plane is running a lower version, the backup is not included in the vCenter file-based backups.

The backup files for the Supervisor control plane capture the state of the following components:

- The etcd state.
- Container images that are used for infrastructure pods to ensure that control plane VMs can be restored post vCenter upgrade.
- The Kubernetes CA certificate and key to ensure that all Kubernetes certificates can be regenerated after a restore from the same CA. This ensures that vSphere Pods and the Spherelet do not have to be reconfigured after restore to trust a new Kubernetes CA.
- All vSphere Namespaces and the state of all the Kubernetes resources that are associated with workloads, such as deployments, pods, VMs, VKS resources, persistent volume claims, and others.

For more information on the vCenter file-based backup and restore, see *File-Based Backup and Restore of vCenter Server* in *vCenter Server Installation and Setup*.

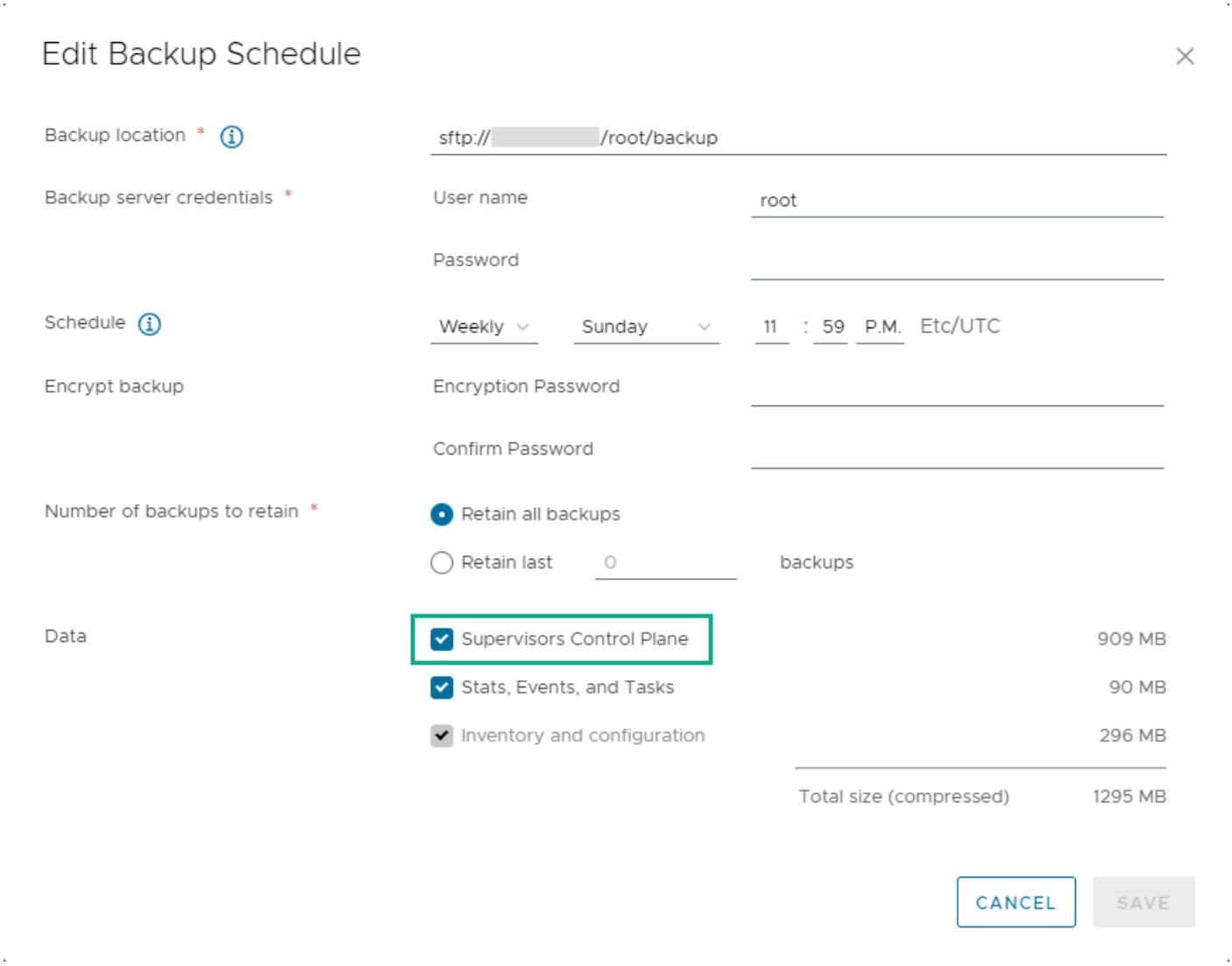
Certain Supervisor settings are stored in vCenter and hence are captured in the vCenter backup:

- Resource pools for vSphere Namespaces.
- General settings such as control plane VM settings, VKS service content libraries, and others.
- Network settings such as network configuration updates of CIDRs, DNS settings, and NTP settings.
- Storage settings such as storage policies.
- Certificates.
- Container registries.
- Identity providers.
- Default CNI.

- Permissions
- vSphere Zones.
- PNID changes.

1. In a web browser, go to the vCenter Management Interface, `https://appliance-IP-address-or-FQDN:5480`.
2. Log in as root.
3. In the vCenter Management Interface, click **Backup**.
4. Click **Edit** if a backup schedule already exists.

   If a backup schedule does not exist, see *Backing Up and Restoring NSX Manager* in the *NSX-T* documentation for information on how to create one.
5. In the Edit Backup Schedule pane, select **Supervisors Control Plane**.



The state of all Supervisors in vCenter is backed up as part of the vCenter backups.

Restore the Supervisor Control Plane

You can restore the Supervisors control pane in vCenter from the backup files of the vCenter system itself.

- Configure recording the Supervisor state from the vCenter management interface file-based backups.

  Note:

  Restoring the Supervisors control plane in vCenter and restoring the state of vCenter are two different workflows. Restoring vCenter does not lead to restoring the Supervisor control plane.

Supervisor settings that are managed by vCenter are captured in the vCenter backup. After the Supervisor backup is created, restoring only the Supervisor state will not revert the settings that are managed by vCenter . To ensure that these configurations are restored to their state at the time of the backup, perform a vCenter restore before restoring the Supervisor state. If you only restore the Supervisor state, the configuration on the Supervisor is overwritten to match the desired configuration on vCenter . When you restore both vCenter and Supervisor, you avoid discrepancies in the Supervisor configuration.

- Resource pools for vSphere Namespaces.
- General settings such as control plane VM settings, VKS service content libraries, and others.
- Network settings such as network configuration updates of CIDRs, DNS settings, and NTP settings.
- Storage settings such as storage policies.
- Certificates.
- Container registries.
- Identity providers.
- Default CNI.
- Permissions
- vSphere Zones.
- PNID changes.

If a vSphere Namespace is deleted by using the vSphere Client, the vSphere Namespace is not created after restoring the Supervisor. To restore the vSphere Namespace, you must restore the vCenter state.

To restore workloads running in vSphere Namespaces, use a backup tool such as Velero or Velero Plug-in for vSphere to restore the workloads. See Considerations for Backing Up and Restoring vSphere Supervisor.

1. In the vSphere Client, go to **Supervisor Management**.
2. Select **Supervisors** and select **Restore**.
3. Enter the backup details.

Option	Description
vCenter	Select the vCenter system that manages the Supervisor.
Backup selection	• Select **Browse backup server folder** to upload the files that are stored in the root folder of the backup file server that is configured with this vCenter system. • Select **Single backup location** to upload a particular backup file, then enter the URL to that backup file. • Select **Use backup location and user name from vCenter backup schedule** to populate the root folder location and user name for the backup location that are configured with vCenter.
Location	Enter the location of the backup root folder.
Username	Enter the user name to access the backups.
password	Enter the password for that user name.

4. Click **Next**.
5. Select a backup file to restore from and click **Next** to start the download of the backup file.
6. Select the Supervisor to restore and click **Next**.
7. Review the settings and click **Finish**.

The Supervisor goes back to configuring state and all the control plane VMs are deleted and re-deployed with the data from the backup file. You can monitor the process by clicking **View** in the **Config Status** column.

Installing and Configuring the Velero Plugin for vSphere on a Supervisor

The Velero Plugin for vSphere provides a solution for backing up and restoring workloads. Once you have the Velero Plugin for vSphere installed and configured on a Supervisor, you can back up and restore VKS cluster workloads and vSphere Pods. For persistent workloads, the Velero Plugin for vSphere lets you take snapshots of the persistent volumes.

Install and Configure the Velero Plugin Version 1.8.x for vSphere

Learn how to install and configure the Velero Plugin for vSphere to back up and restore workloads running on vSphere Pods and VKS clusters.

Before You Begin

Be familiar with the following information and perform the required steps:

- These installation instructions apply only to Velero Plugin for vSphere version 1.8.0-embedded+24696422 or any version with the 1.8 prefix. For information about installing version 1.7.x and earlier, see Install and Configure the Velero Plugin Version 1.7.x in the *vSphere Supervisor 8.0* documentaiton.
- Before installing the new Velero versions, uninstall the current Velero installation. Failing to do so might have unintended consequences. Use the uninstall command that is described in Uninstall the Velero Plugin for vSphere.
- If you use an air-gapped environment, before following these installation instructions, perform the required steps in Install the Velero Plugin for vSphere in an Air-Gapped Environment.

Supported Versions

Velero Plugin for vSphere	Supervisor 8.x and Earlier	vSphere Supervisor 9.0 and Later
1.8.0-embedded+24696422 or any version with the 1.8 prefix	Yes With Supervisor version 8.0 Update 3g and later	Yes
1.7.x and earlier	Yes	No

Using Velero Commands

Velero is installed in the Supervisor Service namespace. All Velero commands must specify the Velero Supervisor Service namespace explicitly. For more details, see https://velero.io/docs/v1.16/namespace/.

If you use the following command, you can avoid specifying the namespace repeatedly.

```
velero client config set namespace=svc-velero-domain-c38
```

Where `svc-velero-domain-c38` is the Velero Supervisor Service namespace.

Prerequisites

- Supervisor is activated.
- A vSphere Namespace is created and configured.
- You must be a member of the VI administrator role. See Using vCenter Roles to Assign Privileges.
- Create a Linux VM where you can run the Velero CLI. Or use an existing Linux jump host where you access Supervisor.
- The Velero version numbers are presented as `X.Y.Z`. Refer to the Velero Compatibility Matrix for the specific versions to use, and substitute accordingly when running the commands.

Step 0 (Optional): Create a Dedicated Network for Backup and Restore Traffic

Although not required, it is recommended that for production environments you separate the backup and restore traffic from the Supervisor control plane management network traffic.

There are two aspects to this:

- Tag the ESX hosts to support network file copy (NFC)
- Configure the backup and restore network using NSX.

To configure ESX hosts to support a dedicated network block device (NBD) transport, you add a VMkernel NIC on each ESX host in the vSphere clusters where the Supervisor runs and set the `vSphereBackupNFC` on that NIC. When the tag `vSphereBackupNFC` is applied to the NIC type for a VMkernel adapter, backup and restore traffic goes through the selected virtual NIC.

To perform this configuration, you use the Virtual Disk Development Kit. See the *Best Practices for NBD Transport* in the *Virtual Disk Development Kit Programming Guide*.

> **Note:**
>
> If the `vSphereBackupNFC` is not enabled on the VMkernel NIC, the backup and restore traffic will be not be sent on the backup and restore network, even if you configure one. If `vSphereBackupNFC` is not enabled, the traffic will travel on the vSphere management network.

Once the `vSphereBackupNFC` tag is enabled, configure the backup and restore network to use NSX by updating the existing vSphere Distributed Switch (VDS) for the cluster as follows.

1. In the vSphere Client, select **Menu** > **Networking**.
2. Select the existing VDS for the cluster.
3. Right-click the VDS and select **Distributed Port Group** > **New Distributed Port Group**.
4. Create a new Distributed Port Group named **BackupRestoreNetwork**.
5. Add a VMkernel adapter to the **BackupRestoreNetwork** Distributed Port Group.
6. Attach all the ESX hosts in the vCenter cluster where the Supervisor is enabled to the **BackupRestoreNetwork** Distributed Port Group.
7. Enable the `vSphereBackupNFC` tag.

Step 1: Create an S3-Compatible Object Store

For backup and restore of persistent volumes, you must provide an S3-compatible object store. Velero supports several object store providers. See official Velero documentation.

To install the Velero Plugin for vSphere, provide the following information about your S3-compatible object store:

Data Item	Example Value
s3Url	http://my-s3-store.example.com
aws_access_key_id	ACCESS-KEY-ID-STRING
aws_secret_access_key	SECRET-ACCESS-KEY-STRING

Create a secrets file name `s3-credentials` with the following information. You reference this file when you install the Velero Plugin for vSphere.

```
[default]
aws_access_key_id = ACCESS-KEY-ID-STRING
aws_secret_access_key = SECRET-ACCESS-KEY-STRING
```

MinIO is an S3-compatible object store that is easy to install and use. Supervisor provides a MinIO Supervisor Service that you can enable. For more information, see Using vSAN Data Persistence Platform with Modern Stateful Services.

Step 2: Install and Configure Data Manager

To facilitate backup and restore using the Velero Plugin for vSphere, you deploy one or more Data Manager VMs to move persistent volume backup data in and out of S3-compatible object storage. Data Manager moves the volume snapshot data from the vSphere volume to the remote durable S3-compatible storage on backup, and from remote S3-compatible storage to a vSphere volume during restore.

> **Warning:**
>
> Data Manager is only functionally tested and it is not meant to work at scale and does not promise any performance expectations. It is not meant to backup business critical applications in production.

In a Supervisor environment, install Data Manager as VM.

1. Using the vSphere Client, right-click the Datacenter where Supervisor is activated and select **Deploy OVF Template**.
2. Download to your local machine the Data Manager OVA file.
3. Select **Local File** and upload the Data Manager OVA to vCenter.
4. Name the virtual machine, for example, `DataManager`.
5. Select the compute resource which is the vSphere cluster where Supervisor is configured.
6. Review the VM deployment details and click **Next**.
7. Accept the license agreements and click **Next**.
8. Select the storage and click **Next**.
9. Select the destination network for the Data Manager VM.
   - Select the **BackupRestoreNetwork** if you configured a dedicated backup and restore network.
   - Select the **Management Network** if you did not configure a dedicated backup and restore network.
10. Confirm the selections and click **Finish** to complete the process.
11. Use the Recent Tasks panel to monitor the progress of the deployment.

    > **Note:**
    >
    > If you receive an error that the OVF descriptor is not available, use the Chrome browser.

12. Once the Data Manager VM is deployed, configure the input parameters for the VM.
13. Right-click the VM and select **Edit Settings**.
14. In the Virtual Hardware tab, for CD/DVD Drive, change from **Host Device** to **Client Device**.

    > **Note:**
    >
    > If you do not do this, you cannot save the required advanced configuration settings.

15. In the **Edit Settings** > **Advanced Parameters** tab, select **Advanced** > **Edit Configuration Parameters**.
16. Configure the input parameters for each of the following settings:

Parameter	Value
`guestinfo.cnsdp.vcUser`	Enter the vCenter user name with sufficient privileges to deploy VMs.
`guestinfo.cnsdp.vcAddress`	Enter the vCenter IP address or FQDN.
`guestinfo.cnsdp.vcPasswd`	Enter the vCenter user password.
`guestinfo.cnsdp.vcPort`	The default is `443`. Do not change this value.
`guestinfo.cnsdp.wcpControlPlaneIP`	Enter the Supervisor floating IP address. Get this value by navigating to the Supervisor in Workload Management, and selecting **Configure** > **Network** > **Management Network** > **Floating IP**

Parameter	Value
`guestinfo.cnsdp.updateKubectl`	The default is `false`. Do not change this value.
`guestinfo.cnsdp.veleroNamespace`	This is the Velero Supervisor service namespace. Leave the default value. Later in the process you will create a vSphere Namespace on the Supervisor with the same name. These two names must match.
`guestinfo.cnsdp.datamgrImage`	If not configured (unset), the system by default pulls the container image from Docker Hub at `vsphereveleroplugin/data-manager-for-plugin:1.1.0`. In an air-gapped environment or if you use private repos, specify the image here.

17. Click **OK** to save the configuration and **OK** again to save the VM settings.

    Note:

    If you did not change the CD/DVD Drive from **Host Device** to **Client Device**, you cannot save the settings. If this is the case, cancel the operation, change the drive, and repeat the advanced configuration settings.

Do not power on the Data Manager VM until after you enable the Velero vSphere Operator.

Step 3: Create a vSphere Namespace for the Velero Plugin for vSphere

Use the vSphere Client to manually create a vSphere Namespace on the Supervisor. This namespace is required for the Velero Plugin for vSphere.

1. Name the vSphere Namespace `velero`.
2. Select the `velero` namespace and configure it.
3. Specify storage for the `velero` namespace.
4. Grant a user with appropriate privileges the edit permission on the **velero** namespace.

Step 4: Configure the Velero vSphere Operator Service on the Supervisor

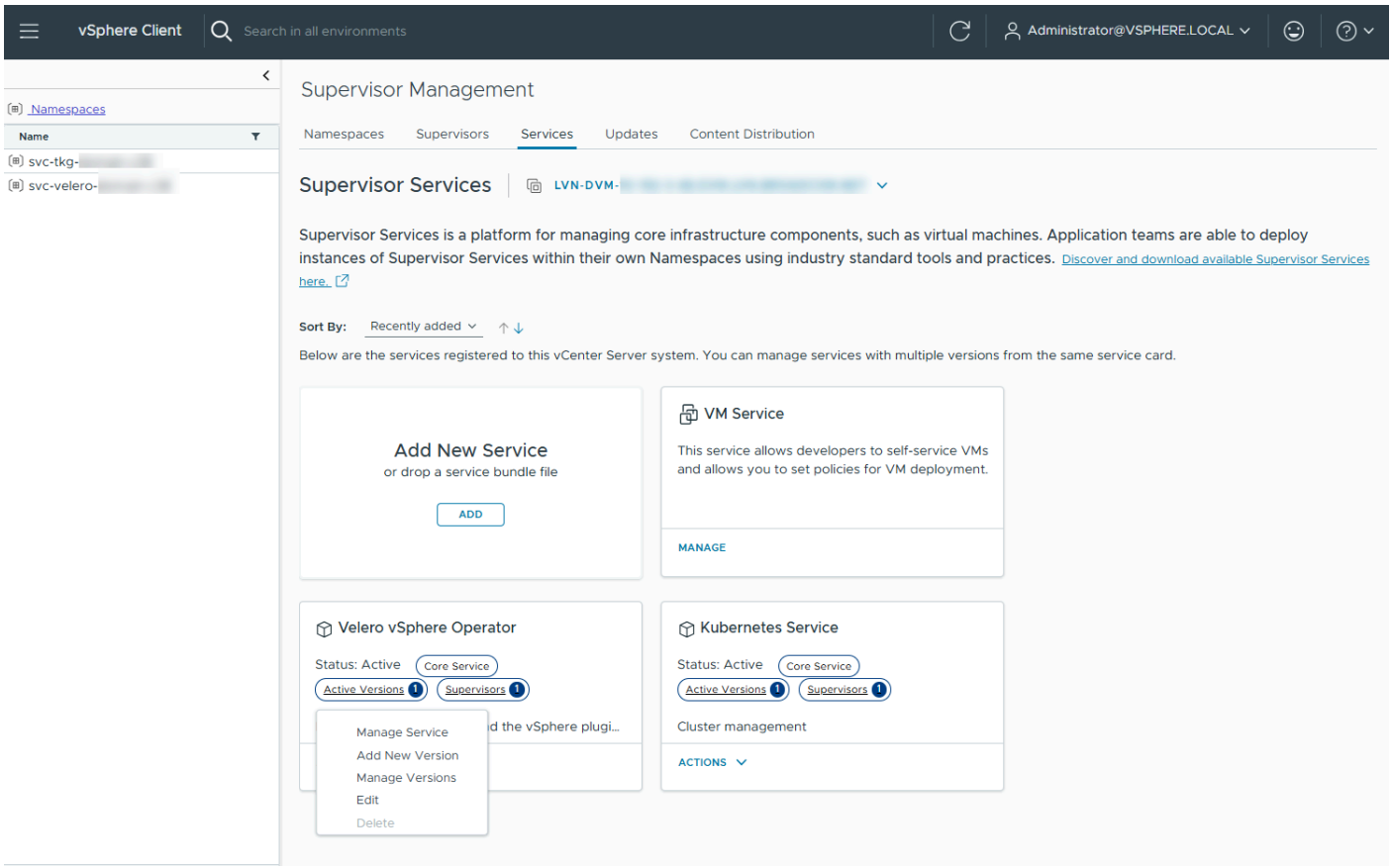
vSphere Supervisor provides the Velero vSphere Operator as a Supervisor Service. The Velero vSphere Operator service works with the Velero Plugin for vSphere to support backup and restore of Kubernetes workloads, including snapshotting persistent volumes.

For more information about Supervisor Services, see Managing Supervisor Services.

The Velero vSphere Operator is a core Supervisor Service, which means that the service operator comes pre-registered with vCenter.

Complete the steps to install the Velero vSphere Operator as a service on a Supervisor.

1. From the vSphere Client home menu, select **Supervisor Management**.
2. Select the **Services** tab.
3. Select the target vCenter from the drop-down menu at the top.
4. From the Velero vSphere Operator card, select **Actions** > **Manage Service**.



5. On the **Configure** page, specify the following items and click **Next**.
   - The target Supervisor where you want to install the service.
   - Make sure that the **Install Version** is 1.8.0-embedded+24696422 or other version with the 1.8 prefix.
6. On the **Review** page, enter the plugin details in the **YAML Service Config (optional)** box.

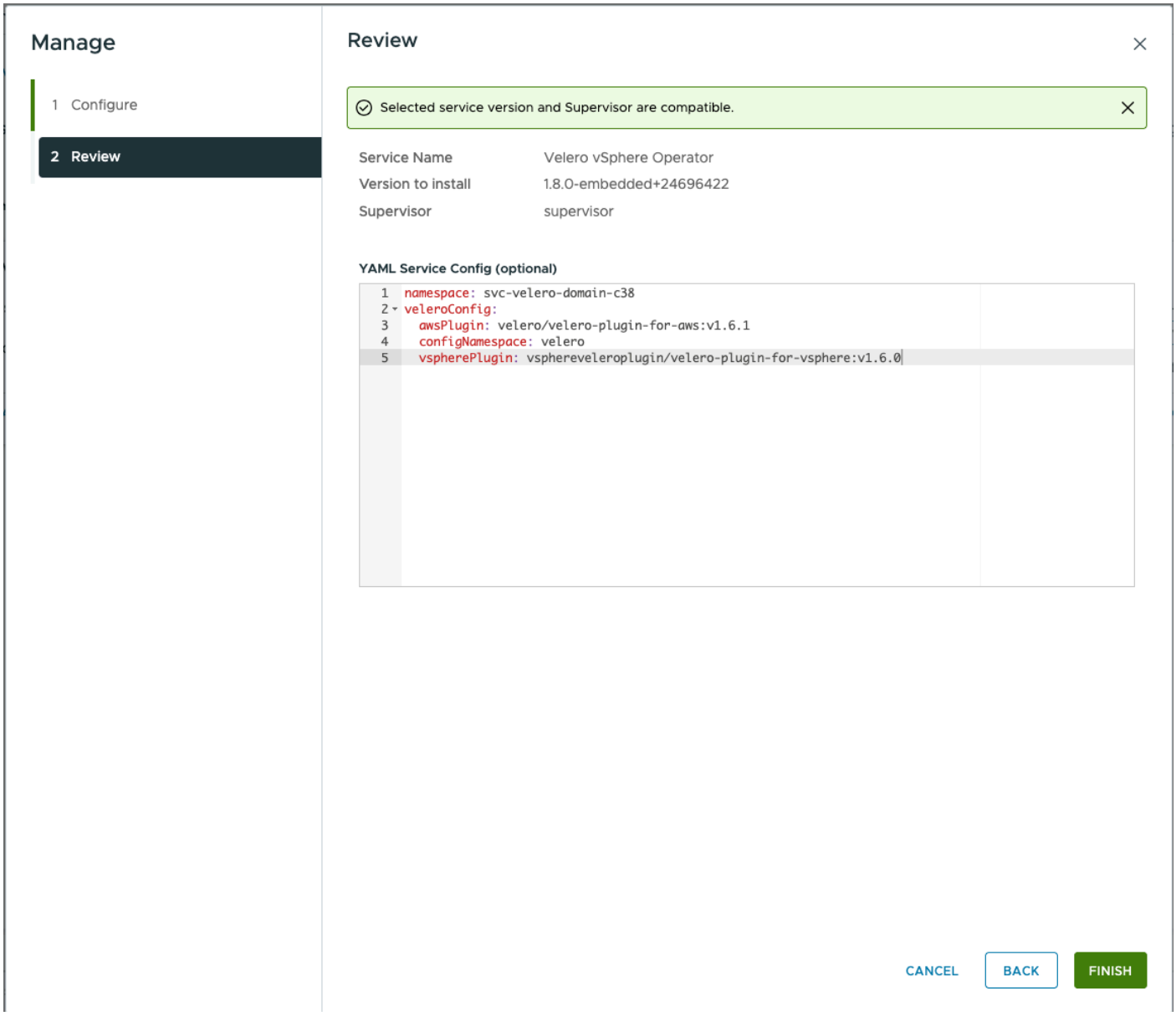
```
namespace: svc-velero-domain-c38
veleroConfig:
  awsPlugin: velero/velero-plugin-for-aws:v1.6.1
  configNamespace: velero
  vspherePlugin: vsphereveleroplugin/velero-plugin-for-vsphere:v1.6.0
```

   Follow these guidelines:

   - Do not modify the `namespace` tag. It represents the Velero vSphere Operator namespace.
   - The `awsPlugin` is the image for the Velero plugin for AWS. If you use a private registry, an air-gapped environment, or any other non-public image, enter the corresponding image here. See Install the Velero Plugin for vSphere in an Air-gapped Environment.
   - The `configNamespace` is the namespace that you created in Step 3: Create a vSphere Namespace for the Velero Plugin for vSphere. Make sure to enter the exact namespace here.
   - The vspherePlugin is the image for the Velero plugin for vSphere. If you use a private registry, an air-gapped environment, or any other non-public image, enter the corresponding image here. See Install the Velero Plugin for vSphere in an Air-gapped Environment.

- The example does not contain `veleroImage` since it is not mandatory. The Velero vSphere Operator picks the correct qualified version of Velero during installation. If you use a private registry, an air-gapped environment, or any other non-public image, enter the corresponding image here. See Install the Velero Plugin for vSphere in an Air-gapped Environment.

Make sure that your screen looks like the following.



7. Click **Finish** to complete the installation of the service.

Step 5: Install the Velero Plugin for vSphere

Now you are ready to install the Velero Plugin for vSphere. To perform the installation, download and run the **velero-vsphere** CLI.

This procedure requires a Linux VM. Download the **velero-vsphere** binary to the Linux jump host where you run the VCF CLI and `kubectl` CLIs.

1. Download the CLI.

   Check the compatibility matrix and download the target version from here: https://github.com/vmware-tanzu/velero-plugin-for-vsphere/releases.

In the commands that follow, replace `X.Y.Z` with the versions of Velero CLI and Plugin that you download. The installation steps mentioned here are for the CLI version 1.8.x.

2. Securely copy the CLI to the Linux jump host.

   For example:

   ```
   pscp -P 22 C:\temp\velero-vsphere-X.Y.Z-linux-amd64.tar.gz ubuntu@10.117.29.131:/home/ubuntu/tanzu
   ```

3. Extract the `velero-vsphere` CLI and make it writable.

   ```
   tar -xf velero-vsphere-X.Y.Z-linux-amd64.tar.gz
   chmod +x velero-vsphere
   ```

4. Add the CLI to your path.

   ```
   export PATH="$(pwd)/velero-vsphere-X.Y.Z-linux-amd64:$PATH"
   ```

5. Create the `s3-credentials` file with the following contents.

   ```
   aws_access_key_id = ACCESS-KEY-ID-STRING
   aws_secret_access_key = SECRET-ACCESS-KEY-STRING
   ```

6. Obtain the region, URL, and bucket name for your S3-compatible object store.
7. Log in to the Supervisor using the VCF CLI.
8. Switch context to the Supervisor.

   See Connect to the Supervisor as a vCenter Single Sign-On User.

9. Run the following `velero-vsphere` CLI command to install the Velero Plugin for vSphere into the **velero** namespace you created.

   Export the values for the AWS `$BUCKET` and `$REGION`. If you deviated from any of the preceding instructions, adjust those values as well, such as the name or location of the secrets file.

   ```
   export BUCKET=example-velero-sv && export REGION=us-east-1

   ./velero-vsphere install \
   --provider aws \
   --bucket $BUCKET \
   --secret-file ~/.aws/credentials \
   --snapshot-location-config region=$REGION \
   --backup-location-config region=$REGION
   ```

   If this installation is in an air-gapped environment, also specify `--use-private-registry` in the command, for example:

   ```
   export BUCKET=example-velero-sv && export REGION=us-east-1

   ./velero-vsphere install \
   --provider aws \
   --bucket $BUCKET \
   --secret-file ~/.aws/credentials \
   --snapshot-location-config region=$REGION \
   --backup-location-config region=$REGION \
   --use-private-registry
   ```

10. Verify your installation.
    a) Make sure that you see a message similar to the following.

       ```
       Send the request to the operator about installing Velero in namespace velero
       ```

b) Run the following command to verify the installation.

Note:

The command assumes that you have the `jq` utility installed, which formats JSON output sent to the terminal. If you do not have `jq` installed, either install it or remove that part of the command (everything after `json`).

```
kubectl -n velero get veleroservice default -o json | jq '.status'
```

You should see the following results that indicate the installation status as `Completed` and show the correct installed version.

```
{
  "enabled": true,
  "installphase": "Completed",
  "version": "v1.13.2_vmware.3"
}
```

c) Verify that the backup driver and Velero pods are up and running in the Velero Supervisor Service namespace.

```
kubectl get pods -n svc-velero-domain-c38
```

Where `svc-velero-domain-c38` is the Velero Supervisor Service namespace.

You should see the output similar to the following:

```
NAME                                          READY   STATUS    RESTARTS   AGE
backup-driver-744c78d64c-885sl                1/1     Running   0          47s
velero-d7b68d494-nzklv                        1/1     Running   0          2m22s
velero-vsphere-operator-7c96f46959-sd4k2      1/1     Running   0          2d1h
velero-vsphere-operator-webhook-57bbf477dd-vfcwr   1/1     Running   0          2d1h
```

11. Power on the Data Manager VMs you created in Step 2: Install and Configure Data Manager.
12. Troubleshoot as necessary.

    If the installation is not successful, remove the installation and try again.

    To remove the installation, follow the steps in Velero Plugin for vSphere.

Uninstall the Velero Plugin for vSphere

Complete these steps to uninstall the Velero Plugin for vSphere.

1. Run the `velero-vsphere` CLI to uninstall the Velero Plugin for vSphere.

   ```
   ./velero-vsphere uninstall -n velero
   ```

2. Verify that the vSphere Pod named `velero` is removed.

   ```
   kubectl get pods -n velero
   ```

   If you see that the pod is `Terminating`, wait for it to be removed before proceeding.

3. Using the vSphere Client, delete the vSphere Namespace named `velero` that you manually created.

   Note:

   Do not proceed with the next step until the namespace deletion is complete. You can use `kubectl` to verify that the `velero` namespace is removed, but do not use `kubectl` to remove the `velero` namespace.

Install the Velero Plugin for vSphere in an Air-gapped Environment

If you use an air-gapped environment, before installing the Velero Plugin for vSphere on a Supervisor, perform these required steps.

If you plan to install the Velero Plugin for vSphere in an air-gapped environment, you must install it with customized images. You must make sure that the matching images of `backup-driver` and `data-manager-for-plugin` of the customized images are available in the expected registry and are accessible from the Kubernetes cluster. In an air-

gapped environment, customized images from a private registry are expected since the released images in docker hub are not accessible.

1. Download the released images of `velero-plugin-for-vsphere`, `backup-driver`, and `data-manager-for-plugin`.
2. Rename the images, that is, tag them with matching `<Registry endpoint and path>` and `<Version tag>` and upload them in customized repositories.
3. Install the plug-in using the `velero-plugin-for-vsphere` image you customized.

   When you install Velero Plugin for vSphere in a vanilla cluster, it deploys two additional components, a `backup-driver` deployment and a `data-manager-for-plugin` DaemonSet in the background. In the Supervisor and VKS clusters, only a `backup-driver` deployment is deployed.

   When you provide the container image of `velero-plugin-for-vsphere`, the matching `backup-driver` and `data-manager-for-plugin` images are parsed using an image parsing mechanism.

   Container images are formalized into the following pattern:

   ```
   <Registry endpoint and path>/<Container name>:<Version tag>
   ```

   When you provide the `velero-plugin-for-vsphere` container image, the corresponding images of `backup-driver` and `data-manager-for-plugin` with matching `<Registry endpoint and path>` and `<Version tag>` are parsed.

   For example, consider the following `velero-plugin-for-vsphere` container image:

   ```
   abc.io:8989/x/y/.../z/velero-plugin-for-vsphere:vX.Y.Z
   ```

   The following matching images of `backup-driver` and `data-manager-for-plugin` are expected to be pulled:

   ```
   abc.io:8989/x/y/.../z/backup-driver:vX.Y.Z
   abc.io:8989/x/y/.../z/data-manager-for-plugin:vX.Y.Z
   ```

4. For a private registry, create a private registry credential secret.

   For information, see Create Private Registry Credential Secret in the *Managing vSphere Kubernetes Service Clusters and Workloads* documentation.

   Use the following as an example.

   ```
   kubectl create secret docker-registry velero-vsphere-regcred --docker-server=<air-gap-hostname> --docker-
   username=<user-name> --docker-password=<token> --namespace=svc-velero-domain-c38
   ```

   The `--namespace` indicates the name of namespace for the Velero vSphere operator Supervisor service. Ensure that you keep the same name `velero-vsphere-regcred`.

   Verify that the Secret is created by using the command:

   ```
   kubectl get secrets -n svc-velero-domain-c38
   ```

   The output is similar to the following:

   ```
   NAME                     TYPE                             DATA   AGE
   velero-vsphere-regcred   kubernetes.io/dockerconfigjson   1      2m39s <==================================
    Created here
   webhook-service-cert     kubernetes.io/tls                3      16h
   ```

5. Continue your installation following the procedures in Install and Configure the Velero Plugin Version 1.8.x for vSphere.

   If the plug-in cannot parse the matching `backup-driver` and `data-manager-for-plugin` images, it falls back to pulling them from the official `velerovspherepluginrepositories` on Docker Hub.

   This fallback is triggered in the following cases:

   - The `velero-plugin-for-vsphereimage` uses an unexpected container name.
     Example: `x/y/velero-velero-plugin-for-vsphere:vX.Y.Z`
   - The Velero deployment name is changed from the default `velero`.

Example: Changing it to `velero-server` in the manifest breaks the image parser, which only recognizes the fixed name `velero`.

Using the Velero Plugin with Third-Party Backup Solutions

Third-party backup solutions can use the Velero Plugin for vSphere to back up and restore VKS clusters.

The third-party solution uses the Velero Plugin to take VKS backups. But backup orchestration and workflows are handled through the vendor's own user interface. This approach allows third-party vendors to integrate with VKS while providing their own backup and restore experience.

For vendor-specific implementation details, refer to the documentation provided by the third-party backup solution.

Backing Up and Restoring VKS Clusters and Workloads

Refer to this section to backup and restore VKS clusters and workloads.

Backing Up and Restoring VKS Cluster Node VMs

To backup and restore VKS cluster nodes, you backup the Supervisor state. Doing so lets you restore vSphere Namespaces objects and VKS cluster node VMs.

You activate the Supervisor backup and restore by using the vCenter backup feature available through the vCenter Management Interface. For details, see Back up the Supervisor State

Note:

You can use Supervisor backups only to restore VKS cluster node VMs. You cannot use Supervisor backups to restore workloads deployed on VKS clusters. You must backup workloads separately, then restore them after the VKS cluster nodes are restored.

Backing Up and Restoring Workloads Running on VKS Clusters

You can backup and restore workloads running on VKS clusters by using Velero. See Backing Up and Restoring VKS Cluster Workloads.

Backing Up and Restoring VM Service VMs in vSphere Supervisor

As a VI administrator, you use backup partner solutions based on VMware vSphere Storage APIs – Data Protection (VADP) to automatically perform a backup, full restore, and registration of a VM Service VM on a Supervisor. If the automatic registration fails due to issues with the underlying infrastructure, you can correct the issues and then invoke the `registerVM` API manually to re-register the VM.

Backing Up a VM Service VM

You can perform automatic backups of VM Service VMs through a backup partner solution, such as Veeam, that uses VMware vSphere Storage APIs – Data Protection.

Typically, the VI administrator uses the partner solution to perform the following tasks:

- Sets up backup infrastructure, including installing backup software and configuring backup storage and repositories.
- Creates a backup job targeting a VM or a group of VMs.
- Initiates backups by triggering the job.

When triggered, the partner software backs up the configuration, data, and Kubernetes state of the VM.

Typically, the backup includes the following items:

- VM's configuration stored in vCenter.
- VMX file.
- Contents of VM data disks that could be static or FCDs.

For VM Service VMs, the backup also includes VM's Kubernetes state and additional resources necessary to bootstrap the virtual machine upon restore.

For more information about VMware vSphere Storage APIs – Data Protection and how to use the partner backup solutions, see the Knowledge Base article 1021175 and your partner documentation.

Restoring a VM Service VM

When necessary, the VI administrator can use the backup to restore the VM. For example, you can restore a VM that fails.

vSphere Supervisor supports full VM restore, which restores an entire VM from a backup file to the latest state of the original VM.

Starting with 9.0 release, Supervisor also supports partial restore, or restoring into an existing VM.

To perform full restore, make sure that the original VM does not exist on the Supervisor and vSphere. If it still exists, use the `kubectl delete vm` command in the Supervisor before triggering the restore job.

After you trigger the restore process from the backup software, the backup software recreates the VM in the resource pool and folder specified during restore. The VM can be restored to the same or a different VM name.

If the process is successful, the system detects and automatically registers the restored VM on the Supervisor in the same vSphere Namespace where it was originally created. During the restore process, a VirtualMachine resource is created on the Supervisor. If applicable, any additional resources such as, a Secret to bootstrap the VM or PersistentVolumeClaims for additional volumes used by the virtual machine, are also created.

For more information about Secrets, see Kubernetes documentation at https://kubernetes.io/docs/concepts/configuration/secret/. For information about Persistent Volumes, see https://kubernetes.io/docs/concepts/storage/persistent-volumes/.

Guidelines and Considerations

When restoring the VM, consider the following:

- Before starting the restore process, make sure to perform the following steps:
  - Verify that the original VM has been removed from the Supervisor and vSphere. If it still exists, use the `kubectl delete vm` command in the Supervisor to delete it.
  - Verify that the underlying infrastructure has not changed between backup and restore. Make sure that all appropriate resources that the original VM used, such as VM classes and storage policies, are intact on the target vSphere Namespace .

    Otherwise, the automatic registration of the VM fails with an error.
- The restored VM is registered on the same target vSphere Namespace where it was originally created.
- Target resource pool and folder name should be the same as those where the original VM existed.
- The name of the restored VM can be the same as the original VM name. You can also use a different VM name.

Manually Register a VM Service VM

If for any reason the automatic registration of the VM does not succeed, you get an error message that notifies you about the exact issues causing the failure. After you correct the issues, you can invoke the `registerVM` API and use the VM's moID to manually register the VM.

The following example uses the Datacenter CLI (DCLI) commands to register the VM.

1. Obtain the moID of the VM to register.

   ```
   # dcli com vmware vcenter vm list
   ```

   The moID, also called MORef ID, is composed of the vm prefix, followed by a numerical ID, for example, vm-123456.

2. Manually register the VM.

   ```
   # dcli com vmware vcenter namespaces instances registervm --namespace my-namespace --vm vm-123456
    +username my-username +password my-password
   ```

   The command returns a task that is similar to the following:

```
task-637:6b051692-7aff-4d59-8a3f-699d114d37e3
```

3. Use the VAPI task service to get the status of the task.

```
# dcli com vmware cis tasks get --task task-637:6b051692-7aff-4d59-8a3f-699d114d37e3 +username my-username
 +password my-password
```

The command returns the status of the task and any error messages if unsuccessful.

Backup and Restore vSphere Pods Using the Velero Plugin for vSphere

You can use the Velero Plugin for vSphere to backup and restore workloads running on vSphere Pods.

Overview

You can use the Velero Plugin for vSphere to backup and restore workloads running on vSphere Pods in the Supervisor. You can back up and restore both stateless and stateful applications running on vSphere Pods. For stateful applications, you use the Velero Plugin for vSphere to take snapshots of the persistent volumes (PVs).

> **Note:**
> You cannot use Velero standalone with Restic to backup and restore vSphere Pods. You must use the Velero Plugin for vSphere installed on the Supervisor.

Prerequisites

Before you can back up and restore vSphere Pods, your VI administrator must install and configure the Velero Plugin for vSphere. See Installing and Configuring the Velero Plug-in for vSphere on a Supervisor.

> **Note:**
> The Velero Plugin for vSphere is not backing up and restoring the state of the Supervisor.

Back Up a vSphere Pod

To backup a stateless vSphere Pod, run the following command:

```
velero backup create <backup name> --include-namespaces=my-namespace
```

The backup is marked `Completed` after all local snapshots have been taken and Kubernetes metadata is uploaded to the object store. However, the backup of volume snapshots occurs asynchronously and may still be occurring in the background and take some time to complete.

You can check the status of volume snapshots by monitoring Snapshot and Upload custom resources.

Snapshot CRD

For each volume snapshot, a Snapshot custom resource is created in the same namespace as the PVC that is snapshotted. You can get all Snapshots in PVC namespace by running the following command.

```
kubectl get -n <pvc namespace> snapshot
```

The Snapshot CRD has several phases for the `status.phase` field, including:

State	Description
New	Not processed yet
Snapshotted	Local snapshot was taken
SnapshotFailed	Local snapshot failed
Uploading	Snapshot is being uploaded
Uploaded	Snapshot is uploaded
UploadFailed	Snapshot failed to be uploaded
Canceling	Upload of snapshot is being cancelled
Canceled	Upload of snapshot is cancelled
CleanupAfterUploadFailed	Cleanup of local snapshot after the upload of snapshot failed

Upload CRD

For each volume snapshot to be uploaded to object store, an Upload CR will be created in the same namespace as Velero. You can get all Uploads in Velero namespace by running the following command.

```
kubectl get -n <velero namespace> upload
```

Upload CRD has several phases for the `status.phase` field, including:

State	Description
New	Not processed yet
InProgress	Upload in progress
UploadError	Upload failed
CleanupFailed	Delete local snapshot failed after the upload Will be retried
Canceling	Upload is being cancelled Can occur if `velero backup delete` is called while snapshot upload is in progress
Canceled	Upload is cancelled

UploadError uploads will be periodically retried. At that point their phase will return to InProgress. After an upload has been successfully completed, its record will remain for a period of time and eventually be removed.

Restore a vSphere Pod

To restore a vSphere Pod workload that has been backed up using the Velero Plugin for vSphere, complete the following steps.

1. Create a vSphere Namespace for the workload you will restore.
2. Configure the storage policy for the namespace.
3. Run the following Velero command to restore the workload:

   ```
   velero restore create --from-backup backup-name
   ```

Velero restore will be marked as `Completed` when volume snapshots and other Kubernetes metadata have been successfully restored to the current cluster. At this point, all tasks of vSphere plugin related to this restore are completed as well. There are no asynchronous data movement tasks behind the scene as that in the case of Velero backup.

Before Velero restore is `Completed`, you can check the status of volume restore by monitoring CloneFromSnapshot/ Download CRs as below.

CloneFromSnapshots CRD

For restore from each volume snapshot, a CloneFromSnapshot CR will be created in the same namespace as the PVC that is originally snapshotted. We can get all CloneFromSnapshots in PVC namespace by running the following command.

```
kubectl -n <pvc namespace> get clonefromsnapshot
```

CloneFromSnapshot CRD has several phases for the `status.phase` field, including:

State	Description
New	Clone from snapshot is not completed
Completed	Clone from snapshot is completed
Failed	Clone from snapshot failed

Download CRD

From each restore of volume snapshot to be downloaded from object store, a Download CR will be created in the same namespace as Velero. We can get all Downloads in Velero namespace by running the following command.

```
kubectl -n <velero namespace> get download
```

The Download CRD has several phases for the `status.phase` field, including:

Status	Description
New	Not processed yet
InProgress	Download in progress
Completed	Download is completed
Retry	Download is retried. When there is any failure during the download of backup data, the download is retried
Failed	Download is failed

Troubleshooting Backup and Restore in vSphere Supervisor Environment

Learn how to troubleshoot backup and restore issues related to Supervisor environment.

Cleaning Up Orphaned Objects After Supervisor Restore from a Backup

After you restore the Supervisor from a backup, all K8s resources that are created after the backup is taken will be deleted after the restore completes. If some of these resources were associated with objects, such as VMs or disks, they will be orphaned in vCenter. You must clean the orphaned objects from vCenter.

1. List all the VMs for a vSphere Namespace.
   a) Retrieve `folderMoId` for the vSphere Namespace by running the following command on a Supervisor control plane VM:

```
root@421c9fa40208448fecc15d277bdca66d [ ~ ]# kubectl get availabilityzone -o json
{
    "apiVersion": "v1",
    "items": [
        {
            "apiVersion": "topology.tanzu.vmware.com/v1alpha1",
            "kind": "AvailabilityZone",
            "metadata": {
                ...
            },
            "spec": {
                "clusterComputeResourceMoIDs": [
                    "domain-c50"
                ],
                "clusterComputeResourceMoId": "domain-c50",
                "namespaces": {
                    "pod-ns": {
                        "folderMoId": "group-v81", <--- this is the folderMoId that you need for next
 step
                        ...
                    },
                    "vmsvc-ns": {
                        "folderMoId": "group-v83", <--- this is the folderMoId that you need for next
 step
                        ...
                    }
                }
            }
        }
    ],
```

```
    "kind": "List",
    "metadata": {
        "resourceVersion": ""
    }
}
```

b) List all existing VMs on the vSphere Namespace by running the following DCLI command:

The example uses the `pod-ns` namespace

```
root@sc2-10-186-199-30 [ ~ ]# dcli +i +username 'Administrator@vsphere.local' +password <password>
Welcome to VMware Datacenter CLI (DCLI)

usage: <namespaces> <command>

To auto-complete and browse DCLI namespaces:   [TAB]
If you need more help for a command:           vcenter vm get --help
If you need more help for a namespace:         vcenter vm --help
To execute dcli internal command: env
For detailed information on DCLI usage visit:  http://vmware.com/go/dcli

dcli> com vmware vcenter vm list --folders group-v81
|---------------|-----|----------------------------------------|-----------|---------|
|memory_size_MiB|vm   |name                                    |power_state|cpu_count|
|---------------|-----|----------------------------------------|-----------|---------|
|512            |vm-84|deployment-before-backup-778449d88d-c9gnc|POWERED_ON |1        |
|512            |vm-85|deployment-before-backup-778449d88d-4jtqj|POWERED_ON |1        |
|512            |vm-86|deployment-before-backup-778449d88d-tqwbh|POWERED_ON |1        |
|512            |vm-91|deployment-after-backup-778449d88d-khkxx |POWERED_OFF|1        |
|512            |vm-92|deployment-after-backup-778449d88d-7dgcc |POWERED_OFF|1        |
|512            |vm-93|deployment-after-backup-778449d88d-sxbcf |POWERED_OFF|1        |
|---------------|-----|----------------------------------------|-----------|---------|
```

.

2. Find orphaned namespaces and clean them up.

   If a namespace is deleted after a Supervisor backup is taken, that namespace will be re-created after Supervisor restore as a Kubernetes resource. You must delete that K8s resource.

   a) Find the orphaned namespace by listing all namespaces in vCenter.

   ```
   dcli> com vmware vcenter namespaces instances list
   ```

   b) List all namespace K8s resources.

   ```
   root@423f9d75bef000dc828a535c6ac0bd4b [ ~ ]# k get ns -A
   ```

   a) Find a diff between the objects resulted from step A and step B and clean up orphaned namespace K8s resources.

   ```
   root@423f9d75bef000dc828a535c6ac0bd4b [ ~ ]# k delete ns test-set-workload-ns
   namespace "test-set-workload-ns" deleted
   ```

3. Find orphaned VMs associated with `VirtualMachine` resources and clean them up.

   `VirtualMachine` Kubernetes resources that are created after Supervisor backup is taken will result in orphaned VMs once the Supervisor is restored from that backup. You must clean these orphaned VMs from the vCenter inventory.

   a) Find orphaned VMs associated with `VirtualMachine` resources.

   The steps that follow use the `vmsvc-ns` namespace as an example.

   1. List all VMs in the vCenter inventory. The example snipped, lists all VMs in vCenter, as `group-96` is associated with the `vmsvc-ns` namespace.

   ```
   dcli> com vmware vcenter vm list --folders group-v96
   |---------------|------|------------|-----------|---------|
   |memory_size_MiB|vm    |name        |power_state|cpu_count|
   ```

```
|---------------|------|------------|-----------|---------|
|2048           |vm-104|vmsvc-after |POWERED_ON |2        |
|2048           |vm-97 |vmsvc-before|POWERED_ON |2        |
|---------------|------|------------|-----------|---------|
```

2. List all `VirtualMachine` K8s resources. Run `kubectl get` to get resources details on any one of control plane VMs, and search for `uniqueID` in the output. In the example, the VM list associated with K8s resources is `vm-97`.

```
root@42344b596f57bfcf9441179fa1ed1a5c [ ~ ]# k get vm -n vmsvc-ns -o json
{
    "apiVersion": "v1",
    "items": [
        {
            "apiVersion": "vmoperator.vmware.com/v1alpha1",
            "kind": "VirtualMachine",
            ...
            "uniqueID": "vm-97",
...
}
```

3. Diff the two lists that resulted from the above steps.
   - The VM list in vCenter: `<vm-104, vm-97>`
   - The VM list associated with k8s resources: `<vm-97>`

   Thus, the orphaned VM list is: `<vm-104>`.

b) Cleanup the orphaned VMs.

```
dcli> com vmware vcenter vm power stop --vm vm-104
dcli> com vmware vcenter vm delete --vm vm-104
```

4. Find orphaned VMs associated with Pod resources and clean them up.

   Pod K8s resources that are created after Supervisor backup is taken will result in orphaned VMs in the vCenter post Supervisor restore. Follow the steps to find and clean them.

   The examples use namespace `pod-ns`.

   a) List all VMs in the vCenter inventory.

   In the example, VM group `group-v83` is associated with namespace `pod-ns`. The VM list is `vm-88`, `vm-89`, `vm-90`, `vm-101`, `vm-102`, and `vm-103`.

```
dcli> com vmware vcenter vm list --folders group-v83
|---------------|------|----------------------------------------|-----------|---------|
|memory_size_MiB|vm    |name                                    |power_state|cpu_count|
|---------------|------|----------------------------------------|-----------|---------|
|512            |vm-101|deployment-after-backup-778449d88d-ldvn8 |POWERED_OFF|1        |
|512            |vm-102|deployment-after-backup-778449d88d-v29dd |POWERED_OFF|1        |
|512            |vm-103|deployment-after-backup-778449d88d-zdbl9 |POWERED_OFF|1        |
|512            |vm-88 |deployment-before-backup-778449d88d-fgq5b|POWERED_ON |1        |
|512            |vm-89 |deployment-before-backup-778449d88d-mp7td|POWERED_ON |1        |
|512            |vm-90 |deployment-before-backup-778449d88d-cjhq6|POWERED_ON |1        |
|---------------|------|----------------------------------------|-----------|---------|
```

   b) List K8s resources.

   Run `kubectl get` to get resource details on any one of the control plane VMs, and search for `vmware-system-vm-moid` in the output. The VM list associated with K8s resources is `vm-88`, `vm-89`, `vm-90`.

```
root@42344b596f57bfcf9441179fa1ed1a5c [ ~ ]# k get pod -n pod-ns -o json
{
    "apiVersion": "v1",
    "items": [
        {
            "apiVersion": "v1",
```

```
                "kind": "Pod",
                "metadata": {
                    "annotations": {
                        ...
                        "vmware-system-vm-moid": "vm-90:5a5198fc-c5cb-4b89-a70f-331025b40539",
                        ...
                    },
                        ...
                        "vmware-system-vm-moid": "vm-88:5a5198fc-c5cb-4b89-a70f-331025b40539",
                        ...
                        "vmware-system-vm-moid": "vm-89:5a5198fc-c5cb-4b89-a70f-331025b40539",
                        ...
}
```

c) Diff the two lists that resulted from the above steps.

- The VM list in vCenter is: <vm-88, vm-89, vm-90, vm-101, vm-102, vm-103>
- The VM list associated with K8s resources is: <vm-88, vm-89, vm-90>

Thus, the orphaned VM list is: <vm-101, vm-102, vm-103>

d) Clean up orphaned VM.

```
dcli> com vmware vcenter vm delete --vm vm-101
dcli> com vmware vcenter vm delete --vm vm-102
dcli> com vmware vcenter vm delete --vm vm-103
```

5. Find orphaned VMs and Resource Pools associated with VKS clusters and clean them up.

VKS clusters that are created after Supervisor backup was taken, will also result in orphaned VMs in the vCenter post Supervisor restore.

Note:

If you encountered any issues while creating VKS clusters post the restore operation, you must clean up orphaned VMs by following the instructions in the current step.

a) Find orphaned VKS cluster list.

Use kubectl to get VKS cluster K8s resource list on any one of the control plane VMs: <test-cluster, test-cluster-e2e-script, tkc-before-backup>.

```
root@4239f4159c7063d5608cf3fc0bdd532e [ ~ ]# k get tkc -A
NAMESPACE             NAME                       CONTROL PLANE   WORKER   TKR NAME                   AGE
   READY   TKR COMPATIBLE   UPDATES AVAILABLE
selfservice-tkc-ns    test-cluster               1               1        v1.23.8---vmware.3-tkg.1   19h
   True    True
test-gc-e2e-demo-ns   test-cluster-e2e-script    3               1        v1.23.8---vmware.3-tkg.1   18h
   False   True
tkc-ns                tkc-before-backup          3               1        v1.23.8---vmware.3-tkg.1   16h
   True    True
```

Use DCLI to get all resource pool associated with namespace or VKS cluster, then get the VKS cluster list in vCenter: <test-cluster, test-cluster-e2e-script, tkc-before-backup, tkc-after-backup>

```
dcli> com vmware vcenter resourcepool list
|-----------------------|-------------|
|name                   |resource_pool|
|-----------------------|-------------|
|Resources              |resgroup-10  |
|Resources              |resgroup-23  |
|Namespaces             |resgroup-56  |
```

```
|selfservice-tkc-ns     |resgroup-62  | <--- this is a namespace
|test-cluster           |resgroup-66  | <--- Tanzu Kubernetes Grid cluster
|test-gc-e2e-demo-ns    |resgroup-70  | <--- this is a namespace
|test-cluster-e2e-script|resgroup-74  | <--- Tanzu Kubernetes Grid cluster
|tkc-ns                 |resgroup-80  | <--- this is a namespace
|tkc-before-backup      |resgroup-89  | <--- Tanzu Kubernetes Grid cluster
|tkc-after-backup       |resgroup-96  | <--- Tanzu Kubernetes Grid cluster
|-----------------------|-------------|
```

b) Diff the two list from above steps, the orphaned VKS cluster list is: `<tkc-after-backup>`

c) Clean up VMs associated with orphaned VKS cluster.

Use DCLI to get all VMs associated with orphaned VKS cluster by using the associated resource pool `<resgroup-96>`:

```
dcli> com vmware vcenter vm list --resource-pools resgroup-96
|---------------|------|---------------------------------------------|-----------|---------|
|memory_size_MiB|vm    |name                                         |power_state|cpu_count|
|---------------|------|---------------------------------------------|-----------|---------|
|2048           |vm-100|tkc-after-backup-zlcdm-wk5xf                 |POWERED_ON |2        |
|2048           |vm-101|tkc-after-backup-zlcdm-76q4h                 |POWERED_ON |2        |
|2048           |vm-98 |tkc-after-backup-zlcdm-9fv2w                 |POWERED_ON |2        |
|2048           |vm-99 |tkc-after-backup-workers-4hdqb-657fb58d45-d7pqg|POWERED_ON |2        |
|---------------|------|---------------------------------------------|-----------|---------|
```

Then delete the VMs one by one:

```
dcli> com vmware vcenter vm power stop --vm vm-100
dcli> com vmware vcenter vm delete --vm vm-100
```

d) Clean up the resource pool associated with orphaned VKS cluster.

```
<dcli> com vmware vcenter resourcepool delete --resource-pool resgroup-96
```

You can also delete the orphaned resource pool from the vSphere Client.

6. Find orphaned FCDs (first class discs) associated with permanent volumes (PVs) and clean them up.

PV K8s resources created after Supervisor backup was taken will result in orphaned FCDs in the vCenter post Supervisor restore. Follow the steps to find and clean them.

a) Find orphaned FCDs associated with PVs.

1. Install govc that you will be using to find orphaned FCDs. Govc is a user-friendly CLI alternative to the UI and well-suited for automation tasks.

```
curl -L -o - "https://github.com/vmware/govmomi/releases/latest/download/govc_$(uname -s)_$(uname -m).tar.gz" | tar -C /usr/local/bin -xvzf - govc
```

More installation options are available at https://github.com/vmware/govmomi/tree/main/govc#installation.

2. Run the below bash script to list PVs that exist on the Supervisor.

```
#!/bin/bash

export GOVC_INSECURE=1
export GOVC_USERNAME='Administrator@vsphere.local'
export GOVC_PASSWORD=<password>
export GOVC_URL=https://<vc ip>/sdk

# datastore path example - /test-vpx-1688432886-30489-wcp.wcp-sanity/datastore/sharedVmfs-0
govc volume.ls -l -ds=<datastore path>
```

And the result:

```
peiyangs@peiyangs-a01 govc % sudo bash orphanedPV.sh
590c8e31-f5bf-4179-9250-5cdd66bf591c    pvc-843c932b-8974-475d-8f8a-9b165137169d    1.0GB
 KUBERNETES  vSphereSupervisorID-7f88d7b3-12ac-4fcf-a101-b80eb76becdf
```

```
37f8ad5b-dfe6-465b-b0f0-11591a2968dc    pvc-77c42590-f0b0-457f-9743-6a3ebca55078    1.0GB
 KUBERNETES  vSphereSupervisorID-7f88d7b3-12ac-4fcf-a101-b80eb76becdf
28a265b8-2e6b-421c-b16d-046ffc7aeea7    pvc-1b88c923-4354-4537-a7cb-a8a6d763d5e7    1.0GB
 KUBERNETES  vSphereSupervisorID-7f88d7b3-12ac-4fcf-a101-b80eb76becdf
```

3. Run the below bash script to list all disks in vCenter:

```
#!/bin/bash

export GOVC_INSECURE=1
export GOVC_USERNAME='Administrator@vsphere.local'
export GOVC_PASSWORD=<password>
export GOVC_URL=https://<vc ip>/sdk

# datastore path example - /test-vpx-1688432886-30489-wcp.wcp-sanity/datastore/sharedVmfs-0
govc disk.ls -l -ds=<datastore path>
```

And the result:

```
peiyangs@peiyangs-a01 govc % sudo bash orphanedPV.sh
28a265b8-2e6b-421c-b16d-046ffc7aeea7  pvc-1b88c923-4354-4537-a7cb-a8a6d763d5e7  1.0G    Jul  4
 02:33:27 <--- this is the disk correspondings to PV
37f8ad5b-dfe6-465b-b0f0-11591a2968dc  pvc-77c42590-f0b0-457f-9743-6a3ebca55078  1.0G    Jul  4
 02:32:41 <--- this is the disk correspondings to PV
3a7517c2-f8c2-46a9-b0d5-18c665759311  vmware-sv-img-cache-domain-c50            26.0M   Jul  4
 02:36:41
590c8e31-f5bf-4179-9250-5cdd66bf591c  pvc-843c932b-8974-475d-8f8a-9b165137169d  1.0G    Jul  4
 02:30:45 <--- this is the disk correspondings to PV
68ba220c-0f83-49eb-b77a-d60471e24844  pvc-92f83ae0-7c2d-46d9-ab85-19858462ddd1  5.0G    Jul  4
 18:27:02 <--- this is the disk correspondings to PV
72dbe8c5-a3b5-4298-8203-ea1cb86116e6  vmware-sv-img-cache-domain-c50            3.0M    Jul  4
 02:38:39
79e233a6-0134-40e7-8ba8-3133442324f9  vmware-sv-img-cache-domain-c50            195.0M  Jul  4
 18:26:12
a1a0a9d7-0baf-4592-9041-8c0feb960246  vmware-sv-img-cache-domain-c50            7.0M    Jul  4
 02:35:37
cec2af09-80af-4086-a069-34140e2480dc  vmware-sv-img-cache-domain-c50            193.0M  Jul  4
 02:31:12
```

4. Diff the two list from above steps.
   - The list of PVs: <590c8e31-f5bf-4179-9250-5cdd66bf591c, 37f8ad5b-dfe6-465b-b0f0-11591a2968dc, 28a265b8-2e6b-421c-b16d-046ffc7aeea7>
   - The list of FCDs: <590c8e31-f5bf-4179-9250-5cdd66bf591c, 37f8ad5b-dfe6-465b-b0f0-11591a2968dc, 28a265b8-2e6b-421c-b16d-046ffc7aeea7, 68ba220c-0f83-49eb-b77a-d60471e24844>

   The orphaned FCDs are: <68ba220c-0f83-49eb-b77a-d60471e24844>

b) Delete the orphaned FCs.

Use govc to delete the orphaned FCDs, example script:

```
#!/bin/bash

export GOVC_INSECURE=1
export GOVC_USERNAME='Administrator@vsphere.local'
export GOVC_PASSWORD=<password>
export GOVC_URL=https://<vc ip>/sdk

# datastore path example - /test-vpx-1688432886-30489-wcp.wcp-sanity/datastore/sharedVmfs-0
govc disk.rm -ds=<datastore path> 68ba220c-0f83-49eb-b77a-d60471e24844
```

And the resut:

```
peiyangs@peiyangs-a01 govc % sudo bash orphanedPV.sh
```

```
[06-07-23 11:36:27] Deleting 68ba220c-0f83-49eb-b77a-d60471e24844...OK
```

Overview of Running vSphere Supervisor on vSAN Stretched Cluster

Learn what are the general topologies and guidelines for deploying a Supervisor on a stretched vSAN cluster. Stretched vSAN cluster brings the capability to run VMs with high availability across a stretched data center environment. Starting from the vSphere 8 Update 3 release, you can also run VKS workloads on a single vSAN stretched cluster that has equal numbers of hosts that are geographically separated. This way, you provide distributed high availability for VKS workloads across a stretched data center environment.

A vSAN stretched cluster is a vSAN cluster that spans across two data sites for faster level of availability and inter-site load balancing. Both sites have equal number of ESX hosts and are part of the same vSphere cluster. Typically, sites part of a vSAN stretched cluster are geographically separated locations and are referred to as vSAN *fault domains*. In most cases, you deploy vSAN stretched clusters in environments where the distance between data centers is limited, such as metropolitan or campus environments. In a stretched vSAN cluster configuration, both data sites are active sites. In case of a site failure, workloads are restarted on the site that is still active. Each vSAN stretched cluster also has a witness node that serves as a tiebreaker when a decision must be made regarding availability of datastore components when the network connection between the two sites is lost.

For more information on vSAN stretched clusters, see the VMware vSAN Documentation and the vSAN Stretched Cluster Guide.

You can deploy a Supervisor on an existing vSAN stretched cluster in active/active mode. When a Supervisor is deployed on a vSAN stretched cluster and settings are applied to provide HA to the Supervisor workloads, this configuration is referred to as a *deployment-mode*.

The supported Supervisor deployment on a vSAN stretched cluster is a single-zone Supervisor, where the underlying vSphere cluster is a vSAN stretched cluster.

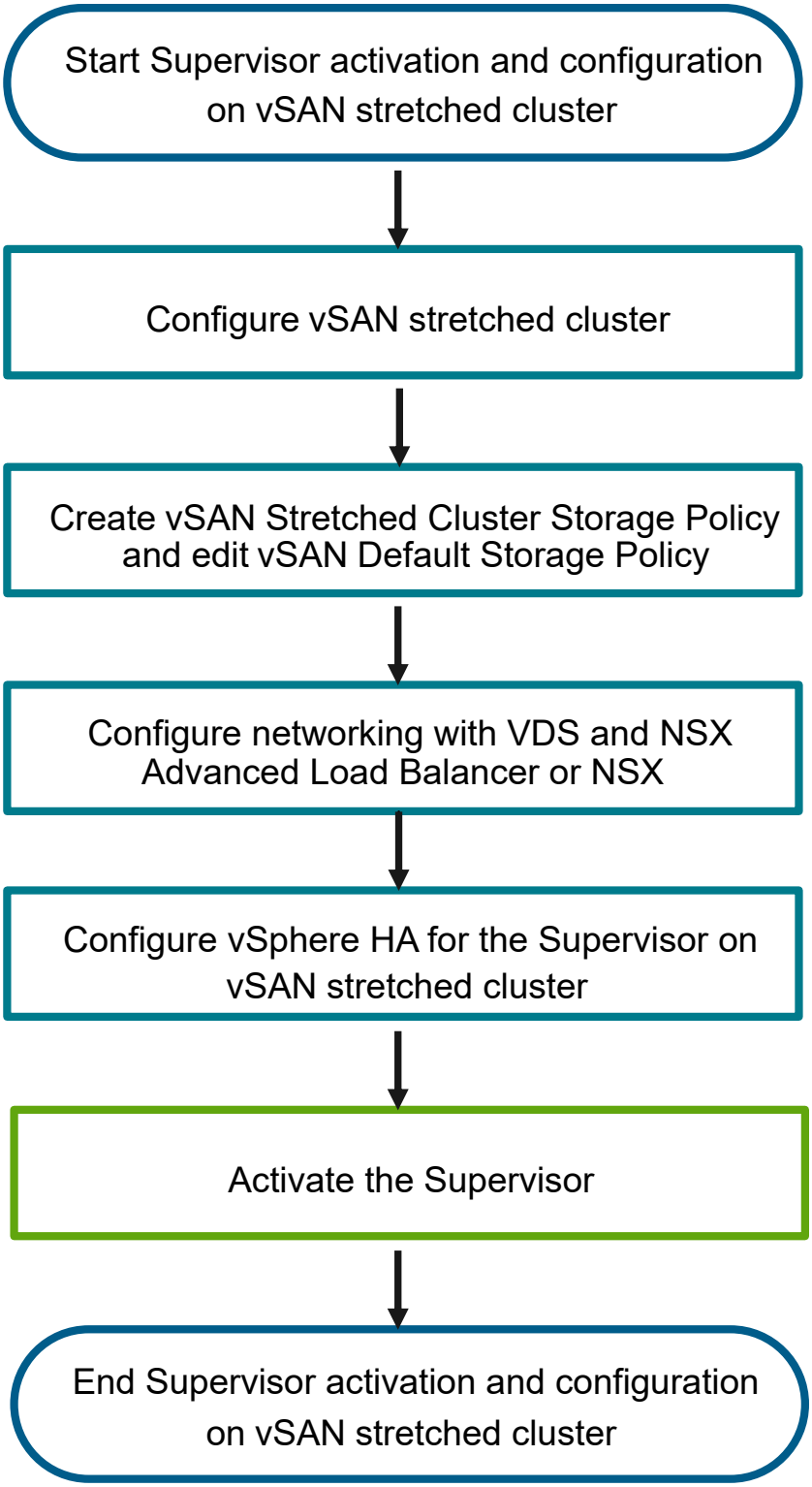
> **Note:**
> You can only use a greenfield deployment for a Supervisor running on a stretched vSAN cluster starting from the vSphere 8 Update 3 release. A greenfield deployment in this case means a Supervisor that is freshly deployed on a vSAN stretched cluster. If the Supervisor is already deployed on a different storage solution or a non-stretched vSAN cluster, you cannot convert the Supervisor to run on a vSAN stretched cluster.

To deploy a vSAN stretched cluster, follow the recommendations and instructions provided in vSAN Stretched Cluster Guide and the VMware vSAN Documentation.

Workflow for Activating and Configuring Supervisor on vSAN Stretched Cluster

Check out what is the workflow for activating a Supervisor on a vSAN stretched cluster and configuring HA for workloads on VKS clusters.

For information on configuring networking for a Supervisor, see Supervisor Networking.


Start Supervisor activation and configuration on vSAN stretched cluster
Configure vSAN stretched cluster
Create vSAN Stretched Cluster Storage Policy and edit vSAN Default Storage Policy
Configure networking with VDS and NSX Advanced Load Balancer or NSX
Configure vSphere HA for the Supervisor on vSAN stretched cluster
Activate the Supervisor
End Supervisor activation and configuration on vSAN stretched cluster

Recommended Environment Settings for Running vSphere Supervisor on vSAN Stretched Cluster

Review the compute, storage, and network requirements for deploying a Supervisor on a vSAN stretched cluster with HA for workloads in VKS clusters.

Table 1799: Compute Requirements

Component	Minimum Value	Description
ESX vSphere 9	3 hosts per site	The minimum number of hosts per site (vSAN fault domain). While three hosts per site is the hard minimum, start with at least four hosts per site. This configuration maintains full policy compliance after a single host failure without manual intervention.
vCenter vSphere 9	1	A vCenter system that manages the vSAN stretched cluster.

Table 1800: Networking Requirements

Component	Minimum Value	Description
Avi Load Balancer	1 deployment	Avi Load Balancer deployed for HA. For guidelines, see Deploying vSphere Supervisor with NSX and Avi Load Balancer.
VMware NSX 4.x	1 deployment	VMware NSX deployed. For guidelines, see Deploying vSphere Supervisor with NSX and Avi Load Balancer.

Table 1801: Storage Requirements

Component	Minimum Value	Description
vSAN stretched cluster vSphere 8 Update 3	1	A vSAN stretched cluster. For more information on deploying a vSAN stretched cluster, follow the recommendations in vSAN Stretched Cluster Guide.

Table 1802: Supervisor Requirements

Component	Minimum Value	Description
Supervisor	1	Single zone Supervisor greenfield deployment. A greenfield deployment in this case means a Supervisor that is freshly deployed on a vSAN stretched cluster. If a Supervisor is already deployed on a different storage solution or a non-stretched vSAN cluster, transition to a vSAN stretched cluster is not supported.
Supervisor Kubernetes version	Version 1.26 for vSphere 8	The Supervisor must run a supported Kubernetes version.
VKr	Version 1.26 for vSphere 8	VKS clusters must run supported Kubernetes versions that are compatible with the VKS service version.

Component	Minimum Value	Description
VKS	Version 3.x	VKS version supported for vSphere 8 Update 3.

Configuring Storage Policy for vSphere Supervisor on vSAN Stretched Cluster

When you run vSphere Supervisor on a vSAN stretched cluster, you can use a default vSAN storage policy or create a storage policy compliant with the requirements of the vSAN stretched cluster.

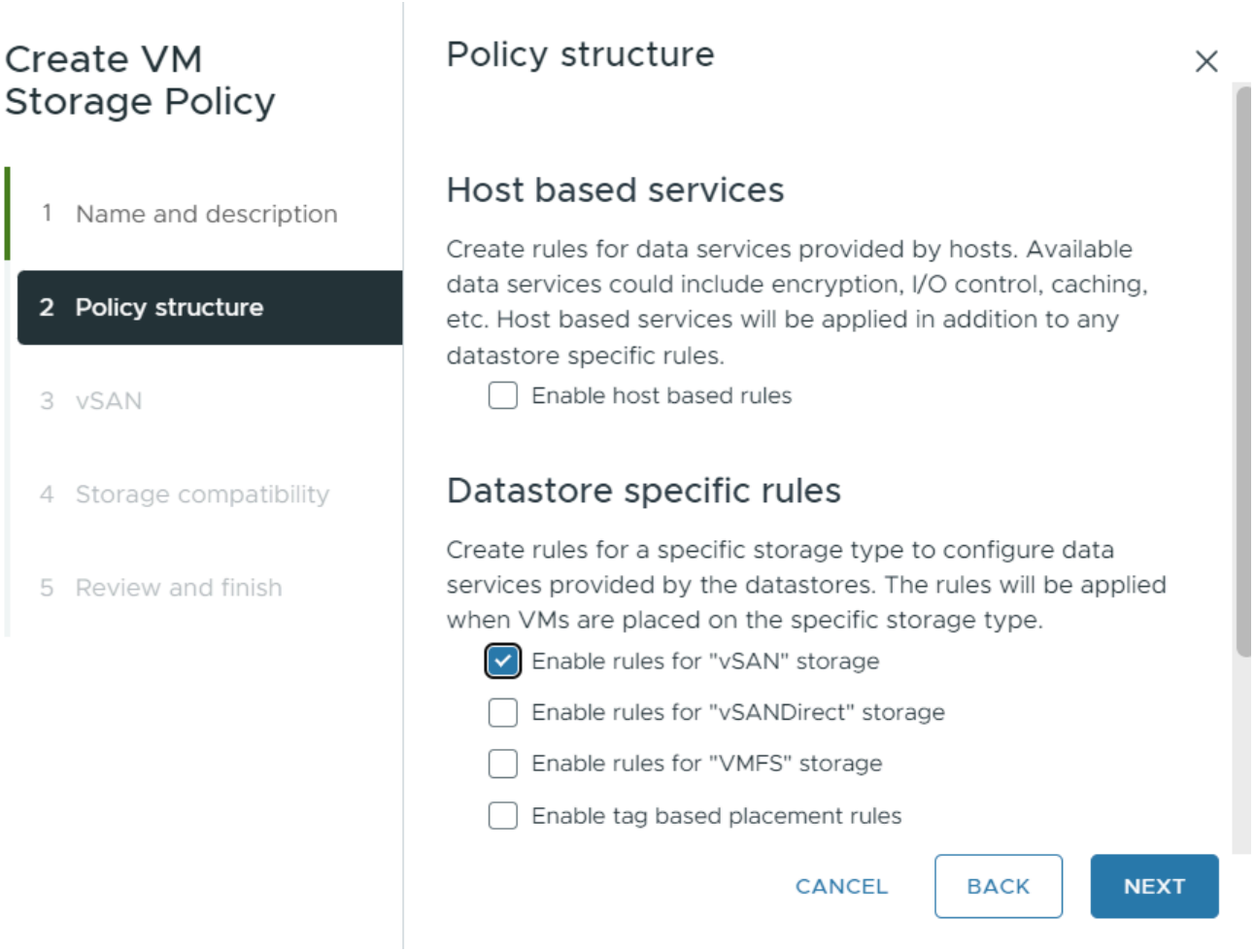
If you decide to create the vSAN stretched cluster storage policy, create it before deploying the Supervisor. During the deployment, this policy is used for the storage placement of Supervisor control plane VMs and should protect the VMs across locations.

After you create the vSAN stretched cluster policy, assign the policy to vSphere Namespaces. Use storage classes associated with this vSAN stretched cluster policy to deploy VKS clusters, persistent volumes, and any other storage objects in the vSphere Namespaces.

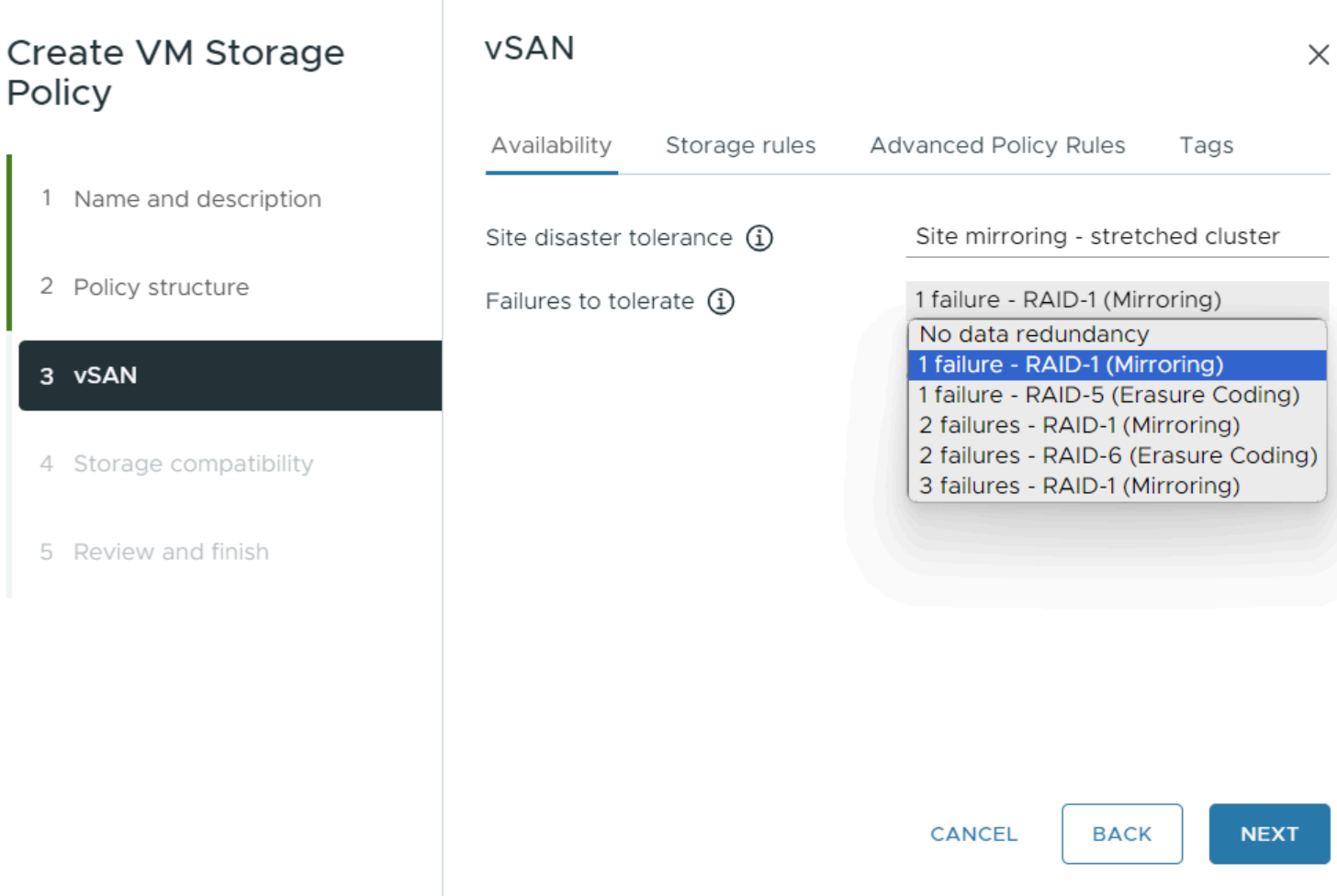
Create a vSAN Stretched Cluster Storage Policy for vSphere Supervisor

When you create the storage policy for vSphere Supervisor, make sure to specify the following settings that apply to the vSAN stretched cluster.

1. On the **Policy structure** page, select **Enable rules for vSAN storage**.



2. On the **vSAN** page, click **Availability**, and set the following parameters.



a) Specify **Site disaster tolerance**.

This setting defines the data redundancy method used by stretched clusters to handle a site failure.

The recommended option is **Site mirroring - stretched cluster**.

This option allows vSAN data to be mirrored, or replicated, across the two sites of the vSAN stretched cluster. When you set this parameter, the vSAN stretched cluster can tolerate one site failure. The data remains accessible even when the site fails.

b) Specify **Failures to tolerate**.

For the stretched cluster, this setting defines the number of disk or host failures a storage object can tolerate within each site.

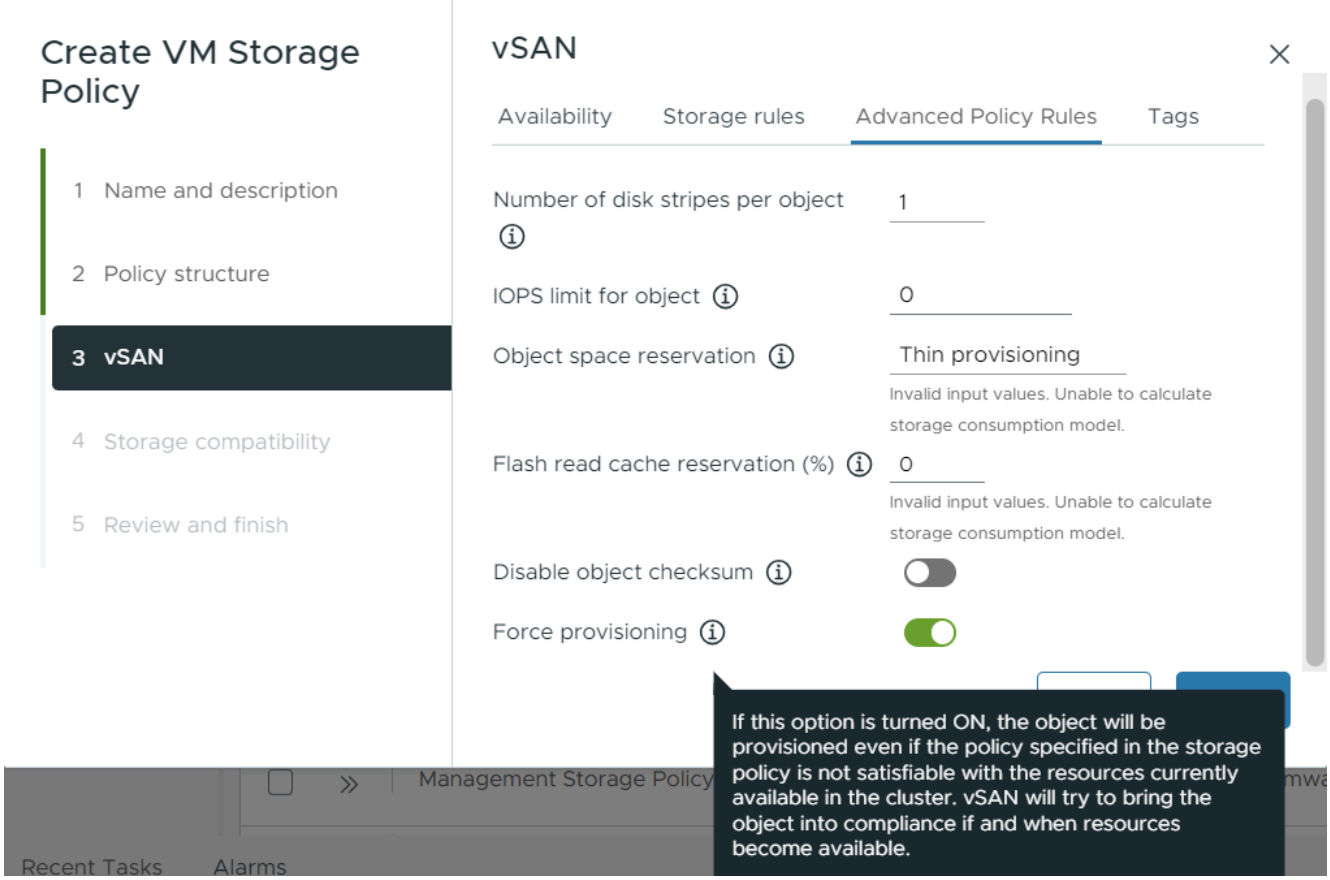
You can select a RAID configuration optimized for either performance (Mirroring) or capacity (Erasure Coding).

With vSAN ESA, it is recommended to use erasure code as it provides the same performance as RAID-1 mirroring.

Table 1803: RAID Configurations, FTT, and Host Requirements

RAID Configuration	Failures to Tolerate (FTT)	Minimum Hosts Required
RAID-1 (Mirroring)	1	2
RAID-5 (Erasure Coding)	1	4
RAID-1 (Mirroring)	2	5
RAID-6 (Erasure Coding)	2	6
RAID-1 (Mirroring)	3	7

3. Click the **Advanced Policy Rules** tab, and enable **Force provisioning**.



If the option is enabled, the vSAN object is provisioned even if other rules defined by the storage policy, such as site disaster tolerance, cannot be satisfied by the datastore. Use this parameter during an outage when standard provisioning is no longer possible.

After you create the vSAN stretched cluster policy, assign the policy to vSphere Namespaces. Use storage classes associated with this vSAN stretched cluster policy to deploy VKS clusters, persistent volumes, and any other storage objects in the vSphere Namespaces.

Configure vSphere HA for Supervisor Running on vSAN Stretched Cluster

Learn how to configure vSphere HA for a Supervisor running on a vSAN stretched cluster so that to provide HA for workloads.

- Enable and configure a vSAN stretched cluster.
- Activate a Supervisor on the vSphere cluster configured as a vSAN stretched cluster.

1. Locate the vSphere cluster where the Supervisor is deployed respectively on a vSAN stretched cluster.
2. Select **Configure** and select **vSphere Availability**.
3. Next to **vSphere HA Turned On**, click **Edit**.
4. Configure **Failure and Responses** settings.

Option	Value	Description
Host monitoring	**Turned on**	Uses network heartbeat to determine the status of hosts participating in the cluster, and whether a corrective action is required, such as restarting a VM on a different host from the cluster
Response for Host Isolation	**Power off restart VMs**	Determines what happens to the VMs on an isolated host, that is a host that cannot communicate to other hosts in the cluster nor reach the isolation response IP address. You configure this setting to **Power o? and restart**, because a clean shutdown is not possible on an isolated host, and the access to the vSAN Datastore and the ability to write to disks is lost.

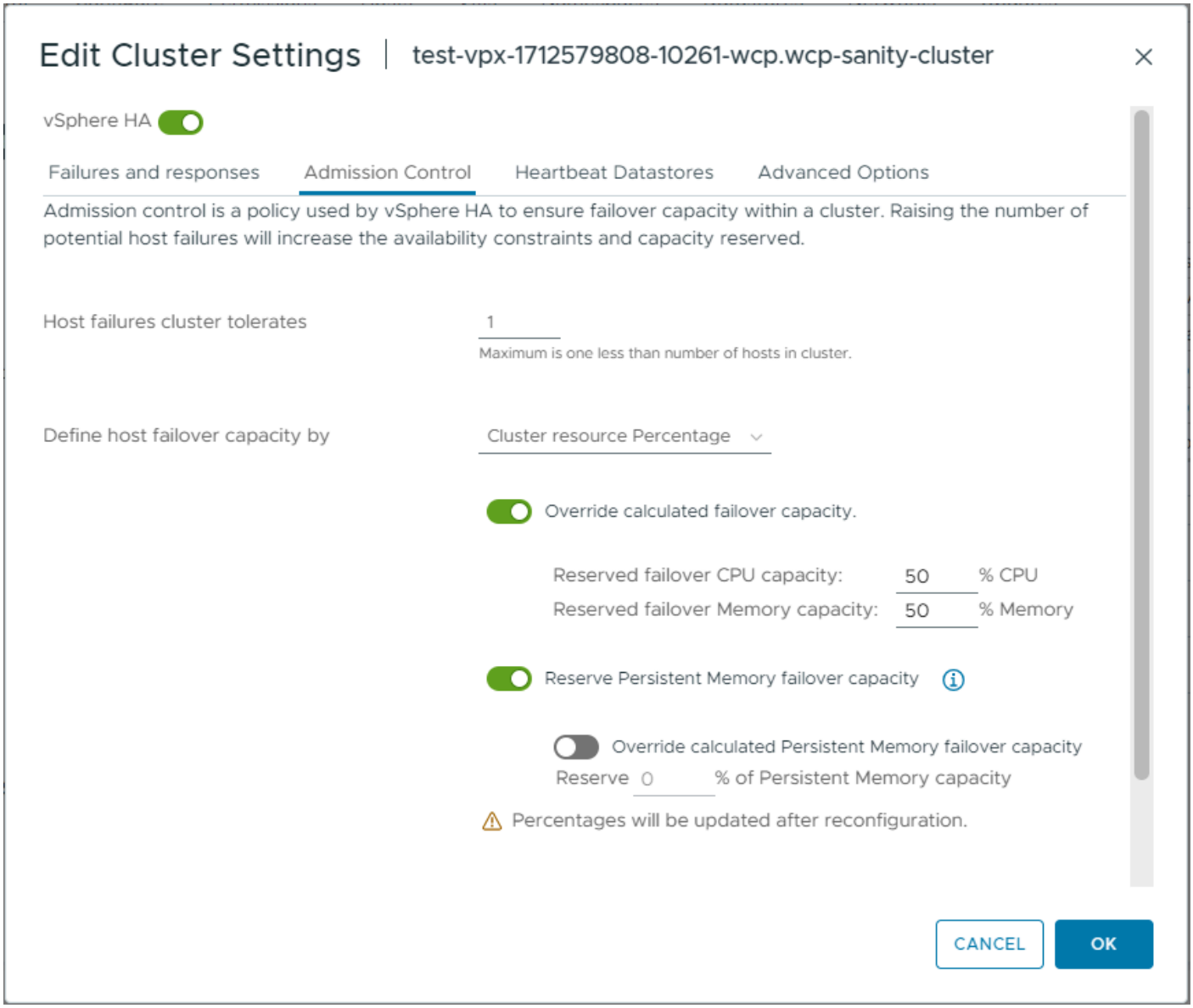
5. Configure **Admission Control**.

   The compute capacity in a vSAN stretched cluster is typically equally divided between the two sites in the cluster. You need to ensure that all the VMs on site 1 can be restarted on site 2 when an entire site goes down. To achieve this, you reserve 50% of the cluster's capacity for failover to allow restart of all the VMs during site failure.

   a) Set **Define host failover capacity by** to **Cluster resource Percentage**.

   b) Enable **Override calculated failover capacity** and set the CPU and memory reservations to 50% each.

   With this setting, vSphere HA ensures that 50% of cluster's aggregate CPU and memory resources are reserved for failover.

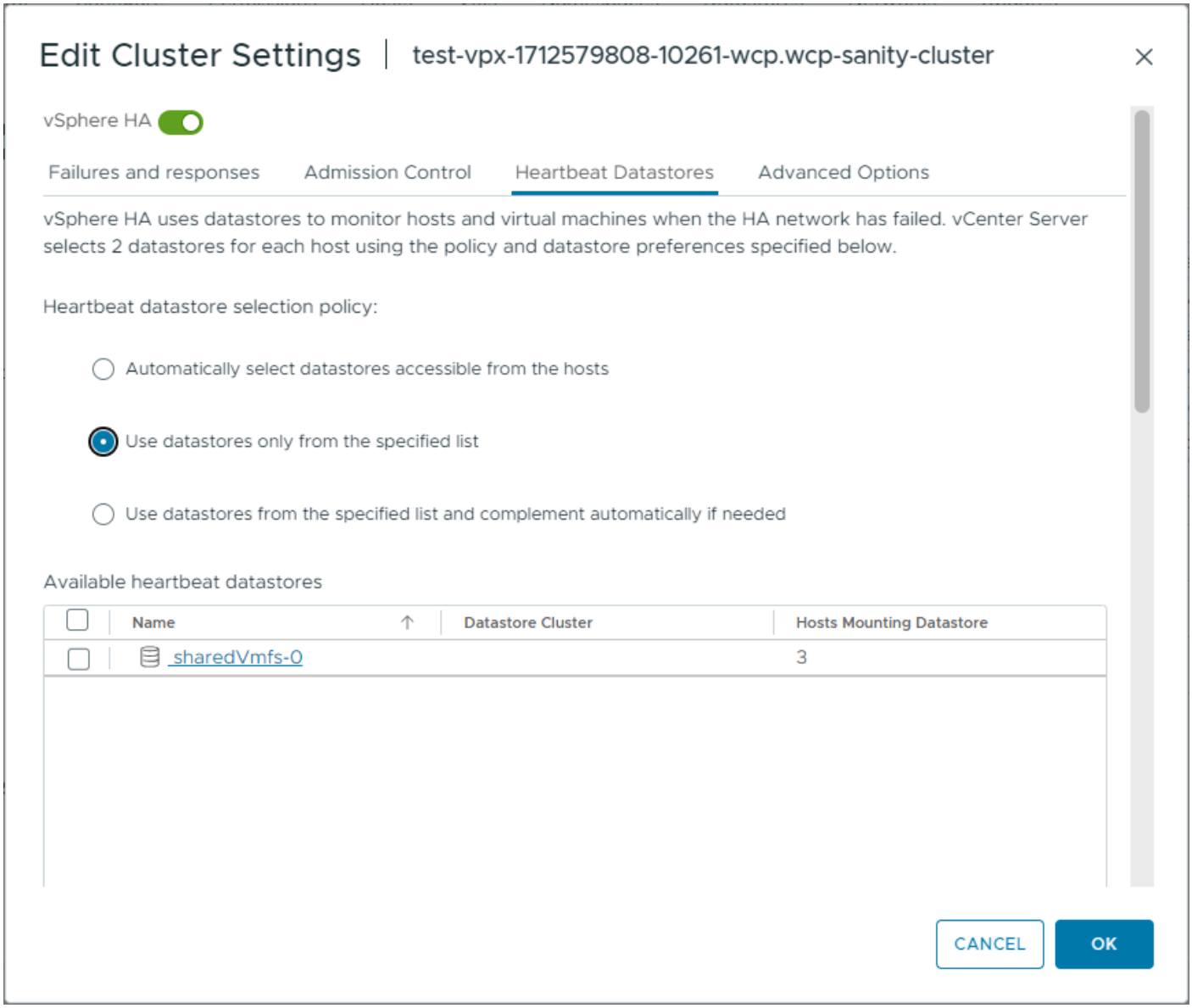


6. Configure **Datastore Heartbeating**.

   In addition to network heartbeating, vSphere HA provides datastore hearbeating for determining the state of the hosts in the cluster. However, a vSAN datastore cannot be used for heartbeating and you must deactivate this function for the vSAN datastore.

   Note:
   If any additional datastore (other than vSAN) is available and if it is accessible by an alternate network path, independent of the vSAN network, then using heartbeat datastores is fully supported.

   To not use datastore heartbeating for the vSAN datastore, select **Use datastores only from the specified list** and make sure that the vSAN datastore is not selected. You will still see an alert that there are less than the required heartbeat datastores, you can disable that alarm by configuring `das.ignoreInsufficientHbDatastore = true`.



7. Configure isolation response addresses.

   When vSphere HA is enabled on a vSAN cluster, HA uses network heartbeat to validate the state of an ESX host.

   In a vSAN environment, vSphere HA uses the vSAN traffic network for communication. This is different from traditional vSphere environments where vSphere HA uses the management network for communication. However, even in a vSAN environment, vSphere HA continues using the default gateway of the management network for isolation detection responses. For this reason, you must configure isolation response IP addresses that reside on the vSAN network to allow HA to react to a vSAN network failure and trigger host isolation response.

   It is recommended that you specify two additional isolation response addresses, and each of these addresses should be site-specific - one isolation address residing in site 1 and the other residing in site 2. This enables vSphere HA to validate host isolation even in the case of network failure between sites. Use the following settings for additional isolation response addresses:

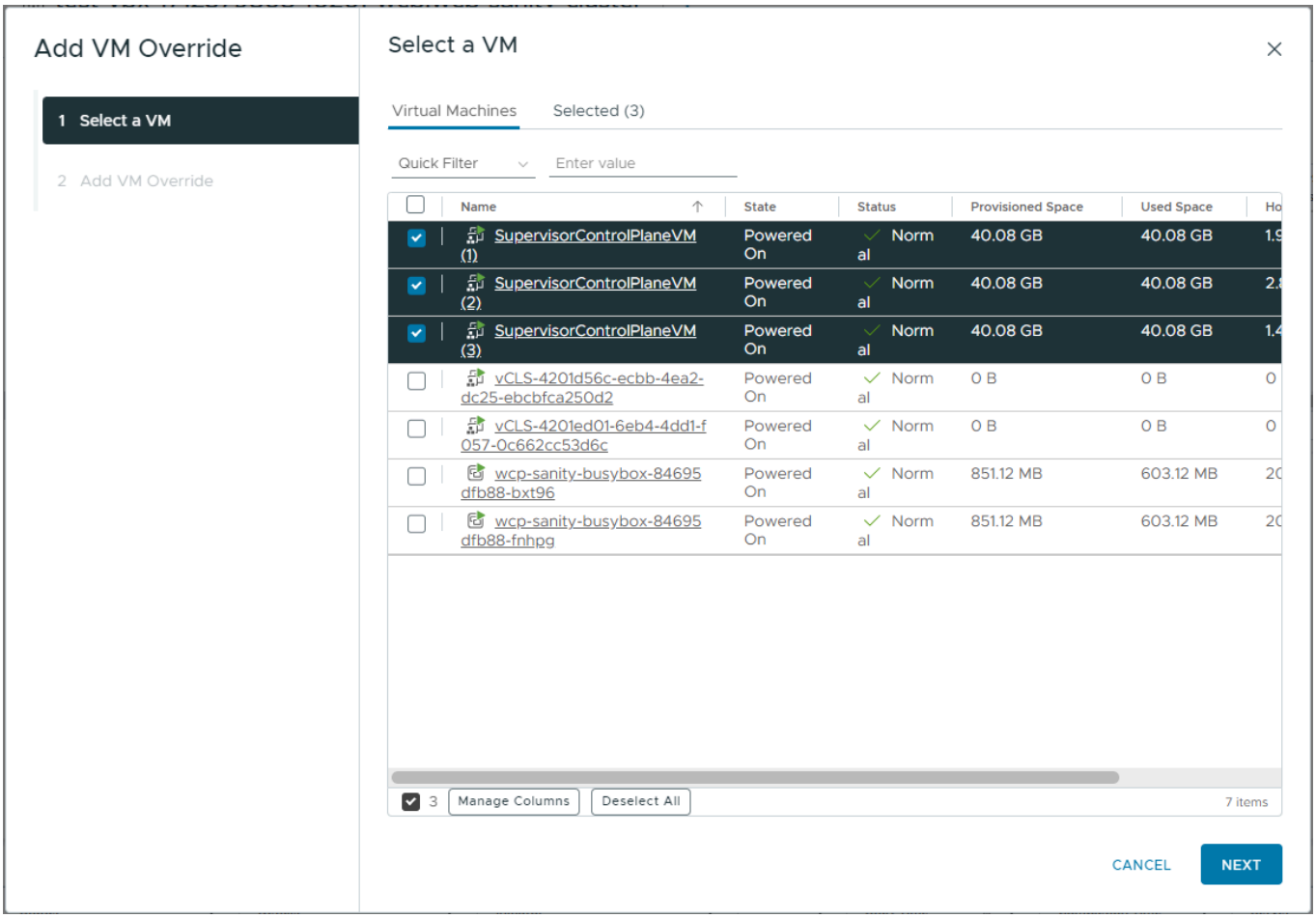
   - `das.isolationaddress0`. Set the value to an IP address on the vSAN network which resides on site 1.
   - `das.isolationaddress1`. Set the value to an IP address on the vSAN network that resides on the site 2.
   - `das.usedefaultisolationaddress`. Set to `false`.

8. Save the settings.
9. Configure HA restart priority for individual VMs.

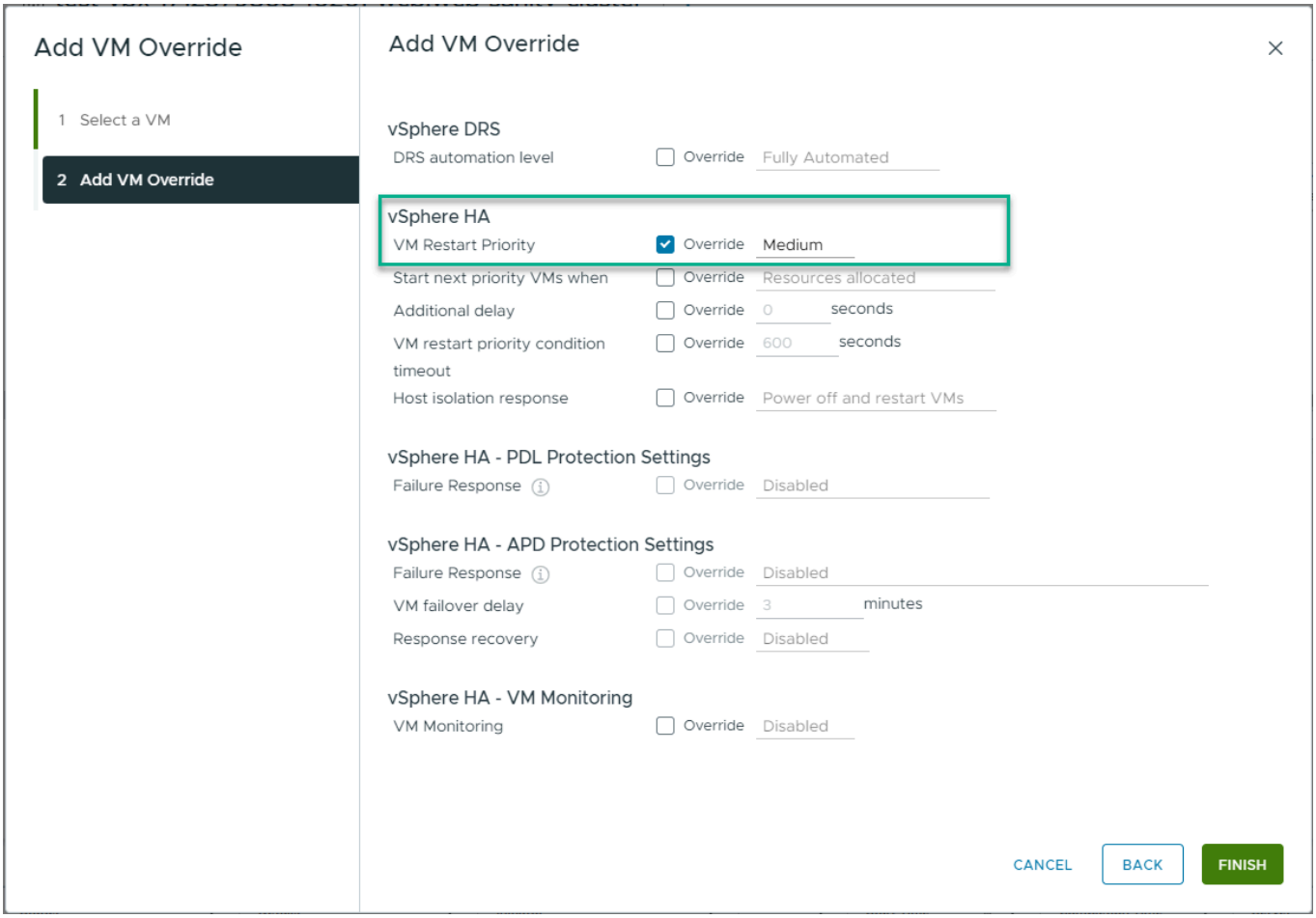
   In case of host or site failure, a possibility exists that the resources are insufficient to restart all the VMs in the vSAN cluster. Therefore, you have to ensure that certain VMs that are of high importance, such as the Supervisor control plane VMs, are restarted first. To do so, configure vSphere HA restart priority to individual VMs, as follows:

   - `Highest` - Supervisor control plan VMs, NSX Edge VMs, and Avi Load Balancer Service engine VMs.
   - `High` - all VKS cluster control plane VMs.
   - `Medium` - all VKS cluster worker VMs.

   a) On the cluster where the Supervisor is enabled, go to **Configure** > **VM Overrides**.
   b) Select VMs from the list, for example all the Supervisor control plane VMs, and click **Next**.



c) Under **vSphere HA**, next to **VM Restart Priority**, select **Override**, and select a priority level, for example **Highest**.



d) Repeat the steps until you set restart priority to all the VMs described from above.

Configure vSphere DRS VM and host groups and rule to affine Supervisor control plane VMs, and VKS cluster control plane and worker nodes.

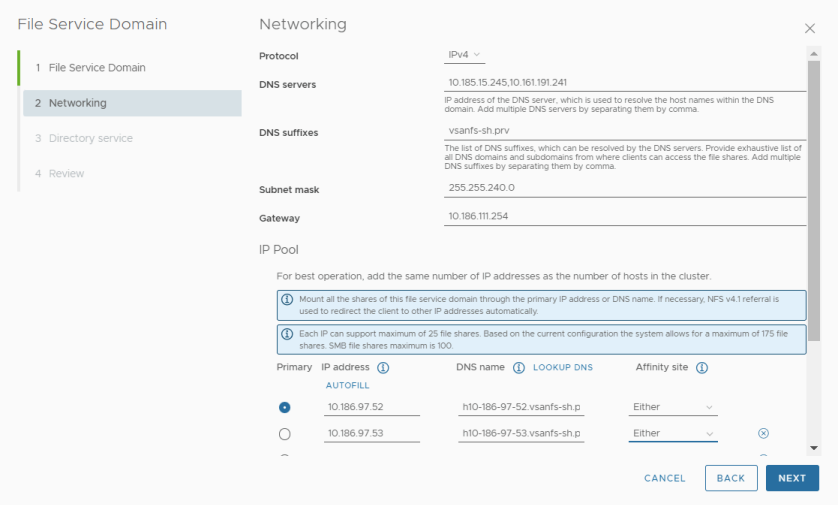
Configuring vSAN File Service for vSphere Supervisor on vSAN Stretched Cluster

To create dynamic ReadWriteMany (RWX) volumes in a VKS cluster, you use vSAN File Service in vSAN environment. When configuring vSAN File Service, use the appropriate affinity site option that applies to vSAN stretched clusters.

For general information about enabling and configuring vSAN File Service and activating file volume support on your Supervisor, see Using vSAN File Service to Create ReadWriteMany Volumes in the *vSphere Services and Standalone Components* documentation.

Affinity Site Option

When configuring File Service Domain for vSAN File Service, make sure to set the **Affinity site** option that is available for vSAN file service on the stretched cluster. Keep the default **Either** value, which indicates that no site affinity rule is applied to the file server.



Troubleshooting vSphere Supervisor

Use the following troubleshooting techniques and best practices for your infrastructure in vSphere Supervisor. This information explains how to approach solving problems by breaking down potential failures in the core Supervisor layers such as storage, networking, identity, life cycle, and others.

How to Approach vSphere Supervisor Issues

Learn what are the practices that might be helpful when you are troubleshooting Supervisor issues.

- Narrow the scope. Determine whether the issue originates from the entire Supervisor, a vSphere Namespace, a Supervisor Service, a VKS cluster, or a workload running in a VKS cluster. By narrowing the scope, you can determine whether to start troubleshooting from the VCF storage, networking, or compute layers, or within the VKS layer. See Troubleshooting VKS Clusters for more information.
- Identify your VCF stack for the Supervisor,.
  - Networking - Possible solutions are NSX VPC, NSX Classic (NSX tier 1), or vSphere Distributed Switch (VDS).
  - Storage - Might be configured with vSAN or other datastores, which are targeted using vSphere storage policies.
  - Load Balancer - The available solutions are Foundation Load Balancer, NSX Load Balancer, Avi, or HA Proxy.
  - Focus the debug on core component layers.
- Get evidence early. If you anticipate opening a Broadcom support case, collect critical diagnostic data immediately.
  - Collect a Supervisor support bundle.
  - Note exactly what changed in the environment, such as recent upgrades, configuration changes, or maintenance windows.
  - Record the exact error messages if possible.

Infrastructure Layer	Functional Scope	Who can debug?
Networking	Connectivity, image pull failures, load balancers, and firewall rules.	Users with `Namespaces.Manage` permission for vCenter, and access to NSX or Avi management interfaces.
Authentication and authorization	Certificates, login, credentials, and user permissions (RBAC).	Users with `Namespaces.Manage` permission for system components. `View` or `Edit` for namespace RBAC or login issues.
Storage	Storage policy mismatches, CNS failures, or capacity issues.	Users with `Namespaces.Manage` permission for vCenter, and storage policy checks.

Infrastructure Layer	Functional Scope	Who can debug?
Lifecycle	Initial enablement or provisioning failures and upgrade pre-check issues.	Users with `Namespaces.Manage` permission. **Note:** If the issue is related to a VKS cluster lifecycle, follow Troublshooring VKS Clusters instead.

Important:

Some of the troubleshooting steps for the Supervisor require SSH access to the Supervisor control plane VMs. Follow the instructions in Broadcom KB 90194.

Supervisor Access Issues

When you cannot reach the Supervisor API, execute kubectl commands, or log in to a Supervisor, the failure typically falls into one of the following categories:

- API Server Down. The API server is unresponsive. Check the Troubleshooting Core Supervisor Issues section.
- Network Reachability. The Supervisor control-plane VIP or load balancer is not responding. Check the The Troubleshooting Networking section.
- Authentication. Issues involving certificates, credentials, or kubeconfig tokens. Check the Troubleshooting Certificate Issues and Troubleshooting Login Issues sections.
- Authorization. The user or service account exists but lacks the explicit RBAC permissions that are needed for the targeted namespace or resource. Check the Troushleshooting Supervisor Login Issues section.

Enablement and Upgrade Issues

Troubleshooting the Supervisor lifecycle involves the initial enablement and configuration and ongoing upgrade and maintenance operations.

- Initial Enablement**.** Focus on ensuring that the fundamental VCF and vSphere infrastructure, including networking, storage, identity, and topology, is configured correctly as outlined in the core VCF and vSphere Supervisor documentation.
- Upgrades. Pre-check failures require a methodical approach that is focused on verifying resource capacity, component health, and compatibility matrixes before the upgrade workflow begins.

When facing issues with initial setup or upgrades, investigate these specific areas:

- Initial enablement or provisioning failures. Start by checking the Supervisor status and consult the documentation for the Supervisor architecture, storage policies, and networking stacks (NSX, VDS, Avi, or HAProxy). Identity configurations such as SSO or external IDPs to ensure that all prerequisite components are correctly deployed and integrated.
- Troubleshooting upgrade precheck failures. Understand pre-check failures and how to remediate them.
- Troubleshooting certificate issues. Check the certificate validity of components in the system.

Networking Issues

Common issues include:

- Access issues to the Supervisor API endpoint.
- Service type `loadbalancer` objects in Kubernetes getting stuck in pending state.
- Certificate errors.
- Communication errors between the Supervisor and vCenter or Supervisor and the NSX controller.

The following sections address troubleshooting networking issues:

- Troubleshoot service type load balancer creation errors. If you are using Avi, follow Troubleshooting Avi Load Balancer.
- Troubleshoot networking errors.
  - When using the NSX Tier 1 or VPC networking type, the health of the Supervisor networking depends on the health of the NSX fabric. Health issues in the NSX tunnel endpoint (TEP) on ESX hosts might cause disruptions to the Supervisor networking. NSX distributed load balancer health issues might cause disruptions to the Kubernetes ClusterIP Service functions in the Supervisor. If you observe node, pod, or ClusterIP Service networking failures in NSX Tier 1 or VPC environments, validate the health status of the NSX fabric in the NSX Management Plane. If health warnings or errors are present, check the NSX Troubleshooting Guide.
- Troubleshoot vSphere Pod or Supervisor Services image pull failures. When Supervisor Services deployment or vSphere Pod initialization fails due to container image retrieval errors, consult the Troubleshooting Failures in Supervisor Service Pod Image Pulls section.

Storage Issues

Key areas to investigate when troubleshooting storage issues in the Supervisor environment include the following:

- Storage policy verification.
  Ensure the storage class that is configured to the Supervisor maps to a valid and available vSphere storage policy. A common problem is a policy mismatch or a storage policy that no longer meets the requirements, such as datastores tagged incorrectly or out of capacity. The vSphere storage policy dictates the capabilities and placement rules for the resulting persistent volumes.
- Namespace configuration check.
  Confirm that the vSphere Namespace has been correctly associated with the necessary storage policies through the vSphere constructs, for example, a vSphere Namespace linked to appropriate storage policies. Incorrect or missing associations prevent persistent volume claims (PVCs) within that namespace from being provisioned correctly.
- CNS (Cloud Native Storage) troubleshooting guide.
  The CNS component is the critical bridge between Kubernetes and vSphere storage. You must check the CNS logs and behavior to understand why a provisioning or de-provisioning operation has failed. Issues often stem from CNS failing to register a newly-created vSphere volume as a first-class disk (FCD) or failing to clean up vSphere objects after a PVC is deleted in Kubernetes.

To pinpoint the exact failure point in the storage provisioning workflow, start by reviewing the relevant vSphere and Kubernetes events and logs, such as the `kube-apiserver` logs on the Supervisor control plane VMs, and the vCenter server logs.

Logs and Support Data

- Make sure that Supervisor logs are forwarded to an external location, such as VCF Operations or an external system. See Forwarding Supervisor Logs to External Monitoring Systems.
- Collect a Supervisor support bundle from the vSphere client, which is helpful in troubleshoot Supervisor issues. You can collect a support bundle even when the Supervisor is in an error or configuring state.

Troubleshooting Core Supervisor Issues

Learn how to troubleshoot core Supervisor issues such as the Supervisor status and API server.

Supervisor Status

The Supervisor activation process comprises multiple steps. They include the deployment and configuration of the Supervisor control plane VMs, configuring ESX hosts as Kubernetes nodes, preparing a virtual IP on the load balancer for the Supervisor API server, and deploying core Supervisor Services.

You can view the Supervisor activation status in the vSphere Client, **Workload Management** > **Supervisor** table view, where the Supervisor activation status is displayed in the **config status** and **host config status** columns. Issues that are encountered in the activation process appear in these columns. Based on the issue description, consult other sections of this troubleshooting guide or Broadcom KBs.

Troubleshooting the Supervisor API Server

If the Supervisor API server is unresponsive or unreachable, remote diagnostics fail. In this scenario, bypass the API and perform low-level node diagnostics directly through control plane SSH.

- Check `kubelet` and node health. When the Supervisor control plane components cannot communicate, check the local node agent status. Run `systemctl status kubelet` to verify if the service is running, and inspect the journal logs for system lifecycle errors, resource starvation, or crashes by using the following command.
  ```
  journalctl -u kubelet -n 100 --no-pager
  ```
- Inspecting the container runtime (`containerd`). If the infrastructure or pod scheduling is stalled, check the container runtime engine to see if underlying pods such as the `kube-apiserver mirror` pod are failing to launch. Run the following command.
  ```
  journalctl -u containerd -n 100
  ```
- Review the vCenter logs. To troubleshoot synchronization mismatches between vSphere constructs and the Supervisor control plane, inspect the Workload Control Plane (WCP) management agent logs on vCenter. These logs help identify why cluster enablement or configuration updates are failing to push downward. Inspect the active sync states by running the following command, noting that you must SSH to vCenter if you are not streaming logs.
  ```
  tail -n 200 /var/log/vmware/wcp/wcpsvc.log
  ```

Troubleshooting Supervisor Certificate Issues

Learn how to address certificate issues within the Supervisor environment.

Certificate Components

- Supervisor API server certificates. On port 443, these certificates serve the VIP certificate, and on port 6443 it serves the Kubernetes certificate.
- Supervisor internal certificates. Kubernetes components use those certificates for internal communication through TLS between themselves and the Supervisor, and the rest of the VCF stack, including vCenter, NSX Manager, Avi controller, and others.
- VKS certificates. Generated by VKS and managed by individual VKS clusters automatically.
- NSX Load Balancer TLS certificate. Used to secure communication transiting or terminating at the NSX load balancer layer. This certificate is only used in network topologies using NSX Tier1 or NSX VPC. The NSX load balancer serves as the entry point provisioning layer 4 virtual servers (VS) to expose Kubernetes services. The associated TLS certificates encrypt communications to these endpoints.

All internally managed certificates are rotated automatically, but if you have an older version and suspect a certificate issue, follow these guidelines.

Check for Expired Certificates

1. First, check the Supervisor VIP certificates.
   a. Navigate to **Supervisor** > **Configure** > **Networking** > **Certificates** to review the expiration dates.
   b. Ensure that no certificates expire within the planned upgrade window.
2. Check internal Kubernetes certificates on the Supervisor.
   a. Download the script hosted on the Supervisor Certificates KB to verify certificates on the Supervisor.
   b. Unzip the script in vCenter, and run these commands to check the Supervisor certificates.

```
# Unzip the script
$ unzip wcp_cert_manager.zip
Archive:  wcp_cert_manager.zip
  inflating: certmgr

# Retrieve the ID of the Supervisor Cluster to check its certificates
$ ./certmgr supervisors

# Use certmgr to list the certificates for the Supervisor Cluster
$ ./certmgr certificates list -c <domain id for the supervisor cluster>
```

3. Check for `cert-manager` managed certificates. In VCF 9 and later, cert-manager manages system and VMware pod certificates separately from the Kubernetes certificates. Therefore, the `certmgr` tool that is used from above does not cover the certificates that are managed by `cert-manager`.

   To debug certificate errors that are reported by the Supervisor system pods, SSH directly to one of the Supervisor control plane VMs.

```
# Check cert-manager pod logs
kubectl get pods -A | grep cert-manager
kubectl logs -n <cert-manager-namespace> <cert-manager-pod>

# Restart cert-manager to force renewal
kubectl rollout restart deploy -n <cert-manager-namespace>

# Confirm certs renewed, then restart affected pods
kubectl rollout restart deploy -n <affected-namespace> <deployment>
```

Check the Certificate Trust Chain

In certain cases, you might identify a repeating error in multiple logs that is related to an x509 certificate that is signed by an unknown authority. This error indicates for a missing certificate authority in the Supervisor trust store that prevents the system from verifying the entire certificate chain.

The CA trust chain must be consistent across five distinct layers. A certificate that is accepted in one layer but missing from another layer is the most common cause of image pull failures. Work through these layers top-down.

1. VCF Automation, if deployed.
2. vCenter.
3. Supervisor control plane.
4. `imageregistry`, connected to the vCenter content libraries.
5. `mgmt-image-proxy` (VCF 9 and later) to pull images from the hosted VCF Automation registry.
6. `containerd` on the Supervisor control plane nodes.
7. `kapp-controller` to pull images for Supervisor Services from hosted or private container registries such as Harbor and others
8. Workloads running on the Supervisor, including load balancers, VKS clusters, and others.
9. VKS cluster nodes if VKS clusters experience image pull issues.
   - `containerd` on VKS cluster nodes.
   - `kapp-controller` on VKS cluster nodes.

Review the following component verification diagnostic matrix to check the trust boundaries.

Layer and Component	Key Checks and Verification Actions
VCF Automation	Use the VCF Operations UI to check the certificates.
vCenter Machine SSL	Check and update the certificate. In the vSphere Client go to**A dministration** > **Certificate Management** > **Machine SSL Certificated** > **Machine SSL Certificate** and click **Replace.** You can also check by using the following command from a command prompt on your local machine: `openssl s_client -connect <vcenter IP>:443`
vCenter trusted roots	Check the trusted roots from the vSphere Client at **Administratio n** > **Certificate Management** > **Trusted Root Certificates**

Layer and Component	Key Checks and Verification Actions
Supervisor content library thumbprint and security policy	View the Supervisor content library, including the content library settings in the vSphere Client : 1. Navigate to **Content Libraries**. 2. Select the Supervisor content library, and select the **Summary** tab. 3. Look for sync errors or the **Security Policy** checkbox state. If you see this error: `cannot authenticate SSL certificate for host wp-content.vmware.com …` Then check the Supervisor content library thumbprint: 1. In the vSphere Client, navigate to **Content Libraries**. 2. Right-click the Supervisor subscribed library and select **Edit Settings**. 3. Click **OK**. A dialog prompts you to accept the new SSL thumbprint 4. Click **Yes** or **Connect anyway**. 5. Trigger a manual sync. A full sync typically is completed within approximately 5 minutes.
Supervisor node `containerd`	Verify that `containerd` is registered properly if you are using a private registry. In the vSphere Client, navigate to the Supervisor, select **Configure**, and select **Container Registries**.

Supervisor to VCF Automation	Check the status of the Supervisor controllers:
	`kubectl get pkgi -n vmware-system-supervisor-services`
	Look for the following errors:
	`tls: failed to verify certificate: x509: certificate is valid for docker-registry.kube-system.svc[...]`
	These errors indicate that the Supervisor cannot verify the TLS connection to the registry that is hosted on VCF Automation.
	Follow the instruction in KB 427460.

Supervisor internal CA	Check if the Supervisor CA certificate is expired, or if it's the correct certificate. The Supervisor root CA is used by all in-cluster components for TLS communications. Use the following command: `kubectl get cm -n kube-system kube-root-ca.crt \ -o jsonpath='{.data.ca\.crt}' | \ openssl x509 -noout -subject -issuer -dates`

Supervisor activation timeout with Avi	Check the Avi controller certificate validation manually by running an open SSL session from a machine that has access to Avi:
	`# Check AVI controller cert expiry`
	`echo | openssl s_client -connect <avi-controller-fqdn>:443 2>/dev/null \ | openssl x509 -noout -dates -ext subjectAltName`
	If the date is pass due the current date, the certificate has expired. See KBs 401804 and 402414.

Layer and Component	Key Checks and Verification Actions
.Supervisor webhooks	Check for errors that are related to Supervisor webhooks. When `cert-manager` auto-renews a CA certificate, it does not automatically refresh child client-server certificates that are signed by the old CA. Admission webhooks failing with `x509 ecdsa verification failure` in the error message can be the symptom. Log in to the Supervisor in to delete stale secrets to force re-creation. See KB 422493.

Supervisor Services	Run the following command:
	`kubectl get pkgi -n vmware-system-supervisor-services → one or more pkgi in ReconcileFailed`
	Check for errors such as:
	`init imagepullbackoff`
	These errors mean that the private registry configured incorrectly. Follow Deploying Supervisor Services from a Private Container Image Registry.
	To check your registry TLS certificate:
	`echo | openssl s_client -showcerts -connect <registry-fqdn>:443 2>/dev/null \ | openssl x509 -noout -subject -issuer -dates -ext subjectAltName`

Kubernetes Services of `type: LoadBalancer` stuck in a Pending state, and no IP assigned	Check the Avi controller certificate by running an open SSL session from a machine that has access to Avi
	`# Check AVI controller cert expiry`
	`echo | openssl s_client -connect <avi-controller-fqdn>:443 2>/dev/null \`
	`| openssl x509 -noout -dates -ext subjectAltName`
	If the date is past today's date, the certificate is expired.

VKS cluster creation with Avi	Check the Avi controller certificate. If the VKS cluster creation fails with: `Timed out waiting for LB service update. This operation is part of the cluster enablement and will be retried` This error means that the Avi controller certificate might not be verified, follow KB 402414.

Troubleshooting Supervisor Login Issues

Learn how to troubleshoot Supervisor login issues. If you are unable to authenticate or log in to the Supervisor, the issue usually stems from permission mismatches, expired credentials, or connectivity gaps between your workstation and the Supervisor API endpoint.

Authentication with the Supervisor depends in the identity provider in use. The Supervisor is integrated with vCenter Single Sign-On by default, and you can also integrate it with an external identity provider. When the Supervisor is registered with VCF Automation, it uses the VCF Automation identity provider.

To learn how to configure login in Supervisor, see:

- Connect to the Supervisor as Single Sign-On User
- Configure Secure Login for vSphere Supervisor
- Connecting to Supervisor and VKS Clusters

To learn how vCenter Single Sign-On maps to Kubernetes permissions, see Understanding how Authentication and Authorization works in vSphere Supervisor.

No matter of the identity provider in use, the troubleshooting steps for resolving Supervisor login issues remain the same.

You connect with the Supervisor by using VCF CLI to create a context, which creates or updates your `kubeconfig` file.

Diagnostics Steps when Login with the Supervisor Fails

Step	Action	Expected Result	Troubleshooting and Failure action
Connectivity	`ping supervisor-vip`	If ICMP is enabled, the VIP should respond on the network. Upon success, this action confirms layer 3 connectivity between your workstation and the Supervisor API endpoint.	Upon failure, ICMP is not enabled or networking route to port 443 is missing. Check your network for the underlying cause.
Login	`vcf context create --server <Supervisor VIP> -u <User>` The syntax must include the full SSO domain, such as `user@vsphere.local`. Check with and without the `--insecure-skip-tls-verify` option to verify the TLS certificate.	Upon success, the command confirms that the user credentials are valid and that the TLS certificates are correct.	Upon failure, check the user credentials and check whether the account that you used is not locked. See KB 326186.
Time Sync	Check that the system time on vCenter matches the workstation from which you are logging in.	The system time on vCenter and the time on the workstation must be identical. Discrepancies between the vCenter and local workstation time settigs might result in generating expired tokens.	Correct the time synchronization settings.
Token Status	Check `~/.kube/config` under your home directory `.kube` config location.	The configuration metadata must be clean.	Delete the old configuration file and re-attempt the login sequence to completely clear cached or invalid tokens.
Supervisor Health	Check the Supervisor in **Workload Management** in the vSphere Client.	The Supervisor must be in **running** state. A `disconnected` or `error` state block all login attempts.	Resolve the Supervisor state in case it's other than `running`.
vCenter Logs	On vCenter check: `/var/log/vmware/wcp/wcpsvc.log`	Log output should contain no errors.	Review specific exceptions or registration errors that are captured in the log output. Collect a support bundle. See Collect the Support Bundle for the Supervisor.
Check `authproxy` logs (VCF and later)	On the Supervisor control plane VMs, check the `wcp-authproxy` pod logs in the `kube-system` namespace. `kubectl -n kube-system logs daemonsets/wcp-authproxy`	For successful login, should see the following log message in the logs: `Incoming request: Method=POST, Path=/wcp/login", "login request completed`	If you experience issues, collect a support bundle. See Collect the Support Bundle for the Supervisor.

Post Supervisor Login Issues

If context creation succeeds but running kubectl commands returns `Forbidden`, `Unauthorized`, or RBAC errors, the root cause is a lack of sufficient user permissions.

To resolve these errors, ensure that your user or group has the required vSphere Namespace roles. Chech the role assignments under **Workload Management** for your target vSphere Namespace. Kubernetes relies on these roles to authorize actions.

You can check the current permission assignments for your user account by using the `can-i` kubectl command. You can check the permissions either for your own user or for another user, such as `<user> @vsphere.local`:

```
$ kubectl auth can-i create services --namespace dev-team-alpha --as user@vsphere.local
```

If answer is `No`, `user@vsphere.local` is authenticated but lacks the necessary `Edit` permission on the `dev-team-alpha` namespace.

Users with `View` permissions can retrieve roles and role bindings to verify access to namespaces:

```
$ kubectl get role,rolebinding --namespace dev-team-alpha
```

Troubleshooting Supervisor Upgrade Pre-check Failures

The upgrade pre-check helps you identify and resolve issues before the Supervisor upgrade begins. If a pre-check fails, it falls into one of five categories, specifically storage, running workloads, Supervisor Services, health, or certificates.

Error Category	What to Check	Resolution
Storage	Datastore free space	Add storage capacity or delete unused files.
VKS clusters	VKS clusters versions	Upgrade the VKS clusters first.
Supervisor Services	Supervisor Services versions	Update Supervisor Services to the required versions.
Health	Etcd and control plane	Resolve unhealthy status in the Supervisor nodes.
Certificates	VIP certificates	Renew any certificates expiring soon.

Resource and Storage Capacity

The upgrade process requires deploying a temporary or new control plane VM. If your environment lacks sufficient resources, the pre-check blocks upgrade.

Check if there is enough free datastore space. If the error indicates insufficient storage space, verify that the physical datastores that are assigned to the Supervisor storage policy have free space of at least 32 GB. Here are some of the possible fixes is the datastores lack sufficient space:

- Reclaim space immediately by deleting unused snapshots or ISOs.
- Add storage capacity.

If you add storage capacity but the vSphere Client fails to reflect the recent additions, restart the `vpxd` service through the vCenter command line can force-refresh the capacity state.

Component Health and Connectivity

The Supervisor relies on a healthy internal state connectivity to external components such as NSX, if this is the networking stack of choice.

- Etcd Health. If the Etcd cluster is reported as unhealthy, the internal database of your Kubernetes environment is out of sync. Follow Troubleshooting ETCD Issues in vSphere Supervisor
- Network compatibility. Ensure that your NSX version is on the compatible list that is provided in the error message. An incompatible version of NSX can prevent the new Supervisor version from communicating with the underlying network.

Version and Service Compatibility

The target Supervisor version to which you want to upgrade might be incompatible with the currently installed VKS or Supervisor Services versions.

- VKS version**.** If a pre-check indicates `VKS clusters Compatibility`, one or more VKS clusters are running a VKS version that is incompatible with the target Supervisor version.
  Resolution: Upgrade VKS to version that is compatible with the target Supervisor version and retry the Supervisor upgrade. See Installing and Updating VKS.
- Supervisor Services versions. Supervisor Services such as Harbor or ArgoCD might be running versions that are not supported with the target Supervisor version. The error message specifies which versions are required.
  Resolution: Upgrade these services from the vSphere Client, see Upgrade a Supervisor Service to a Newer Version.

NSX Load Balancer Certificate Validity

NSX expired or near-expiry certificates might cause the upgrade to hang or networking to fail post-upgrade.

In the vSphere Client, navigate to **Supervisor** > **Configure** > **Certificates** to verify that all entries are valid and not set to expire during the upgrade window.

Troubleshooting Failures in Supervisor Services Pod Image Pulls

Most of the Supervisor Services run as vSphere Pods and issues might accrue in pod image pulls.

Before a vSphere Pod is deployed, its container image must be:

1. Downloaded from its container image repository.
2. Stored on the ESX hostas an image disk.

A failure at either of these steps blocks the vSphere Pod deployment.

Quick Remediation Flow

When a vSphere Pod is stuck in `ErrImagePull` or `ImagePullBackOff`:

1. Run `kubectl get images -n <namespace>` and find the image that is in `Failed` state.
2. Run `kubectl get image <name> -n <namespace> -o yaml` and check the `status.lastError`.
3. Classify the message in `lastError` by using the table in Step 3:
   - `dial tcp: lookup ...`, indicates a DNS failure. Check if the CoreDNS is reachable.
   - `dial tcp <ip>:<port> ...`, indicates a network failure. Check the routes and firewall to the container registry.

Step 0: Get Access to the Service Namespace

Request that your user account is assigned at least the `View` permissions on the Supervisor Service namespace. Creating a context for this namespace fails if you do not have access to that namespace.

Step 1: Check the vSphere Pods Status

1. Identify the failing pod and inspect its status:

   ```
   kubectl get pods -n <namespace>
   ```

2. Map the pod status to a likely cause:

Pod Status	Meaning
`ErrImagePull`	The image download from the repository has failed.
`ImagePullBackOff`	Repeated `ErrImagePull`; kubelet is preventing retries.

Possible root causes for image pull failures are the following:

- A DNS resolution failure, cannot reach the registry hostname.

- The image storage disk on the host has no sufficient free space.
- The registry is unreachable due to network or firewall restriction..
- The registry returned an error, such as *image not found* or *manifest invalid*.

Step 2: Check the Image Object

An `Image` custom resource tracks each pod image. Inspect the `Image` custom resource to find the exact failure reason.

1. List images in the Supervisor Service namespace:
   ```
   kubectl get images -n <namespace>
   ```
2. Find the image that the failing pod is referencing. If its `STATE` is `Failed`, dump the full object:
   ```
   kubectl get image <image-name> -n <namespace> -o yaml
   ```

The following example shows a failed `Image` output. The `status.lastError field` contains the actionable reason.

```
apiVersion: imagecontroller.vmware.com/v1
kind: Image
metadata:
  name: ...
  namespace: ...
  ...
spec:
  imageURI: ...
  ...
status:
  state: Failed
  lastError: 'Failed to resolve on node <esx hostname>
    Reason: Http request failed. Code 400: ErrorType(2) failed to do request: Head
    "https://<registry domain name:port>/<manifest url>":
    dial tcp: lookup <registry domain name>: i/o timeout'
```

Step 3: Classify the Failure from `lastError`

Match the error text against the table from below to identify the problem.

Error pattern in `status.lastError`	Classification	Likely cause
`dial tcp: lookup <host>: i/o timeout`	DNS resolution failure	The image fetcher cannot reach the DNS server (CoreDNS).

`dial tcp <ip>:<port>: i/o timeout`
or
`connection refused`
	Network reachability failure	The registry IP is reachable from the host but the port is closed or the route missing.

Step 4: DNS Resolution Failures

A `dial tcp: lookup ..` error indicates that the image fetcher cannot reach the DNS server.

CoreDNS in the Supervisor is the DNS server that is used by image fetchers. If CoreDNS cannot recognize a DNS name, CoreDNS acts as a DNS relay to the Workload DNS Server that is specified as part of the Supervisor configuration. How image fetchers reach CoreDNS depends on the Supervisor network topology. If you observe a `dial tcp: lookup … i/o timeout` error, check the connectivity to CoreDNS.

Topology	How image fetchers reach CoreDNS
VPC	CoreDNS is exposed through a load-balancer Virtual IP (VIP) that is behind the `kube-dns` ClusterIP Service.
NSX Tier 1	CoreDNS is exposed through the `kube-dns` ClusterIP Service.

Topology	How image fetchers reach CoreDNS
VDS	CoreDNS is reached over the workload network through the `kube-dns` ClusterIP Service.

Note:

Image fetchers and Supervisor control plane VMs use different network paths to reach CoreDNS and name servers.

- Supervisor control plane VMs**.** Name lookups connect to CoreDNS directly through localhost.
- Image fetchers. These routing paths bypass `localhost` and use a separate network route.

Because of this routing difference, a DNS lookup can succeed inside a control plane VM but still fail during an image pull.

Storage Best Practices and Troubleshooting

You can use best practices and troubleshooting techniques for your storage environment in vSphere Supervisor.

Place Control Plane VMs on Separate Datastores

When you use datastores other than vSAN in your cluster with vSphere Supervisor, place the three control plane VMs on separate datastores for availability reasons. Spreading control plane VMs across datastores ensures that a single datastore failure does not affect more than one control plane VM.

Verify that at least three shared datastores are available and attached to all hosts in the Supervisor cluster.

On vSAN, the platform handles VM placement automatically. This procedure applies to VMFS and NFS datastores only.

1. In the vSphere Client, create a datastore cluster.
   a) Navigate to data centers.
   b) Right-click the data center object and select **New Datastore Cluster**.
   c) Name your datastore cluster and make sure **Turn ON Storage DRS** is enabled.
   d) Set the cluster automation level to **No Automation (Manual Mode)**.
   e) Leave the Storage DRS runtime settings as default.
   f) Select the ESX cluster enabled with vSphere Supervisor.
   g) Select all shared datastores to add to the datastore cluster.
   h) Click **Finish**.
2. Define Storage DRS rules for control Plane VMs.
   a) Navigate to the datastore cluster.
   b) Click the **Configure** tab and click **Rules** under **Configuration**.
   c) Click the **Add** icon and enter a name for the rule.
   d) Make sure **Enable rule** is enabled.
   e) Set **Rule type** to **VM anti-affinity**.
   f) Click the **Add** icon and select the three supervisor control plane VMs.
   g) Click **OK** to finish your configuration.
3. Create VM overrides.
   a) Navigate to the datastore cluster.
   b) Click the **Configure** tab and click **VM Overrides** under **Configuration**.
   c) Click the **Add** icon and select the three control plane VMs.
   d) To enable the Storage DRS automation level, select the **Override** check box and set the value to **Fully Automated**.
   e) Click **Finish**.

This task enables Storage DRS for the control plane VMs only and rebalances the VMs to be on different datastores.

Once the Storage vMotion takes place, you can remove the SDRS rules and overrides, disable Storage DRS, and remove the datastore cluster.

Restore a Removed Storage Class

When you use the vSphere Client to remove the storage policy from vCenter or a namespace in the Supervisor, its matching storage class remains in the Kubernetes environment, but cannot be used. Recreating the policy with the same name restores the storage class and resolves the issue.

If you run the `kubectl get sc` command, the output continues to list the storage class as available in the namespace. However, the storage class cannot be used. For example, attempts of a DevOps engineer to use the storage class for a new persistent volume claim fail.

If the storage class is already used by a Kubernetes deployment, the deployment might behave unpredictably.

1. To verify which storage classes exist in the namespace, run the `kubectl describe namespace` *`namespace_name`* command.

   The output for this command does not list the storage class if its matching storage policy is removed.

2. If the storage class is already used by a deployment, restore the storage class.
   a) Use the vSphere Client to create a new storage policy with the same name as the policy you removed.

      For example, if you deleted the *`Gold`* policy, name the new policy *`Gold`*. See Create Storage Policies for Supervisor.
   b) Assign the policy to the namespace.

      See Change Storage Settings on a vSphere Namespace.

      After you assign the policy to the namespace, vSphere Supervisor deletes the old storage class and creates a matching storage class with the same name. Existing PVCs in Pending state resume binding once the storage class is available.

Use External Storage with vSAN Direct

When you use vSAN Direct in the vSphere Supervisor environment, you can use external shared storage to store management internal VMs and other metadata.

When you deploy a homogeneous vSAN Direct cluster, you must create a replicated vSAN datastore on each ESX host in the cluster to store Supervisor control plane VMs and other metadata. The vSAN datastore consumes space, requires an additional I/O controller on each host, and limits hardware configuration on which vSAN Direct can be supported.

Instead of configuring the vSAN datastore, you can use external shared storage for storing management internal VMs and other metadata.

1. If vSAN or vSAN Direct was deployed on the ESX hosts in the cluster, clear the hosts from any configuration.
   a) Remove any disks assigned to vSAN or vSAN Direct. See *Remove Disk Groups or Devices from vSAN* in *Administering VMware vSAN*.
   b) Optional: Use the script to tag disks on the hosts for vSAN Direct. See Use Script to Tag Storage Devices for vSAN Direct in *vSphere Supervisor Services and Standalone Components* documentation.
2. Use VMware Cloud Foundation to create a workload domain with external storage.

   Make sure to select one of the storage options, such as NFS, vVols, or Fibre Channel. Only one of these options can be selected.

   For more information, see Managing VCF Domains in VMware Cloud Foundation.

   This step deploys a workload domain with vCenter and specified ESX hosts. The external storage is mounted on all hosts and added to the default cluster.
3. Enable vSAN.

   Make sure that no disks are claimed for vSAN.

   For more information, see *Enable vSAN on an Existing Cluster* in *Administering VMware vSAN*.

This step creates a zero byte vSAN datastore with vSAN network. No local disks are used for vSAN.

4. Claim local disks on the hosts for vSAN Direct.

   For information, see Create a vSAN Direct Datastore in *vSphere Supervisor Services and Standalone Components* documentation.

   For each device you claim, vSAN Direct creates a separate datastore.

5. Create storage policies for vSAN Direct.

   For information, see Create vSAN Direct Storage Policy in *vSphere Supervisor Services and Standalone Components* documentation.

6. Enable the Supervisor.

   For information, see the *Installing and Configuring vSphere Supervisor* documentation.

Identify and Remove Orphaned Volumes

Orphaned volumes are storage objects that remain on the datastore after their corresponding Kubernetes PersistentVolume or first class disk (FCD) catalog entry is gone. You can use `govc` to investigate discrepancies between the Kubernetes volume inventory and the vSphere layer, and to identify and remove orphaned volumes.

Orphaned volumes consume datastore capacity and indicate a gap in the cleanup sequence. Common causes include:

- A VKS cluster deleted without first removing its PVCs.
- A StatefulSet deleted without explicitly deleting its PVCs.
- Finalizers manually removed from PVCs during troubleshooting, allowing Kubernetes objects to be cleaned up while leaving the backing FCD on the datastore.
- vCenter or FCD operations returning `not found` errors while the virtual disk remains on the datastore.
- A Supervisor restore from backup deleted Kubernetes resources created after the backup was taken, leaving their associated virtual disks on the datastore.

  For cleanup after a Supervisor restore from backup, see Cleaning Up Orphaned Objects After Supervisor Restore from a Backup.

For more information about `govc` and for installation instructions, see https://github.com/vmware/govmomi/blob/main/govc/README.md

1. Identify orphaned volumes.
   a) Run `govc volume.ls` to list all volumes tracked by vCenter and cross-reference the output against the Kubernetes PersistentVolume list.

      ```
      govc volume.ls
      kubectl get pv
      ```

      Any FCD visible in the vCenter inventory that has no corresponding PersistentVolume is a candidate orphan.
   b) To confirm, retrieve the disk metadata and check whether the `cns.k8s.pv[].referredEntity.pvc.name` field matches an active PVC.

      ```
      govc disk.metadata.ls -json=true <fcd-id> | jq '.'
      ```
   c) To verify that no active snapshot is bound to the volume before removing it, list snapshots at the vSphere layer and cross-reference against `kubectl get volumesnapshotcontent`.

      ```
      govc volume.snapshot.ls <volume-id>
      ```

2. Remove orphaned volumes.

   Before removing a volume, confirm it has no active PersistentVolume binding and no running workloads attached. Then run the following command:

   ```
   govc volume.rm <fcd-id>
   ```

Note:

Never delete VMDKs directly from the datastore. Deleting a virtual disk outside the FCD API creates an inconsistency in the vCenter inventory that the system cannot repair automatically. Always remove volumes through the Kubernetes API or vCenter.

Reconcile the First Class Disk Catalog

The vSphere managed virtual disk inventory can become temporarily inconsistent with datastore backing metadata. This issue occurs after transient conditions such as I/O errors or brief datastore unavailability. Or when virtual disks are created or removed using tools other than the vSphere Client or vSphere API.

When the first class disk (FCD) inventory appears incorrect, for example, volumes visible on the datastore do not appear in the vCenter volume list, invoke the FCD reconciliation task. Use the vSphere Web Services API. The reconciliation task reviews the managed virtual disk inventory and attempts to correct any inconsistencies.

After reconciliation completes, verify the FCD and vCenter volume lists are consistent before resuming normal operations.

For details, see the Broadcom KB article 321994.

Monitor Datastore Capacity

Insufficient free space on a datastore causes volume provisioning failures and can affect running workloads that rely on thin-provisioned disks or snapshots. Follow these guidelines to avoid problems.

For information about decommissioning a datastore, see Maintenance Mode for a Datastore Used by Supervisor.

For information about monitoring volumes on a datastore, see Monitoring and Migrating Persistent Volumes in Supervisor Environment.

For VMFS and NFS Datastores

- Maintain 15–20% free space at all times.
- Do not allow utilization to exceed 90%.

  Note:

  Redo-log based snapshots can grow to the full size of the original virtual disk, increasing the risk of a datastore-full condition.
- Enable datastore utilization alarms in vCenter to alert before thresholds are exceeded.
- Where the datastore supports it, enable Reserved Capacity to prevent thin-provisioned disks from exhausting all available space without warning.

For vSAN Datastores

- Enable the following vSAN health alarms:
  Storage space — alerts when vSAN datastore capacity or vSAN managed datastore usage exceeds configured thresholds.
- Enable automatic rebalancing to redistribute data when the cluster becomes unbalanced after adding or removing hosts or capacity devices.
  By default, vSAN rebalancing is disabled. When enabled, vSAN moves components from over-utilized disks when imbalance exceeds the configured threshold. Default is 30%.
  Uneven data distribution across vSAN disk groups can cause provisioning latency for Supervisor control plane VMs and VKS workload PVCs, even when overall datastore capacity appears sufficient. Review rebalancing status as part of routine capacity checks, particularly after adding or removing hosts or changing storage policies.

Troubleshooting Networking

You can troubleshoot networking and load balancer problems that you might encounter when you enable the Supervisor.

Troubleshooting NSX

Refer to this section to troubleshoot NSX issues that you might encounter.

Troubleshooting NSX VPC Networking with the Supervisor

Learn how to troubleshoot NSX VPC networking issues with the Supervisor.

Verify NSX Distributed Load Balancer Health

In a Supervisor that is configured with NSX VPC networking, the Supervisor Kubernetes ClusterIP Services are implemented by using NSXDistributed LoadBalancer. The overall health of the NSX fabric and the health of the Distributed Load Balancer determine the proper function of NSX VPC.

To verify that the NSX Distributed Load Balancer is healthy:

1. Log in to NSX Manager and search for `Supervisor Instance ID`.
   The search results might include multiple objects that are grouped by type.
2. Click the **Load Balancers** tab.
3. Locate the NSX Distributed Load Balancer instance.
   One NSX Distributed Load Balancer instance is available for per Supervisor.
4. Check the status of the load balancer.
   Any status other than `Success` might indicate that the NSX Distributed Load Balancer is unhealthy or misconfigured.

Verify that NSX Load Balancer is Healthy

For a Supervisor using the NSX Load Balancer, the health of the NSX Load Balancer instance serving the CoreDNS Virtual IP (VIP) is critical. If NSX Load Balancer is compromised, CoreDNS becomes unreachable from the VPC attempting the image pulls.

To verify that the NSX Load Balancer is healthy:

1. Log in to NSX Manager and search for `Supervisor Instance ID`.
   The search results might include multiple objects that are grouped by type.
2. Click the **Load Balancers** tab.
3. Locate the NSX Load Balancer instance.
   Multiple load balancer instances might exist for the Supervisor The name of the load balancer serving the CoreDNS VIP starts with the `kube-system*` prefix.
4. Check the status of the load balancer.
   Any status other than `Success` might indicate that the NSX Load Balancer is unhealthy or misconfigured.

Register vCenter with NSX Manager

You might need to re-register vCenter OIDC with NSX Manager in certain situations, for example when the FQDN/PNID of vCenter changes.

1. Connect to the vCenter Appliance through SSH.
2. Run the command `shell`.
3. To get the vCenter thumbprint, run the follwoing command:

   ```
   - openssl s_client -connect vcenterserver-FQDN:443 </dev/null 2>/dev/null | openssl x509 -fingerprint -
   sha256 -noout -in /dev/stdin
   ```

   The thumbprint is displayed. For example,
   `08:77:43:29:E4:D1:6F:29:96:78:5F:BF:D6:45:21:F4:0E:3B:2A:68:05:99:C3:A4:89:8F:F2:0B:EA:3A:BE:9`
4. Copy the SHA256 thumbprint and remove colons.
   `08774329E4D16F2996785FBFD64521F40E3B2A680599C3A4898FF20BEA3ABE9D`

5. To update the OIDC of vCenter, run the following command:

```
curl --location --request POST 'https://<NSX-T_ADDRESS>/api/v1/trust-management/oidc-uris' \
    --header 'Content-Type: application/json' \
    --header 'Authorization: Basic <AUTH_CODE>' \
    --data-raw '{
 "oidc_type": "vcenter",
     "oidc_uri": "https://<VC_ADDRESS>/openidconnect/vsphere.local/.well-known/openid-configuration",
     "thumbprint": "<VC_THUMBPRINT>"
    }'
```

Unable to Change NSX Appliance Password

You might be unable to change the NSX appliance password for the `root`, `admin`, or `audit` users.

Attempts to change the NSX appliance password for the `root`, `admin`, or `audit` users. through the vSphere Client may fail.

During the installation of the NSX Manager, the procedure only accepts one password for all three roles. Attempts to change this password later may fail.

Use the NSX APIs to change the passwords.

For more information, see https://kb.vmware.com/s/article/70691 and the *NSX Administration Guide*.

Troubleshooting Failed Workflows and Unstable NSX Edges

If your workflows fail or the NSX Edges are unstable, you can perform troubleshooting steps.

When you change the distributed portgroup configuration on the vSphere Client, workflows might fail and the NSX Edges might become unstable.

Removal or modification of the distributed port groups for overlay and uplink that were created during the NSX Edge cluster setup of cluster configuration, is not allowed by design.

If you require to change the VLAN or IP Pool configuration of NSX Edges, you must first remove elements of NSX and the vSphere Supervisor configuration from the cluster.

For information about removing elements of NSX, see the *NSX Installation Guide*.

Collect Support Bundles for Troubleshooting NSX

You can collect support bundles on registered cluster and fabric nodes for troubleshooting and download the bundles to your machine or upload them to a file server.

If you choose to download the bundles to your machine, you get a single archive file consisting of a manifest file and support bundles for each node. If you choose to upload the bundles to a file server, the manifest file and the individual bundles are uploaded to the file server separately.

1. From your browser, log in with admin privileges to an NSX Manager.
2. Select **System** > **Support Bundle**.
3. Select the target nodes.
   The available types of nodes are **Management Nodes**, **Edges**, **Hosts**, and **Public Cloud Gateways**.
4. Optional: Specify log age in days to exclude logs that are older than the specified number of days.
5. Optional: Toggle the switch that indicates whether to include or exclude core files and audit logs.

   Note:
   Core files and audit logs might contain sensitive information such as passwords or encryption keys.
6. Optional: Select the check box to upload the bundles to a file server.
7. Click **Start Bundle Collection** to start collecting support bundles.
   The number of log files for each node determines the time taken for collecting support bundles.

8. Monitor the status of the collection process.
   The **Status** tab shows the progress of collecting support bundles.
9. Click **Download** to download the bundle if the option to send the bundle to a file server was not set.

Collect Log Files for NSX

You can collect logs that are in the vSphere Supervisor and NSX components to detect and troubleshoot errors. The log files might be requested by VMware Support.

1. Log in to the vCenter using the vSphere Client .
2. Collect the following log files.

Log File	Description
`/var/log/vmware/wcp/wcpsvc.log`	Contains information related to vSphere Supervisor enablement.
`/var/log/vmware/wcp/nsxd.log`	Contains information related to the NSX components configuration.

3. Log in to NSX Manager.
4. Collect the `/var/log/proton/nsxapi.log` for information on the error that the NSX Manager returns when a specific vSphere Supervisor operation has failed.

Restart the WCP Service If the NSX Management Certificate, Thumbprint, or IP Address Changes

If the NSX Management certificate, thumbprint or IP address changes after you have installed vSphere Supervisor, you must restart the `WCP` service.

Restart the `WCP` Service If the NSX Certificate Changes

Currently, vSphere Supervisor requires that if the NSX certificate or thumbprint, or if the NSX IP address changes, you must restart the `WCP` service for the change to take effect. If either change occurs without a restart of the service, communication between vSphere Supervisor and NSX fails and certain symptoms can arise, such as NCP entering into CrashLoopBackoff stage or Supervisor resources becoming undeployable.

To restart the `WCP` service, use the `vmon-cli`.

1. SSH to the vCenter Server and log in as the root user.
2. Run the command `shell`.
3. Run the command `vmon-cli -h` to view usage syntax and options.
4. Run the command `vmon-cli -l` to view the `wcp` process.
   You see the `wcp` service at the bottom of the list.
5. Run the command `vmon-cli --restart wcp` to restart the `wcp` service.
   You see the message `Completed Restart service request`.
6. Run the command `vmon-cli -s wcp` and verify that the `wcp` service is started.
   For example:

```
root@localhost [ ~ ]# vmon-cli -s wcp
Name: wcp
Starttype: AUTOMATIC
RunState: STARTED
RunAsUser: root
CurrentRunStateDuration(ms): 22158
HealthState: HEALTHY
FailStop: N/A
MainProcessId: 34372
```

Troubleshooting Avi Load Balancer

Refer to the Avi documentation to troubleshoot Avi Load Balancer issues that you might encounter.

Troubleshooting Network Topology Upgrade

When you install vSphere Supervisor version 7.0 Update 1c or upgrade the Supervisor from version 7.0 Update 1 to version 7.0 Update 1c, the networking topology is upgraded from a single tier-1 gateway topology to a topology that has a tier-1 gateway for each namespace within the Supervisor.

You can troubleshoot problems that you might encounter during the upgrade.

Upgrade Precheck Fails Due to Insufficient Edge Load Balancer Capacity

The upgrade precheck fails and the error message indicates insufficient load balancer capacity.

The upgrade precheck process fails with an error message that indicates that the load balancer capacity is less than the capacity required by the Supervisor.

Perform one of the following steps to resolve the problem:

- Force the upgrade by clicking on the **Force Upgrade** button in the error message or use the vCenter Server command line with the `--ingnore-precheck-warnings true` flag.

  Note:
  This solution is recommended only if the Edge cluster can support the existing namespace workloads. Otherwise, these workloads might be skipped during upgrade.
- Delete unused workloads.
- Add additional Edge nodes to the cluster.

Supervisor Workload Namespaces Skipped During Upgrade

During the Supervisor upgrade some namespace workloads are not upgraded.

The Supervisor upgrade succeeds but some namespace workloads are skipped during the upgrade. The Kubernetes resources indicate insufficient resources and the newly created tier-1 gateway is in `ERROR` state.

The load balancer capacity is insufficient to support the workloads.

Perform one of the following steps to resolve the problem:

- Delete unused workloads, restart NCP, and run the upgrade again.
- Add additional Edge nodes to the cluster and trigger a re-allocation to the tier-1 gateway. Restart NCP and run the upgrade again.

Load Balancer Service Skipped During Upgrade

During the Supervisor upgrade some load balancer services are not upgraded.

The Supervisor upgrade suceeds but some Kubernetes load balancer services are skipped during the upgrade.

The number of Kubernetes load balancer type services in the Supervisor workloads and associated VKS cluster exceed the NSX Edge virtual server limit.

Delete unused workloads, restart NCP, and run the upgrade again.

Shut Down and Start Up the vSphere Supervisor Workload Domain

To avoid data loss and maintain the components and workloads of your vSphere Supervisor environment operational, you must follow the specified order when shutting down or starting up the components.

Typically, you perform the shutdown and startup operations after you apply a patch, upgrade, or restore your vSphere Supervisor environment.

The vSphere Supervisor solution, including VKS clusters provisioned by the , is part of the vSphere software-defined data center (SDDC). As such, you must consider the entire vSphere infrastructure stack when performing a shutdown and startup of your vSphere Supervisor environment. Refer to the following validated set of procedures for shutdown and start up of the vSphere SDDC including vSphere Supervisor:

- vSphere SDDC including vSphere Supervisor shutdown procedure. See the *VMware Cloud Foundation Product Documentation* for more information.
- vSphere SDDC including vSphere Supervisor startup procedure. See the *VMware Cloud Foundation Product Documentation* for more information.

Collect the Support Bundle for a Supervisor

Checkout how to collect a support bundle for a Supervisor. You can collect a support bundle even in the Supervisor is in an error or configuring state.

- Your user account must have the **Global** > **Diagnostics** privilege.

1. Log in to your vSphere Supervisor environment using the vSphere Client.
2. Select **Menu** > **Workload Management**.
3. Select the **Supervisors** tab.
4. Select the target Supervisor.
5. Click **Export Logs**.